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WCAG in a Book

Understanding the Web Content Accessibility Guidelines

Compiled by Jamal Mazrui with Claude AI

April 2026

Title Page

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This book contains material derived from the following W3C documents:

- *Web Content Accessibility Guidelines (WCAG) 2.2*, W3C Recommendation, 5 October 2023. Copyright © 2023 World Wide Web Consortium. <https://www.w3.org/TR/WCAG22/>
- *Understanding WCAG 2.2*, W3C Accessibility Guidelines Working Group. Copyright © 2023 World Wide Web Consortium. <https://www.w3.org/WAI/WCAG22/Understanding/>

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Preface

[Editorial — Jamal Mazrui]

This book brings together the complete set of Web Content Accessibility Guidelines (WCAG) 2.2 success criteria along with the official W3C Understanding documents that explain each one. It is intended as a convenient, readable reference for web developers, designers, accessibility specialists, policy makers, and anyone who wants to understand what web accessibility requires and why.

What Is WCAG?

WCAG are international standards published by the World Wide Web Consortium (W3C) that define how to make web content more accessible to people with disabilities. WCAG 2.2 was published as a W3C Recommendation on 5 October 2023 and is also approved as ISO/IEC 40500:2025.

The Four Principles (POUR)

- **Perceivable:** Information and user interface components must be presentable to users in ways they can perceive.
- **Operable:** User interface components and navigation must be operable.
- **Understandable:** Information and the operation of the user interface must be understandable.
- **Robust:** Content must be robust enough that it can be interpreted by a wide variety of user agents, including assistive technologies.

Conformance Levels

- **Level A** is the minimum level. Failure creates serious barriers for users with disabilities.

- **Level AA** is the recommended target for most websites, required by many accessibility laws worldwide.
- **Level AAA** is the highest level, recommended where practical but not required for entire sites.

What This Book Contains

Part One presents all 87 WCAG 2.2 success criteria in official numbered order, each with its normative definition and all content from the corresponding W3C Understanding document.

Part Two introduces the broader W3C accessibility ecosystem: the essential components model; WAI-ARIA; ATAG; UAAG; ACT Rules; the relative luminance formula; WCAG2ICT; web accessibility laws; and conformance.

The Glossary appendix contains the official WCAG 2.2 term definitions from the normative specification. Techniques documents are not included as they are technology-specific and frequently updated; current techniques are at <https://www.w3.org/WAI/WCAG22/Techniques/>

A Note on How This Book Was Made

This project is partly an exercise in self-reference and transparency. The subject of the book is web accessibility guidelines. The book itself was produced by an AI assistant (Claude, by Anthropic) working with Jamal Mazrui, using a Python program that attempts to conform each output format to the accessibility requirements applicable to that format type.

The production process is intentionally contained in a single script, `wcag_in_a_book.py`. Running that script downloads the official W3C WCAG GitHub repository, parses it, and writes six output files: a Markdown source file, a Word document, a cover image, an EPUB3 ebook, a tagged PDF, and a TOML metadata

guide for KDP submission. The self-contained design means a developer can read the entire pipeline in one file. A developer wishing to change how content is extracted, restructured, or formatted can edit the script directly or upload it to an AI assistant and ask for a different approach — for example, a different chapter structure, alternative Part Two topics, or a different output format entirely.

This project is also a practical test of how well AI-assisted Python tooling can produce accessibility-conformant documents at scale. Specifically: the DOCX manuscript follows WCAG guidance for Word documents, including sequential heading hierarchy, language metadata, underlined hyperlinks, and tagged structure for screen readers. The EPUB3 file is generated with an embedded stylesheet, a semantic navigation document (`nav.xhtml`), and language declarations on the root element. The PDF is exported with PDF/UA structure tags using the Microsoft Word COM API when available on Windows, requiring Microsoft Word 2408 or later on Windows.

The project is published at
https://github.com/jamalmazrui/wcag_in_a_book

Limitations are real and worth noting. Automated accessibility checking catches structural issues but cannot evaluate meaning, reading comprehension, or the adequacy of descriptions. The W3C Understanding documents contain placeholder images with empty source attributes, which this program detects and removes. Repeated sub-heading text such as "In brief" and "Resources" — appearing once per criterion — will trigger duplicate-heading warnings in automated checkers designed for web pages, where such repetition would be a navigation problem; in a book with 87 distinct sections, these headings are structurally correct.

A Note on the Text

Sections extracted from W3C source documents carry a *[Source: W3C]* marker and are reproduced with only light editorial cleanup: whitespace normalization and HTML tag escaping. No changes were made to substance or meaning of any W3C text. Editorial sections written by the compiler carry an *[Editorial — Jamal Mazrui]* marker. The Preface, How WCAG 2.2 Is Organized, Part Two introductions, and the About section are editorial. All criteria text, Understanding content, Part Two source descriptions, and the Glossary are W3C source material.

How WCAG 2.2 Is Organized

WCAG 2.2 uses a four-level hierarchy: principles, guidelines, success criteria, and conformance levels. Success criteria are numbered Principle.Guideline.Criterion — for example, 2.4.7 is Principle 2 (Operable), Guideline 4 (Navigable), Criterion 7.

This book mirrors that hierarchy in its heading structure. Each of the four WCAG principles (Perceivable, Operable, Understandable, Robust) appears as a top-level section heading. Each of the 13 guidelines appears as a sub-section under its principle. Each of the 87 success criteria appears as a sub-section under its guideline, followed by sub-sections drawn from the official W3C Understanding document for that criterion: Intent, Benefits, Examples, Resources, and Techniques.

New criteria added in WCAG 2.2 were appended to the end of their guidelines rather than inserted in level order. Conformance level cannot be inferred from position — only the Level A, AA, or AAA label on each criterion indicates its level.

Note on 4.1.1 Parsing: This criterion was rendered obsolete in WCAG 2.2. It is included for completeness and for organizations reporting against WCAG 2.0 or 2.1, where it remains active.

Part One: WCAG 2.2 Success Criteria

[Source: W3C — all criteria and Understanding content in this part is extracted from the W3C WCAG 2.2 GitHub repository and reproduced under the W3C Document License.]

Principle 1: Perceivable

*[Source: W3C WCAG 2.2 —
<https://www.w3.org/TR/WCAG22/#principle1>]*

Information and user interface components must be presentable to users in ways they can perceive.

Guideline 1.1: Text Alternatives

*[Source: W3C WCAG 2.2 —
<https://www.w3.org/TR/WCAG22/#guideline11>]*

Provide text alternatives for any non-text content so that it can be changed into other forms people need, such as large print, braille, speech, symbols or simpler language.

1.1.1 Non-text Content (Level A)

Source: WCAG 2.2 / Understanding 1.1.1 — Copyright © 2023 World Wide Web Consortium

All non-text content that is presented to the user has a text alternative that serves the equivalent purpose, except for the situations listed below.

Controls, Input

If non-text content is a control or accepts user input, then it has a name that describes its purpose. (Refer to Success Criterion 4.1.2 for additional requirements for controls and content that accepts user input.)

Time-Based Media

If non-text content is time-based media, then text alternatives at least provide descriptive identification of the non-text content. (Refer to Guideline 1.2 for additional requirements for media.)

Test

If non-text content is a test or exercise that would be invalid if presented in text , then text alternatives at least provide descriptive identification of the non-text content.

Sensory

If non-text content is primarily intended to create a specific sensory experience , then text alternatives at least provide descriptive identification of the non-text content.

CAPTCHA

If the purpose of non-text content is to confirm that content is being accessed by a person rather than a computer, then text alternatives that identify and describe the purpose of the non-text content are provided, and alternative forms of CAPTCHA using

output modes for different types of sensory perception are provided to accommodate different disabilities.

Decoration, Formatting, Invisible

If non-text content is pure decoration , is used only for visual formatting, or is not presented to users, then it is implemented in a way that it can be ignored by assistive technology .

In brief

Goal

Non-text information is available to more people.

What to do

Create a text alternative for visual and auditory content.

Why it's important

People who can't fully see or hear content can understand it.

Intent of Non-text Content

The intent of this success criterion is to make information conveyed by non-text content accessible through the use of a text alternative. Text alternatives are a primary way for making information accessible because they can be rendered through any sensory modality (for example, visual, auditory or tactile) to match the needs of the user. Providing text alternatives allows the information to be rendered in a variety of ways by a variety of user agents. For example, people who cannot see a picture can have the text alternative read aloud using synthesized speech. People who cannot hear an audio file can have the text alternative displayed so that they can read it. In the future, text alternatives will also allow information to be more easily translated into sign language or into a simpler form of the same language.

Note on CAPTCHA

CAPTCHAs are a controversial topic in the accessibility community. As is described in the

paper Inaccessibility of CAPTCHA , CAPTCHAs intrinsically push the edges of human abilities in an attempt to defeat

automated processes. Every type of CAPTCHA will be unsolvable by users with certain

disabilities. However, they are widely used, and the Web Content Accessibility Guidelines

Working Group believes that if CAPTCHAs were forbidden outright, websites would choose

not to conform to WCAG rather than abandon CAPTCHA. This would create barriers for

a great many more users with disabilities. For this reason the Working Group has chosen

to structure the requirement about CAPTCHA in a way that meets the needs of most people

with disabilities, yet is also considered adoptable by sites.

Requiring two different

forms of CAPTCHA on a given site ensures that most people with disabilities will find

a form they can use.

Because some users with disabilities will still not be able to access sites that meet the minimum requirements, the Working Group provides recommendations for additional steps. Organizations motivated to conform to WCAG should be aware of the importance of this topic and should go as far beyond the minimum

requirements of the guidelines as possible. Additional recommended steps include:

- Providing more than two modalities of CAPTCHAs
- Providing access to a human customer service representative who can bypass CAPTCHA
- Not requiring CAPTCHAs for authorized users

Note on alternatives matching the language of content

Text alternatives and equivalents should match the human language of the original content (normally the default human language of the page). The [5.2 Conformance Requirements](#) section, through the defined terms used there, states that success criteria be met through accessibility-supported ways (5.2.4), where the technology is used “in the human language of the content.” Where an alternative version is used (5.2.1), it is defined as something that “provides all of the same information and functionality in the same human language.”

Additional information

Non-text content can take a number of forms, and this success criterion specifies how each is to be handled.

For non-text content that is not covered by one of the other situations listed below, such as charts, diagrams, audio recordings, pictures, and animations, text alternatives can make the same information available in a form that can be rendered through any modality (for example, visual, auditory or tactile). Short and long text alternatives can be used as needed to convey the information in the non-text content. Note that **prerecorded audio-only** and **prerecorded video-only** files are covered here. **Live-audio-only** and **Live-video-only** files are covered below (see 3rd paragraph following this one).

For non-text content that is a control or accepts user input , such as images used as submit buttons, image maps or complex animations, a name is provided to describe the purpose of the non-text content so that the person at least knows what the non-text content is and why it is there.

****Non-text content that is time-based media**** is made accessible through 1.2: Time-Based Media . However, it is important that users know what it is when they encounter it on a page so they can decide what action if any they want to take with it. A text alternative

that describes the time-based media and/or gives its title is therefore provided.

For Live Audio-only and live video-only content , it can be much more difficult to provide text alternatives that provide equivalent information as live audio-only and live video-only content. For these types of non-text content, text alternatives provide a descriptive label.

Sometimes a test or exercise must be partially or completely presented in non-text format. Audio or visual information is provided that cannot be changed to text because the test or exercise must be conducted using that sense. For example, a hearing test would be invalid if a text alternative were provided. A visual skill development exercise would similarly make no sense in text form. And a spelling test with text alternatives would not be very effective. For these cases, text alternatives should be provided to describe the purpose of the non-text content; of course, the text alternatives would not provide the same information needed to pass the test.

Sometimes content is primarily intended to create a specific sensory experience that words cannot fully capture.

Examples include a symphony performance, works of visual art etc. For such content, text alternatives at least identify the non-text content with a descriptive label and where possible, additional descriptive text. If the reason for including the content in the page is known and can be described it is helpful to include that information.

Sometimes there are non-text exercises that are used to prove you are human. To avoid spam robots and other software from gaining access to a site a device called a CAPTCHA is used. These usually involve visual or auditory tasks that are beyond the current capabilities of web robots. Providing a text alternative to them would however make them operable by Robots, thus defeating their purpose. In this case a text alternative would describe the purpose of the CAPTCHA, and alternate forms using different modalities would be provided to address the needs of people with different disabilities.

Sometimes there is non-text content that really is not meant to be seen or understood by the user. Transparent images used to move text over on a page; an invisible image that is used to track usage statistics; and a swirl in the corner that conveys no information but just fills up a blank space to create an aesthetic effect are all examples of this. Putting alternative text on such items just distracts people using screen readers from the content on the page. Not marking the content in any way, though, leaves users guessing what the non-text content is and what information they may have missed (even though they have not missed anything in reality). This type of non-text content, therefore, is marked or implemented in a way that assistive technologies (AT) will ignore it and not present anything to the user.

Benefits of Non-text Content

- This success criterion helps people who have difficulty perceiving visual content. Assistive technology can read text aloud, present it visually, or convert it to braille.
- Text alternatives may help some people who have difficulty understanding the meaning of photographs, drawings, and other images (e.g., line drawings, graphic designs, paintings, three-dimensional representations), graphs, charts, animations, etc.
- People who are deaf, are hard of hearing, or who are having trouble understanding audio information for any reason can read the text presentation. Research is ongoing regarding automatic translation of text into sign language.
- People who are deaf-blind can read the text in braille.
- Additionally, text alternatives support the ability to search for non-text content and to repurpose content in a variety of ways.

Examples of Non-text Content

A data chart

A bar chart compares how many widgets were sold in June, July, and August. The short label says, "Figure one - Sales in June, July and August." The longer description identifies the type of chart, provides a high-level summary of the data, trends and implications comparable to those available from the chart. Where possible and practical, the actual data is provided in a table.

An audio recording of a speech

The link to an audio clip says, "Chairman's speech to the assembly." A link to a text transcript is provided immediately after the link to the audio clip.

An animation that illustrates how a car engine works

An animation shows how a car engine works. There is no audio and the animation is part of a tutorial that describes how an engine works. Since the text of the tutorial already provides a full explanation, the image is an alternative for text and the text alternative includes only a brief description of the animation and refers to the tutorial text for more information.

A traffic web camera

A website allows users to select from a variety of web cameras positioned throughout a major city. After a camera is selected, the image updates every two minutes. A short text alternative identifies the web camera as "traffic web camera." The site also provides a table of travel times for each of the routes covered by the web cameras. The table is also updated every two minutes.

A photograph of an historic event in a news story

A photograph of two world leaders shaking hands accompanies a news story about an international summit meeting. The text alternative says, "President X of Country X shakes hands with Prime Minister Y of country Y."

A photograph of a historic event in content discussing diplomatic relationships

The same image is used in a different context intended to explain nuances in diplomatic encounters. The image of the president shaking hands with the prime minister appears on a website discussing intricate diplomatic relationships. The first text alternative reads, "President X of country X shakes hands with Prime Minister Y of country Y on January 2, 2009." An additional text alternative describes the room where the leaders are standing as well as the expressions on the leaders' faces, and identifies the other people in the room. The additional description might be included on the same page as the photograph or in a separate file

associated with the image through a link or other standard programmatic mechanism.

An audio recording of a press conference

A web page includes a link to an audio recording of a press conference. The link text identifies the audio recording. The page also links to a text transcript of the press conference. The transcript includes a verbatim record of everything the speakers say. It identifies who is speaking as well as noting other significant sounds that are part of the recording, such as applause, laughter, questions from the audience, and so on.

An e-learning application

An e-learning application uses sound effects to indicate whether or not the answers are correct. The chime sound indicates that the answer is correct and the beep sound indicates that the answer is incorrect. A text description is also included so that people who can't hear or understand the sound understand whether the answer is correct or incorrect.

A linked thumbnail image

A thumbnail image of the front page of a newspaper links to the home page of the "Smallville Times". The text alternative says "Smallville Times".

The same image used on different sites

Different alternatives for an image of the world: An image of the world that is used on a travel site as a link to the International Travel section has the text alternative "International Travel". The same image is used as a link on a university website with the text alternative "International Campuses".

An image map

An image of a building floor plan is interactive, allowing the user to select a particular room and navigate to a page containing information about that room. The short text alternative describes the image and its interactive purpose: "Building floor plan. Select a room for more information."

Resources for Non-text Content

- [Excerpts from the NBA Tape Recording Manual, Third Edition](#)
- [Inaccessibility of CAPTCHA](#)
- [All That Malarkey: Accessible Alternatives](#)
- [456 Berea Street: The Alt and Title Attributes](#)
- [The Alt and Accessibility](#)
- [Better Connected, Better Results: Alt Text](#)

Guideline 1.2: Time-based Media

[Source: W3C WCAG 2.2 —

<https://www.w3.org/TR/WCAG22/#guideline12>]

Provide alternatives for time-based media.

1.2.1 Audio-only and Video-only (Prerecorded) (Level A)

Source: [WCAG 2.2](#) / [Understanding 1.2.1](#) — Copyright © 2023 World Wide Web Consortium

For prerecorded audio-only and prerecorded video-only media, the following are true, except when the audio or video is a media alternative for text and is clearly labeled as such:

Prerecorded Audio-only

An alternative for time-based media is provided that presents equivalent information for prerecorded audio-only content.

Prerecorded Video-only

Either an alternative for time-based media or an audio track is provided that presents equivalent information for prerecorded video-only content.

In brief

Goal

Audio and video-only content can be understood by more people.

What to do

Provide an alternative when content is perceivable with only one sense.

Why it's important

People who can't fully see or hear content can understand it.

Intent of Audio-only and Video-only (Prerecorded)

The intent of this success criterion is to make information conveyed by prerecorded audio-only and prerecorded video-only content available to all users. Alternatives for time-based media that are text based make information accessible because text can be rendered through any sensory modality (for example, visual,

auditory or tactile) to match the needs of the user. In the future, text could also be translated into symbols, sign language or simpler forms of the language (future).

An example of pre-recorded video with no audio information or user interaction is a silent movie. The purpose of the transcript is to provide an equivalent to what is presented visually. For prerecorded video content, authors have the option to provide an audio track. The purpose of the audio alternative is to be an equivalent to the video. This makes it possible for users with and without vision impairment to review content simultaneously. The approach can also make it easier for those with cognitive, language and learning disabilities to understand the content because it would provide parallel presentation.

A text equivalent is not required for audio that is provided as an equivalent for video with no audio information. For example, it is not required to caption audio description that is provided as an alternative to a silent movie.

See also [1.2.9 Audio-only \(Live\)](#)

Benefits of Audio-only and Video-only (Prerecorded)

- This success criterion helps people who have difficulty perceiving visual content. Assistive technology can read text alternatives aloud, present them visually, or convert them to braille.
- Alternatives for timed-based media that are text based may help some people who have difficulty understanding the meaning of prerecorded video content.
- People who are deaf, are hard of hearing, or who are having trouble understanding audio information for any reason can read the text presentation. Research is ongoing regarding automatic translation of text into sign language.

- People who are deaf-blind can read the text in braille.
- Additionally, text supports the ability to search for non-text content and to repurpose content in a variety of ways.

Examples of Audio-only and Video-only (Prerecorded)

An audio recording of a speech

The link to an audio clip says, "Chairman's speech to the assembly." A link to a text transcript is provided immediately after the link to the audio clip.

An audio recording of a press conference

A web page includes a link to an audio recording of a press conference that identifies the audio recording. The page also links to a text transcript of the press conference. The transcript includes a verbatim record of everything the speakers say. It identifies who is speaking as well as noting other significant sounds that are part of the recording, such as applause, laughter, questions from the audience, and so on.

An animation that illustrates how a car engine works

An animation shows how a car engine works. There is no audio and the animation is part of a tutorial that describes how an engine works. Since the text of the tutorial already provides a full explanation, the media is an alternative for text and the text alternative includes only a brief description of the animation and refers to the tutorial text for more information.

A video-only file with an audio track

A silent movie includes an audio track which includes a description of the action in the video.

Resources for Audio-only and Video-only (Prerecorded)

- [Making Audio and Video Media Accessible, W3C Web Accessibility Initiative \(WAI\)](#)
- [Overcoming the challenge of podcast transcription](#)
- [uiAccess list of transcription services](#)
- [Transcripts on the Web: Getting people to your podcasts and videos](#)

1.2.2 Captions (Prerecorded) (Level A)

Source: WCAG 2.2 / Understanding 1.2.2 — Copyright © 2023 World Wide Web Consortium

Captions are provided for all prerecorded audio content in synchronized media , except when the media is a media alternative for text and is clearly labeled as such.

In brief

Goal

Videos can be played with captions.

What to do

Provide synchronized text for audio content in existing videos.

Why it's important

People who are deaf or hard of hearing can understand audio in videos.

Intent of Captions (Prerecorded)

The intent of this success criterion is to enable people who are deaf or hard of hearing to watch synchronized media presentations. Captions provide the part of the content available via the audio track. Captions not only include dialogue, but identify who is speaking and include non-speech information conveyed through sound, including meaningful sound effects.

It is acknowledged that at the present time there may be difficulty in creating captions for time-sensitive material and this may result in the author being faced with the choice of delaying the information until captions are available, or publishing time-sensitive content that is inaccessible to the deaf, at least for the interval until captions are available. Over time, the tools for

captioning as well as building the captioning into the delivery process can shorten or eliminate such delays.

Captions are not needed when the synchronized media is, itself, an alternate presentation of information that is also presented via text on the web page. For example, if information on a page is accompanied by a synchronized media presentation that presents no more information than is already presented in text, but is easier for people with cognitive, language, or learning disabilities to understand, then it would not need to be captioned since the information is already presented on the page in text or in text alternatives (e.g., for images).

See also [1.2.4 Captions \(Live\)](#) .

Benefits of Captions (Prerecorded)

- People who are deaf or have a hearing loss can access the auditory information in the synchronized media content through captions.

Examples of Captions (Prerecorded)

- A captioned tutorial A video clip shows how to tie a knot. The captions read, "(music) Using rope to tie knots was an important skill for the likes of sailors, soldiers and woodsmen.." From Sample Transcript Formatting by Whit Anderson.
- A complex legal document contains synchronized media clips for different paragraphs that show a person speaking the contents of the paragraph. Each clip is associated with its corresponding paragraph. No captions are provided for the synchronized media.
- An instruction manual containing a description of a part and its necessary orientation is accompanied by a synchronized

media clip showing the part in its correct orientation. No captions are provided for the synchronized media clip.

- An orchestra provides captions for videos of performances. In addition to capturing dialog and lyrics verbatim, captions identify non-vocal music by title, movement, composer, and any information that will help the user comprehend the nature of the audio. For instance captions read, "[Orchestral Suite No. 3.2 in D major, BWV 1068, Air] [Johann Sebastian Bach, Composer] ♪ Calm melody with a slow tempo ♪" Style guides for captions may differ among different languages.

Style guides for captions may differ among different languages.

Guides to Captioning

- [Captions/Subtitles](#) , in [Making Audio and Video Media Accessible](#) , W3C Web Accessibility Initiative (WAI)
- [BBC: Subtitle Guidelines](#)
- [Captioning Key: Guidelines and Preferred Techniques](#)
- [Best Practices in Online Captioning](#)

SMIL Resources

- [Synchronized Multimedia Integration Language \(SMIL\) 3.0](#)
- [Accessibility Features of SMIL](#)

Other Captioning Resources

- [National Center for Accessible Media](#)
- [WebAIM: Captioning Resource List](#)

1.2.3 Audio Description or Media Alternative (Prerecorded) (Level A)

Source: WCAG 2.2 / Understanding 1.2.3 — Copyright © 2023 World Wide Web Consortium

An alternative for time-based media or audio description of the prerecorded video content is provided for synchronized media , except when the media is a media alternative for text and is clearly labeled as such.

In brief

Goal

Prerecorded videos can be understood by more people.

What to do

Provide a description of the visual content in videos.

Why it's important

People who are blind or who cannot understand the visual content can have it described.

Intent of Audio Description or Media Alternative (Prerecorded)

The intent of this success criterion is to provide people who are blind or visually

impaired access to the visual information in a synchronized media presentation in the same human language as the video or page on which it appears. This

success criterion describes two approaches, either of which can be used.

One approach is to provide audio description of the video content. The audio description augments the audio portion of the presentation with the information needed when the video portion

is not available. During existing pauses in dialogue, audio description provides information about actions, characters, scene changes, and on-screen text that are important and are not described or spoken in the main sound track.

The second approach involves providing all of the information in the synchronized media (both visual and auditory) in text form. An alternative for time-based media provides a running description of all that is going on in the synchronized media content. The alternative for time-based media reads something like a screenplay or book. Unlike audio description, the description of the video portion is not constrained to just the pauses in the existing dialogue. Full descriptions are provided of all visual information, including visual context, actions and expressions of actors, and any other visual material. In addition, non-speech sounds (laughter, off-screen voices, etc.) are described, and transcripts of all dialogue are included. The sequence of description and dialogue transcripts are the same as the sequence in the synchronized media itself. As a result, the alternative for time-based media can provide a much more complete representation of the synchronized media content than audio description alone.

If there is any interaction as part of the synchronized media presentation (e.g., "press now to answer the question") then the alternative for time-based media would provide hyperlinks or whatever is needed to provide the same functionality.

For 1.2.3, 1.2.5, and 1.2.7, if all of the important information in the video track is already conveyed in the audio track, no additional audio description is necessary.

1.2.3 Audio Description or Media Alternative (Prerecorded), 1.2.5 Audio Description (Prerecorded), and 1.2.8 Media Alternative (Prerecorded) overlap somewhat with each other. This is to give

the author some choice at the minimum conformance level, and to provide additional requirements at higher levels. At Level A in Success Criterion 1.2.3, authors do have the choice of providing either an audio description or a full text alternative. If they wish to conform at Level AA, under Success Criterion 1.2.5 authors must provide an audio description - a requirement already met if they chose that alternative for 1.2.3, otherwise an additional requirement. At Level AAA under Success Criterion 1.2.8 they must provide an extended text description. This is an additional requirement if both 1.2.3 and 1.2.5 were met by providing an audio description only. If 1.2.3 was met, however, by providing a text description, and the 1.2.5 requirement for an audio description was met, then 1.2.8 does not add new requirements.

See also [1.2.5 Audio Description \(Prerecorded\)](#) , [1.2.7 Extended Audio Description \(Prerecorded\)](#) and [1.2.8 Media Alternative \(Prerecorded\)](#) .

Benefits of Audio Description or Media Alternative (Prerecorded)

- This success criterion may help some people who have difficulty watching video or other synchronized media content, including people who have difficulty perceiving or understanding moving images.

Examples of Audio Description or Media Alternative (Prerecorded)

A movie with audio description

Describer: A title, "Teaching Evolution Case Studies. Bonnie Chen." A teacher shows photographs of birds with long, thin beaks. Bonnie Chen: "These photos were all taken at the Everglades." Describer: The teacher hands each student two flat, thin wooden sticks. Bonnie Chen: "Today you will pretend to be a species of wading bird that has a beak like this." Describer: The teacher holds two of the sticks to her mouth making the shape of a

beak. Transcript of audio based on the first few minutes of "Teaching Evolution Case Studies, Bonnie Chen " (copyright WGBH and Clear Blue Sky Productions, Inc.)

An alternative for time-based media for a training video

A company purchases a Training video for use by its employees and puts it on the companies intranet. The video involves explaining use of a new technology and has a person talking and showing things at the same time. Since there is no place to insert audio description of the visual demonstrations during gaps in dialogue, the company provides an alternative for time-based media that all employees, including those who cannot see the demonstrations, can use to better understand what is being presented.

Resources for Audio Description or Media Alternative (Prerecorded)

- [Making Audio and Video Media Accessible, W3C Web Accessibility Initiative \(WAI\)](#)
- [GBH - Integrate audio descriptions into multimedia presentations using SMIL](#)
- [Standard Techniques in Audio Description](#)
- [Synchronized Multimedia Integration Language \(SMIL\) 1.0](#)
- [Synchronized Multimedia Integration Language \(SMIL 2.0\)](#)
- [Accessibility Features of SMIL](#)

1.2.4 Captions (Live) (Level AA)

Source: [WCAG 2.2](#) / [Understanding 1.2.4](#) — Copyright © 2023 World Wide Web Consortium

Captions are provided for all live audio content in synchronized media .

In brief

Goal

Live videos have captions.

What to do

Provide synchronized text for audio content in real-time videos.

Why it's important

People who are deaf or hard of hearing can understand audio in real-time video content.

Intent of Captions (Live)

The intent of this success criterion is to enable people who are deaf or hard of hearing to watch *real-time* presentations. Captions provide the part of the content available via the audio track. Captions not only include dialogue, but also identify who is speaking and notate sound effects and other significant audio.

This success criterion was intended to apply to broadcast of synchronized media and is not intended to require that two-way multimedia calls between two or more individuals through web apps must be captioned regardless of the needs of users.

Responsibility for providing captions would fall to the content providers (the callers) or the “host” caller, and not the application.

Benefits of Captions (Live)

- People who are deaf or have a hearing loss can access the auditory information in the synchronized media content through captions.

Examples of Captions (Live)

A web cast

A news organization provides a live, captioned web cast.

A music web cast

An orchestra provides Communication Access Realtime Translation (CART) captioning of each real-time web performance. The CART service captures lyrics and dialog as well as identifies non-vocal music by title, movement, composer, and any information that will help the user comprehend the nature of the audio.

Resources for Captions (Live)

- See [1.2.2 Captions \(Prerecorded\)](#) .

1.2.5 Audio Description (Prerecorded) (Level AA)

Source: [WCAG 2.2](#) / [Understanding 1.2.5](#) — Copyright © 2023 World Wide Web Consortium

Audio description is provided for all prerecorded video content in synchronized media .

In brief

Goal

Videos can be played with audio descriptions.

What to do

Provide a synchronized spoken description of the visual content in videos.

Why it's important

People who cannot see or understand the visual content can hear about it while playing videos.

Intent of Audio Description (Prerecorded)

The intent of this success criterion is to provide people who are blind or visually

impaired access to the visual information in a synchronized media presentation in the same human language as the video or page on which it appears. The

audio description augments the audio portion of the presentation with the information

needed when the video portion is not available. During existing pauses in dialogue,

audio description provides information about actions, characters, scene changes, and

on-screen text that are important and are not described or spoken in the main sound

track.

For 1.2.3, 1.2.5, and 1.2.7, if all of the important information in the video track is already conveyed in the audio track, no additional audio description is necessary.

1.2.3 Audio Description or Media Alternative (Prerecorded), 1.2.5 Audio Description (Prerecorded), and 1.2.8 Media Alternative (Prerecorded) overlap somewhat with each other. This is to give the author some choice at the minimum conformance level, and to provide additional requirements at higher levels. At Level A in Success Criterion 1.2.3, authors do have the choice of providing either an audio description or a full text alternative. If they wish to conform at Level AA, under Success Criterion 1.2.5 authors must provide an audio description - a requirement already met if they chose that alternative for 1.2.3, otherwise an additional requirement. At Level AAA under Success Criterion 1.2.8 they must provide an extended text description. This is an additional requirement if both 1.2.3 and 1.2.5 were met by providing an audio description only. If 1.2.3 was met, however, by providing a text description, and the 1.2.5 requirement for an audio description was met, then 1.2.8 does not add new requirements.

Benefits of Audio Description (Prerecorded)

- People who are blind or have low vision as well as those with cognitive limitations who have difficulty interpreting visually what is happening benefit from audio description of visual information.

Examples of Audio Description (Prerecorded)

A movie with audio description

Describer: A title, "Teaching Evolution Case Studies. Bonnie Chen." A teacher shows photographs of birds with long, thin beaks. Bonnie Chen: "These photos were all taken at the Everglades." Describer: The teacher hands each student two flat, thin wooden sticks. Bonnie Chen: "Today you will pretend to be a species of wading bird that has a beak like this." Describer: The teacher holds two of the sticks to her mouth making the shape of a beak. Transcript of audio based on the first few minutes of "Teaching Evolution Case Studies, Bonnie Chen " (copyright WGBH and Clear Blue Sky Productions, Inc.)

Resources for Audio Description (Prerecorded)

- Description of Visual Information , in Making Audio and Video Media Accessible , W3C Web Accessibility Initiative (WAI)
- GBH - Integrate audio descriptions into multimedia presentations using SMIL
- Standard Techniques in Audio Description
- Synchronized Multimedia Integration Language (SMIL) 1.0
- Synchronized Multimedia Integration Language (SMIL 2.0)
- Accessibility Features of SMIL

1.2.6 Sign Language (Prerecorded) (Level AAA)

Source: WCAG 2.2 / Understanding 1.2.6 — Copyright © 2023 World Wide Web Consortium

Sign language interpretation is provided for all prerecorded audio content in synchronized media .

In brief

Goal

Videos can be accompanied by sign language.

What to do

Provide sign language interpretation for audio content in existing videos.

Why it's important

People who are deaf or hard of hearing have more ways to understand multimedia content.

Intent of Sign Language (Prerecorded)

The intent of this success criterion is to enable people who are deaf or hard of hearing and who are fluent in a sign language to understand the content of the audio track of synchronized media presentations. Written text, such as that found in captions, is often a second language. Because sign language provides the ability to provide intonation, emotion and other audio information that is reflected in sign language interpretation, but not in captions, sign language interpretation provides richer and more equivalent access to synchronized media. People who communicate extensively in sign language are also faster in sign language and synchronized media is a time-based presentation.

Benefits of Sign Language (Prerecorded)

- People whose human language is a sign language sometimes have limited reading ability. These individuals may not be able to read and comprehend the captions and thus require a sign language interpretation to gain access to the synchronized media content.

Examples of Sign Language (Prerecorded)

- Example 1. A corporation is making an important announcement to all of its employees. The announcement will be held in the main headquarters and later streamed to the web. A sign language interpreter is provided at the announcement location for the employees that are present in the meeting room. For the web version of the announcement, the sign language interpreter is shown/superimposed in the corner of the display.
- Example 2. A university is providing an online version of a particular lecture by creating a synchronized media presentation of the professor delivering the lecture. The presentation includes video of the professor speaking and demonstrating a science experiment. A sign language interpretation of the lecture is created after the lecture and presented on the web with the synchronized media version.

Resources for Sign Language (Prerecorded)

- Sign Languages , in Making Audio and Video Media Accessible , W3C Web Accessibility Initiative (WAI)
- Synchronized Multimedia Integration Language (SMIL) 1.0
- Synchronized Multimedia Integration Language (SMIL 2.0)
- Accessibility Features of SMIL
- GBH - Integrate audio descriptions into multimedia presentations using SMIL

- National Institute on Deafness and other Communication Disorders : Information on American Sign Language
- Techniques for filming sign language interpreters
- Royal National Institute for Deaf People (RNID)

1.2.7 Extended Audio Description (Prerecorded) (Level AAA)

Source: [WCAG 2.2](#) / [Understanding 1.2.7](#) — Copyright © 2023 World Wide Web Consortium

Where pauses in foreground audio are insufficient to allow audio descriptions to convey the sense of the video, extended audio description is provided for all prerecorded video content in synchronized media .

In brief

Goal

Videos can be played with more detailed audio descriptions.

What to do

Provide extended spoken descriptions of the visual content in videos.

Why it's important

People who are blind or who cannot understand the visual content can have it described to them while playing videos.

Intent of Extended Audio Description (Prerecorded)

The intent of this success criterion is to provide people who are blind or visually impaired access to a synchronized media presentation beyond that which can be provided by standard audio description. This is done by periodically freezing the synchronized media presentation and playing additional audio description. The synchronized media presentation is then resumed.

Because it disrupts viewing for those who do not need the additional description, techniques that allow you to turn the feature on and off are often provided. Alternately, versions with and without the additional description can be provided.

Benefits of Extended Audio Description (Prerecorded)

- People who are blind, people with low vision who cannot see the screen, as well as those with cognitive limitations who have difficulty interpreting visually what is happening, often use audio description of the visual information. However, if there is too much dialogue the audio description is insufficient. Extended audio description can provide the additional information needed to understand the video.

Examples of Extended Audio Description (Prerecorded)

- Example 1. Video of a lecture. A physics professor is giving a lecture. The professor makes freehand sketches on the whiteboard, speaking rapidly while drawing. As soon as discussion on one problem is finished, the professor erases the drawing and makes another sketch while continuing to speak and gesture. The video is paused between problems, and extended audio description of the professor's drawings and gestures is provided; the video is then resumed.

Resources for Extended Audio Description (Prerecorded)

- [Synchronized Multimedia Integration Language \(SMIL\) 1.0](#)
- [Synchronized Multimedia Integration Language \(SMIL\) 2.0](#)
- [Accessibility Features of SMIL](#)
- [GBH - Integrate audio descriptions into multimedia presentations using SMIL](#)
- [Standard Techniques in Audio Description](#)

1.2.8 Media Alternative (Prerecorded) (Level AAA)

Source: [WCAG 2.2](#) / [Understanding 1.2.8](#) — Copyright © 2023 World Wide Web Consortium

An alternative for time-based media is provided for all prerecorded synchronized media and for all prerecorded video-only media.

In brief

Goal

Prerecorded videos can be understood by more people.

What to do

Provide a text equivalent for all content in videos.

Why it's important

More people, including those who are deaf-blind, can better understand the content at their own pace.

Intent of Media Alternative (Prerecorded)

The intent of this success criterion is to make audio visual material available to

individuals whose vision is too poor to reliably read captions and whose hearing is too poor to reliably hear dialogue and audio description . This is done by providing an alternative for time-based media in the same human language as the video or page on which it appears.

This approach involves providing all of the information in the synchronized media (both visual and auditory) in text form. An alternative for time-based media provides a running description of all that is going on in the synchronized media content. The alternative for time-based media reads something like a book. Unlike audio description, the description of the video portion is

not constrained to just the pauses in the existing dialogue. Full descriptions are provided of all visual information, including visual context, actions and expressions of actors, and any other visual material. In addition, non-speech sounds (laughter, off-screen voices, etc.) are described, and transcripts of all dialogue are included. The sequence of descriptions and dialogue transcripts is the same as the sequence in the synchronized media itself. As a result, the alternative for time-based media can provide a much more complete representation of the synchronized media content than audio description alone.

If there is any interaction as part of the synchronized media presentation (e.g., "press now to answer the question") then the alternative for time-based media would provide hyperlinks or whatever is needed to provide parallel functionality.

Individuals whose vision is too poor to reliably read captions and whose hearing is too poor to reliably hear dialogue can access the alternative for time-based media by using a refreshable braille display.

1.2.3 Audio Description or Media Alternative (Prerecorded), 1.2.5 Audio Description (Prerecorded), and 1.2.8 Media Alternative (Prerecorded) overlap somewhat with each other. This is to give the author some choice at the minimum conformance level, and to provide additional requirements at higher levels. At Level A in Success Criterion 1.2.3, authors do have the choice of providing either an audio description or a full text alternative. If they wish to conform at Level AA, under Success Criterion 1.2.5 authors must provide an audio description - a requirement already met if they chose that alternative for 1.2.3, otherwise an additional requirement. At Level AAA under Success Criterion 1.2.8 they must provide an extended text description. This is an additional requirement if both 1.2.3 and 1.2.5 were met by providing an audio

description only. If 1.2.3 was met, however, by providing a text description, and the 1.2.5 requirement for an audio description was met, then 1.2.8 does not add new requirements.

Benefits of Media Alternative (Prerecorded)

- People who are deaf-blind, who cannot see well or at all, and who also cannot hear well or at all can get access to information in audio-visual presentations.

Examples of Media Alternative (Prerecorded)

Example 1. alternative for time-based media for a training video

A community center purchases a Training video for use by its clients and puts it on the center's intranet. The video involves explaining use of a new technology and has a person talking and showing things at the same time. The community center provides an alternative for time-based media that all clients, including those who can neither see the demonstrations nor hear the explanations in the synchronized media, can use to better understand what is being presented.

Resources for Media Alternative (Prerecorded)

- Transcripts , in Making Audio and Video Media Accessible , W3C Web Accessibility Initiative (WAI)
- uiAccess list of transcription services
- Transcripts on the Web: Getting people to your podcasts and videos

1.2.9 Audio-only (Live) (Level AAA)

Source: [WCAG 2.2](#) / [Understanding 1.2.9](#) — Copyright © 2023 World Wide Web Consortium

An alternative for time-based media that presents equivalent information for live audio-only content is provided.

In brief

Goal

Live audio can be understood by more people.

What to do

Provide a text equivalent for live audio-only content.

Why it's important

People who cannot hear or understand real-time audio can read an equivalent.

Intent of Audio-only (Live)

The intent of this success criterion is to make information conveyed by live audio, such as web-based audio conferencing, live speeches and radio Webcasts, accessible through the use of a text alternative. A live text caption service will enable live audio to be accessible to people who are deaf or hard of hearing, or who cannot otherwise hear the audio. Such services use a trained human operator who listens in to what is being said and uses a special keyboard to enter the text with only a small delay. They are able to capture a live event with a high degree of fidelity, and also to insert notes on any non spoken audio which is essential to understanding the event. A transcript is sometimes a possibility if the live audio is following a set script; but a live caption service is preferred because it plays out at the same pace as the audio itself, and can adapt to any deviations from the script that might occur.

Using untrained operators, or providing a transcript which differs markedly from what actually happens would not be considered meeting this success criterion.

This success criterion was intended to apply to broadcast of audio and is not intended to require that two-way audio calls between two or more individuals through web apps must be captioned regardless of the needs of users. Responsibility for providing captions would fall to the content providers (the callers) or the “host” caller, and not the application.

Examples of Audio-only (Live)

- A public relations firm uses web-based caption services to cover live events; the output from the service is incorporated in a sub frame of the web page which includes the streaming audio control.
- A live radio play of a fringe theatre group is being broadcast to the web. As the actors stick largely to a set script, and the budget for the program is small, the producers provide a link (with the playwright's permission) to the script of the play.
- A streaming audio server uses a technology which can also accommodate text and graphics, such as HTML. A stenographer is used to create live captions at an event, and these are mixed on the fly to produce live captions in the media stream which can be viewed by the media player.
- A CEO is to give a press release by telephone to the media in response to a breaking news story, the audio is being recorded and streamed over the internet, but due to time constraints a web captioning service cannot be set up in time. As the press release is a set statement which the CEO will be reading out, the company simultaneously provides the transcript of the release.

Resources for Audio-only (Live)

- WebAIM Real time captioning resource
- uiAccess list of transcription services
- Transcripts on the Web: Getting people to your podcasts and videos

Guideline 1.3: Adaptable

[Source: W3C WCAG 2.2 —

<https://www.w3.org/TR/WCAG22/#guideline13>]

Create content that can be presented in different ways (for example simpler layout) without losing information or structure.

1.3.1 Info and Relationships (Level A)

Source: [WCAG 2.2](#) / [Understanding 1.3.1](#) — Copyright © 2023 World Wide Web Consortium

Information, structure, and relationships conveyed through presentation can be programmatically determined or are available in text.

In brief

Goal

Information about content structure is available to more people.

What to do

Use code to reinforce relationships and information conveyed through presentation.

Why it's important

People can adapt the presentation to suit their needs while preserving the original meaning.

Intent of Info and Relationships

The intent of this success criterion is to ensure that information and relationships that are implied by visual or auditory formatting are preserved when the presentation format changes. For example, the presentation format changes when the content is read by a screen reader or when a user style sheet is substituted for the style sheet provided by the author.

Sighted users perceive structure and relationships through various visual cues — headings are often in a larger, bold font separated from paragraphs by blank lines; list items are preceded by a bullet and perhaps indented; paragraphs are separated by a blank line; items that share a common characteristic are organized into tabular rows and columns; form fields may be positioned as

groups that share text labels; a different background color may be used to indicate that several items are related to each other; words that have special status are indicated by changing the font family and /or bolding, italicizing, or underlining them; items that share a common characteristic are organized into a table where the relationship of cells sharing the same row or column and the relationship of each cell to its row and/or column header are necessary for understanding; and so on. Having these structures and these relationships programmatically determined or available in text ensures that information important for comprehension will be perceivable to all.

Auditory cues may be used as well. For example, a chime might indicate the beginning of a new section; a change in voice pitch or speech rate may be used to emphasize important information or to indicate quoted text; etc.

When such relationships are perceivable to one set of users, those relationships can be made to be perceivable to all. One method of determining whether or not information has been properly provided to all users is to access the information serially in different modalities.

If links to glossary items are implemented using anchor elements (or the proper link element for the technology in use) and identified using a different font face, a screen reader user will hear that the item is a link when the glossary term is encountered even though they may not receive information about the change in font face. An on-line catalog may indicate prices using a larger font colored red. A screen reader or person who cannot perceive red, still has the information about the price as long as it is preceded by the currency symbol.

Some technologies do not provide a means to programmatically determine some types of information and relationships. In that

case then there should be a text description of the information and relationships. For instance, "all required fields are marked with an asterisk (*)". The text description should be near the information it is describing (when the page is linearized), such as in the parent element or in the adjacent element.

There may also be cases where it may be a judgment call as to whether the relationships should be programmatically determined or be presented in text. However, when technologies support programmatic relationships, it is strongly encouraged that information and relationships be programmatically determined rather than described in text.

It is not required that color values be programmatically determined. The information

conveyed by color cannot be adequately presented simply by exposing the value. Therefore, [Success Criterion 1.4.1 Use of Color](#) addresses the specific case of color, rather than Success Criterion 1.3.1 Info and Relationships.

Benefits of Info and Relationships

- This success criterion helps people with different disabilities by allowing user agents to adapt content according to the needs of individual users.
- Users who are blind (using a screen reader) benefit when information conveyed through color is also available in text (including text alternatives for images that use color to convey information).
- Users who are deaf-blind using braille (text) refreshable displays may be unable to access color-dependent information.

Examples of Info and Relationships

A form with required fields

A form contains several required fields. The labels for the required fields are displayed in red. In addition, at the end of each label is an asterisk character, . *The instructions for completing the form indicate that "all required fields are displayed in red and marked with an asterisk "*, followed by an example.

A form that uses color and text to indicate required fields

A form contains both required and optional fields. Instructions at the top of the form explain that required fields are labeled with red text and also with an icon whose text alternative says, "Required." Both the red text and the icon are programmatically associated with the appropriate form fields so that assistive technology users can determine the required fields.

A bus schedule table where the headers for each cell can be programmatically determined

A bus schedule consists of a table with the bus stops listed vertically in the first column and the different buses listed horizontally across the first row. Each cell contains the time when the bus will be at that bus stop. The bus stop and bus cells are identified as headers for their corresponding row or column so that assistive technology can programmatically determine which bus and which bus stop are associated with the time in each cell.

A form where the labels for the checkboxes can be programmatically determined

In a form, the labels for each checkbox can be programmatically determined by assistive technology.

A text document

A simple text document is formatted with double blank lines before titles, asterisks to indicate list items and other standard formatting conventions so that its structure can be programmatically determined.

Resources for Info and Relationships

- WebAIM: Semantic Structure
- Heading Tags

1.3.2 Meaningful Sequence (Level A)

Source: [WCAG 2.2](#) / [Understanding 1.3.2](#) — Copyright © 2023 World Wide Web Consortium

When the sequence in which content is presented affects its meaning, a correct reading sequence can be programmatically determined .

In brief

Goal

The order of content can be understood by more people.

What to do

Use code to preserve meaningful content order.

Why it's important

Assistive technology can present content to users in the proper order.

Intent of Meaningful Sequence

The intent of this success criterion is to enable a user agent to provide an alternative presentation of content while preserving the reading order needed to understand the meaning. It is important that it be possible to programmatically determine at least one sequence of the content that makes sense. Content that does not meet this Success Criterion may confuse or disorient users when assistive technology reads the content in the wrong order, or when alternate style sheets or other formatting changes are applied.

A sequence is *meaningful* if the order of content in the sequence cannot be changed without affecting its meaning. For example, if a page contains two independent articles, the relative order of the articles may not affect their meaning, as long as they are not interleaved. In such a situation, the articles themselves may have

meaningful sequence, but the container that contains the articles may not have a meaningful sequence.

The semantics of some elements define whether or not their content is a meaningful sequence. For instance, in HTML, text is always a meaningful sequence. Tables and ordered lists are meaningful sequences, but unordered lists are not.

The order of content in a sequence is not always meaningful. For example, the relative order of the main section of a web page and a navigation section does not affect their meaning. They could occur in either order in the programmatically determined reading sequence. As another example, a magazine article contains several callout sidebars. The order of the article and the sidebars does not affect their meaning. In these cases there are a number of different reading orders for a web page that can satisfy the success criterion.

For clarity:

- Providing a particular linear order is only required where it affects meaning.
- There may be more than one order that is "correct" (according to the WCAG 2 definition).
- Only one correct order needs to be provided.

Benefits of Meaningful Sequence

- This success criterion may help people who rely on assistive technologies that read content aloud. The meaning evident in the sequencing of the information in the default presentation will be the same when the content is presented in spoken form.

Examples of Meaningful Sequence

- Example 1: In a multi-column document, the linear presentation of the content flows from the top of a column to the bottom of the column, then to the top of the next column.

- Example 2: CSS is used to position a navigation bar, the main story on a page, and a side story. The visual presentation of the sections does not match the programmatically determined order, but the meaning of the page does not depend on the order of the sections.

1.3.3 Sensory Characteristics (Level A)

Source: [WCAG 2.2](#) / [Understanding 1.3.3](#) — Copyright © 2023 World Wide Web Consortium

Instructions provided for understanding and operating content do not rely solely on sensory characteristics of components such as shape, color, size, visual location, orientation, or sound.

For requirements related to color, refer to Guideline 1.4 .

In brief

Goal

Instructions are understandable by more people.

What to do

Describe controls by name, not just by appearance or location.

Why it's important

People who are blind or have low vision need non-visual instructions.

Intent of Sensory Characteristics

The intent of this success criterion is to ensure that all users can access instructions for using the content, even when they cannot perceive shape or size or use information about spatial location or orientation. Some content relies on knowledge of the shape or position of objects that are not available from the structure of the content (for example, "round button" or "button to the right"). Some users with disabilities are not able to perceive shape or position due to the nature of the assistive technologies they use. This success criterion requires that additional information be provided to clarify instructions that are dependent on this kind of information.

Providing information using shape and/or location, however, is an effective method for many users including those with cognitive limitations. This provision should not discourage those types of cues as long as the information is also provided in other ways.

In some languages, it is commonly understood that "above" refers to the content previous to that point in the content and "below" refers to the content after that point. In such languages, if the content being referenced is in the appropriate place in the reading order and the references are unambiguous, statements such as "choose one of the links below" or "all of the above" would conform to this success criterion.

WCAG was designed to apply only to controls that were displayed on a web page. The intent was to avoid describing controls solely via references to visual or auditory cues. When applying this to instructions for operating physical hardware controls (e.g. a web kiosk with dedicated content), tactile cues on the hardware might be described (e.g. the arrow shaped key, the round key on the right side). This success criterion is not intended to prevent the use of tactile cues in instructions.

Benefits of Sensory Characteristics

- People who are blind and people who have low vision may not be able to understand instructions if they rely only on a description of the shape and/or location of content. Providing additional information in any instructions other than shape and/or location will allow users to understand the instructions even if they cannot perceive shape and/or location.

Examples of Sensory Characteristics

Example 1: Instructions for interpreting a schedule of competitive events references colored icons in different shapes to indicate the venue for each event

A table presents a list of times across the top row and a list of events in the first vertical column and instructions are provided under the table: "Events marked with a blue diamond are played on field A and events marked with a green circle are played on field B." The instructions rely on color and shape only and result in a failure of this criterion.

Example 2: An online multi-page survey

An online multi-page survey uses a link implemented as a green arrow icon placed in the lower right hand corner of the content to move from one survey page to the next. The arrow is clearly labeled with "Next" and the instructions state, "To move to the next section of the survey, select the green arrow icon labeled 'Next' in the lower right corner below the last survey question." The instruction uses positioning and color to help identify the icon; the instruction does not rely on these sensory characteristics since it also refers to the label, so it passes this criterion.

1.3.4 Orientation (Level AA)

Source: [WCAG 2.2](#) / [Understanding 1.3.4](#) — Copyright © 2023 World Wide Web Consortium

Content does not restrict its view and operation to a single display orientation, such as portrait or landscape, unless a specific display orientation is essential .

Examples where a particular display orientation may be essential are a bank check, a piano application, slides for a projector or television, or virtual reality content where parts of the image (for instance, a status bar or heads-up display) always remain at a fixed orientation relative to the headset screens (and the user's eyes) regardless of the physical orientation of the headset.

In brief

Goal

Devices can be used in any orientation.

What to do

Don't lock content to either portrait or landscape presentation.

Why it's important

Wheelchair users and others may have devices mounted in a fixed orientation.

Intent of this Success Criterion

The intent of this success criterion is to ensure that content displays in the orientation (portrait or landscape) preferred by the user. Some websites and applications automatically set and restrict the screen to a particular display orientation and expect that users will respond by rotating their device to match, but this can create problems. Some users have their devices mounted in a fixed orientation (e.g. on the arm of a power wheelchair).

Therefore, websites and applications need to support both orientations by not restricting the orientation. Changes in content or functionality due to the size of display are not covered by this criterion which is focused on restrictions of orientation.

Historically, devices tended to have a fixed-orientation display, and all content was created to match that display orientation. Today, most handhelds and many other devices (e.g., monitors) have a hardware-level ability to dynamically adjust default display orientation based on sensor information. The goal of this success criterion is that authors should never restrict content's orientation, thus ensuring that it always match the device display orientation.

Locking a device to an orientation

It is important to distinguish between an author's responsibility not to restrict content to a specific orientation, and device-specific settings governing the locking of display orientation.

Many handheld devices offer a mechanical switch or a system setting (or both) to allow the user to lock the device display to a specific orientation. Where a user decides to lock their entire device to an orientation, all applications are expected to pick up that setting and to display content accordingly.

This success criterion complements device "lock orientation" settings. A web page that does not restrict its display orientation will always support the system-level orientation setting, since the system setting is picked up by the user agent. Alternatively, where a device-level orientation lock is not in place, the user agent should display the page according to the orientation of the device (either its default, or the current orientation determined by any device sensors).

The exception for things considered essential is aimed at situations where the content would only be understood in a particular orientation, or where the technology restricts the possible orientations. If content is aimed at a specific environment which is only available in one orientation (such as a television) then the content can restrict the orientation. Technologies such as virtual reality use screens within goggles that cannot change orientation relative to the user's eyes.

Benefits

- Users with dexterity impairments, who have a mounted device will be able to use the content in their fixed orientation.
- Users with low-vision will be able to view content in the orientation that works best for them, for example to increase the text size by viewing content in landscape.

Examples

- Example 1: Online video site A video is shown in either portrait or in landscape based on the user's chosen orientation.
- Example 2: Messaging website A messaging website can display messages in both portrait and landscape orientations.
- Example 3: eReader web app An eReader web app can display the contents of a book in both portrait and landscape orientation.
- Example 4: Check deposit in banking app An example where orientation is essential could be a banking app that requires the device be in landscape mode to easily and accurately capture an image of a check for deposit. These paper forms are typically about twice as wide as they are high.
- Example 5: Piano app An example where orientation is essential could be a piano app that requires the device to be in landscape mode, allowing the piano keys to have enough room

to be functionally usable. Since a piano app is emulating a physical piano keyboard that needs to retain relative physical characteristics between keys, either too few keys would be available, or the keys would be much too narrow.

Related Resources

- [Managing Screen Orientation](#)
- [Using Media Queries](#)
- [Media Queries for Standard Devices](#)
- [Orientation Lock](#)
- [Responsive Design for Landscape Orientation](#)
- [The Orientation Descriptor](#)
- [The Screen Orientation API](#)

1.3.5 Identify Input Purpose (Level AA)

Source: [WCAG 2.2](#) / [Understanding 1.3.5](#) — Copyright © 2023 World Wide Web Consortium

The purpose of each input field collecting information about the user can be programmatically determined when:

- The input field serves a purpose identified in the Input Purposes for user interface components section ; and
- The content is implemented using technologies with support for identifying the expected meaning for form input data.

In brief

Goal

It is easier to fill out forms.

What to do

Use code to indicate the purpose of common inputs, where technology allows.

Why it's important

Some people with cognitive disabilities may not understand the input's purpose from the label alone.

Intent of this Success Criterion

The intent of this success criterion is to ensure that the purpose of a form input collecting information about the user can be programmatically determined, so that user agents can extract and present this purpose to users using different modalities. The ability to programmatically declare the specific kind of data expected in a particular field makes filling out forms easier, especially for people with certain cognitive disabilities.

Appropriate visible labels and instruction can help users understand the purpose of form input fields, but users may benefit from having fields that collect specific types of information be rendered in an unambiguous, consistent, and possibly customized way for different modalities - either through defaults in their user agent, or through the aid of assistive technologies.

For some input fields, the `type` attribute already offers a way to broadly specify the intention of the input field, for example, `<input type="tel">`, `<input type="email">`, or `<input type="password">`. However, these are only very broad categories, describing the type of input, but not necessarily its purpose, especially as it relates to user-specific input fields. As an example, `type="email"` indicates that the field is for an email address but does not clarify if the purpose is for entering the user's email address or some other person's email.

This success criterion defines the types of user interface component input purposes, found in Section 7 of the WCAG 2.1 Recommendation, that must be programmatically identifiable. When these user input purposes are present, and if the technology supports doing so, the field purpose must be programmatically identifiable.

The HTML `autocomplete` attribute only accepts a certain number of specific well-defined fixed values. This allows a more fine-grained definition or identification of purpose than the `type` attribute, for example, by allowing the author to specify a specific type of name: `Name ( autocomplete="name" )`, `Given Name ( autocomplete="given-name" )`, `Family Name ( autocomplete="family-name" )`, as well as `Username ( autocomplete="username" )`, and `Nickname ( autocomplete="nickname" )`.

By adopting and repurposing this predefined taxonomy of definitions, user agents and assistive technologies can now present the purpose of the inputs to users in different modalities. For example, assistive technologies may display familiar icons next to input fields to help users who have difficulties reading. An icon of a birthday cake may be shown in front of an input field with `autocomplete="bday"` , or the icon of a telephone in front of an input field with `autocomplete="tel"` .

In addition to repurposing this taxonomy, when the autocomplete attribute technique is used to meet this Success Criterion, browsers and other user agents can suggest and 'autofill' the right content by autocompleting these fields based on past user input stored in the browser. By defining more granular definitions of common input purposes, for example “Birthday” (`autocomplete="bday"`), browsers can store personalized values for each of these fields (the user's birthday date). The user is relieved of having to type the information and can instead confirm or, if needed, change the value of the field, a significant benefit for users with memory issues, dyslexia, and other disabilities. Because the `autocomplete` values are independent of language, users that may not be familiar with the text used to visually identify user input fields (the label) can still have that purpose consistently identified to them due to the fixed taxonomy of terms.

If an input field accepts two different types of input purpose (as in combined user name/user email fields) and the technology used does not allow multiple purpose values to be defined, it is valid to provide either one or the other value or leave out the designation of input purpose altogether.

When the user agent and assistive technology support for other metadata formats matures, metadata schemes like the WAI-Adapt: Symbols Module may be used in addition or instead of the HTML

autocomplete attribute to identify the purpose of input fields. They can also support automated adaptations that identify and match author-provided input labels to defined vocabularies or symbols that are used instead for labelling inputs.

This success criterion is specifically scoped to inputs collecting *information about the user*. The list of Input Purposes includes a few values that may not strictly be interpreted as directly relating to a user – most prominently, `transaction-amount`. An input field for information that is not *about the user* does not need to programmatically expose its purpose, even if that purpose is included in the Input Purposes list.

The term input is used here as a generic way to refer to form controls that accept user input. For instance, in HTML, it is not limited to the `<input>` element, but also covers other controls such as `<textarea>` and `<select>`.

Specific Benefits of Success Criterion 1.3.5

- People with language and memory related disabilities or disabilities that affects executive function and decision-making benefit from the browser auto-filling personal information (such as name or address) when the autocomplete attribute is used to meet this Success Criterion, which means information does not need to be remembered by the user.
- People with cerebral palsy, stroke, head injury, motor neuron disease or learning disability sometimes prefer images for communication. They can employ assistive technology which adds icons to input fields to communicate the purpose of the fields visually.
- People with motor impairments also benefit from reducing the need for manual input when filling out forms.

Examples of Success Criterion 1.3.5

A contact form using autofill

A contact form auto-fills in the fields for name, street, post code, city, telephone number and email address from autofill values stored in the user's browser. Assistive technology can offer a customized way of identifying particular input fields, for example drawing on a set of symbols / icons that is familiar to the user, to communicate the purpose of the fields visually.

An order form with separate billing and shipping address

A product order form fills in the address fields for billing address and a separate set of address fields for the shipping address, using the autofill detail tokens 'billing' and 'shipping'

A contact form using icons

A browser plugin to add icons inserts icons representing the person's name, home address, telephone number and email address to identify the input purpose visually.

Related Resources

- [COGA Gap Analysis Table 3: Entering Data, Error Prevention, & Recovery](#)

1.3.6 Identify Purpose (Level AAA)

Source: [WCAG 2.2](#) / [Understanding 1.3.6](#) — Copyright © 2023 World Wide Web Consortium

In content implemented using markup languages, the purpose of user interface components , icons, and regions can be programmatically determined .

In brief

Goal

It is easier to operate and navigate content.

What to do

Use code to indicate the meaning of all controls and other key information, where available.

Why it's important

Some people with cognitive disabilities may not understand a control's purpose from the name alone.

Intent

The intent of this success criterion is to ensure that the purpose of many elements on a page can be programmatically determined, so that user agents can extract and present that purpose to users using different modalities.

Many users with limited vocabularies rely on familiar terms or symbols in order to use the web. However, what is familiar to one user may not be familiar to another. When authors indicate the purpose, users can take advantage of personalization and user preferences to load a set of symbols or vocabulary familiar to them.

This success criterion requires the author to programmatically associate the purpose of icons, regions and components (such as

buttons, links, and fields) so that user agents can determine the purpose of each and adapt indicators or terminology to make them understandable for the user. It is achieved by adding semantics or metadata that provide this context. It is similar to adding role information (as required by 4.1.2) but instead of providing information about what the UI component is (such as an image) it provides information about what the component represents (such as a link to the home page).

Identifying regions of the page allows people to remove or highlight regions with their user agent.

Products for people who are non-vocal often use symbols to help users communicate. These symbols are in fact people's language. Unfortunately, many of these symbols are both subject to copyright and not interoperable. That means end users can only use one device, and cannot use content, apps, or assistive technologies that have not been made by a single company.

This Success Criterion enables symbols to be interoperable so that symbol users can understand different content that was not just made by one company. When users' symbols are mapped to the same nodes, then user agents can load the user-understandable symbol. People can then buy the symbols and use them across different devices or applications. (Note that the symbols would still be proprietary, but they could then be interoperable.)

Benefits

People who benefit have many different cognitive disabilities including:

- Memory
- Focus and attention
- Language-related

- Executive function and decision making.

Meeting this success criterion helps users who need extra support or a familiar interface, including the need for:

- Symbols and graphics with which users are familiar
- Fewer features and less cognitive overload
- Keyboard shortcuts

Examples

- A website uses ARIA landmarks to identify the regions of the page, and users can hide areas that do not have a role of `main` .
- The links in the navigation of a website are marked up so that users can add their own icons.
- Icons on a website are marked up so that users can substitute their own icon sets into the page.

Resources

- [Making Content Usable for People with Cognitive and Learning Disabilities](#)
- An [WAI-Adapt Overview, enabling users to adapt content presentation](#)
- [Personalization and User Preferences](#)
- [The coga.personalisation project](#)

Guideline 1.4: Distinguishable

[Source: W3C WCAG 2.2 —

<https://www.w3.org/TR/WCAG22/#guideline14>]

Make it easier for users to see and hear content including separating foreground from background.

1.4.1 Use of Color (Level A)

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Color is not used as the only visual means of conveying information, indicating an action, prompting a response, or distinguishing a visual element.

This success criterion addresses color perception specifically. Other forms of perception are covered in Guideline 1.3 including programmatic access to color and other visual presentation coding.

In brief

Goal

Color is not the only way of distinguishing information.

What to do

Use information in addition to color, such as shape or text, to convey meaning.

Why it's important

Not everyone sees colors or sees them the same way.

Intent of Use of Color

The intent of this success criterion is to ensure that all sighted users can access information that is conveyed by color differences, that is, by the use of color where each color has a meaning assigned to it. If the information is conveyed through color differences in an image (or other non-text format), the color may not be seen by users with color deficiencies. In this case, providing the information conveyed with color through another visual means ensures users who cannot see color can still perceive the information.

Color is an important asset in the design of web content, enhancing its aesthetic appeal, its usability, and its accessibility. However, some users have difficulty perceiving color. People with partial sight often experience limited color vision, and many older users do not see color well. In addition, people using limited-color or monochrome displays and browsers will be unable to access information that is presented only in color.

Examples of information conveyed by color differences: “required fields are red”, “error is shown in red”, and “Mary's sales are in red, Tom's are in blue”. Examples of indications of an action include: using color to indicate that a link will open in a new window or that a database entry has been updated successfully. An example of prompting a response would be: using highlighting on form fields to indicate that a required field had been left blank.

This should not in any way discourage the use of color on a page, or even color coding if it is complemented by other visual indication.

If content is conveyed through the use of colors that differ not only in their hue, but that also have a significant difference in lightness, then this counts as an additional visual distinction, as long as the difference in relative luminance between the colors leads to a contrast ratio of 3:1 or greater. For example, a light green and a dark red differ **both** by color (hue) **and** by lightness, so they would pass if the contrast ratio is at least 3:1. Similarly, if content is distinguished by inverting an element's foreground and background colors, this would pass (again, assuming that the foreground and background colors have a sufficient contrast).

However, if content relies on the user's ability to accurately perceive or differentiate a particular color an additional visual indicator will be required regardless of the contrast ratio between

those colors. For example, knowing whether an outline is green for valid or red for invalid.

This criterion does not apply to situations where color has *not* been used to convey information, indicate an action, prompt a response or distinguish a visual element. For instance, a hyperlink which has been styled to appear no different than neighboring static text would not fail this success criterion, as there would be no color differentiation between the actionable hyperlink text and the adjacent static text. Such lack of styling differentiation could result in poor usability for anyone looking at the interface, regardless of disability.

This criterion does not directly address the needs of users with assistive technologies.

It aims to ensure that sighted users who cannot distinguish between some colors can still understand content.

How information is conveyed to assistive technology users is covered separately in other

criteria, including (but not limited to) [1.1.1 Non-text Content](#) , [1.3.1 Info and Relationships](#) , and [4.1.2 Name, Role, Value](#) .

Conversely, even if information that is conveyed by color differences is appropriately conveyed to assistive technologies, it does not necessarily pass this criterion, as sighted users who cannot distinguish between certain color may not necessarily be using any assistive technologies. This criterion requires a visible alternative to color.

Most user agents provide users with a color-only cue that a link has been previously activated by them ("visited"). However, several technical constraints result in authors having very limited

control over these color-only indications of visited links. The technical constraints are as follows:

- User agents constrain the exposure of a link's visited state due to privacy concerns . Author queries to user agents will indicate all links have not been visited.
- Any available information on the visited state of a link would be inaccurate since it is both user and browser-dependent. Even if an author could accurately get information on previously activated links by a certain user, the author would be constrained to the current browser's preserved history, and would be unable to determine if the user had visited the page using a different browser (or if the history was not preserved, either from cache clearing or use of private sessions).
- Authors can only use color to modify the :visited CSS pseudoclass style. The technology constrains any non-color styling. Due to palette limitations, an author cannot use color alone to achieve 3:1 contrast between link and non-link text as well as between visited and unvisited links while also achieving 4.5:1 contrast for all link and non-link text.
- Authors also cannot set the visited state of links. The anchor element does not include a "visited" attribute; therefore the author has no ability to alter the state through an attribute setting. As such, authors cannot achieve [1.3.1 Info and Relationships](#) or [4.1.2 Name, Role, Value](#) in regard to visited links.

For these reasons, setting or conveying a link's visited status is not an author responsibility. Where color alone distinguishes between visited and unvisited links, it does not result in a failure of this Success Criterion, even where the contrast between the two link colors is below 3:1. Note that authors must continue to ensure that

all text links meet contrast minimums against the page background (SC 1.4.3).

Benefits of Use of Color

- Users with partial sight often experience limited color vision.
- Some older users may not be able to see color well.
- Users with color vision deficiency who cannot distinguish between certain colors benefit when information conveyed by color is available in other visual ways.
- People using limited color monochrome displays may be unable to access color-dependent information.
- Users who have problems distinguishing between colors can look or listen for text cues.

Examples of Use of Color

A form that uses color and text to indicate required fields

A form contains both required and optional fields. Instructions at the top of the form explain that required fields are labeled with red text and also with an icon. Users who cannot perceive the difference between the optional field labels and the red labels for the required fields will still be able to see the icon next to the red labels.

An examination

Students view an SVG image of a chemical compound and identify the chemical elements present based both on the colors used, as well as numbers next to each element. A legend shows the color and number for each type of element. Sighted users who cannot perceive all the color differences can still understand the image by relying on the numbers.

Resources for Use of Color

- [Vischeck](#)
- [AWARE Color Laboratory](#)
- [Wikipedia: Color vision deficiency](#)
- [Microsoft: Verify that a page is usable by people with color blindness](#)
- [Causes and Incidence of Colorblindness](#)
- [How to make figures and presentations that are friendly to Colorblind people](#)
- [The Color Tutor application](#)

1.4.2 Audio Control (Level A)

Source: [WCAG 2.2](#) / [Understanding 1.4.2](#) — Copyright © 2023 World Wide Web Consortium

If any audio on a web page plays automatically for more than 3 seconds, either a mechanism is available to pause or stop the audio, or a mechanism is available to control audio

volume independently from the overall system volume level.

Since any content that does not meet this success criterion can interfere with a user's

ability to use the whole page, all content on the web page (whether or not it is used

to meet other success criteria) must meet this success criterion. See Conformance Requirement 5: Non-Interference .

In brief

Goal

A page that plays music or sounds doesn't disrupt people.

What to do

If you play audio content automatically, let people turn it down or off.

Why it's important

Sound distracts some people, and also interferes with screen readers.

Intent of Audio Control

Individuals who use screen reading software can find it hard to hear the speech output if there is other audio playing at the same time. This difficulty is exacerbated when the screen reader's speech output is software based (as most are today) and is

controlled via the same volume control as the sound. Therefore, it is important that the user be able to turn off the background sound.

Having control of the volume includes being able to reduce its volume to zero. Muting the system volume is not "pausing or stopping" the autoplay audio. Both the "pause or stop" and control of audio volume need to be independent of the overall system volume.

Playing audio automatically when landing on a page may affect a screen reader user's ability to find the mechanism to stop it because they navigate by listening and automatically started sounds might interfere with that navigation. Therefore, we discourage the practice of automatically starting sounds (especially if they last more than 3 seconds), and encourage that the sound be *started* by an action initiated by the user after they reach the page, rather than requiring that the sound be *stopped* by an action of the user after they land on the page.

See also [1.4.7 Low or No Background Audio](#) .

In the context of this success criterion, "plays automatically" broadly refers to audio that is not started/played as a direct result of a user's intentional activation. For example, selecting a link or button with clear labelling or context that it will start an experience where audio will play is an example of starting in response to a user's intended action. This criterion is also not intended to apply to a conference call or other interaction where two-way voice communication may take place; the potential for any participant to speak during a conference call is not equivalent to audio that "plays automatically for more than 3 seconds." However, a mechanism to control the volume of conference call output independently from the overall system volume would be a best practice.

Benefits of Audio Control

- Individuals who use screen reading technologies can hear the screen reader without other sounds playing. This is especially important for those who are hard of hearing and for those whose screen readers use the system volume (so they cannot turn sound down and screen reader up).
- This success criterion also benefits people who have difficulty focusing on visual content (including text) when audio is playing.

Examples of Audio Control

- An audio file begins playing automatically when a page is opened. However, the audio can be stopped by the user by selecting a "silent" link at the top of the page.

1.4.3 Contrast (Minimum) (Level AA)

Source: [WCAG 2.2](#) / [Understanding 1.4.3](#) — Copyright © 2023 World Wide Web Consortium

The visual presentation of text and images of text has a contrast ratio of at least 4.5:1, except for the following:

Large Text

Large-scale text and images of large-scale text have a contrast ratio of at least 3:1;

Incidental

Text or images of text that are part of an inactive user interface component, that are pure decoration, that are not visible to anyone, or that are part of a picture that contains significant other visual content, have no contrast requirement.

Logotypes

Text that is part of a logo or brand name has no contrast requirement.

In brief

Goal

Text can be seen by more people.

What to do

Provide sufficient contrast between text and its background.

Why it's important

Some people cannot read faint text.

Intent of Contrast (Minimum)

The intent of this success criterion is to provide enough contrast between text and its background, so that it can be read by people

with moderately low vision or impaired contrast perception, without the use of contrast-enhancing assistive technology.

For all consumers of visual content, adequate light-dark contrast is needed between the relative luminance of text and its background for good readability. Many different visual impairments can substantially impact contrast sensitivity, requiring more light-dark contrast, regardless of color (hue). For people with color vision deficiency who are not able to distinguish certain shades of color, hue and saturation have minimal or no effect on legibility as assessed by reading performance. Further, the inability to distinguish certain shades of color does not negatively affect light-dark contrast perception. Therefore, in the recommendation, contrast is calculated in such a way that color (hue) is not a key factor.

Text that is decorative and conveys no information is excluded. For example, if random words are used to create a background and the words could be rearranged or substituted without changing meaning, then it would be decorative and would not need to meet this criterion.

Text that is larger and has wider character strokes is easier to read at lower contrast.

The contrast requirement for larger text is therefore lower. This allows authors to

use a wider range of color choices for large text, which is helpful for design of

pages, particularly titles. 18 point text or 14 point bold text is judged to be large

enough to require a lower contrast ratio. (See The American Printing House for the

Blind Guidelines for Large Printing and The Library of Congress Guidelines for Large

Print under Resources). "18 point" and "bold" can both have different meanings in

different fonts but, except for very thin or unusual fonts, they should be sufficient. Since there

are so many different fonts, the general measures are used and a note regarding thin or unusual

fonts is included in the definition for large-scale text.

When evaluating this Success Criterion, the font size in points should be obtained

from the user agent or calculated on font metrics in the way that user agents do.

Point sizes are based on the CSS `pt` size as defined in CSS3 Values . The ratio between

sizes in points and CSS pixels is $1\text{pt} = 1.333\text{px}$, therefore 14pt and 18pt are equivalent to approximately 18.5px and 24px .

Because different image editing applications default to different pixel densities (e.g., 72ppi or 96ppi), specifying point sizes for fonts from within an image editing application can be unreliable when it comes to presenting text at a specific size. When creating images of large-scale text, authors should ensure that the text in the resulting image is roughly equivalent to 1.2 and 1.5 em or to 120% or 150% of the default size for body text. For example, for a 72ppi image, an author would need to use approximately 19pt and 24pt font sizes in order to successfully present images of large-scale text to a user.

The 3:1 and 4.5:1 contrast ratios referenced in this success criterion are intended to be treated as threshold values. When comparing the computed contrast ratio to the Success Criterion ratio, the computed values should not be rounded (e.g., 4.499:1 would not meet the 4.5:1 threshold).

Because authors do not have control over user settings for font smoothing/anti-aliasing, when evaluating this Success Criterion, refer to the foreground and background colors obtained from the user agent, or the underlying markup and stylesheets, rather than the text as presented on screen.

Due to anti-aliasing, particularly thin or unusual fonts may be rendered by user agents with a much fainter color than the actual text color defined in the underlying CSS. This can lead to situations where text has a contrast ratio that nominally passes the Success Criterion, but has a much lower contrast in practice. In these cases, best practice would be for authors to choose a font with stronger/thicker lines, or to aim for a foreground/background color combination that exceeds the normative requirements of this success criterion.

The contrast requirements for text also apply to images of text (text that has been rendered into pixels and then stored in an image format) - see [Success Criterion 1.4.5 Images of Text](#) .

This requirement applies to situations in which images of text were intended to be understood as text. Incidental text, such as in photographs that happen to include a street sign, are not included. Nor is text that for some reason is designed to be invisible to all viewers. Stylized text, such as in corporate logos, should be treated in terms of its function on the page, which may or may not warrant including the content in the text alternative. Corporate visual

guidelines beyond logo and logotype are not included in the exception.

In this provision there is an exception that reads "that are part of a picture that contains significant other visual content,". This exception is intended to separate pictures that have text in them from images of text that are done to replace text in order to get a particular look.

Images of text do not scale as well as text because they tend to pixelate. It is also harder to change foreground and background contrast and color combinations for images of text, which is necessary for some users. Therefore, we suggest using text wherever possible, and when not, consider supplying an image of higher resolution.

This success criterion applies to text in the page, including placeholder text and text that is shown when a pointer is hovering over an object or when an object has keyboard focus. If any of these are used in a page, the text needs to provide sufficient contrast.

Although this success criterion only applies to text, similar issues occur for content presented

in charts, graphs, diagrams, and other non-text-based information, which is covered by [Success Criterion 1.4.11 Non-Text Contrast](#) .

See also [1.4.6 Contrast \(Enhanced\)](#) .

Rationale for the Ratios Chosen

A contrast ratio of 3:1 is the minimum level recommended by [[ISO-9241-3]] and [[ANSI-HFES-100-1988]] for standard text and vision. The 4.5:1 ratio is used in this success criterion to account for the loss in contrast that results from moderately low

visual acuity, congenital or acquired color deficiencies, or the loss of contrast sensitivity that typically accompanies aging.

The rationale is based on a) adoption of the 3:1 contrast ratio for minimum acceptable contrast for normal observers, in the ANSI standard, and b) the empirical finding that in the population, visual acuity of 20/40 is associated with a contrast sensitivity loss of roughly 1.5 [[ARDITI-FAYE]]. A user with 20/40 would thus require a contrast ratio of $3 * 1.5 = 4.5 \pm 1$. Following analogous empirical findings and the same logic, the user with 20/80 visual acuity would require contrast of about 7:1. This ratio is used in Success Criterion 1.4.6.

Hues are perceived differently by users with color vision deficiencies (both congenital and acquired) resulting in different colors and relative luminance contrasts than for normally sighted users. Because of this, effective contrast and readability are different for this population. However, color deficiencies are so diverse that prescribing effective general use color pairs (for contrast) based on quantitative data is not feasible. Requiring good luminance contrast accommodates this by requiring contrast that is independent of color perception. Fortunately, most of the luminance contribution is from the mid and long wave receptors which largely overlap in their spectral responses. The result is that effective luminance contrast can generally be computed without regard to specific color deficiency, except for the use of predominantly long wavelength colors against darker colors (generally appearing black) for those who have protanopia. (We provide an advisory technique on avoiding red on black for that reason). For more information see [[ARDITI-KNOBLAUCH-1994]] [[ARDITI-KNOBLAUCH-1996]] [[ARDITI]].

Some people with cognitive disabilities require color combinations or hues that have low contrast, and therefore we allow and

encourage authors to provide mechanisms to adjust the foreground and background colors of the content. Some of the combinations that could be chosen may have contrast levels that will be lower than those specified here. This is not a violation of this Success Criterion, provided there is a mechanism that will return to the required values set out here.

The contrast ratio of 4.5:1 was chosen for level AA because it compensated for the loss in contrast sensitivity usually experienced by users with vision loss equivalent to approximately 20/40 vision. (20/40 calculates to approximately 4.5:1.) 20/40 is commonly reported as typical visual acuity of elders at roughly age 80. [[GITTINGS-FOZARD]]

The contrast ratio of 7:1 was chosen for level AAA because it compensated for the loss in contrast sensitivity usually experienced by users with vision loss equivalent to approximately 20/80 vision. People with more than this degree of vision loss usually use assistive technologies to access their content (and the assistive technologies usually have contrast enhancing, as well as magnification capability built into them). The 7:1 level therefore generally provides compensation for the loss in contrast sensitivity experienced by users with low vision who do not use assistive technology and provides contrast enhancement for color deficiency as well.

Calculations in [[ISO-9241-3]] and [[ANSI-HFES-100-1988]] are for body text. A relaxed contrast ratio is provided for text that is much larger.

Notes on formula

Conversion from nonlinear to linear RGB values is based on IEC/4WD 61966-2-1 [[IEC-4WD]].

The formula (L_1/L_2) for contrast is based on [[ISO-9241-3]] and [[ANSI-HFES-100-1988]] standards.

The ANSI/HFS 100-1988 standard calls for the contribution from ambient light to be included in the calculation of L_1 and L_2 . The .05 value used is based on Typical Viewing Flare from [[IEC-4WD]].

This success criterion and its definitions use the terms "contrast ratio" and "relative luminance" rather than "luminance" to reflect the fact that web content does not emit light itself. The contrast ratio gives a measure of the relative luminance that would result when displayed. (Because it is a ratio, it is dimensionless.)

Refer to related resources for a list of tools that utilize the contrast ratio

to analyze the contrast of web content.

See also [2.4.7 Focus Visible](#) for techniques for indicating keyboard focus.

Inactive User Interface Components

User Interface Components that are not available for user interaction (e.g., a disabled control in HTML) are not required to meet contrast requirements. An inactive user interface component is visible but not currently operable. An example would be a submit button at the bottom of a form that is visible but cannot be activated until all the required fields in the form are completed.

Benefits of Contrast (Minimum)

- People with low vision often have difficulty reading text that does not contrast with its background. This can be exacerbated if the person has a color vision deficiency that lowers the contrast even further. Providing a minimum luminance

contrast ratio between the text and its background can make the text more readable even if the person does not see the full range of colors. It also works for the rare individuals who see no color.

Resources for Contrast (Minimum)

- [Colour Contrast Analyser application](#)
- [Luminosity Colour Contrast Ratio Analyser](#)
- [Colour Contrast Check](#)
- [Contrast Ratio Calculator](#)
- [Adobe Color - Color Contrast Analyzer Tool](#)
- [Atypical colour response](#)
- [Colors On the Web Color Contrast Analyzer](#)
- [Tool to convert images based on color loss](#) so that contrast is restored as luminance contrast when there was only color contrast (that was lost due to color deficiency)
- [List of color contrast tools](#)
- [The American Printing House for the Blind Guidelines for Large Printing](#)
- [National Library Service for the Blind and Physically Handicapped \(NLS\), The Library of Congress reference guide on large print materials](#)
- [Types of Color Vision Deficiency, National Eye Institute \(NEI\), National Institutes of Health \(NIH\), U.S. Department of Health and Human Services \(HHS\)](#)
- [Effects of chromatic and luminance contrast on reading, Knoblauch et al., 1991](#)
- [Achromatic luminance contrast sensitivity in X-linked color-deficient observers: an addition to the debate, Márta Janáky et al., 2013](#)

- Contrast sensitivity of patients with congenital color vision deficiency, Cagri Ilhan et al., 2018

1.4.4 Resize Text (Level AA)

Source: WCAG 2.2 / Understanding 1.4.4 — Copyright © 2023 World Wide Web Consortium

Except for captions and images of text, text can be resized without assistive technology up to 200 percent without loss of content or functionality.

In brief

Goal

Text can be enlarged.

What to do

Ensure text can be doubled in size.

Why it's important

Some people can only read text when it is bigger.

Intent of Resize Text

The intent of this success criterion is to ensure that visually rendered text, including controls and labels using text, can be made larger so that it can be read more easily by people with milder visual impairments, without requiring the use of assistive technology (such as a screen magnifier). Users may benefit from scaling all content on the web page, but text is most critical.

The scaling of content is primarily a user agent responsibility. User agents that

satisfy UAAG 1.0 Checkpoint 4.1 allow users to configure text scale through a number of mechanisms - including zoom (of the entire page's content),

magnification, text-only resizing, and allowing the user to configure a size for rendered text.

The author's responsibility is to create web content that does not prevent the user agent from scaling the content effectively.

Authors may satisfy this success criterion by verifying that content does not interfere

with user agent support for resizing text, including text-based controls, or by providing

direct support for resizing text or changing the layout. An example of direct support

might be via server-side script that can be used to assign different style sheets.

Content satisfies the success criterion if it can be scaled up to 200% using at least one text scaling mechanism supported by user agents.

If the author is using a technology whose user agents do not provide support for specific text scaling mechanisms, the author is responsible for providing this type of functionality directly, or providing content that works with the type of functionality provided by the user agent. For instance, if the user agent doesn't provide full-page zoom functionality, but does let the the user change the text size, the author is responsible for ensuring that the content remains usable when the text is resized.

Some user interface components that function as a label and require activation by the user to access content are not wide enough to accommodate the label's content. For example, in web mail applications the subject column may not be wide enough to accommodate every possible subject header, but activating the subject header takes the user to the full message with the full subject header. In Web-based spreadsheets, cell content that is too long to be displayed in a column can be truncated, and the full

content of the cell is available to the user when the cell receives focus. The content of a user interface component may also become too wide in user interfaces where the user can resize the column width. In this type of user interface component, line wrapping is not required; truncation is acceptable if the component's full content is available on focus or after user activation and an indication that this information can be accessed, is provided to the user in some way besides the fact that it is truncated.

Content satisfies the success criterion if it can be scaled up to 200% - that is, up to twice the width and height. Authors may support scaling beyond that limit, however, as scaling becomes more extreme, adaptive layouts may introduce usability problems. For example, words may be too wide to fit into the horizontal space available to them, causing them to be truncated; layout constraints may cause text to overlap with other content when it is scaled larger; or only one word of a sentence may fit on each line, causing the sentence to be displayed as a vertical column of text that is difficult to read.

If the resizing mechanism (whether it's provided by the user agent, or implemented by the author) offers incremental resizing steps between 100% and 200%, authors must ensure that there is no loss of content or functionality at these incremental steps.

The working group feels that 200% is a reasonable accommodation that can support a wide range of designs and layouts, and complements older screen magnifiers that provide a minimum magnification of 200%. Above 200%, zoom (which resizes text, images, and layout regions and creates a larger canvas that may require both horizontal and vertical scrolling) may be more effective than text resizing. Assistive technology dedicated to zoom support would usually be used in such a situation, and may

provide better accessibility than attempts by the author to support the user directly.

Images of text do not scale as well as text because they tend to pixelate, and therefore

we suggest using text wherever possible. It is also harder to change foreground and

background contrast and color combinations for images of text, which are necessary

for some users. See also Success Criterion 1.4.5 Images of Text .

As with most other Success Criteria, this criterion applies to each variation of the page that is automatically presented for various screen sizes (e.g. media query variations in a responsive site). In an implementation where text does not consistently increase its size as people zoom in (such as when it is transformed based on a media query to adapt to small-screen usage), it must still be possible to get to 200% enlargement in order to satisfy the criterion.

For example, if at the default browser setting of 100% zoom, text is set at 20px when the window is 1280 CSS pixels wide, at 200% zoom it will visually appear at twice the size. After zooming by 400% the page reflows to fit within the 320 CSS pixel viewport, the author may decide to set the page's text size to 10px. The text is now half the original size in CSS pixels, but as it has been enlarged to 400%, it is still displayed at twice the size compared to the default browser setting at 100% zoom. It is not required to achieve 200% text enlargement while remaining inside a specific breakpoint (as zooming may result in the variation for a new breakpoint becoming active), but it should still be possible to get 200% text enlargement in some way compared to the default 100% zoom.

See also [1.4.8 Visual Presentation](#) .

Benefits of Resize Text

- This success criterion helps people with low vision by letting them increase text size in content so that they can read it.

Examples of Resize Text

- A user with vision impairments increases the text size on a web page in a browser from 1 em to 1.2 ems. While the user could not read the text at the smaller size, they can read the larger text. All the information on the page is still displayed when the larger font is used for the text.
- A web page contains a control for changing the scale of the page. Selecting different settings changes the layout of the page to use the best design for that scale.
- A user changes the scale of the content with the browser's full-page zoom function. All the content scales uniformly, and the browser provides scroll bars, if necessary.

Resources for Resize Text

- [CSS 2 Box Model](#)
- [CSS 2 Visual formatting Model](#)
- [CSS 2 Visual formatting Model Details](#)
- [About fluid and fixed width layouts](#)
- [Accessible CSS](#)

1.4.5 Images of Text (Level AA)

Source: [WCAG 2.2](#) / [Understanding 1.4.5](#) — Copyright © 2023 World Wide Web Consortium

If the technologies being used can achieve the visual presentation, text is used to convey information rather than images of text except for the following:

Customizable

The image of text can be visually customized to the user's requirements;

Essential

A particular presentation of text is essential to the information being conveyed.

Logotypes (text that is part of a logo or brand name) are considered essential.

In brief

Goal

Users can adjust how text is presented.

What to do

Use text instead of pictures of text.

Why it's important

People cannot alter how text looks in images.

Intent of Images of Text

The intent of this success criterion is to encourage authors, who are using technologies which are capable of achieving their desired default visual presentation, to enable people who require a particular visual presentation of text to be able to adjust the text presentation as needed. This includes people who require the text

in a particular font size, foreground and background color, font family, line spacing or alignment.

If authors can use text to achieve the same visual effect, they should present the information as text rather than using an image. If for any reason, the author cannot format the text to get the same effect, the effect won't be reliably presented on the commonly available user agents, or using a technology to meet this criterion would interfere with meeting other criteria such as 1.4.4, then an image of text can be used. This includes instances where a particular presentation of text is essential to the information being conveyed, such as type samples, logotypes, branding, etc. Images of text may also be used in order to use a particular font that is either not widely deployed or which the author doesn't have the right to redistribute, or to ensure that the text would be anti-aliased on all user agents.

Images of text can also be used where it is possible for users to customize the image of text to match their requirements.

The definition of images of text contains the note: This does not include text that is part of a picture that contains significant other visual content. Examples of such pictures include graphs, screenshots, and diagrams which visually

convey important information through more than just text.

The success criterion is intended to address situations where images of text are used *rather than* text. Where images of text are used *in addition to* text to convey the same information, and where both are presented to the user, this success criterion is met. This allows authors to convey content using any styling they desire, while also presenting the information in text, which can then be manipulated by users to make it more distinguishable. This is in contrast to [1.4.9 Images of Text \(No Exception\)](#) , which

applies to all images of text, regardless of whether or not they are used *in addition to* text.

Techniques for satisfying this success criterion are the same as those for Success Criterion 1.4.9, except that they only need to apply if the visual presentation can be achieved with the technologies that the author is using. For Success Criterion 1.4.9, the sufficient techniques would be applied only when the user can customize the output.

See also [1.4.9 Images of Text \(No Exception\)](#) .

Benefits of Images of Text

- People with low vision (who may have trouble reading the text with the authored font family, size and/or color).
- People with visual tracking problems (who may have trouble reading the text with the authored line spacing and/or alignment).
- People with cognitive disabilities that affect reading.

Examples of Images of Text

Styled Headings

Rather than using bitmap images to present headings in a specific font and size, an author uses CSS to achieve the same result.

Dynamically Generated Images

A web page uses server-side scripting to present text as an an image. The page includes controls that allow the user to adjust the font size and foreground and background colors of the generated image.

A quote

A web page contains a quote. The quote itself is presented as italicized text, indented from the left margin. The name of the

person to whom the quote is attributed is below the quote with 1.5x the line space and further indented from the left margin. CSS is used to position the text; set the spacing between lines; as well as display the text's font family, size, color and decoration.

Navigation items

A web page contains a menu of navigation links that have both an icon and text to describe their target. CSS is used to display the text's font family, size and foreground and background colors; as well as the spacing between the navigation links.

A logo containing text

A website contains the organization's logo in the top left corner of each web page. The logo contains logotype (text as part, or all, of the logo). The visual presentation of the text is essential to the identity of the logo and is included as a gif image which does not allow the text characteristics to be changed. The image has a text alternative.

Representation of a font family

A web page contains information about a particular font family. Substituting the font family with another font would defeat the purpose of the representation. The representation is included as a jpeg image which does not allow the text characteristics to be changed. The image has a text alternative.

A representation of a letter

A web page contains a representation of an original letter. The depiction of the letter in its original format is essential to information being conveyed about the time period in which it was written. The letter is included as a gif image which does not allow the text characteristics to be changed. The image has a text alternative.

Symbolic text characters

A form allows users to enter blocks of text. The form provides a number of buttons, including functions to style the text and check spelling. Some of the buttons use text characters that do not form a sequence that expresses something in human language. For example "B" to increase font weight, "I" to italicize the text and "ABC" to check the spelling. The symbolic text characters are included as gif images which do not allow the text characteristics to be changed. The buttons have text alternatives.

Customizable font settings in images of text

A website allows users to specify font settings and all images of text on the site are then provided based on those settings.

The text in an image is also provided as text.

A user has to upload an event poster image, which includes text, to their website's events calendar. The site's CMS (content management system) is limited, and won't allow them to create a custom HTML/CSS/SVG recreation of the poster. However, in addition to the image, they can add regular text to the calendar entry, so they post both the poster and the text contained in the image. This text is shown next to the poster image on the site's calendar page.

Resources for Images of Text

- [Fundamental text and font styling](#)
- [Web fonts \(MDN\)](#)

1.4.6 Contrast (Enhanced) (Level AAA)

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The visual presentation of text and images of text has a contrast ratio of at least 7:1, except for the following:

Large Text

Large-scale text and images of large-scale text have a contrast ratio of at least 4.5:1;

Incidental

Text or images of text that are part of an inactive user interface component, that are pure decoration, that are not visible to anyone, or that are part of a picture that contains significant other visual content, have no contrast requirement.

Logotypes

Text that is part of a logo or brand name has no contrast requirement.

In brief

Goal

Text can be seen by people who need strong contrast.

What to do

Strongly contrast text against its background.

Why it's important

Some people cannot read text with minimum contrast.

Intent of Contrast (Enhanced)

The intent of this success criterion is to provide enough contrast between text and its background so that it can be read by people with moderately low vision (who do not use contrast-enhancing

assistive technology). For people without color deficiencies, hue and saturation have minimal or no effect on legibility as assessed by reading performance (Knoblauch et al., 1991). Color deficiencies can affect luminance contrast somewhat. Therefore, in the recommendation, the contrast is calculated in such a way that color is not a key factor so that people who have a color vision deficit will also have adequate contrast between the text and the background.

Text that is decorative and conveys no information is excluded. For example, if random words are used to create a background and the words could be rearranged or substituted without changing meaning, then it would be decorative and would not need to meet this criterion.

Text that is larger and has wider character strokes is easier to read at lower contrast.

The contrast requirement for larger text is therefore lower. This allows authors to

use a wider range of color choices for large text, which is helpful for design of

pages, particularly titles. 18 point text or 14 point bold text is judged to be large

enough to require a lower contrast ratio. (See The American Printing House for the

Blind Guidelines for Large Printing and The Library of Congress Guidelines for Large

Print under Resources). "18 point" and "bold" can both have different meanings in

different fonts but, except for very thin or unusual fonts, they should be sufficient. Since there

are so many different fonts, the general measures are used and a note regarding thin or unusual

fonts is included in the definition for large-scale text.

When evaluating this Success Criterion, the font size in points should be obtained

from the user agent or calculated on font metrics in the way that user agents do.

Point sizes are based on the CSS `pt` size as defined in CSS3 Values . The ratio between

sizes in points and CSS pixels is $1\text{pt} = 1.333\text{px}$, therefore 14pt and 18pt are equivalent to approximately 18.5px and 24px .

Because different image editing applications default to different pixel densities (e.g., 72ppi or 96ppi), specifying point sizes for fonts from within an image editing application can be unreliable when it comes to presenting text at a specific size. When creating images of large-scale text, authors should ensure that the text in the resulting image is roughly equivalent to 1.2 and 1.5 em or to 120% or 150% of the default size for body text. For example, for a 72ppi image, an author would need to use approximately 19pt and 24pt font sizes in order to successfully present images of large-scale text to a user.

The 7:1 and 4.5:1 contrast ratios referenced in this success criterion are intended to be treated as threshold values. When comparing the computed contrast ratio to the Success Criterion ratio, the computed values should not be rounded (e.g., 4.499:1 would not meet the 4.5:1 threshold).

Because authors do not have control over user settings for font smoothing/anti-aliasing, when evaluating this Success Criterion, refer to the foreground and background colors obtained from the user agent, or the underlying markup and stylesheets, rather than the text as presented on screen.

Due to anti-aliasing, particularly thin or unusual fonts may be rendered by user agents with a much fainter color than the actual text color defined in the underlying CSS. This can lead to situations where text has a contrast ratio that nominally passes the Success Criterion, but has a much lower contrast in practice. In these cases, best practice would be for authors to choose a font with stronger/thicker lines, or to aim for a foreground/background color combination that exceeds the normative requirements of this success criterion.

The contrast requirements for text also apply to images of text (text that has been rendered into pixels and then stored in an image format) - see [Success Criterion 1.4.5 Images of Text](#) .

This requirement applies to situations in which images of text were intended to be understood as text. Incidental text, such as in photographs that happen to include a street sign, are not included. Nor is text that for some reason is designed to be invisible to all viewers. Stylized text, such as in corporate logos, should be treated in terms of its function on the page, which may or may not warrant including the content in the text alternative. Corporate visual guidelines beyond logo and logotype are not included in the exception.

In this provision there is an exception that reads "that are part of a picture that contains significant other visual content,". This exception is intended to separate pictures that have text in them

from images of text that are done to replace text in order to get a particular look.

Images of text do not scale as well as text because they tend to pixelate. It is also

harder to change foreground and background contrast and color combinations for images

of text, which is necessary for some users. See [1.4.5 Images of Text](#) .

This success criterion applies to text in the page, including placeholder text and text that is shown when a pointer is hovering over an object or when an object has keyboard focus. If any of these are used in a page, the text needs to provide sufficient contrast.

Although this success criterion only applies to text, similar issues occur for content presented

in charts, graphs, diagrams, and other non-text-based information, which is covered by [Success Criterion 1.4.11 Non-Text Contrast](#) .

Rationale for the Ratios Chosen

A contrast ratio of 3:1 is the minimum level recommended by [[ISO-9241-3]] and [[ANSI-HFES-100-1988]] for standard text and vision. The 4.5:1 ratio is used in Success Criterion 1.4.3 Contrast (Minimum) to account for the loss in contrast that results from moderately low visual acuity, congenital or acquired color deficiencies, or the loss of contrast sensitivity that typically accompanies aging.

The rationale is based on a) adoption of the 3:1 contrast ratio for minimum acceptable contrast for normal observers, in the ANSI

standard, and b) the empirical finding that in the population, visual acuity of 20/40 is associated with a contrast sensitivity loss of roughly 1.5 [\[\[ARDITI-FAYE\]\]](#). A user with 20/40 would thus require a contrast ratio of $3 * 1.5 = 4.5$ to 1. Following analogous empirical findings and the same logic, the user with 20/80 visual acuity would require contrast of about 7:1. This ratio is used in this success criterion.

Hues are perceived differently by users with color vision deficiencies (both congenital and acquired) resulting in different colors and relative luminance contrasts than for normally sighted users. Because of this, effective contrast and readability are different for this population. However, color deficiencies are so diverse that prescribing effective general use color pairs (for contrast) based on quantitative data is not feasible. Requiring good luminance contrast accommodates this by requiring contrast that is independent of color perception. Fortunately, most of the luminance contribution is from the mid and long wave receptors which largely overlap in their spectral responses. The result is that effective luminance contrast can generally be computed without regard to specific color deficiency, except for the use of predominantly long wavelength colors against darker colors (generally appearing black) for those who have protanopia. (We provide an advisory technique on avoiding red on black for that reason). For more information see [\[\[ARDITI-KNOBLAUCH-1994\]\]](#) [\[\[ARDITI-KNOBLAUCH-1996\]\]](#) [\[\[ARDITI\]\]](#).

Some people with cognitive disabilities require color combinations or hues that have low contrast, and therefore we allow and encourage authors to provide mechanisms to adjust the foreground and background colors of the content. Some of the combinations that could be chosen may have contrast levels that will be lower than those specified here. This is not a violation of

this Success Criterion, provided there is a mechanism that will return to the required values set out here.

The contrast ratio of 4.5:1 was chosen for level AA because it compensated for the loss in contrast sensitivity usually experienced by users with vision loss equivalent to approximately 20/40 vision. (20/40 calculates to approximately 4.5:1.) 20/40 is commonly reported as typical visual acuity of elders at roughly age 80. [[GITTINGS-FOZARD]]

The contrast ratio of 7:1 was chosen for level AAA because it compensated for the loss in contrast sensitivity usually experienced by users with vision loss equivalent to approximately 20/80 vision. People with more than this degree of vision loss usually use assistive technologies to access their content (and the assistive technologies usually have contrast enhancing, as well as magnification capability built into them). The 7:1 level therefore generally provides compensation for the loss in contrast sensitivity experienced by users with low vision who do not use assistive technology and provides contrast enhancement for color deficiency as well.

Calculations in [[ISO-9241-3]] and [[ANSI-HFES-100-1988]] are for body text. A relaxed contrast ratio is provided for text that is much larger.

Notes on formula

Conversion from nonlinear to linear RGB values is based on IEC/4WD 61966-2-1 [[IEC-4WD]].

The formula (L_1/L_2) for contrast is based on [[ISO-9241-3]] and [[ANSI-HFES-100-1988]] standards.

The ANSI/HFS 100-1988 standard calls for the contribution from ambient light to be included in the calculation of L_1 and L_2 . The

.05 value used is based on Typical Viewing Flare from [[IEC-4WD]].

This success criterion and its definitions use the terms "contrast ratio" and "relative luminance" rather than "luminance" to reflect the fact that web content does not emit light itself. The contrast ratio gives a measure of the relative luminance that would result when displayed. (Because it is a ratio, it is dimensionless.)

Refer to related resources for a list of tools that utilize the contrast ratio

to analyze the contrast of web content.

See also [2.4.7 Focus Visible](#) for techniques for indicating keyboard focus.

Inactive User Interface Components

User Interface Components that are not available for user interaction (e.g., a disabled control in HTML) are not required to meet contrast requirements. An inactive user interface component is visible but not currently operable. An example would be a submit button at the bottom of a form that is visible but cannot be activated until all the required fields in the form are completed.

Benefits of Contrast (Enhanced)

- People with low vision often have difficulty reading text that does not contrast with its background. This can be exacerbated if the person has a color vision deficiency that lowers the contrast even further. Providing a minimum luminance contrast ratio between the text and its background can make the text more readable even if the person does not see the full range of colors. It also works for the rare individuals who see no color.

Resources for Contrast (Enhanced)

- [Colour Contrast Analyser application](#)
- [Luminosity Colour Contrast Ratio Analyser](#)
- [Colour Contrast Check](#)
- [Contrast Ratio Calculator](#)
- [Adobe Color - Color Contrast Analyzer Tool](#)
- [Atypical colour response](#)
- [Colors On the Web Color Contrast Analyzer](#)
- [Reading with Dyslexia - Fonts that can help alleviate visual stress.](#)
- [Good Fonts for Dyslexia - An Experimental Study](#)

1.4.7 Low or No Background Audio (Level AAA)

Source: [WCAG 2.2](#) / [Understanding 1.4.7](#) — Copyright © 2023 World Wide Web Consortium

For prerecorded audio-only content that (1) contains primarily speech in the foreground, (2) is not an audio CAPTCHA or audio logo, and (3) is not vocalization intended to be primarily musical expression

such as singing or rapping, at least one of the following is true:

No Background

The audio does not contain background sounds.

Turn Off

The background sounds can be turned off.

20 dB

The background sounds are at least 20 decibels lower than the foreground speech content, with the exception of occasional sounds that last for only one or two seconds. Per the definition of "decibel," background sound that meets this requirement will be approximately four times quieter than the foreground speech content.

In brief

Goal

Prerecorded speech is not disrupted by background sound.

What to do

Avoid or lessen background sound, or let users turn it off.

Why it's important

People who are hard of hearing may have difficulty distinguishing speech from music and other sounds.

Intent of Low or No Background Audio

The intent of this success criterion is to ensure that any non-speech sounds are low enough that a user who is hard of hearing can separate the speech from background sounds or other noise foreground speech content.

The value of 20 dB was chosen based on Large area assistive listening systems (ALS): Review and recommendations [[LAALS]] and In-the-ear measurements of interference in hearing aids from digital wireless telephones [[HEARING-AID-INT]]

Benefits of Low or No Background Audio

- People who are hard of hearing often have great difficulty separating speech from background sound.

Resources for Low or No Background Audio

- [About Decibels](#)

1.4.8 Visual Presentation (Level AAA)

Source: [WCAG 2.2](#) / [Understanding 1.4.8](#) — Copyright © 2023 World Wide Web Consortium

For the visual presentation of blocks of text , a mechanism is available to achieve the following:

- Foreground and background colors can be selected by the user.
- Width is no more than 80 characters or glyphs (40 if CJK).
- Text is not justified (aligned to both the left and the right margins).
- Line spacing (leading) is at least space-and-a-half within paragraphs, and paragraph spacing is at least 1.5 times larger than the line spacing.
- Text can be resized without assistive technology up to 200 percent in a way that does not require the user to scroll horizontally to read a line of text on a full-screen window .

Content is not required to use these values. The requirement is that a mechanism is available for users to change these presentation aspects. The mechanism can be provided by the browser or other user agent. Content is not required to provide the mechanism.

Writing systems for some languages use different presentation aspects to improve readability and legibility. If a presentation aspect in this success criterion is not used in a writing system, content in that writing system does not need to use that presentation setting and can conform without it. Authors are encouraged to follow guidance for improving readability and legibility of text in their writing system.

In brief

Goal

Text appearance can be altered by users to meet preferences.

What to do

Meet text display requirements or allow users to adjust them.

Why it's important

Some text formats are more readable for people with cognitive disabilities and low vision.

Intent of Visual Presentation

The intent of this success criterion is to ensure that visually rendered text is presented in such a manner that it can be perceived without its layout interfering with its readability. People with some cognitive, language and learning disabilities and some low vision users cannot perceive the text and/or lose their reading place if the text is presented in a manner that is difficult for them to read.

People with some visual or cognitive disabilities need to be able to select the color of text and the color of the background. They sometimes choose combinations that seem unintuitive to someone without that disability. Sometimes these combinations have very low contrast. Sometimes only very specific color combinations work for them. Control of color or other aspects of text presentation makes a huge difference to their comprehension.

For people with some reading or vision disabilities, long lines of text can become a significant barrier. They have trouble keeping their place and following the flow of text. Having a narrow block of text makes it easier for them to continue on to the next line in a block. Lines should not exceed 80 characters or glyphs (40 if CJK), where glyphs are the element of writing in the writing system for the text. Studies have shown that Chinese, Japanese and Korean (CJK) characters are approximately twice as wide as non-CJK

characters when both types of characters are displayed with characteristics that achieve the same readability, so the maximum line width for CJK characters is half that of non-CJK characters.

People with some cognitive disabilities find it difficult to track text where the lines are close together. Providing extra space between lines and paragraphs allows them to better track the next line and to recognize when they have reached the end of a paragraph. It is best if there are several different options, for instance, space-and-a-half and double spacing for line spacing. By space and a half within paragraphs we mean that top of one line is 150% further from the top of the line below it than would be true when the text is 'single spaced' (the default spacing for the font). By Paragraph spacing that is 1.5 times larger than the line spacing we mean that the spacing from the top of the last line of 1 paragraph is 250% farther from the Top of the first line of the next paragraph (i.e., that there is a blank line between the two paragraphs that is 150% of the single space blank line).

People with certain cognitive disabilities have problems reading text that is both left and right justified. The uneven spacing between words in fully justified text can cause "rivers of white" space to run down the page making reading difficult and in some cases impossible. Text justification can also cause words to be spaced closely together, so that it is difficult for them to locate word boundaries.

The resizing provision ensures that visually rendered text, including controls and labels using text, can be made larger without requiring the user to then scroll left and right to see all of the content. When the content has been authored so that this is possible, the content is said to reflow. This permits people with low vision and people with cognitive disabilities to increase the size of the text without becoming disoriented.

The scaling of content is primarily a user agent responsibility. User agents that

satisfy UAAG 1.0 Checkpoint 4.1 allow users to configure text scale. The author's

responsibility is to create web content that does not prevent the user agent from

scaling the content and that allows the reflow of the content within the current

width of the viewport. See [1.4.4 Resize Text](#) for additional discussion of resizing text.

The horizontal scrolling requirement is not intended to apply to small-screen devices where long words may be displayed on a single line and require users to scroll horizontally. For the purposes of this requirement, authors should ensure that content meets this requirement on standard desktop/laptop displays with the browser window maximized. Since people generally keep their computers for several years, it is best not to rely on the latest desktop/laptop display resolutions but to consider the common desktop/laptop display resolutions over the course of several years when making this evaluation.

Wrapping should always be possible as long as words are not so long that a single word is more than half the width of a full screen. Very long URIs may run off the side of an enlarged screen, but they would not be considered text for "reading" and, therefore, would not violate this provision.

This provision does not mean that a user would never need to use horizontal scrolling. It only means that they would not need to use horizontal scrolling back and forth to read a line of text. For example, if a page consisted of two equal sized columns of text, it

would automatically meet this provision. Enlarging the page would mean that the first column was completely on screen and the user could just scroll vertically down the page to read it. To read the second column, they would horizontally scroll to the right, where the right hand column would then fit entirely within the width of the screen, and read that column without further horizontal scrolling.

Benefits of Visual Presentation

This success criterion helps low vision users by letting them see text without distracting presentational features. It lets them configure text in ways that will be easier for them to see by letting them control the color and size of blocks of text.

This success criterion helps people with cognitive, language and learning disabilities perceive text and track their location within blocks of text.

- People with some cognitive disabilities can read text better when they select their own foreground and background color combinations.
- People with some cognitive disabilities can track their locations more easily when blocks of text are narrow and when they can configure the amount of space between lines and paragraphs.
- People with some cognitive disabilities can read text more easily when the spacing between words is regular.

Examples of Visual Presentation

Examples of glyphs include "A", "→" (an arrow symbol), and "さ" (a Japanese character).

Resources for Visual Presentation

- [CSS 2 Box Model](#)

- [CSS 2 Visual formatting Model](#)
- [CSS 2 Visual formatting Model Details](#)
- [About fluid and fixed width layouts](#)
- [Accessible CSS](#)
- [Practical Typography - Line Length](#)
- [Developing sites for users with Cognitive disabilities and learning difficulties](#)
- [RDFA Primer](#)
- [MULTIFUNK: Bringing computer-supported reading one step further](#) , Date: April 2002, ISBN: 82-539-0491-6, Author: Gjertrud W. Kamstrup, Eva Mjøvik,

Anne-Lise Rygvold og Bjørn Gunnar Saltnes

- [Effective Monitor Display Design](#) on the ERIC web portal
- [Cognitive difficulties and access to information systems - an interaction design perspective](#) ", Peter Gregor and Anna Dickinson, Applied Computing, University of Dundee
- Legge, G.E., Pelli, D.G., Rubin, G.S., & Schleske, M.M.:Psychophysics of reading. I. Normal Vision,Vision Research, 25, 239-252, 1985.
- Legge, G.E., Rubin, G.S., Pelli, D.G., & Schleske, M.M.:Psychophysics of reading. II. Low Vision,Vision Research, 25, 253-266, 1985.
- Osaka,N. and Oda, K. (1991). Effective visual field size necessary for vertical reading during Japanese text processing. Bulletin of Psychonomic Society,29(4),345-347.
- Beckmann, P.J. & Legge, G.E. (1996). Psychophysics of reading. XIV. The page-navigation problem in using magnifiers. Vision Research, 36, 3723-3733.
- 川嶋英嗣・小田浩一(2003).読書におけるスクロール方向とウィンドウ幅の影響 日本心理学会第 67 回大会, 502.

- 小田浩一・今橋真理子(1995). 文字認知の閾値と読みの閾値. VISION, 7, 165-168.
- Osaka,N. (1994). Size of saccade and fixation duration of eye movements during reading: psychophysics of Japanese text processing. Journal of Optical Society of America A, 9(1), 5-13.
- 山中今日子・小田浩一 (2007). 漢字の画数と書体のウェイトが視認性に及ぼす影響. 視覚学会 2007 年夏季大会ポスター 1p1 Vision, P.167.
- Line Length, Volume, and Density
- Guidance on accessible publishing
- An Accessibility Frontier: Cognitive disabilities and learning difficulties
- Cognitive/Perceptual Difference And Good Web Design
- 6 Surprising Bad Practices That Hurt Dyslexic Users
- Design for Dyslexics
- Web Design for Dyslexia

1.4.9 Images of Text (No Exception) (Level AAA)

Source: [WCAG 2.2](#) / [Understanding 1.4.9](#) — Copyright © 2023 World Wide Web Consortium

Images of text are only used for pure decoration or where a particular presentation of text is essential to the information being conveyed.

Logotypes (text that is part of a logo or brand name) are considered essential.

In brief

Goal

Users can always adjust how text is presented.

What to do

Do not use pictures of text unless there is no other way to present information.

Why it's important

People cannot alter how text looks in images.

Intent of Images of Text (No Exception)

The intent of this success criterion is to enable people who require a particular visual presentation of text to be able to adjust the text presentation as required. This includes people who require the text in a particular font size, foreground and background color, font family, line spacing or alignment.

This means implementing the text in a manner that allows its presentation to be changed or providing a mechanism by which users can select an alternate presentation. Using images of text is an example of an implementation that does not allow users to alter the presentation of the text within it.

In some situations, a particular visual presentation of the text is essential to the information being conveyed. This means that information would be lost without that particular visual presentation. In this case implementing the text in a manner that allows its presentation to be changed is not required. This includes text that demonstrates a particular visual aspect of the text, such as a particular font family, or text that conveys an identity, such as text within a company logo.

Text that is decorative does not require implementing the text in a manner that allows its presentation to be changed.

The definition of image of text contains the note: Note: This does not include text that is part of a picture that contains significant other visual content. Examples of such pictures include graphs, screenshots, and diagrams which visually convey important information through more than just text.

Benefits of Images of Text (No Exception)

- People with low vision (who may have trouble reading the text with the authored font family, size and/or color).
- People with visual tracking problems (who may have trouble reading the text with the authored line spacing and/or alignment).
- People with cognitive disabilities that affect reading.

Examples of Images of Text (No Exception)

A quote

A web page contains a quote. The quote itself is presented as italicized text, indented from the left margin. The name of the person to whom the quote is attributed is below the quote with 1.5x the line space and further indented from the left margin. CSS

is used to position the text; set the spacing between lines; as well as display the text's font family, size, color and decoration.

Navigation items

A web page contains a menu of navigation links that have both an icon and text to describe their target. CSS is used to display the text's font family, size and foreground and background colors; as well as the spacing between the navigation links.

A logo containing text

A website contains the organization's logo in the top left corner of each web page. The logo contains logotype (text as part, or all, of the logo). The visual presentation of the text is essential to the identity of the logo and is included as a gif image which does not allow the text characteristics to be changed. The image has a text alternative.

Representation of a font family

A web page contains information about a particular font family. Substituting the font family with another font would defeat the purpose of the representation. The representation is included as a jpeg image which does not allow the text characteristics to be changed. The image has a text alternative.

A representation of a letter

A web page contains a representation of an original letter. The depiction of the letter in its original format is essential to information being conveyed about the time period in which it was written. The letter is included as a gif image which does not allow the text characteristics to be changed. The image has a text alternative.

Symbolic text characters

A form allows users to enter blocks of text. The form provides a number of buttons, including functions to style the text and check spelling. Some of the buttons use text characters that do not form a sequence that expresses something in human language. For example "B" to increase font weight, "I" to italicize the text and "ABC" to check the spelling. The symbolic text characters are included as gif images which do not allow the text characteristics to be changed. The buttons have text alternatives.

Resources for Images of Text (No Exception)

- [CSS @ Ten: The Next Big Thing \(A List Apart\)](#)
- [Weblog comments: WebKit now supports CSS @font-face rules](#)
- [Creating Cross Browser Compatible CSS Text Shadows](#)
- [CSS and text](#)

1.4.10 Reflow (Level AA)

Source: [WCAG 2.2](#) / [Understanding 1.4.10](#) — Copyright © 2023 World Wide Web Consortium

Content can be presented without loss of information or functionality, and without requiring scrolling in two dimensions for:

- Vertical scrolling content at a width equivalent to 320 CSS pixels ;
- Horizontal scrolling content at a height equivalent to 256 CSS pixels .

Except for parts of the content which require two-dimensional layout for usage or meaning.

320 CSS pixels is equivalent to a starting viewport width of 1280 CSS pixels wide at 400% zoom. For web content which is designed to scroll horizontally (e.g., with vertical text), 256 CSS pixels is equivalent to a starting viewport height of 1024 CSS pixels at 400% zoom.

Examples of content which requires two-dimensional layout are images required for understanding (such as maps and diagrams), video, games, presentations, data tables (not individual cells), and interfaces where it is necessary to keep toolbars in view while manipulating content. It is acceptable to provide two-dimensional scrolling for such parts of the content.

In brief

Goal

Content can be enlarged without increasing line length.

What to do

Make lines of text reflow within the viewport.

Why it's important

People who need bigger text find it difficult if they must scroll to read long lines.

Intent of this Success Criterion

The intent of this success criterion is to let users enlarge text and other related content without having to scroll in two dimensions to read. When lines of text extend beyond the edge of a viewport , users will be forced to scroll back-and-forth to read line by line. This can cause them to lose their place and can significantly increase both physical and cognitive effort. Therefore, most sections of content are expected to reflow within the appropriate sizing requirement defined by this success criterion.

A section of content that requires two-dimensional layout for understanding or functionality, such as a table or map, has an exception to this success criterion. However, sections of content within the two-dimensional layout, such as each cell within a table, would still need to meet this success criterion. Although there is an exception for sections of content that require two-dimensional layout for understanding or functionality, authors can improve the user's experience by making efforts to reduce scrolling for that type of content.

Basic text reflow

The most basic form of reflow occurs on a text-only page. By default, a long line of text will wrap to fit within the available viewport. For text written in left-to-right (LTR) and right-to-left (RTL) horizontal languages, it will wrap within the width of the viewport. For languages that are written vertically, the text will wrap within the height of the viewport.

Users who need to enlarge (zoom) text to read it benefit significantly when it reflows. Regardless of the size of the text, it

continues to wrap within the visible viewport. A whole line of text is visible, making it easier for the eye to track from the end of one line to the start of the next. Users only need to scroll in the direction of reading to read additional lines of text.

This criterion requires non-excepted sections of content that are written in horizontal languages reflow when narrowed to a width equivalent to 320 CSS pixels. Similarly, sections of content that are written in vertical languages (which are not as commonly used on the web and so will be used much less frequently in examples) need to be able to reflow when rendered at a height equivalent to 256 CSS pixels.

Reflowing websites and web applications

Modern websites and applications commonly employ responsive web design best practices to adjust or relocate sections of content to fit within smaller viewports. Neither adjusting or relocating content is considered a loss of information or functionality, so long as users are still able to access the content. When content can fit within smaller viewports it not only helps those using mobile devices to read content. It also helps people who need to resize or zoom in a web interface on larger devices, as is the intent of this success criterion.

Adjusting content into a single column

A common way for many article-driven web pages to meet the Reflow success criterion is to ensure that the presentation of a web page can adjust to a single column of content, fitting into a 320 CSS pixel wide viewport and only requiring a user scroll vertically to read the content of the web page.

A "stacking into a single column" approach will work for many web pages whose primary objective is to present sections of content for people to read, such as articles with tangential

sidebars, or home pages that present distinct sections of teaser content in a visual grid or masonry layout - where these layouts are not essential to the understandability of the content. However, this approach may not work for all instances of common, but complex, widgets and layouts which would become more difficult to visually understand or interact with if they were simply stacked to fit in a single column layout.

Content that can benefit from two-dimensional layout for meaning or functionality

Building complex web interfaces will often involve incorporating complex layouts and widgets which often need modified layout, functional adjustments or both when content is enlarged.

Conforming to this success criterion doesn't mean all sections of content need to stack in a single column within smaller viewports. Nor does it require all sections of content scroll in a single direction, so long as each can be read without two-dimensional scrolling to read the lines of text they present.

Although the following examples could be designed to work without horizontal scrolling, they have been designed to demonstrate conforming to the success criterion.

Lists and blocks of code often rely on indentation of text to visually convey hierarchy and meaning. If the indentation were to be removed, then meaning or functionality could suffer. For instance, the nesting of unordered list hierarchies would become more difficult to notice if indentation were removed. Additionally, maintaining indentations and non-breaking lines can be important to help understand code patterns, if not even necessary for the code to function.

See technique: [Accounting for meaningful text indentation and Reflow](#) for additional details.

Within vertically scrolling web pages, implementations of carousels can allow for horizontal scrolling of their content. As the carousel scrolls, a new carousel panel or panels are displayed within the visible viewport. As long as each individual panel within the carousel can fit within a 320 CSS pixel viewport, then a user need only scroll in a single direction to read an individual panel's content.

See technique: [Section panels that scroll horizontally are designed to fit within a width of 320 CSS pixels on a vertically scrolling page](#) for additional details.

Being able to review differences in code, or text, side-by-side can be an instance where a scrolling two-column layout can benefit users.

Content that meets exceptions for Reflow

For many traditional web pages, where the page's primary goal is for users to read distinct [sections of content](#) containing multi-line paragraphs of text, the web page commonly scrolls vertically (when written in western languages). People with low vision might need to zoom in the web page to read its content. When they do this using browser zooming the content becomes larger, but the viewport size decreases.

Not all content can fully reflow into smaller viewports without degradation of information or functionality. As already mentioned, this includes tabular data, graphics, maps, presentations, interfaces that require persistent toolbars for use, and two-dimensional layouts where consistent orientation to related sections of content is important for understandability, functionality, or both.

If a section of content meets the exception for two-dimensional scrolling, the exception only applies to that section. When a

section of content is excepted from Reflow, the exception does not automatically extend to other content that doesn't need two-dimensional scrolling for understanding or functionality.

A data table relies on two-dimensional layout for understanding, but by presenting the table in its own scrollable container it allows other content which does not meet a two-dimensional layout exception to reflow as its containing element adjusts. for example, due to browser resizing, or zooming.

There can be instances where two-dimensional excepted content extends beyond the visible viewport, causing the entire page to scroll. At the same time, other non-excepted content on the same page reflows within the visible viewport. Unlike the prior table example with the paragraph not reflowing, such a situation does pass. Reflow does not prohibit web pages from presenting both horizontal and vertical scrollbars for individual sections of content. Nor does it disallow the use of bidirectional scrollbars at the page (viewport) level in order to support the viewing of excepted content, so long as the non-excepted content only needs scrolling in one direction.

Whenever possible it is in the best interest of the user to limit two-dimensional scrolling only to the individual sections of content which necessitate such scrolling, rather than allowing the page at large to scroll in two dimensions. For example, an unnecessary horizontal scrollbar could lead a user to believe there is content existing off-screen for them to scroll to. But if that scrollbar appears only because of one instance of excepted content, a user could expend effort looking for *other* non-reflowing content which doesn't exist.

Graphics and video are by their nature two-dimensional. Slicing up a graphic (photograph, drawing, graph, etc.) and stacking the blocks would make the graphic difficult to understand, if not

rendering it unintelligible. However, it is possible to have these elements stay within the bounds of the viewport even as other content zooms to 400%.

Please review the advisory technique [Using CSS max-width and height to fit images](#) for more information.

Data tables and grids have a two-dimensional relationship between column and row headers and their data cells. This success criterion therefore has exceptions for data tables and grids from needing to display without scrolling in the direction of text. However, individual cells would still need to meet Reflow - unless the cell contains content that also requires two-dimensional layout for usage or meaning.

The exception for tables and grids covers content beyond what typically is considered "tabular data". For example, an "[electronic program guide](#)" is a type of grid used to display media programs to stream online. It might be presented alongside other content outside of its grid-based layout. The individual cells of the program guide containing information about how to watch and poster art would be expected to fit within the required width or height.

Other content that is related to the table or grid, such as a preceding heading, a search field, or an accompanying pagination to load different sets of data are not excepted from meeting Reflow. For instance, while a table may require two-dimensional scrolling to maintain its understandability, a heading or paragraph that introduce the table, or a search field and pagination component that allow users to search for or navigate to different "pages" of table content are not extended this exception. They would still be expected to adjust to smaller viewports, so as to reduce the need for users to unnecessarily scroll the UI to find and interact with the non-excepted content.

Interfaces which provide toolbars to edit content need to show both the content and the toolbar in the viewport. Depending on the number of toolbar buttons, the toolbar may need to scroll in the direction of text, or might even need to remain fully visible and scroll along with the rich text content or editable canvas area that it provides features for editing.

Complex web application interfaces allow users to create content set to defined dimensions. For instance, presentations or documents can be based on physical printing or display sizes. Complex interfaces often allow for content to be created that is directly related content to, but not a part of, the primary author-created content. For instance, presentation notes, editing comments/suggestions for a text document, dynamic messaging presented along with a message (chat) interface, etc. Each of these related sections of content often need to be presented in parallel to each other, and their understandability can diminish if their overall location were to change.

Finally, interfaces which represent the creation of or rendering of presentations created by users (for instance, creating or presenting slide decks) often cannot simply reflow content, as doing so could easily result in breakages to the presentation of the author's content.

Instead, it would be expected that such interfaces provided content authors with a means to create an alternative presentation that does meet the requirements of this success criterion.

Why specifically 320px and 256px?

The value of 320 CSS pixels for vertically scrolling content was chosen as a reasonable minimum size that authors can achieve. This value aligned with reported viewport width of small displays of common mobile devices that were available when this criterion

was originally drafted. The 320 CSS pixels generally corresponds to a desktop browser window set to a width of 1280px and the browser viewport then zoomed to 400%. It should be noted that 400% applies to the dimension, not the area. It means four times the default zoom level viewport width and four times the default zoom level height.

When we read, the size of the print is not as important as the image it projects on the retina of our eye. Phones are designed for close viewing while desktops are designed for viewing farther away. As a consequence, 16px print on a phone is physically smaller than 16px print on a desktop. This is generally not a problem because both print sizes cast the same image on our retina if they are viewed at their intended distance. The CSS reference pixel, the px unit used on the web, reflects this. The CSS reference unit is an angular measurement (1.278 arc minutes), not a length measurement. Most older people, and some people with low vision hold their phone closer than is typical, but again, by using the CSS reference pixel, developers (and designers and content authors) have stable metrics for their design choices.

Acknowledging viewport vs resolution discrepancies

As technology and user interaction with technology evolves, adhering to Reflow becomes less about if there is a specific resolution or browser zoom level that neatly aligns with the sizing requirements of this success criterion, but rather that sections of content **can** be presented in a viewport equivalent to those sizing requirements. For instance, someone with a browser window docked to one half of their display could have a viewport width of 960 CSS pixels. A user could then zoom the browser 300% and achieve a viewport of 320 CSS pixels wide.

Additionally, not all users who can benefit from Reflow will be viewing web pages at a screen resolution that neatly zooms to

either of Reflow's sizing requirements. Rather, an operating system's display resolution and the size of one's browser viewport will commonly not align. For instance, if one were to set their desktop computer's display resolution to 1280 by 1024 pixels, the browser's viewport would be smaller than these dimensions since the operating system and browser UI would detract from the available screen real estate. Some operating systems might not even provide a display resolution that neatly scales to the 320 or 256 CSS pixel dimensions, regardless of zoom level.

Not all user agents will support this SC's normative sizing requirements

An important factor in being able to support Reflow is for a user agent to allow the adjustment of content within specific windows of content, if not the size of the window itself. When Reflow was originally drafted, most mobile operating system browsers handled content zooming and reflow differently than on traditional desktop browsers. For instance, the content of a web page could reflow only when the mobile device's orientation changed. This was because for many mobile browsers at the time, zooming a web page was more of a magnification feature. “Pinch zoom” gestures or double-tap to enlarge would magnify the web page content, allowing a user to then swipe or drag the web page so portions of it could be visible by the magnified screen.

These days, more browsers on mobile devices or other devices that allow users to open applications at fixed dimensions (e.g., full screen or split screen) allow for scaling features that are more in-line with traditional desktop browser zooming. Or, these devices have incorporated zooming features at the OS level that can allow at least some level of reflowing of content for web pages. Many of these features, however, do not commonly adjust the fixed-dimension browser viewport to the 320 CSS pixel width for

vertically scrolling content, or 256 CSS pixel height for horizontally scrolling content.

While one might regard the inability to resize fixed-dimension browsers to the 320 CSS pixel or 256 CSS pixel sizes used by this criterion a user agent support issue, it is not the intent of this criterion to mandate devices need to support these specific sizes for presenting content, nor is it to be assumed that if content cannot be presented within such sizes on specific devices that the web page would pass or fail Reflow. If a web page can be viewed on a device where the sizing requirements of this criterion can be accurately tested, then it is on such devices where a web page's ability to meet Reflow is best to be determined.

Resize Text

If zooming in a web page by 200% results in a viewport that is smaller than the sizing requirements of Reflow, then two-dimensional scrolling may occur. This would not be a failure of Reflow, as the sizing of the viewport would be smaller than what Reflow requires.

Focus Not Obscured (Minimum)

Beyond this sticky advertisement example, commonly toolbars, menubars, navigation and other "sidebar" content may be presented with sticky or fixed positions at larger viewport sizes. It is strongly suggested that at smaller viewport sizes that such components are modified to have static positioning, or their display can be toggled by the user. Doing so will help ensure the zoomed in content can be read by users, as the sticky components will no longer obstruct the view of the web page's content.

Examples

- One column view in responsive design Note that as the zoom percentage increases, the navigation changes first to hide options behind a "More" dropdown menu. As zooming continues, most navigation options are eventually behind a "hamburger" menu button. All the information and functionality is still available from this web page. There is no horizontal scrolling.
- PDF offering reflow In a PDF created to conform to PDF/Universal Accessibility (ISO 14289), the content can be reflowed and zoomed in to make reading possible for someone with low-vision.
- Alternative presentations to truncating content A web page presents long strings of text which are truncated to save space. E.g., user generated content that does not fit into the space allocated for the interface's design, or authentication keys which do not wrap, etc.. The content is presented as truncated, but a link is provided to a web page where the content is fully visible without truncation, or a mechanism is provided on the web page to reveal the truncated content.
- Preformatted text conveys meaning The presentation of text where the layout has specific meaning, such as code indentation for Python or "ascii art" as just two examples, would lose meaning if the layout were not presented correctly. This success criterion does not apply where that meaning would be lost. However, this is not the case for most other instances of text where text wrapping can be applied without loss of meaning. Additionally, for instances of indentation that convey meaning, consider reducing the size of the indentation at zoomed in levels. As the text will be bigger, the reduced indentation width can still be noticeable.

Resources

- Reading with Low Vision in a Digital Setting: by Wayne Dick, PhD.
- Operational Overhead Caused by Horizontal Scrolling Text by Wayne Dick, 2017. The study shows the impact of horizontal scrolling on reading effort
- Accessibility Requirements for People with Low Vision . W3C First Public Working Draft 17 March 2016
- Responsive design resources from MDN web docs
- Responsive web design basics tutorial from Google

1.4.11 Non-text Contrast (Level AA)

Source: [WCAG 2.2](#) / [Understanding 1.4.11](#) — Copyright © 2023 World Wide Web Consortium

The visual presentation of the following have a contrast ratio of at least 3:1 against adjacent color(s):

User Interface Components

Visual information required to identify user interface components and states , except for inactive components or where the appearance of the component is determined by the user agent and not modified by the author;

Graphical Objects

Parts of graphics required to understand the content, except when a particular presentation of graphics is essential to the information being conveyed.

In brief

Goal

Important visual information meets the same minimum contrast required for larger text.

What to do

Ensure meaningful visual cues achieve 3:1 against the background.

Why it's important

Some people cannot see elements with low contrast.

Intent

The intent of this success criterion is to ensure that user interface components (i.e., controls) and meaningful graphics are distinguishable by people with moderately low vision. The requirements and rationale are similar to those for large text in

[1.4.3 Contrast \(Minimum\)](#) . Note that this requirement does not apply to **inactive** user interface components.

Low contrast controls are more difficult to perceive, and may be completely missed by people with a visual impairment. Similarly, if a graphic is needed to understand the content or functionality of the web page then it should be perceivable by people with low vision or other impairments without the need for contrast-enhancing assistive technology.

The 3:1 contrast ratios referenced in this success criterion is intended to be treated as threshold values. When comparing the computed contrast ratio to the success criterion ratio, the computed values should not be rounded (e.g. 2.999:1 would not meet the 3:1 threshold).

Because authors do not have control over user settings for font smoothing and anti-aliasing, when evaluating this Success Criterion, refer to the colors obtained from the user agent, or the underlying markup and stylesheets, rather than the non-text elements as presented on screen.

Due to anti-aliasing, particularly thin lines and shapes of non-text elements may be rendered by user agents with a much fainter color than the actual color defined in the underlying CSS. This can lead to situations where non-text elements have a contrast ratio that nominally passes the Success Criterion, but have a much lower contrast in practice. In these cases, best practice would be for authors to avoid particularly thin lines and shapes, or to use a combination of colors that exceeds the normative requirements of this success criterion.

User Interface Components

Unless the control is *inactive* , any visual information provided that is necessary for a user to identify that a control is present and

how to operate it must have a minimum 3:1 contrast ratio with the adjacent colors. Also, any visual information necessary to indicate state, such as whether a component is selected or focused must also ensure that the information used to identify the control in that state has a minimum 3:1 contrast ratio.

This success criterion does not require that changes in color that differentiate between states of an individual component meet the 3:1 contrast ratio when they do not appear next to each other. For example, there is not a new requirement that visited links contrast with the default color, or that mouse hover indicators contrast with the default state. However, the component must not lose contrast with the adjacent colors, and non-text indicators such as the check in a checkbox, or an arrow graphic indicating a menu is selected or open must have sufficient contrast to the adjacent colors.

This success criterion does not require that controls have a visual boundary indicating the hit area, but if the visual indicator of the control is the only way to identify the control, then that indicator must have sufficient contrast. If text (or an icon) within a button or placeholder text inside a text input is visible and there is no visual indication of the hit area then the success criterion is passed. If a button with text also has a colored border, since the border does not provide the only indication there is no contrast requirement beyond the text contrast ([1.4.3 Contrast \(Minimum\)](#)). Note that for people with cognitive disabilities it is recommended to delineate the boundary of controls to aid in the recognition of controls and therefore the completion of activities.

For user interface components 'adjacent colors' means the colors adjacent to the component. For example, if an input has a white internal background, dark border, and white external background

the 'adjacent color' to the component would be the white external background.

If components use several colors, any color which does not interfere with identifying the component can be ignored for the purpose of measuring contrast ratio. For example, a 3D drop-shadow on an input, or a dark border line between contrasting backgrounds is considered to be subsumed into the color closest in brightness (perceived luminance).

The following example shows an input that has a light background on the inside and a dark background around it. The input also has a dark grey border which is considered to be subsumed into the dark background. The border does not interfere with identifying the component, so the contrast ratio is taken between the white background and dark blue background.

For visual information required to identify a state, such as the check in a checkbox or the thumb of a slider, that part might be within the component so the adjacent color might be another part of the component.

It is possible to use a flat design where the status indicator fills the component and does not contrast with the component, but does contrast with the colors adjacent to the component.

The [Use of Color](#) success criterion addresses changing **only the color** (hue) of an object or text without otherwise altering the object's form. The principle is that contrast ratio (the difference in brightness) can be used to distinguish text or graphics. For example, [G183](#) is a technique to use a contrast ratio of 3:1 with surrounding text to distinguish links and controls. In that case the Working Group regards a link color that meets the 3:1 contrast ratio relative to the non-linked text color as satisfying the Success

Criterion [1.4.1 Use of color](#) since it is relying on contrast ratio as well as color (hue) to convey that the text is a link.

Non-text information within controls that uses a change of hue alone to convey the value or state of an input, such as a 1-5 star indicator with a black outline for each star filled with either yellow (full) or white (empty) is likely to fail the Use of color criterion rather than this one.

Using a change of contrast for focus and other states is a technique to differentiate the states. This is the basis for G195: Using an author-supplied, highly visible focus indicator , and more techniques are being added.

In combination with [2.4.7 Focus Visible](#) , the visual focus indicator for a component *must* have sufficient contrast against the adjacent background when the component is focused, except where the appearance of the component is determined by the user agent and not modified by the author.

Most focus indicators appear outside the component - in that case it needs to contrast with the background that the component is on. Other cases include focus indicators which are:

- only inside the component and need to contrast with the adjacent color(s) within the component.
- the border of the component (inside the component and adjacent to the outside) and need to contrast with both adjacent colors.
- partly inside and partly outside, where either part of the focus indicator can contrast with the adjacent colors.

If an indicator is partly inside and partly outside the component, either part of the indicator could provide contrast.

If the focus indicator changes the border of the component within the visible boundary it must contrast with the component.

Typically an outline goes around (outside) the visible boundary of the component, in this case changing the border is just inside the visible edge of the component.

Note that this success criterion does not directly compare the focused and unfocused states of a control - if the focus state relies on a change of color (e.g., changing *only* the background color of a button), this success criterion does not define any requirement for the difference in contrast between the two states.

Hover states

The language of Non-text Contrast specifically calls out "visual information required to identify...states." When users talk about a hover state, they are normally referring to a visual effect that takes place when the pointer is positioned over a control. However, there are a number of HTML components (such as buttons, checkboxes, radio buttons, and selects) which do not by default display any additional visual effects when the user moves a pointer control over them. The pointer itself, via its location, is the indicator of whether the user is hovering on a component.

Therefore, additional author-supplied visual treatments for hover are not "required to identify" the hover state. Those treatments can be considered supplemental and do not themselves need to contrast 3:1 against the background.

This is not to say that other hover effects are discouraged. For instance, some native components alter the shape of the pointer when it is hovering over a control; the pointer becomes an I-beam when it hovers over text inputs and text areas. There will be cases where some users may benefit from additional visual hover effects, such as bolding text or use of drop shadows. However, other users may find strong hover effects distracting. The key consideration

for any hover effect is that it does not cause a component itself to lose sufficient contrast against adjacent colors, or cause the visual indicators for other states, such as focus or selection, to lose sufficient contrast.

User Interface Component Examples

For designing focus indicators, selection indicators and user interface components that need to be perceived clearly, the following are examples that have sufficient contrast.

Type | Description | Examples

Link text | The browser's default link text color is covered by 1.4.3 Contrast (Minimum) . Since the underline is the same color as the text, which must meet at least 3:1 to pass, the default underline will always pass the requirements of Non-text Contrast. | {% include "./img/1.4.11-ntc-default-link-text.svg" %}

Default focus style | Links are required to have a visible focus indicator by 2.4.7 Focus Visible . Where the focus style of the user agent is not adjusted on interactive controls (such as links, form fields or buttons) by the website (author), the default focus style is exempt from contrast requirements (but must still be visible). | {% include "./img/1.4.11-ntc-default-link-focus.svg" %}

Buttons | A button which has a distinguishing indicator such as position, text style, or context does not need a contrasting visual indicator to show that it is a button, although some users are likely to identify a button with an outline that meets contrast requirements more easily. | {% include "./img/1.4.11-ntc-button-background.svg" %}

Text input (minimal) | Where a text-input has a visual indicator to show it is an input, such as a bottom border (#767676), that

indicator must meet 3:1 contrast ratio. | {% include "./img/1.4.11-ntc-text-input-minimal.svg" %}

Text input | Where a text-input has an indicator such as a complete border (#767676), that indicator must meet 3:1 contrast ratio. | {% include "./img/1.4.11-ntc-text-input-default.svg" %}

Text input focus style | A focus indicator is required. While in this case the additional gray (#CCC) outline has an insufficient contrast of 1.6:1 against the white (#FFF) background, the cursor/caret which is displayed when the input receives focus does provide a sufficiently strong visual indication. | {% include "./img/1.4.11-ntc-text-input-focus.svg" %}

Text input using background color | Text inputs that have no border and are differentiated only by a background color must have a 3:1 contrast ratio to the adjacent background (#043464). | {% include "./img/1.4.11-ntc-text-input-background.svg" %}

Toggle button | The toggle button's internal background (#070CD5) has a good contrast with the external white background. Also, the round toggle within (#7AC2FF) contrasts with the internal background. | {% include "./img/1.4.11-ntc-toggle.svg" %}

Dropdown indicator | The down-arrow is required to understand that there is drop-down functionality, it has a contrast of 4.7:1 for the white icon on dark gray (#6E747B). | {% include "./img/1.4.11-ntc-dropdown.svg" %}

Dropdown indicator | The down-arrow is required to understand that there is drop-down functionality, it has a contrast of 21:1 for the black icon on white. | {% include "./img/1.4.11-ntc-dropdown2.svg" %}

Checkbox - empty | A black border on a white background indicates the checkbox. | {% include "./img/1.4.11-ntc-checkbox-example1.svg" %}

Checkbox - checked | A black border on a white background indicates the checkbox, the black tick shape indicates the state of checked. | {% include "./img/1.4.11-ntc-checkbox-example2.svg" %}

Checkbox - Subtle hover style | A checkbox is visually identified by its black border against a white background, but when the mouse pointer hovers on the checkbox, a subtle grey background is added (#DEDEDE). The black border has a 15:1 contrast ratio with the grey background, so the checkbox continues to have good contrast. Note that the grey hover effect does not itself need to contrast 3:1 with the page background, since the pointer position is the primary indicator of the hover state. | {% include "./img/1.4.11-ntc-checkbox-example4.svg" %}

The following are examples that have insufficient contrast.

Type | Description | Examples

Colored underline is the only indicator of a link | Link and non-link text are both white on an almost-black (#0D0F13) background. The link's custom underline (#B1262B) is the only way to identify the link. The red underline contrasts less than 3:1 with the background color. | {% include "./img/1.4.11-ntc-link-underline-only.svg" %}

Checkbox - border color | The grey border color of the checkbox (#9D9D9D) has a contrast ratio of 2.7:1 with the white background, which is not sufficient for the visual information required to identify the checkbox. | {% include "./img/1.4.11-ntc-checkbox-example3.svg" %}

Checkbox - subtle focus style | A focus indicator is required. If the focus indicator is styled by the author, it must meet the 3:1 contrast ratio with adjacent colors. In this case, the gray (#AAA) indicator has an insufficient ratio of 2.3:1 with the white (#FFF) adjacent background. | {% include "./img/1.4.11-ntc-checkbox-example5.svg" %}

Text input - subtle border and customized caret | The text input's border color (#CCC) has contrast lower than 3:1 against its adjacent white background. Additionally, the caret / cursor has been given a customized light-blue color (#00C7FD) which has a contrast lower than 3:1 against the interior white background color of the input. |

User Interface Components that are not available for user interaction (e.g., a disabled control in HTML) are not required to meet contrast requirements. An inactive user interface component is visible but not currently operable. An example would be a submit button at the bottom of a form that is visible but cannot be activated until all the required fields in the form are completed.

Inactive components, such as disabled controls in HTML, are not available for user interaction. The decision to exempt inactive controls from the contrast requirements was based on a number of considerations. Although it would be beneficial to some people to discern inactive controls, a one-size-fits-all solution has been very difficult to establish. A method of varying the presentation of disabled controls, such as adding an icon for disabled controls, based on user preferences is anticipated as an advancement in the future.

Graphical Objects

The term "graphical object" applies to stand-alone icons such as a print icon (with no text), and the important parts of a more

complex diagram such as each line in a graph. For simple graphics such as single-color icons the entire image is a graphical object. Images made up of multiple lines, colors and shapes will be made of multiple graphical objects, some of which are required for understanding.

Not every graphical object needs to contrast with its surroundings - only those that are required for a user to understand what the graphic is conveying. Gestalt principles such as the "law of continuity" can be used to ignore minor overlaps with other graphical objects or colors.

Image | Notes

| The phone icon is a simple shape within the orange (#E3660E) circle. The meaning can be understood from that icon alone, the background behind the circle is irrelevant. The orange background and the white icon have a contrast ratio greater than 3:1, which passes. The graphical object is the white phone icon.

| A magnet can be understood by the "U" shape with lighter colored tips. Therefore to understand this graphic you should be able to discern the overall shape (against the background) and the lighter colored tips (against the rest of the U shape and the background). The graphical objects are the "U" shape (by outline or by the solid red color #D0021B), and each tip of the magnet.

| The symbol to show a currency (the £) going down can be understood with recognition of the shape (down arrow) and the currency symbol (pound icon with the shape which is part of the graphic). To understand this graphic you need to discern the arrow shape against the white background, and the pound icon against the yellow background (#F5A623). The graphical objects are the shape and the currency symbol.

| In order to understand the graph you need to discern the lines and shapes for each condition. To perceive the values of each line along the chart you need to discern the grey lines marking the graduated 100 value increments. The graphical objects are the lines in the graph, including the background lines for the values, and the colored lines with shapes. The lines should have 3:1 contrast against their background, but as there is little overlap with other lines they do not need to contrast with each other or the graduated lines. (See the testing principles below.)

| To understand the pie chart you have to discern each slice of the pie chart from the others. The graphical objects are the slices of the pie (chart). Note: If the values of the pie chart slices were also presented in a conforming manner (see the Pie Charts example for details), the slices would not be required for understanding.

Taking the magnet image above as an example, the process for establishing the graphical object(s) is to:

- Assess what part of each image is needed to understand what it represents. The magnet's "U" shape can be conveyed by the outline or by the red background (either is acceptable). The white tips are also important (otherwise it would be a horseshoe), which needs to contrast with the red background.
- Assume that the user could only see those aspects. Do they contrast with the adjacent colors? The outline of the magnet contrasts with the surrounding text (black/white), and the red and white between the tips also has sufficient contrast.

Due to the strong contrast of the red and white, it would also be possible to only put the outline around the white tips of the magnet and it would still conform.

The term "required for understanding" is used in the success criterion as many graphics do not need to meet the contrast

requirements. If a person needs to perceive a graphic, or part of a graphic (a graphical object) in order to understand the content it should have sufficient contrast. However, that is not a requirement when:

- A graphic with text embedded or overlaid conveys the same information, such as labels and values on a chart. Text within a graphic must meet 1.4.3 Contrast (Minimum) .
- The graphic is for aesthetic purposes that does not require the user to see or understand it to understand the content or use the functionality.
- The information is available in another form, such as in a table that follows the graph, which becomes visible when a "Long Description" button is pressed.
- The graphic is part of a logo or brand name (which is considered "essential" to its presentation).

Gradients can reduce the apparent contrast between areas, and make it more difficult to test. The general principle is to identify the graphical object(s) required for understanding, and take the central color of that area. If you remove the adjacent color which does not have sufficient contrast, can you still identify and understand the graphical object?

Some graphics may have interactions that either vary the contrast, or display the information as text when you mouseover/tap/focus each graphical object. In order for someone to discern the graphics exist at all, the unfocused default version must already have sufficiently contrasting colors or text. For the area that receives focus, information can then be made available dynamically as pop-up text, or be foregrounded dynamically by increasing the contrast.

Infographics can mean any graphic conveying data, such as a chart or diagram. On the web it is often used to indicate a large graphic with lots of statements, pictures, charts or other ways of conveying data. In the context of graphics contrast, each item within such an infographic should be treated as a set of graphical objects, regardless of whether it is in one file or separate files.

Infographics often fail to meet several WCAG level AA criteria including:

- [1.1.1 Non-text Content](#)
- [1.4.1 Use of Color](#)
- [1.4.3 Contrast \(Minimum\)](#)
- [1.4.5 Images of Text](#)

An infographic can use text which meets the other criteria to minimize the number of graphical objects required for understanding. For example, using text with sufficient contrast to provide the values in a chart. A long description would also be sufficient because then the infographic is not relied upon for understanding.

When text characters are used as symbols – used for their visual appearance, rather than expressing something in human language – they fall under the definition of non-text content .

Graphical objects do not have to meet the contrast requirements when "a particular presentation of graphics is essential to the information being conveyed". The Essential exception is intended to apply when there is no way of presenting the graphic with sufficient contrast without undermining the meaning. For example:

- Logotypes and flags : The brand logo of an organization or product is the representation of that organization and

therefore exempt. Flags may not be identifiable if the colors are changed to have sufficient contrast.

- Sensory : There is no requirement to change pictures of real life scenes such as photos of people or scenery.
- **Representing other things** : If you cannot represent the graphic in any other way, it is essential. Examples include: Screenshots to demonstrate how a website appeared. Diagrams of medical information that use the colors found in biology (example medical schematic from Wikipedia). color gradients that represent a measurement, such as heat maps (example heatmap from Wikipedia).

Testing Principles

A summary of the high-level process for finding and assessing non-text content on a web page:

- Identify each user-interface component (link, button, form control) on the page and: Identify the visual (non-text) indicators of the component that are required to identify that a control exists, and indicate the current state. In the default (on page load) state, test the contrast ratio against the adjacent colors. Test those contrast indicators in each state.
- Identify each graphic on the page that includes information required for understanding the content (i.e., excluding graphics which have visible text for the same information, or are decorative) and: Check the contrast of the graphical object against its adjacent colors; If there are multiple colors and/or a gradient, choose the least contrasting area to test; If it passes, move to the next graphical object; If the least-contrasting area is less than 3:1, assume that area is invisible, is the graphical object still understandable? If there is enough of the graphical object to understand, it passes, else fail.

The techniques below each have testing criteria, and the related criteria for [Focus visible \(2.4.7\)](#) , [Use of color \(1.4.1\)](#) , and [Contrast minimum](#) also have techniques.

Benefits

People with low vision often have difficulty perceiving graphics that have insufficient contrast. This can be exacerbated if the person has a color vision deficiency that lowers the contrast even further. Providing a [relative luminance](#) (lightness difference) of 3:1 or greater can make these items more distinguishable when the person does not see a full range of colors.

Graphical Object Examples

- Status icons on an application's dashboard (without associated text) have a 3:1 minimum contrast ratio.
- A text input has a dark border around the white editable area.
- A graph uses a light background and ensures that the colors for each line have a 3:1 contrast ratio against the background.

Pie Charts

Pie charts make a good case study for the graphical objects part of this success criterion, the following pie charts are intended to convey the proportion of market share each browser has. Please Note: The actual figures are made up, these are not actual market shares.

Infographics

The graphical objects are the circles (measured against the background) and the icons in each circle (measured against the circle's background).

There are many possible solutions to ensuring contrast, the example shows the use of borders. Other techniques are to use

darker colors for the circle backgrounds, or to add text labels & values for each item.

Resources

- [Accessibility Requirements for People with Low Vision](#) .
- [Smith Kettlewell Eye Research Institute](#) - "If the text is better understood with the graphics, they should be equally visible as the text".
- [Gordon Legge](#) - "Contrast requirements for form controls should be equivalent to contrast requirements for text".

1.4.12 Text Spacing (Level AA)

Source: [WCAG 2.2](#) / [Understanding 1.4.12](#) — Copyright © 2023 World Wide Web Consortium

In content implemented using markup languages that support the following text style properties , no loss of content or functionality occurs by setting all of the following and by changing no other style property:

- Line height (line spacing) to at least 1.5 times the font size;
- Spacing following paragraphs to at least 2 times the font size;
- Letter spacing (tracking) to at least 0.12 times the font size;
- Word spacing to at least 0.16 times the font size.

Exception: Human languages and scripts that do not make use of one or more of these text style properties in written text can conform using only the properties that exist for that combination of language and script.

Content is not required to use these text spacing values. The requirement is to ensure that when a user overrides the authored text spacing, content or functionality is not lost.

Writing systems for some languages use different text spacing settings, such as paragraph start indent. Authors are encouraged to follow locally available guidance for improving readability and legibility of text in their writing system.

In brief

Goal

Users can adjust text spacing to make it easier to read.

Author task

Ensure content adapts to user-defined text settings.

Why it's important

Some people need text with different spacing or font characteristics.

Intent

The intent of this success criterion (SC) is to ensure that when people override author-specified text spacing to improve their reading experience, content is still readable and operable. Each of the requirements stipulated in the SC's four bullets helps ensure text styling can be adapted by the user to suit their needs.

The metrics set a target for a minimum for text spacing that must be met. Starting from the author's presentation, changing these four style properties to the specified values should not result in a loss of content or functionality.

This SC focuses on the adaptability of content to a change in spacing between lines, words, letters, and paragraphs. Any combination of these may assist a user with effectively reading text. As well, ensuring that content correctly adapts when users override author settings for spacing also significantly increases the likelihood other style preferences can be set by the user. For example, a user may need to change to a wider font family than the author has set in order to effectively read text.

Author Responsibility

This SC does not dictate that authors must set all their content to the specified metrics, or provide a mechanism to do so. Rather, it specifies that an author's content has the ability to be set to those metrics without loss of content or functionality. The author requirement is both to not interfere with a user's ability to override the author settings, and to ensure that content thus modified does not break content in the manners shown in figures 1 through 3 in *Effects of Not Allowing for Spacing Override*. The

values in the SC are a baseline. Authors are encouraged to allow spacing to surpass the values specified, not see them as a ceiling.

It is beneficial for users if authors use any locally available guidance for improving readability in the local language or writing system. Conversely, in some human languages and scripts, some of the metrics specified by the SC are inapplicable. For example, languages such as Japanese do not use spacing following paragraphs, meaning that users are unlikely to make any paragraph spacing changes in practice. The exception in this SC allows authors to ignore text style properties which are inapplicable to the combination of language and script being used.

User Responsibility

The user may choose to exceed the spacing adjustments in the SC. The ability to read and derive meaning from the overridden spacing rests with the user. If large changes in spacing reduce readability, or cause loss of content or functionality, the user can adjust or return to spacing within the bounds of the SC. How such changes are achieved is up to the user, who may choose a user stylesheet, bookmarklet, extension, or application. Regardless, the user needs the flexibility to adjust spacing – and within the bounds set in the SC, without loss of content or functionality.

Further, this SC does not require that content implement its own mechanisms to allow users to do this. It is not a failure of the content if a user agent or platform does not provide a way for users to do this. Content does not fail this SC if the method chosen by the user - for instance, the use of an extension or bookmarklet - fails to correctly set the line height and spacing text properties on the content (provided that the content is not actively and purposely preventing the properties from being added).

Applicability

If the markup-based technologies being used are capable of overriding text to the Success Criterion's metrics, then this SC is applicable. For instance Cascading Style Sheet/HTML technologies are quite able to allow for the specified spacing metrics. Plugin technologies would need to have a built-in ability to modify styles to the specified metrics. Currently, this SC does not apply to PDF, as the portable document format is not implemented using markup.

Examples of text typically not affected by style properties and not expected to adapt are:

- Video captions embedded directly into the video frames and not provided as an associated caption file
- Images of text

For this SC, canvas implementations of text are considered to be images of text .

There may be regions of a page where text containers cannot expand due to design constraints (such as a maximum width for the left navigation or table column headers). A common convention if text exceeds its space is to replace truncated text with an ellipsis. Where ellipses appear as a result of modifying text style properties, the page can still meet the Text Spacing requirements, so long as the content is still available. For example:

- a mechanism is provided to reveal the truncated text on the page (for instance, the text appears on focus or on activation)
- where the ellipsis is part of a section of content which includes a link, the truncated text is revealed on the linked page

Where text is not truncated but it is when text is spaced, if there is no mechanism to show the truncated text, it fails this success criterion.

Effects of Not Allowing for Spacing Override

The following images show some types of failures when authors do not take into consideration that users may override spacing to the metrics specified in this success criterion.

The bottom portion of the words "Your Needs" is cut off in a heading making that text unreadable in Figure 1. It should read "We Provide a Mobile Application Service to Meet Your Needs."

In Figure 2 the last portion of text is cut off in three side-by-side headings. The first heading should read "A cog in the wheel" but it reads "A cog in the whe". Only half of the second "e" is visible and the letter "l" is completely missing. The second heading should read "A penny for your thoughts" but it reads "A penny for your". The third should read "Back to the drawing board." but it reads "Back to the drawi".

In Figure 3 the last three words "Groups and Programs" of the heading "Technologists Seeking Input from Groups and Programs" overlap the following sentence. That sentence should read, "You are invited to share ideas and areas of interest related to the integration of technology from a group or program perspective." However, the words "You are invited to share ideas" are obscured and unreadable.

Benefits

- People with low vision who require increased space between lines, words, and letters are able to read text.
- People with dyslexia may increase space between lines, words, and letters to increase reading speed.

- Although not required by this SC, white space between blocks of text can help people with cognitive disabilities discern sections and call-out boxes.

Examples

When spacing is being overridden to the SC's metrics:

- Text fits within the bounds of its containing box without being cut off.
- Text fits within the bounds of its containing box without overlapping other boxes.

Research

The grounds for this SC are based on research . The metrics chosen as measures are based on the McLeish study. She ran from .04 to .25 em tests. McLeish found an increasing curve in reading speed of actual materials up to .25, but it started to flatten at .20. Previous studies that reported no improvement started at .5em. Right at the flat point. Wayne E. Dick, Ph.D. analyzed the McLeish study and translated from points. Dr. Dick recommended the metrics that the Working Group adopted.

Roughly 480 different languages and scripts have been tested . Maximum spacing adjustments allowed by the SC were set on the following 3 pages:

- Languages in their own writing systems
- Online Encyclopedia of writing systems and languages – language names
- Universal Declaration of Human Rights

No adverse effects occurred. The following are the specific findings:

Character Spacing

Individual characters in words remained intact though they were spaced a bit further apart.

Word Spacing

Words were spaced further apart. In languages that typically do not include spaces between words (e.g., Japanese), applying word spacing had no effect. This is expected.

Line Height

Changing line height did not separate diacritics from characters, nor did it adversely impact ascenders or descenders.

As previously discussed, the ability to read text with adjusted spacing is a user responsibility. This is true no matter the language.

The SC's exception addresses cases where a text style property is not used in a language or script. In such cases, authors are only required to ensure relevant properties do not break the layout.

Other references

- Allan, Kirkpatrick, Lawton Henry, Editors. (2017). Accessibility Requirements for People with Low Vision (3.4 Spacing for Reading) . World Wide Web Consortium.
- Stylus Team (2012). Stylus browser extension (Firefox, Chrome, and Opera) (compatible with Userstyles.org material).
- Campbell, Alastair. (2017). Text Adaptation Bookmarklet . GitHub.
- Chung, Susana T. L. (2012). Dependence of Reading Speed on Letter Spacing in Central Vision Loss . Optom Vis Sci.
- Chung, Susana T. L. (2002). The Effect of Letter Spacing on Reading Speed in Central and Peripheral Vision (PDF) . IOVS ARVO Journals.

- Mcleish, Eve. (2007). A study of the effect of letter spacing on the reading speed of young readers with low vision (PDF) . The British Journal of Visual Impairment 25.2: 133-43.
- Rello, L., & Baeza-Yates, R. A. (2017). How to present more readable text for people with dyslexia . Universal Access in the Information Society, 16(1), 29-49.
- Sjoblom, A.M., Eaton, E. and Stagg, S.D., (2016). The effects of letter spacing and coloured overlays on reading speed and accuracy in adult dyslexia . British Journal of Educational Psychology, 86(4), pp. 630-639).
- Zorzi, Marco et, al. (2012). Extra-large letter spacing improves reading in dyslexia . Proceedings of the National Academy of Sciences.

1.4.13 Content on Hover or Focus (Level AA)

Source: WCAG 2.2 / Understanding 1.4.13 — Copyright © 2023 World Wide Web Consortium

Where receiving and then removing pointer hover or keyboard focus triggers additional content to become visible and then hidden, all of the following are true:

Dismissible

A mechanism is available to dismiss the additional content without moving pointer hover or keyboard focus, unless the additional content communicates an input error or does not obscure or replace other content;

Hoverable

If pointer hover can trigger the additional content, then the pointer can be moved over the additional content without the additional content disappearing;

Persistent

The additional content remains visible until the hover or focus trigger is removed, the user dismisses it, or its information is no longer valid.

Exception: The visual presentation of the additional content is controlled by the user agent and is not modified by the author.

Examples of additional content controlled by the user agent include browser tooltips created through use of the HTML [title attribute](#) `[[HTML]]`.

Custom tooltips, sub-menus, and other nonmodal popups that display on hover and focus are examples of additional content covered by this criterion.

This criterion applies to content that appears in addition to the triggering component itself. Since hidden components that are made visible on keyboard focus (such as links used to skip to another part of a page) do not present additional content they are not covered by this criterion.

In brief

Goal

More users can perceive and dismiss non-persistent content.

What to do

If hover or focus causes content changes, ensure interaction is predictable.

Why it's important

Unpredictable temporary content can be hard for some to consume and may disrupt others.

Intent

Additional content that appears and disappears in coordination with keyboard focus or pointer hover often leads to accessibility issues. Reasons for such issues include:

- the user may not have intended to trigger the interaction
- the user may not know new content has appeared
- the new content may interfere with a user's ability to do a task

Examples of such interactions can include custom tooltips, sub-menus and other non-modal popups which display on hover and focus. The intent of this success criterion is to ensure that authors who cause additional content to appear and disappear in this manner must design the interaction in such a way that users can:

- perceive the additional content AND

- dismiss it without disrupting their page experience.

There are usually more predictable and accessible means of adding content to the page, which authors are recommended to employ. If an author *does* choose to make additional content appear and disappear in coordination with hover and keyboard focus, this success criterion specifies three conditions that must be met:

- dismissible
- hoverable
- persistent

Each of these is discussed in a separate section.

Dismissible

The intent of this condition is to ensure that the additional content does not interfere with viewing or operating the page's original content. When magnified, the portion of the page visible in the viewport can be significantly reduced. Mouse users frequently move the pointer to pan the magnified viewport and display another portion of the screen. However, almost the entire portion of the page visible in this restricted viewport may trigger the additional content, making it difficult for a user to pan without re-triggering the content. A keyboard means of dismissing the additional content provides a workaround.

Alternatively, low vision users who can only navigate via the keyboard do not want the small area of their magnified viewport cluttered with hover text. They need a keyboard method of dismissing something that is obscuring the current focal area.

Two methods may be used to satisfy this condition and prevent such interference:

- Position the additional content so that it does not obscure any other content including the trigger, with the exception of white space and purely decorative content, such as a background graphic which provides no information.
- Provide a mechanism to easily dismiss the additional content, such as by pressing Escape.

For most triggers of relatively small size, it is desirable for both methods to be implemented. If the trigger is large, noticing the additional content may be of concern if it appears away from the trigger. In those cases, only the second method may be appropriate.

The success criterion allows for input error messages to persist as there are cases that require attention, explicit confirmation or remedial action.

Hoverable

The intent of this condition is to ensure that additional content which may appear on hover of a target may also be hovered itself. Content which appears on hover can be difficult or impossible to perceive if a user is required to keep their mouse pointer over the trigger. When the added content is large, magnified views may mean that the user needs to scroll or pan to completely view it, which is impossible unless the user is able to move their pointer off the trigger without the additional content disappearing.

Another common situation is when large pointers have been selected via platform settings or assistive technology. Here, the pointer can obscure a significant area of the additional content. A technique to view the content fully in both situations is to move the mouse pointer directly from the trigger onto the new content. This capability also offers significant advantages for users who utilize screen reader feedback on mouse interactions. This

condition generally implies that the additional content overlaps or is positioned adjacent to the target.

Persistent

The intent of this condition is to ensure users have adequate time to perceive the additional content after it becomes visible. Users with disabilities may require more time for many reasons, such as to change magnification, move the pointer, or simply to bring the new content into their visual field. Once it appears, the content should remain visible until:

- The user removes hover or focus from the trigger and the additional content, consistent with the typical user experience;
- The user dismisses the additional content via the mechanism provided to satisfy the Dismissible condition; or
- The information conveyed by the additional content becomes invalid, such as a 'busy' message that is no longer valid.

Additional Notes

- This criterion does not attempt to solve such issues when the appearance of the additional content is completely controlled by the user agent. A prominent example is the common behavior of browsers to display the title attribute in HTML as a small tooltip.
- Modal dialogs are out of scope for this criterion because they must take keyboard focus and thus should not appear on hover or focus. Refer to [Success Criterion 3.2.1, On Focus](#) .
- Content which can be triggered via pointer hover should also be able to be triggered by keyboard focus. Refer to [Success Criterion 2.1.1, Keyboard](#) .

Benefits

- Users with low vision who view content under magnification will be better able to view content on hover or focus without reducing their desired magnification.
- Users who increase the size of mouse cursors via platform settings or assistive technology will be able to employ a technique to view obscured content on hover.
- Users with low vision or cognitive disabilities will have adequate time to perceive additional content appearing on hover or focus and to view the trigger content with less distraction.
- users with low pointer accuracy will be able to more easily dismiss unintentionally-triggered additional content

Resources

- [Tooltip design described in WAI-ARIA Authoring Practices](#)

Principle 2: Operable

[Source: W3C WCAG 2.2 — <https://www.w3.org/TR/WCAG22/#principle2>]

User interface components and navigation must be operable.

Guideline 2.1: Keyboard Accessible

[Source: W3C WCAG 2.2 — <https://www.w3.org/TR/WCAG22/#guideline21>]

Make all functionality available from a keyboard.

2.1.1 Keyboard (Level A)

Source: [WCAG 2.2](#) / [Understanding 2.1.1](#) — Copyright © 2023 World Wide Web Consortium

All functionality of the content is operable through a keyboard interface without requiring specific timings for individual keystrokes, except where the underlying

function requires input that depends on the path of the user's movement and not just

the endpoints.

This exception relates to the underlying function, not the input technique. For example, if using handwriting to enter text, the input technique (handwriting) requires path-dependent input but the underlying function (text input) does not.

This does not forbid and should not discourage providing mouse input or other input methods in addition to keyboard operation.

In brief

Goal

Everything can be done with a keyboard except freehand movements.

What to do

Ensure pointer actions have a keyboard equivalent.

Why it's important

Many people rely on the keyboard interface, including blind and some mobility impaired people.

Intent of Keyboard

The intent of this success criterion is to ensure that, wherever possible, content can be operated through a keyboard or keyboard

interface (so an alternate keyboard can be used). When content can be operated through a keyboard or alternate keyboard, it is operable by people with no vision (who cannot use devices such as mice that require eye-hand coordination) as well as by people who must use alternate keyboards or input devices that act as keyboard emulators. Keyboard emulators include speech input software, sip-and-puff software, on-screen keyboards, scanning software and a variety of assistive technologies and alternate keyboards. Individuals with low vision also may have trouble tracking a pointer and find the use of software much easier (or only possible) if they can control it from the keyboard.

Examples of "specific timings for individual keystrokes" include situations where a user would be required to repeat or execute multiple keystrokes within a short period of time or where a key must be held down for an extended period before the keystroke is registered.

The phrase "except where the underlying function requires input that depends on the path of the user's movement and not just the endpoints" is included to separate those things that cannot reasonably be controlled from a keyboard.

Most actions carried out by a pointing device can also be done from the keyboard (for example, clicking, selecting, moving, sizing). However, there is a small class of input that is done with a pointing device that cannot be done from the keyboard in any known fashion without requiring an inordinate number of keystrokes. Free hand drawing, or watercolor painting require path dependent input. Drawing straight lines, regular geometric shapes, re-sizing windows and dragging objects to a location (when the path to that location is not relevant) do not require path dependent input.

The use of MouseKeys would not satisfy this success criterion because it is not a keyboard equivalent to the application; it is a mouse equivalent (i.e., it looks like a mouse to the application).

It is assumed that the design of user input features takes into account that operating system keyboard accessibility features may be in use. For example, modifier key locking may be turned on. Content continues to function in such an environment, not sending events that would collide with the modifier key lock to produce unexpected results.

Platforms and user agents usually have conventions for how web content or applications are controlled with a keyboard interface. If content does not follow the platform/user agent conventions it may be difficult to use, as users will need to learn different interaction methods. As a *best practice*, content should follow the platform/user agent conventions. However, deviating from these conventions does *not* fail the normative requirement of this success criterion.

For instance, buttons that have focus can generally be activated using both the Enter key and the Space bar. If a custom button control in a web application instead only reacts to Enter (or even a completely custom key or key combination), this still **satisfies** the requirements of this success criterion.

This success criterion does not require that every visible control that can be activated using a mouse or touch screen must also be focusable and actionable using the keyboard. The normative requirement is only that there must be a way for keyboard interface users to perform the same, or comparable, actions and to operate the content. Generally, the easiest way to achieve this is to provide controls that can be operated with all possible input devices; however, if a web application implements a separate mode of operation for keyboard interface users, it will **not** fail the

success criterion. However, authors are advised to consider how users will discover any keyboard equivalents which are available.

Benefits of Keyboard

- People who are blind (who cannot use devices such as mice that require eye-hand coordination)
- People with low vision (who may have trouble finding or tracking a pointer indicator on screen)
- Some people with hand tremors find using a mouse very difficult and therefore usually use a keyboard

Examples of Keyboard

Example 1: A drawing Program

A drawing program allows users to create, size, position and rotate objects from the keyboard.

Example 2: A drag and Drop Feature

An application that uses drag and drop also supports "cut" and "paste" or form controls to move objects.

Example 3: Moving between and connecting discrete points

A connect-the-dots program allows the user to move between dots on a screen and use the spacebar to connect the current dot to the previous one.

Example 4: Exception - Painting Program

A watercolor painting program passes as an exception because the brush strokes vary depending on the speed and duration of the movements.

Example 5: Exception - Model helicopter flight training simulator

A model helicopter flight training simulator passes as an exception because the nature of the simulator is to teach real-time behavior of a model helicopter.

Example 6: A PDA with an optional keyboard

A PDA device that is usually operated via a stylus has an optional keyboard that can be attached. The keyboard allows full web browsing in standard fashion. The web content is operable because it was designed to work with keyboard-only access.

Example 7: Simple search form with pointer-operable submit button

A search form includes a text input field followed by a submit button. The submit button itself has been coded so that it does not receive focus, and can only be activated using a pointer input. However, since keyboard users can submit the search by pressing Enter in the text input after typing their search terms, the form passes this success criterion.

2.1.2 No Keyboard Trap (Level A)

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If keyboard focus can be moved to a component of the page using a keyboard interface, then focus can be moved away from that component using only a keyboard interface,

and, if it requires more than unmodified arrow or tab keys or other standard exit

methods, the user is advised of the method for moving focus away.

Since any content that does not meet this success criterion can interfere with a user's

ability to use the whole page, all content on the web page (whether it is used to

meet other success criteria or not) must meet this success criterion. See Conformance Requirement 5: Non-Interference .

In brief

Goal

Keyboard users don't get stuck.

What to do

Ensure users always know how to navigate away from components.

Why it's important

People who rely on the keyboard often have no other means to navigate.

Intent of No Keyboard Trap

The intent of this success criterion is to ensure that that content does not "trap" keyboard focus within subsections of content on a web page. This is a common problem when multiple formats are combined within a page and rendered using plug-ins or embedded applications.

There may be times when the functionality of the web page restricts the focus to a subsection of the content, as long as the user knows how to leave that state and "untrap" the focus.

Benefits of No Keyboard Trap

- People who rely on a keyboard or keyboard interface to use the web including people who are blind and people with physical disabilities.

Examples of No Keyboard Trap

A calendar widget

A calendar widget allows users to add, remove or update items in their calendar using the keyboard. The controls in the widget are part of the tab order within the web page, allowing users to tab through the controls in the widget as well as to any links or controls that follow.

A puzzle applet

Once a user tabs into an applet, further tabs and other keystrokes are handled by the applet. Instructions describing the keystroke used to exit the applet are provided prior to the applet as well as within the applet itself.

A modal dialog box

A web application brings up a dialog box. At the bottom of the dialog are two buttons, Cancel and OK. When the dialog has been

opened, all web page content outside the dialog becomes inert and cannot receive focus (though, depending on implementation, the focus cycle might still include user agent controls). The dialog is dismissed by activating the Cancel button or the OK button.

2.1.3 Keyboard (No Exception) (Level AAA)

Source: [WCAG 2.2](#) / [Understanding 2.1.3](#) — Copyright © 2023 World Wide Web Consortium

All functionality of the content is operable through a keyboard interface without requiring specific timings for individual keystrokes.

In brief

Goal

Everything can be done with a keyboard.

What to do

Ensure all pointer actions have a keyboard equivalent.

Why it's important

People who can only use the keyboard interface need to be able to accomplish everything.

Intent of Keyboard (No Exception)

The intent of this success criterion is to ensure that **all** content is operable from the keyboard. This is the same as Success Criterion 2.1.1, except that no exceptions are allowed. This does not mean that content where the underlying function requires input that depends on the path of the user's movement and not just the endpoints (excluded from the requirements of 2.1.1) must be made keyboard accessible. Rather, it means that content that uses path-dependent input cannot conform to this success criterion and therefore cannot meet Guideline 2.1 at Level AAA.

Platforms and user agents usually have conventions for how web content or applications are controlled with a keyboard interface. If content does not follow the platform/user agent conventions it may be difficult to use, as users will need to learn different

interaction methods. As a *best practice*, content should follow the platform/user agent conventions. However, deviating from these conventions does *not* fail the normative requirement of this success criterion.

For instance, buttons that have focus can generally be activated using both the Enter key and the Space bar. If a custom button control in a web application instead only reacts to Enter (or even a completely custom key or key combination), this still **satisfies** the requirements of this success criterion.

This success criterion does not require that every visible control that can be activated using a pointer (such as a mouse or touch screen input) must also be focusable and actionable using the keyboard. The normative requirement is only that there must be a way for keyboard interface users to perform the same, or comparable, actions and to operate the content. Generally, the easiest way to achieve this is to provide controls that can be operated with all possible input devices; however, if a web application implements a separate mode of operation for keyboard interface users, it will **not** fail the success criterion.

2.1.4 Character Key Shortcuts (Level A)

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If a keyboard shortcut is implemented in content using only letter (including upper- and lower-case letters), punctuation, number, or symbol characters, then at least one of the following is true:

Turn off

A mechanism is available to turn the shortcut off;

Remap

A mechanism is available to remap the shortcut to include one or more non-printable keyboard keys (e.g., Ctrl, Alt);

Active only on focus

The keyboard shortcut for a user interface component is only active when that component has focus.

In brief

Goal

Reduce accidental activation of keyboard shortcuts.

What to do

Ensure character-only shortcut keys can be turned off or modified.

Why it's important

Character-key shortcuts are easy to accidentally trigger, especially with speech input.

Intent

The intent of this success criterion is to reduce accidental activation of keyboard shortcuts. Character key shortcuts work well for many keyboard users. However, they can be inappropriate and frustrating for speech input users, whose dictation is

interpreted as strings of letters, and for keyboard users who are prone to accidentally hit keys. To rectify this issue, authors need to allow users to turn off or reconfigure shortcuts that are made up of only character keys.

Even though this success criterion refers to character keys , note that it's not relevant whether a shortcut can be activated using a single physical key on a keyboard, or if it requires a combination of keys to be pressed. For instance, on most full-size US and UK keyboard, the ? (question mark) symbol is accessed using Shift + / (forward slash key next to the right-hand Shift key). On a UK keyboard, in Windows, the é (lowercase "e" with an acute accent) requires the use of AltGr + e . The specific key combination required for certain characters will also vary depending on the user's keyboard layout. However, shortcuts that use these characters still fall under the requirements of this success criterion. What matters is that a shortcut relies on a printable character (letters, punctuation, numbers, symbol characters), and not the number of physical keyboard keys that users need to press to trigger it.

The success criterion also applies to situations where a shortcut is based on a *sequence* of character keys – for example, pressing G and then A in quick succession to trigger an action. While the individual character key presses don't immediately trigger the action, overall these types of shortcuts still rely on a series of *character keys* .

This success criterion doesn't affect components such as listboxes and drop-down menus. Although these components contain values (words) that may be selected by one or more character keys, the shortcuts are only active when the components have focus. Other components such as menus may be accessed or opened with a single non-character shortcut (e.g., Alt or Alt + F) before pressing

a single character key to select an item. This makes the full path to invoking a menu a two-step shortcut that includes a non-printable key. Access keys are also not affected because they are generally (depending on the user agent) activated using modifier keys.

Background on the mechanics of speech input

Speech input users generally work in a single mode where they can use a mix of dictation and speech commands. This works well because the user knows to pause before and after commands, and commands are usually at least two words long. So, for instance, a user might say a bit of dictation, such as "the small boat", then pause, and say a command to delete that dictation, such as "Delete Line". In contrast, if the user were to say the two phrases together without a pause, the whole phrase would come out as dictation (i.e., "the small boat delete line"). Although speech input programs often include modes that listen only for dictation or only for commands, most speech users use the all-encompassing mode all the time because it is a much more efficient workflow. It could decrease command efficiency significantly if users were to change to command mode and back before and after issuing each command.

Speech users can also speak most keyboard commands (e.g., "press Control Foxtrot") without any problems. If the website or app is keyboard enabled, the speech user can also write a native speech macro that calls the keyboard command, such as "This Print" to carry out Ctrl + P .

Single-key shortcuts are the exception. While using single letter keys as controls might be appropriate and efficient for many keyboard users, single-key shortcuts are disastrous for speech users. The reason for this is that when only a single key is used to trip a command, a spoken word can become a barrage of single-

key commands if the cursor focus happens to be in the wrong place.

For example, a speech-input user named Kim has her cursor focus in the main window of a web mail application that uses common keyboard shortcuts to navigate (k), archive (y) and mute messages (m). A coworker named Mike enters her office and says "Hey Kim" and her microphone picks that up. The Y of "hey" archives the current message. K in "Kim" moves down one conversation and M mutes a message or thread. And, if Kim looks up and says "Hey Mike" without remembering to turn off the microphone, the same three things happen in a different sequence.

A user interacting with a web page or web app that doesn't use single-character shortcuts doesn't have this problem. Inadvertent strings of characters from the speech application are not interpreted as shortcuts if a modifier key is required. A speech user filling in a text input form may find that a phrase that is accidentally picked up by the speech microphone results in stray text being entered into the field, but that is easily seen and undone. The Resources section of this page contains links to videos demonstrating these types of issues.

Benefits

- Speech users will be able to turn off single-key shortcuts so they can avoid accidentally firing batches of them at once. This will allow speech users to make full use of programs that offer single-key shortcuts to keyboard users.
- Keyboard-only users who have dexterity challenges can also be prone to accidentally hitting keys. Those users would be able to avoid problematic single character shortcuts by turning them off or modifying them to include at least one non-character key.

- Allowing all shortcut keys to be remapped can help users with some cognitive disabilities, since the same shortcuts can be assigned to perform the same actions across different applications.

Disable Shortcuts

A mechanism is provided to allow users to disable character-key shortcuts. The character key shortcuts are not the only way to carry out these commands. A speech user disables the shortcuts and can prevent words that are picked up by the microphone from triggering single-key shortcuts.

Alternate Control

Keyboard-only users are in a long issues thread. While reading the thread they accidentally hit the S key, which moves focus to the search bar at the top of the document. This causes them to lose their place and train of thought. However, a mechanism is provided to allow users to change character-key shortcuts. They change the shortcut to include another key so they can avoid future interruptions.

Resources

Web apps that use character-key shortcuts and allow users to disable and/or change these shortcuts:

- Gmail
- WordPress

Videos of speech user trouble with single character key shortcuts:

- [Single character key shortcuts affecting speech input – example 1](#)
- [Single character key shortcuts affecting speech input – example 2](#)

Guideline 2.2: Enough Time

[Source: W3C WCAG 2.2 —

<https://www.w3.org/TR/WCAG22/#guideline22>]

Provide users enough time to read and use content.

2.2.1 Timing Adjustable (Level A)

Source: [WCAG 2.2](#) / [Understanding 2.2.1](#) — Copyright © 2023 World Wide Web Consortium

For each time limit that is set by the content, at least one of the following is true:

Turn off

The user is allowed to turn off the time limit before encountering it; or

Adjust

The user is allowed to adjust the time limit before encountering it over a wide range that is at least ten times the length of the default setting; or

Extend

The user is warned before time expires and given at least 20 seconds to extend the time limit with a simple action (for example, "press the space bar"), and the user is allowed to extend the time limit at least ten times; or

Real-time Exception

The time limit is a required part of a real-time event (for example, an auction), and no alternative to the time limit is possible; or

Essential Exception

The time limit is essential and extending it would invalidate the activity; or

20 Hour Exception

The time limit is longer than 20 hours.

This success criterion helps ensure that users can complete tasks without unexpected

changes in content or context that are a result of a time limit. This success criterion

should be considered in conjunction with Success Criterion 3.2.1 , which puts limits on changes of content or context as a result of user action.

In brief

Goal

Users have adequate time to complete tasks.

What to do

Let users turn off, adjust, or extend time limits.

Why it's important

People with disabilities may need more time to complete activities.

Intent of Timing Adjustable

The intent of this success criterion is to ensure that users with disabilities are given adequate time to interact with web content whenever possible. People with disabilities such as blindness, low vision, dexterity impairments, and cognitive limitations may require more time to read content or to perform functions such as filling out on-line forms. If web functions are time-dependent, it will be difficult for some users to perform the required action before a time limit occurs. This may render the service inaccessible to them. Designing functions that are not time-dependent will help people with disabilities succeed at completing these functions. Providing options to disable time limits, customize the length of time limits, or request more time before a time limit occurs helps those users who require more time than expected to successfully complete tasks. These options are listed in the order that will be most helpful for the user. Disabling time limits is

better than customizing the length of time limits, which is better than requesting more time before a time limit occurs.

Any process that happens without user initiation after a set time or on a periodic basis is a time limit. This includes partial or full updates of content (for example, page refresh), changes to content, or the expiration of a window of opportunity for a user to react to a request for input.

It also includes content that is advancing or updating at a rate beyond the user's ability to read and/or understand it. In other words, animated, moving or scrolling content introduces a time limit on a users ability to read content.

This success criterion is generally not applicable when the content repeats or is synchronized with other content, so long as the information and data is adjustable or otherwise under the control of the end user. Examples of time limits for which this success criterion is not applicable include scrolling text that repeats, captioning, and carousels . These are situations which do include time limits, but the content is still available to the user because it has controls for accessing it, as specified in [2.2.2 Pause, Stop, Hide](#)

.

In some cases, however, it is not possible to change the time limit (for example, for an auction or other real-time event) and exceptions are therefore provided for those cases.

Content that operates on a timer does not need to be time adjustable if there is an alternative that does not rely on a timer. For example, a web application such as an email client provides notification of new email arriving with a temporary message (such as a 'toast' message) in the lower right-hand side of the interface, and the message disappears after 5 seconds. Users are able to identify the arrival of email through other means, such as viewing

the Inbox, so the disappearance of the message does not set a time limit on their ability to determine if new mail has arrived. If the user has no other means of discovering the same information (or performing the same function), then each message would need to meet this success criterion in order to provide users with sufficient time to access the information.

Notes regarding time limits

- Timed server redirects can be found below under Common Failures.
- Non-timed server redirects (e.g., 3xx response codes) are not applicable because there is no time limit: they work instantly.
- This success criterion applies only to time limits that are set by the content itself.

For example, if a time limit is included in order to address security concerns, it

would be considered to have been set by the content because it is designed to be

part of the presentation and interaction experience for that content. Time limits

set externally to content, such as by the user agent or by factors intrinsic to the

Internet are not under the author's control and not subject to WCAG conformance requirements.

Time limits set by web servers should be under the author's/organization's control

and are covered. Success Criteria [2.2.3 No Timing](#) , [2.2.4 Interruptions](#) , and [2.2.5 Re-Authenticating](#) may also apply.

- Certain time limits implemented for security reasons, such as time-based / time-limited

two-factor authentication tokens, can be considered *essential* , and may be

exempt from this criterion. However, other criteria may apply, such as [3.3.7 Redundant Entry](#) , [3.3.8 Accessible Authentication \(Minimum\)](#) , and [3.3.9 Accessible Authentication \(Enhanced\)](#) may also apply.

- Ten times the default was chosen based on clinical experience and other guidelines. For example, if 15 seconds is allowed for a user to respond and hit a switch, 150 seconds would be sufficient to allow almost all users to hit a switch even if they had trouble.
- 20 seconds was also based on clinical experience and other guidelines. 20 seconds to hit 'any switch' is sufficient for almost all users including those with reduced motion. Some would fail, but some would fail all lengths of time. A reasonable period for requesting more time is required since an arbitrarily long time can provide security risks to all users, including those with disabilities, for some applications. For example, with kiosks or terminals that are used for financial transactions, it is quite common for people to walk away without signing off. This leaves them vulnerable to those walking up behind them. Providing a long period of inactivity before asking, and then providing a long period for the person to indicate that they are present can leave terminals open for abuse. If there is no activity the system should ask if the user is there. It should then ask for an indication that a person is there ('hit any key') and then wait long enough for almost anyone to respond. For "hit any key," 20 seconds would meet

this. If the person indicates that they are still present, the device should return the user to the exact condition that existed before it asked the question.

- 20 hours was chosen as an upper limit because it is longer than a full waking day.

In cases where timing is not an intrinsic requirement but giving users control over timed events would invalidate the outcome, a third party can control the time limits for the user (for example, granting double time on a test).

See also [2.2.3 No Timing](#) .

Benefits of Timing Adjustable

- People with physical disabilities often need more time to react, to type and to complete activities. People with low vision need more time to locate things on screen and to read. People who are blind and using screen readers may need more time to understand screen layouts, to find information and to operate controls. People who have cognitive or language limitations need more time to read and to understand. People who are deaf and communicate in sign language may need more time to read information printed in text (which may be a second language for some).
- In circumstances where a sign-language interpreter may be relating audio content to a user who is deaf, control over time limits is also important.
- People with reading disabilities, cognitive limitations, and learning disabilities who may need more time to read or comprehend information can have additional time to read the information by pausing the content.

Examples of Timing Adjustable

- A website uses a client side time limit to help protect users who may step away from their computer. After a period of inactivity the web page asks if the user needs more time. If it doesn't get a response – it times out.
- A web page has a field that automatically updates with the latest headlines in a rotating fashion. There is an interactive control that allows the user to extend the length of time between each update to as much as ten times the default. The control can be operated with either a mouse or a keyboard.
- A web page includes an animation which includes text that appears and disappears throughout. In some cases, the text is scrolling across the screen and in others, it is only displayed for a short time before it fades into the background. The page includes a pause button so that users who have trouble reading the text before it disappears can read it.
- In an auction, there is a time limit on the amount of time a user has to submit a bid. Since the time limit applies to all users who want to bid on a particular item, it would be unfair to extend the time limit for any one particular user. Therefore, a time limit is required for this type of activity and no extension, adjustment, or deactivation of the time limit is required by this success criterion.
- An on-line ticket-purchasing site gives the user two minutes to confirm a purchase before the seats are returned to the general pool. Because tickets on such sites can sell out quickly, holding a ticket longer than that may invalidate the nature of the site, so this is a case in which the timing is essential and cannot be extended without invalidating the activity. However, the site does move as much of the process out of the time-critical period as possible, for instance allowing users to provide

necessary information like name, payment method, etc., before entering the time-critical stage.

- A ticket-purchasing site allows the user two minutes to confirm purchase of selected seats, but warns the user when their time is almost out and allows the user to extend this time limit some number of times with a simple action such as clicking a "Extend time limit" button.

2.2.2 Pause, Stop, Hide (Level A)

Source: [WCAG 2.2](#) / [Understanding 2.2.2](#) — Copyright © 2023 World Wide Web Consortium

For moving, blinking , scrolling, or auto-updating information, all of the following are true:

Moving, blinking, scrolling

For any moving, blinking or scrolling information that (1) starts automatically, (2) lasts more than five seconds, and (3) is presented in parallel with other content, there is a mechanism for the user to pause , stop, or hide it unless the movement, blinking, or scrolling is part of an activity where it is essential ; and

Auto-updating

For any auto-updating information that (1) starts automatically and (2) is presented in parallel with other content, there is a mechanism for the user to pause, stop, or hide it or to control the frequency of the update unless the auto-updating is part of an activity where it is essential.

For requirements related to flickering or flashing content, refer to Guideline 2.3 .

Since any content that does not meet this success criterion can interfere with a user's

ability to use the whole page, all content on the web page (whether it is used to

meet other success criteria or not) must meet this success criterion. See Conformance Requirement 5: Non-Interference .

Content that is updated periodically by software or that is streamed to the user agent is not required to preserve or present information that is generated or received between the initiation of the pause and resuming presentation, as this may not be

technically possible, and in many situations could be misleading to do so.

An animation that occurs as part of a preload phase or similar situation can be considered essential if interaction cannot occur during that phase for all users and if not indicating progress could confuse users or cause them to think that content was frozen or broken.

In brief

Goal

Fewer users are distracted by content that updates or moves.

What to do

Let users control content changes that occur in parallel with other content.

Why it's important

Some people with cognitive disabilities and attention deficits are distracted by continuous movement.

Intent of Pause, Stop, Hide

The intent of this success criterion is to avoid distracting users during their interaction with a web page.

In the context of this Success Criterion, "starts automatically" broadly refers to animations/updates that are not the direct result of a user's intentional activation, for example, selecting a link or button.

"Moving, blinking and scrolling" refers to content in which the visible content conveys a sense of motion. Common examples include motion pictures, synchronized media presentations, animations, real-time games, and scrolling stock tickers. "Auto-updating" refers to content that updates or disappears based on a

preset time interval. Common time-based content includes automatically updated weather information, news, stock price updates, and auto-advancing presentations and messages. The requirements for moving, blinking and scrolling content and for auto-updating content are the same except that:

- authors have the option of providing the user with a means to control the frequency of updates when content is auto-updating and
- there is no five second exception for auto-updating since it makes little sense to auto-update for a few seconds and then stop

Content that moves or auto-updates can be a barrier to anyone who has trouble reading stationary text quickly as well as anyone who has trouble tracking moving objects. It can also cause problems for screen readers.

Moving content can also be a severe distraction for some people. Certain groups, particularly those with attention deficit disorders, find blinking content distracting, making it difficult for them to concentrate on other parts of the web page. Five seconds was chosen because it is long enough to get a user's attention, but not so long that a user cannot wait out the distraction if necessary to use the page.

Content that is paused can either resume in real-time or continue playing from the point in the presentation where the user left off.

- Pausing and resuming where the user left off is best for users who want to pause to read content and works best when the content is not associated with a real-time event or status. See 2.2.1 Timing Adjustable for additional requirements related to time-limits for reading and interactions.

- Pausing and jumping to current display (when pause is released) is better for information that is real-time or "status" in nature. For example, weather radar, a stock ticker, a traffic camera, or an auction timer, would present misleading information if a pause caused it to display old information when the content was restarted. Hiding content would have the same result as pausing and jumping to current display (when pause is released).

See [2.2.1 Timing Adjustable](#) for additional requirements related to time-limits for reading and interactions.

Hiding content would have the same result as pausing and jumping to current display (when pause is released).

For a mechanism to be considered "a mechanism for the user to pause," it must provide the user with a means to pause that does not tie up the user or the focus so that the page cannot be used. The word "pause" here is meant in the sense of a "pause button" although other mechanisms than a button can be used. Having an animation stop only so long as a user has focus on it (where it restarts as soon as the user moves the focus away) would not be considered a "mechanism for the user to pause" because it makes the page unusable in the process and would not meet this SC.

This success criterion is specifically concerned with moving, blinking, scrolling, and

auto-updating visual content. For audio content that starts automatically, refer to [1.4.2 Audio Control](#) .

Moving, blinking, scrolling content that starts automatically because of a general user interaction (such as focusing/hovering over an element,

or scrolling the page), rather than as a result of an intentional *activation* (such as activating a button),

and which doesn't provide a way to Pause, Stop, or Hide, will fail this Criterion, and potentially [2.3.3 Animation from Interaction](#) .

It is important to note that the terms "blinking" and "flashing" can sometimes refer to the same content.

- "Blinking" refers to content that causes a distraction problem. Blinking can be allowed for a short time as long as it stops (or can be stopped)
- "Flashing" refers to content that can trigger a seizure (if it is more than 3 per second and large and bright enough). This cannot be allowed even for a second or it could cause a seizure. And turning the flash off is also not an option since the seizure could occur faster than most users could turn it off.
- Blinking usually does not occur at speeds of 3 per second or more, but it can. If blinking occurs faster than 3 per second, it would also be considered a flash.

"Flashing" content that starts automatically will need to be evaluated against [2.3.2 Three Flashes](#) and [2.3.1 Three Flashes or Below Threshold](#) .

Benefits of Pause, Stop, Hide

- Providing content that stops blinking after five seconds or providing a mechanism for users to stop blinking content allows people with certain disabilities to interact with the web page.
- One use of content that blinks is to draw the visitor's attention to that content. Although this is an effective technique for all users with vision, it can be a problem for some users if it persists. For certain groups, including people with low literacy,

reading and intellectual disabilities, and people with attention deficit disorders, content that blinks may make it difficult or even impossible to interact with the rest of the web page.

Examples of Pause, Stop, Hide

An essential animation can be paused without affecting the activity

A website helps users understand 'how things work' through animations that demonstrate processes. Animations have "pause" and "restart" buttons.

A stock ticker

A stock ticker has "pause" and "restart" buttons. Pausing the ticker causes it to pause on the currently displayed stock. Restarting causes the ticker to resume from the stopped point but with a notice that the display is delayed. Since the intent of the stock ticker is usually to provide realtime information, there might also be a button that would advance the ticker to the most recently traded stock.

A game is designed so that users take turns rather than competing in real-time

One party can pause the game without invalidating the competitive aspect of it.

A web advertisement

An advertisement blinks to get viewers attention but stops after 5 seconds

A form prompt

A form blinks an arrow near the submit button if a user finishes filling out the form but does not activate the submit button. The blinking stops after 5 seconds.

An animation

An animation runs in the upper portion of the page but has a "freeze animation" button near the bottom of the animation.

A "loading" animation

A preloader animation is shown on a page which requires a certain percentage of a large file to be downloaded before playback can begin. The animation is the only content on the page and instructs the user to please wait while the video loads. Because the moving content is not presented in parallel with other content, no mechanism to pause, stop or hide it needs to be provided, even though the animation may run for more than 5 seconds for users with slower connections.

A full-page advertisement

A site requires that all users view a 15 second advertisement before they can access free content available from their site. Because viewing the advertisement is a requirement for all users and because it is not presented in parallel with other content, no mechanism to pause, stop or hide it needs to be provided.

2.2.3 No Timing (Level AAA)

Source: [WCAG 2.2](#) / [Understanding 2.2.3](#) — Copyright © 2023 World Wide Web Consortium

Timing is not an essential part of the event or activity presented by the content, except for non-interactive synchronized media and real-time events .

In brief

Goal

Users do not face time limits.

What to do

Do not use time limits, except for video and live events.

Why it's important

People with disabilities often need more time to complete actions.

Intent of No Timing

The intent of this success criterion is to minimize the occurrence of content that requires timed interaction. This enables people with blindness, low vision, cognitive limitations, or motor impairments to interact with content. This differs from the Level A success criterion in that the only exception is for real-time events.

Video only, such as sign language, is covered in [Guideline 1.1](#) .

Benefits of No Timing

- People with physical disabilities often need more time to react, to type and to complete activities. People with low vision need more time to locate things on screen and to read. People who are blind and using screen readers may need more time to understand screen layouts, to find information and to operate controls. People who have cognitive or language limitations

need more time to read and to understand. People who are deaf and communicate in sign language may need more time to read information printed in text (which may be a second language for some).

- In circumstances where a sign-language interpreter may be relating audio content to a user who is deaf, control over time limits is also important.

Examples of No Timing

A test is designed so that time to complete the test does not affect the scoring

Rather than calibrating an on-line test using a time limit, the test is calibrated based on scores when users have no time limits.

A game is designed so that users take turns rather than competing in real-time

One party can pause the game without invalidating the competitive aspect of it.

2.2.4 Interruptions (Level AAA)

Source: [WCAG 2.2](#) / [Understanding 2.2.4](#) — Copyright © 2023 World Wide Web Consortium

Interruptions can be postponed or suppressed by the user, except interruptions involving an emergency .

In brief

Goal

Users are not interrupted.

What to do

Let people delay or turn off updates, except in emergencies.

Why it's important

Updates distract and disrupt assistive technology users and people with attention deficits.

Intent of Interruptions

The intent of this success criterion is to allow users to turn off updates from the author/server except in emergencies.

Emergencies would include civil emergency alert messages or any other messages that warn of danger to health, safety, or property, including data loss, loss of connection, etcetera.

This allows access by people with cognitive limitations or attention disorders by enabling them to focus on the content. It also allows users who are blind or have low vision to keep their "viewing" focus on the content they are currently reading.

Benefits of Interruptions

- Individuals with attention deficit disorders can focus on content without distraction.

- Individuals with low vision or who use screen readers will not have content updated while they are viewing it (which can lead to discontinuity and misunderstanding if they start reading in one topic and finish in another).

Examples of Interruptions

Example 1. Setting user preferences

The preferences page of a web portal includes an option to postpone all updates and alerts until the end of the current session, except for alerts concerning emergencies.

2.2.5 Re-authenticating (Level AAA)

Source: [WCAG 2.2](#) / [Understanding 2.2.5](#) — Copyright © 2023 World Wide Web Consortium

When an authenticated session expires, the user can continue the activity without loss of data after re-authenticating.

In brief

Goal

Users do not lose information or context due to reauthentication.

What to do

Preserve users' prior activity and data through reauthentication.

Why it's important

Some people may require additional time to complete an activity.

Intent of Re-authenticating

The intent of this success criterion is to allow all users to complete authenticated transactions that have inactivity time limits or other circumstances that would cause a user to be logged out while in the midst of completing the transaction.

For security reasons, many sites implement an authentication time limit after a certain period of inactivity. These time limits may cause problems for persons with disabilities because it may take longer for them to complete the activity.

Other sites will log a person out of a session if a person logs in on the website from another computer or if other activities arise that make the site suspicious of whether the person is still the same legitimate person who logged in originally. When users are logged out while still in the midst of a transaction - it is important that they be given the ability to re-authenticate and continue with the transaction without the loss of any data already entered.

Sites that implement session time limits and re-authentication requests are

still subject to the requirements of other criteria, such as [2.2.1 Timing Adjustable](#) .

Benefits of Re-authenticating

- This success criterion benefits people who may require additional time to complete an activity. People with cognitive limitations may read slowly and require additional time to read and respond to a questionnaire. Users interacting via a screen reader may need extra time to navigate and complete a complicated form. A person with motor impairments or who navigates with an alternative input device may require additional time to navigate through or complete input within a form.

Examples of Re-authenticating

A shopping site checkout

A user is logged into a shopping site. While in the middle of the checkout process, the user is interrupted and has to leave their computer. While they are away, the site prompts the user that the process is about to time out, and offers the ability to extend the timeout – but with the user away, the timeout is not extended and the user is logged out. When the user returns to the computer, they have to re-authenticate. After the user logs in, the check out process is restored with the same information and at the same stage. The user did not lose any data because the server had temporarily accepted and stored the submission even though the session had timed out and restored the user to the same state after re-authentication was completed.

Authentication in an email program

An email program has an authentication time-out after 30 minutes. The program prompts the user several minutes before the time-out occurs and provides a link to open a new window in order to re-authenticate. The original window with the in-progress email remains intact and, after re-authentication, the user may send that data.

2.2.6 Timeouts (Level AAA)

Source: [WCAG 2.2](#) / [Understanding 2.2.6](#) — Copyright © 2023 World Wide Web Consortium

Users are warned of the duration of any user inactivity that could cause data loss, unless the data is preserved for more than 20 hours when the user does not take any actions.

Privacy regulations may require explicit user consent before user identification has been authenticated and before user data is preserved. In cases where the user is a minor, explicit consent may not be solicited in most jurisdictions, countries or regions. Consultation with privacy professionals and legal counsel is advised when considering data preservation as an approach to satisfy this success criterion.

In brief

Goal

Users do not lose data due to unknown timeouts.

Author task

Tell users how long their session can be inactive before they may lose information.

Why it's important

People with disabilities may need more time to complete actions.

Intent

The intent of this success criterion is to ensure that when a timeout is used, users know what duration of inactivity will cause the page to time out and result in lost data. The use of timed events can present significant barriers for users with cognitive disabilities, as these users may require more time to read content or to perform functions, such as completing an online form.

During the completion of an online process, such as to reserve a hotel room or purchase a plane ticket, a user with a cognitive impairment may become overwhelmed with lengthy instructions and data input required to complete the process. The user may not be able to complete the process in one sitting and may need to take a break. Users should be able to leave a process without losing their current place within the process, and without losing information that has already been entered. If users cannot take a break and check their work, many will often be unable to complete a task correctly.

This success criterion works in tandem with Success Criterion 2.2.1 Timing Adjustable, but is specifically focused on notification of timeouts related to user inactivity.

The best way to conform to this success criterion is to keep the user data for at least 20 hours. This enables the user with disabilities and the aging community to start and finish a task, taking breaks as needed. However, when it is not practical to save the user data the author must warn the user about the duration of inactivity which will result in a timeout. Timeouts should be displayed to the user once at the beginning of the related task or process and not at each step.

This success criterion only applies to timeouts that are within the content provider's knowledge or control. For example, if the user closes a web browser or device and loses content in an open page that has not yet been submitted, the success criterion has not been violated.

Examples of privacy regulations mentioned in the success criterion note, and related compliance standards, are PCI DSS (Payment Card Industry Data Security Standard) and HIPAA (Health Insurance Portability and Accountability Act of 1996).

Benefits

This success criterion helps users by ensuring they are notified about timeouts related to inactivity.

When a user knows how much time they are allowed for a task, they will know whether they can take a needed break and resume their work without needing to start again. This enables many users to complete tasks online that they otherwise could not do. If a situation exists where a timeout is necessary, the user is warned at the start of the task about the length of inactivity that would generate a timeout. The user can then decide if they can manage this task or not in the given time, or if they need to prepare materials in advance of starting a process. This will reduce the frustration of losing work due to a timeout.

This success criterion helps people with many different cognitive disabilities, including people with:

- language-related disabilities;
- memory-related disabilities;
- focus-and-attention-related disabilities; and
- disabilities that affect executive function and decision making.

Examples

- While making a purchase on an e-commerce website, the information input by the user is stored for more than 20 hours. This helps ensure that if they need to stop working for a while that they are more likely to be able to continue the purchase when they return.
- A web application allowing people to file tax returns provides a notice that the application will time out for security purposes. The notice indicates that a lack of activity for a continuous period of time that is greater than an hour will trigger initiate the time out process.

- An online contact form does not implement any type of time out process. Information entered into the contact form can be submitted at any time and would only be lost if the user closes their browser window.

Related Resources

- [User Needs Table 3: Entering data, error prevention & recovery](#)
- [SSA Best Practices Library: Timeouts](#)

Guideline 2.3: Seizures and Physical Reactions

[Source: W3C WCAG 2.2 —

<https://www.w3.org/TR/WCAG22/#guideline23>]

Do not design content in a way that is known to cause seizures or physical reactions.

2.3.1 Three Flashes or Below Threshold (Level A)

Source: [WCAG 2.2](#) / [Understanding 2.3.1](#) — Copyright © 2023 World Wide Web Consortium

Web pages do not contain anything that flashes more than three times in any one second period,

or the flash is below the general flash and red flash thresholds .

Since any content that does not meet this success criterion can interfere with a user's

ability to use the whole page, all content on the web page (whether it is used to

meet other success criteria or not) must meet this success criterion. See Conformance Requirement 5: Non-Interference .

In brief

Goal

Content does not trigger seizures.

What to do

Avoid content that flashes, or keep it under thresholds.

Why it's important

Flashing content can cause migraines, dizziness, nausea, and seizures.

Intent of Three Flashes or Below Threshold

The intent of this success criterion is to allow users to access the full content of a site without inducing seizures due to photosensitivity.

Individuals who have photosensitive seizure disorders can have a seizure triggered by content that flashes at certain frequencies for

more than a few flashes. People are even more sensitive to red flashing than to other colors, so a special test is provided for saturated red flashing. These guidelines were originally based on guidelines for the broadcasting industry as adapted for desktop monitors, where content is viewed from a closer distance (using a larger angle of vision).

Flashing can be caused by the display, the computer rendering the image or by the content being rendered. The author has no control of the first two. They can be addressed by the design and speed of the display and computer. The intent of this criterion is to ensure that flicker that violates the flash thresholds is not caused by the content itself. For example, the content could contain a video clip or animated image of a series of strobe flashes, or close-ups of rapid-fire explosions.

This success criterion replaces a much more restrictive criterion in WCAG 1.0 that did not allow any flashing (even of a single pixel) within a broad frequency range (3 to 50 Hz). This success criterion is based on existing specifications in use in the UK and by others for television broadcast and has been adapted for computer display viewing. In WCAG 2.0, a 1024 x 768 screen was used as the reference screen resolution for the evaluation. The 341 x 256 pixel block represents a 10 degree viewport at a typical viewing distance. (The 10 degree field is taken from the original specifications and represents the central vision portion of the eye, where people are most susceptible to photo stimuli.)

With the proliferation of devices of varying screen sizes (from small handhelds to large living room displays), as well as the adoption of CSS pixels as a density-independent unit of measurement, the prior assessment criteria may seem outdated. However, an image of a consistent size uses up relatively the same percentage of a user's visual field on any device. On a large screen,

the image takes up less size, but the large screen takes up a larger part of the visual field. On a mobile screen, the image may take up most or all of the screen; however, the mobile screen itself takes up a smaller portion of the user's visual field. So the same dimension of the flashing content, represented in CSS pixels can still provide a consistent means of assessment. Substituting CSS pixels for the original pixel block means that the combined area of flashing becomes 341 x 256 CSS pixels, or a flashing area of 87,296 CSS pixels.

Content should be analyzed at the largest scale at which a user may view the content, and at the standard zoom level of the user agent. For example, with a video that may play in an area of a web page and also at full screen, the video should be analyzed for risks at full screen.

Where video content is provided in color spaces other than sRGB, the version provided with the highest dynamic range should be tested. In such cases the industry standard definition of a general flash is a change in luminance of 20 cd/m² or more where the darker image is below 160 cd/m². ([ITU-R BT.1702](#) .) This is applicable for standard dynamic range (SDR) and high dynamic range (HDR) content. For HDR content when the darker state is 160 cd/m² or more, a general flash is one where the Michelson contrast is 1/17 or greater — where the Michelson contrast is calculated as $(L_{High} - L_{Low}) / (L_{High} + L_{Low})$, and where L_{High} and L_{Low} are the luminance of the high and low luminance states, respectively.

For short clips that might be looped (such as GIF animations), the content should be analyzed while looping.

The specification cannot account for the actual viewing distance that a person chooses. Users that are closer to their screens than the idealized viewing distance will be affected by flashing areas

that normatively pass. The same problem applies to users who rely on zoom or screen magnification. Conversely, users who are further away from the screen than the idealized distance should be able to tolerate flashing areas that are larger than the threshold.

The combined area of flashes occurring concurrently and contiguously means the total area that is actually flashing at the same time. It is calculated by adding up the contiguous area that is flashing simultaneously within any 10 degree angle of view.

The terms "blinking" and "flashing" can sometimes refer to the same content.

- "Blinking" refers to content that causes a distraction problem. Blinking can be allowed for a short time as long as it stops (or can be stopped)
- "Flashing" refers to content that can trigger a seizure (if it is more than 3 per second and large and bright enough). This cannot be allowed even for a second or it could cause a seizure. And turning the flash off is also not an option since the seizure could occur faster than most users could turn it off.
- Blinking usually does not occur at speeds of 3 per second or more, but it can. If blinking occurs faster than 3 per second, it would also be considered a flash.

The new working definition in the field for **"pair of opposing transitions involving a saturated red"** is a pair of opposing transitions where, one transition is either to or from a state with a value $R/(R + G + B)$ that is greater than or equal to 0.8, and the difference between states is more than 0.2 (unitless) in the CIE 1976 UCS chromaticity diagram. [ISO 9241-391]

The chromaticity difference is calculated as:

- $\text{SQRT}((u'1 - u'2)^2 + (v'1 - v'2)^2)$

where $u'1$ and $v'1$ are chromaticity coordinates of State 1 and $u'2$ and $v'2$ are chromaticity coordinates of State 2. The 1976 UCS chromaticity coordinates of u' and v' are calculated as:

- $u' = 4 X / (X + 15 Y + 3 * Z)$
- $v' = 9 Y / (X + 15 Y + 3 * Z)$

where X, Y, and Z are the tristimulus values of a color in the CIE XYZ colorspace, which can be calculated as:

- $X = 0.4124564 R + 0.3575761 G + 0.1804375 * B$
- $Y = 0.2126729 R + 0.7151522 G + 0.0721750 * B$
- $Z = 0.0193339 R + 0.1191920 G + 0.9503041 * B$

where R, G, & B are values that range from 0-1 as specified in “relative luminance” definition.

Benefits of Three Flashes or Below Threshold

- Individuals who have seizures when viewing flashing material will be able to view all of the material on a site without having a seizure and without having to miss the full experience of the content by being limited to text alternatives. This includes people with photosensitive epilepsy as well as other photosensitive seizure disorders.

Examples of Three Flashes or Below Threshold

- A website has video of muzzle flash of machine gun fire, but limits the size of the flashing image to a small portion of the screen below the flash threshold size.
- A movie with a scene involving very bright lightning flashes is edited so that the lightning only flashes three times in any one second period.

Resources for Three Flashes or Below Threshold

- [Harding FPA Web Site](#)
- [Trace Center Photosensitive Epilepsy Analysis Tool \(PEAT\)](#)
- [Information about Photosensitive Seizure Disorders](#)
- [Epilepsy Action](#)
- [Epilepsy Foundation - Photosensitivity and Seizures](#)

2.3.2 Three Flashes (Level AAA)

Source: [WCAG 2.2](#) / [Understanding 2.3.2](#) — Copyright © 2023 World Wide Web Consortium

Web pages do not contain anything that flashes more than three times in any one second period.

In brief

Goal

Content does not trigger seizures.

What to do

Do not flash content more than 3 times a second.

Why it's important

Flashing content can cause migraines, dizziness, nausea, and seizures.

Intent of Three Flashes

The purpose of this success criterion is to further reduce the chance of seizures. Seizures cannot be completely eliminated since some people are so sensitive. However, by eliminating all 3-per-second flashing over any area of the screen, the chances of a person having a seizure are further reduced than when just meeting the measures ordinarily used today in standards internationally, as we do at Level A.

Compared to [Success Criterion 2.3.1 Three Flashes or Below Threshold](#) – which allows flashing if it is dim enough or has a small enough area – this criterion

does not allow any flashing that occurs at a frequency greater than 3 per second, regardless

of brightness or size. As a result, even a single flashing pixel would violate this criterion. The

intent is to guard against flashing larger than a single pixel, but since an unknown

amount of magnification or high contrast setting may be applied, the prohibition is

against any flashing.

In some cases, what we refer to as "blinking" and what we refer to as "flashing" may

overlap slightly. We are using different terms for the two because "blinking" causes

a distraction problem which you can allow for a short time as long as it stops (or

can be stopped) whereas "flashing" is a seizure trigger and cannot be allowed or it

will cause a seizure. The seizure would occur faster than most users could turn it

off. "Blink" therefore refers to slow repeating changes that would distract. "Flash"

refers to changes that could cause a seizure if they were bright enough or persisted

long enough. Blinking usually doesn't occur at speeds of 3 per second or more so blink

and flash do not overlap. However, blinking can occur faster than 3 per second so

there could be an overlap. See [2.2.2 Pause, Stop, Hide](#) for more information on blink.

Benefits of Three Flashes

- Individuals who have seizures when viewing flashing material will be able to view all of the material on a site without having a seizure and without having to miss the full experience of the content by being limited to text alternatives. This includes people with photosensitive epilepsy as well as other photosensitive seizure disorders.

Examples of Three Flashes

- A movie with a scene involving very bright lightning flashes is edited so that the lightning only flashes three times in any one second period.

Resources for Three Flashes

- [Harding FPA Web Site](#)
- [Trace Center Photosensitive Epilepsy Analysis Tool \(PEAT\)](#)
- [Information about Photosensitive Seizure Disorders](#)
- [Epilepsy Action](#)
- [Epilepsy Foundation - Photosensitivity and Seizures](#)

2.3.3 Animation from Interactions (Level AAA)

Source: [WCAG 2.2](#) / [Understanding 2.3.3](#) — Copyright © 2023 World Wide Web Consortium

Motion animation triggered by interaction can be disabled, unless the animation is essential to the functionality or the information being conveyed.

In brief

Goal

Users are not harmed or distracted by motion.

What to do

Support user preferences for motion, and eliminate unnecessary motion effects.

Why it's important

People can get sick from motion effects.

Intent

The intent of this success criterion is to allow users to prevent animation from being displayed on web pages. Some users experience distraction or nausea from animated content. For example, if scrolling a page causes elements to move (other than the essential movement associated with scrolling) it can trigger vestibular disorders. Vestibular (inner ear) disorder reactions include dizziness, nausea and headaches. Another animation that is often non-essential is parallax scrolling. Parallax scrolling occurs when backgrounds move at a different rate to foregrounds. Animation that is essential to the functionality or information of a web page is allowed by this success criterion.

"Animation from interactions" applies when a user's interaction initiates non-essential animation. In contrast, [2.2.2 Pause, Stop,](#)

[Hide](#) applies when the web page initiates animation "automatically" that is not in response to an intentional user activation. There may be situations where a particular animation may fail *both* success criteria.

The impact of animation on people with vestibular disorders can be quite severe. Triggered reactions include nausea, migraine headaches, and potentially needing bed rest to recover.

How can a website reduce the chances of triggering a vestibular disorder? Choose any one of the following solutions. Avoid using unnecessary animation. Provide a control for users to turn off non-essential animations from user interaction. Take advantage of the reduce motion feature in the user agent or operating system.

What about movement caused by a user scrolling a page? Moving new content into the viewport is essential for scrolling. The user controls the essential scrolling movement so it is allowed. Only add non-essential animation to the scrolling interaction in a responsible way. Always give users the ability to turn off unnecessary movement.

Benefits

- **Vestibular Disorder** People with vestibular disorders need control over movement triggered by interactions. Non-essential movement can trigger vestibular disorder reactions. Vestibular (inner ear) disorder reactions include distraction, dizziness, headaches and nausea. Persona Quote: "Stop that extra movement! You are making me so dizzy I cannot concentrate. Now I have to turn off my computer and go lie down."

Examples

Parallax scrolling with option to turn off unnecessary motion globally

A site includes extra animations when the user scrolls. Decorative elements move in and out of view horizontally when the essential page content is scrolled vertically. A control at the top of each page allows the user to turn off unnecessary animations. The ability to turn off non-essential animations is a site-wide setting.

Transitions that support the reduce motion preference

A site includes a non-essential transition when loading new content. The transition is a page-flipping animation that respects the prefers-reduced-motion CSS media query. When the user enables the reduce motion preference, the page-flipping animation is turned off.

Essential animation

A web application provides a feature to author animated sequences. As part of this tool the author needs to preview the animation.

Resources

- [Mozilla documentation for prefers-reduced-motion](#)
- [Demonstration of prefers-reduced-motion in Webkit](#)
- [An Introduction to the Reduced Motion Media Query](#)
- [Designing Safer Web Animations for Motion Sensitivity](#)
- [iOS: Reduce Motion on iPhone, iPad or iPod touch](#)
- [Mac: Reduce Motion](#)
- [Windows 10: Reduce motion](#)

Guideline 2.4: Navigable

[Source: W3C WCAG 2.2 —

<https://www.w3.org/TR/WCAG22/#guideline24>]

Provide ways to help users navigate, find content, and determine where they are.

2.4.1 Bypass Blocks (Level A)

Source: [WCAG 2.2](#) / [Understanding 2.4.1](#) — Copyright © 2023 World Wide Web Consortium

A mechanism is available to bypass blocks of content that are repeated on multiple web pages .

In brief

Goal

Users can more easily navigate by keyboard.

What to do

Provide a means of skipping repeating content.

Why it's important

Users reliant on the keyboard interface can move around pages efficiently.

Intent of Bypass Blocks

The intent of this success criterion is to allow people who navigate sequentially through content more direct access to the primary content of the web page. Web pages and applications often have content that appears on other pages or screens. Examples of repeated blocks of content include but are not limited to navigation links, header content, and advertising frames. Small repeated sections such as individual words, phrases or single links are not considered blocks for the purposes of this provision.

Users who navigate sequentially through content will generally have to navigate through repeated content on each page. This is in contrast to a sighted user's ability to ignore the repeated material either by focusing on the center of the screen (where main content usually appears) or a mouse user's ability to select a link with a

single mouse click rather than encountering every link or form control that comes before the item they want.

It is not the intent of this success criterion to require authors to provide methods that are redundant to functionality provided by the user agent. Most web browsers provide keyboard shortcuts to move the user focus to the top of the page, so if a set of navigation links is provided at the bottom of a web page providing a "skip" link may be unnecessary.

Although this success criterion deals with blocks of content that are repeated on multiple pages, we also strongly promote structural markup on individual pages as per Success Criteria 1.3.1.

Although the success criterion does not specifically use the term “within a set of web pages”, the concept of the pages belonging to a set is implied. An author would not be expected to avoid any possible duplication of content in any two pages that are not in some way related to each other, and are not "web pages that share a common purpose and that are created by the same author, group or organization” (the definition of set of web pages).

Even for web pages that are not in a set, if a web page has blocks of text that are repeated within the page it may be helpful (but not required) to provide a means to skip over them.

Benefits of Bypass Blocks

When this success criterion is not satisfied, it may be difficult for people with some disabilities to reach the main content of a web page quickly and easily:

- Screen reader users who visit several pages on the same site can avoid having to hear all header content and dozens of

navigation links on every page before the main content is spoken.

- People who use only the keyboard or a keyboard interface can reach content with fewer keystrokes. Otherwise, they might have to make dozens of keystrokes before reaching a link in the main content area. This can take a long time and may cause severe physical pain for some users.
- People who use screen magnifiers do not have to search through the same header content or other blocks of information to find where the main content begins each time they enter a new page.
- People with cognitive limitations as well as people who use screen readers may benefit when links are grouped into lists

Examples of Bypass Blocks

- A news organization's home page contains a main story in the middle of the page, surrounded by many blocks and sidebars for advertising, searching, and other services. There is a link at the top of the page that jumps to the main story. Without using this link, a keyboard user needs to tab through approximately 40 links to reach the main story; the screen reader user has to listen to 200 words; and the screen magnifier user must search around for the location of the main body.
- An e-commerce website includes a long list of filters prior to the search results listing. A link above the list enables users to skip the filters and get to the product results quickly.

Resources for Bypass Blocks

- [WebAIM: Semantic Structure](#)
- [Heading Tags](#)

2.4.2 Page Titled (Level A)

Source: [WCAG 2.2](#) / [Understanding 2.4.2](#) — Copyright © 2023 World Wide Web Consortium

Web pages have titles that describe topic or purpose.

In brief

Goal

Each web page has a meaningful title.

What to do

Provide a descriptive page title using appropriate technology.

Why it's important

Page titles help users identify and distinguish different pages.

Intent of Page Titled

The intent of this success criterion is to help users find content and orient themselves within it by ensuring that each web page has a descriptive title. Titles identify the current location without requiring users to read or interpret page content. When titles appear in site maps or lists of search results, users can more quickly identify the content they need. User agents make the title of the page easily available to the user for identifying the page. For instance, a user agent may display the page title in the window title bar or as the name of the tab containing the page.

In cases where the page is a document or a web application, the name of the document or web application would be sufficient to describe the purpose of the page. Note that it is not required to use the name of the document or web application; other things may also describe the purpose or the topic of the page.

In cases such as Single Page Applications (SPAs), where various distinct pages/views are all nominally served from the same URI

and the content of the page is changed dynamically, the title of the page should also be changed dynamically to reflect the content or topic of the current view.

[Success Criteria 2.4.4 Link Purpose \(In Context\)](#) and [2.4.9 Link Purpose \(Link Only\)](#) deal with the purpose of

links, many of which are links to web pages. Here also,

the name of a document or web application being linked to would be sufficient to describe

the purpose of the link. Having the link and the title agree, or be very similar,

is good practice and provides continuity between the link 'clicked on' and the web

page that the user lands on.

Benefits of Page Titled

- This criterion benefits all users in allowing users to quickly and easily identify whether the information contained in the web page is relevant to their needs.
- People with visual disabilities will benefit from being able to differentiate content when multiple web pages are open.
- People with cognitive disabilities, limited short-term memory and reading disabilities also benefit from the ability to identify content by its title.
- This criterion also benefits people with severe mobility impairments whose mode of operation relies on audio when navigating between web pages.

Examples of Page Titled

An HTML web page

The descriptive title of an HTML web page is marked up with the `<title>` element so that it will be displayed in the title bar of the user agent.

A set of web pages

The title of the landing page is "ARIA Authoring Practices Guide | APG | WAI | W3C" The patterns list page has the title "Patterns | APG | WAI | W3C" Specific patterns are pages titled "X Pattern | APG | WAI | W3C" (e.g., "Alert and Message Dialogs Pattern | APG | WAI | W3C") Specific examples for each pattern are pages titled "X Example | APG | WAI | W3C" (e.g., "Alert Dialog Example | APG | WAI | W3C") Practices page has the title "Practices | APG | WAI | W3C" Index page has the title "Index | APG | WAI | W3C"

A web application

A banking application lets users inspect their bank accounts, view past statements, and perform transactions. The web application dynamically generates titles for each web page, e.g., "Bank XYZ, accounts for Alex Smith" "Bank XYZ, December 2005 statement for Account 1234-5678".

Resources for Page Titled

- `<title>`: The Document Title element: Usage notes (Accessibility) .
- Guidelines for Accessible and Usable Web Sites: Observing Users Who Work With Screen Readers (PDF) . Theofanos, M.F., and Redish, J. (2003). Interactions, Volume X, Issue 6, November-December

2003, pages 38-51,

<https://dl.acm.org/doi/10.1145/947226.947227>

2.4.3 Focus Order (Level A)

Source: [WCAG 2.2](#) / [Understanding 2.4.3](#) — Copyright © 2023 World Wide Web Consortium

If a web page can be navigated sequentially and the navigation sequences affect meaning or operation, focusable components receive

focus in an order that preserves meaning and operability.

In brief

Goal

Keyboard users navigate content in a correct order.

What to do

Elements receive focus in an order that preserves meaning.

Why it's important

Navigating a site or application with only the keyboard will make sense.

Intent of Focus Order

The intent of this success criterion is to ensure that when users navigate sequentially through content, they encounter information in an order that is consistent with the meaning of the content and can be operated from the keyboard. This reduces confusion by letting users form a consistent mental model of the content. There may be different orders that reflect logical relationships in the content. For example, moving through components in a table one row at a time or one column at a time both reflect the logical relationships in the content. Either order may satisfy this success criterion.

The way that sequential navigation order is determined in web content is defined by

the technology of the content. For example, simple HTML defines sequential navigation

via the notion of tabbing order. Dynamic HTML may modify the navigation sequence using

scripting along with the addition of a `tabindex` attribute to allow focus to additional

elements. If no scripting or `tabindex` attributes are used, the navigation order is

the order that components appear in the content stream. (See [Focus in the HTML Living Standard](#)).

An example of keyboard navigation that is not the sequential navigation addressed by this success criterion is using arrow key navigation to traverse a tree component. The user can use the up and down arrow keys to move from tree node to tree node.

Pressing the right arrow key may expand a node, then using the down arrow key, will move into the newly expanded nodes. This navigation sequence follows the expected sequence for a tree control - as additional items get expanded or collapsed, they are added or removed from the navigation sequence.

The focus order may not be identical to the programmatically determined reading order

(see [Success Criterion 1.3.2 Meaningful Sequence](#))

as long as the user can still understand and operate

the web page. Since there may be several possible logical reading orders for the content,

the focus order may match any of them. However, when the order of a particular presentation

differs from the programmatically determined reading order, users of one of these

presentations may find it difficult to understand or operate the web page. Authors

should carefully consider all these users as they design their web pages.

For example, a screen reader user interacts with the programmatically determined reading order, while a sighted keyboard user interacts with the visual presentation of the web page. Care should be taken so that the focus order makes sense to both of these sets of users and does not appear to either of them to jump around randomly.

Focus order needs to allow the user to navigate focusable elements in a logical order, and that order needs to preserve any meaning or operation that the page is conveying. Focus order does not necessarily need to follow the visual layout of the web page, as long as the order in which elements receive focus is logical, and the hierarchy and relationship of content implied by the visual presentation is preserved. For instance, in a two-column layout, the assumption in western left-to-right systems may be that focus moves through the elements in the left-hand column first, from the top to the bottom of the column, and then proceed to the right-hand column. However, if the two columns are independent of each other, and meaning/operation are not affected, it is not a failure if elements in the right-hand column receive focus first, followed by the elements in the left-hand column. As a best practice, though, make sure the focus order reinforces the reading order implied by the visual layout.

This requirement does not specify what should or should not receive focus, but rather assesses the order in which elements

receive focus. This success criterion does not prohibit making non-operable content (e.g., static text) focusable, or programmatically moving focus to such content. However, it is a failure of Focus Order if items receive focus in an order that impedes the meaning or operation of content, or creates confusing or illogical focus orders — for example, a control appearing to receive focus multiple times due to the use of nested focusable elements.

```
<div tabindex="0"><button>...</button></div>
```

As a best practice, avoid having focusable elements which cannot be operated or actioned, as these are likely to make operation tedious for keyboard users.

For clarity:

- Focusable components need to receive focus in an order that preserves meaning and operability.
- Focus order does not necessarily need to follow the visual presentation or layout.
- If there is more than one order that preserves meaning and operability, only one of them needs to be provided.
- Static/non-interactive elements can receive focus, as long as they don't impede operation of the content, or result in confusing or illogical focus order.

Benefits of Focus Order

These techniques benefit keyboard users who navigate documents sequentially and expect the focus order to be consistent with the sequential reading order.

- People with mobility impairments who must rely on keyboard access for operating a page benefit from a logical, usable focus order.

- People with disabilities that make reading difficult can become disoriented when tabbing takes focus someplace unexpected. They benefit from a logical focus order.
- People with visual impairments can become disoriented when tabbing takes focus someplace unexpected or when they cannot easily find the content surrounding an interactive element.
- Only a small portion of the page may be visible to an individual using a screen magnifier at a high level of magnification. Such a user may interpret a field in the wrong context if the focus order is not logical.

Examples of Focus Order

- On a web page that contains a tree of interactive controls, the user can use the up and down arrow keys to move from tree node to tree node. Pressing the right arrow key expands a node, then using the down arrow key moves into the newly expanded nodes.
- A web page implements non-modal dialogs. When the trigger button is activated, a dialog opens. The interactive elements in the dialog are inserted in the focus order immediately after the button. When the dialog is open, the focus order goes from the button to the elements of the dialog, then to the interactive element following the button. When the dialog is closed, the focus order goes from the button to the following element.
- A web page implements modal dialogs. When the trigger button is activated, a dialog opens and focus is set within the dialog. As long as the dialog is open, all web page content outside the dialog becomes inert and cannot receive focus (though, depending on implementation, the focus cycle might still include user agent controls). When the dialog is dismissed,

focus returns to the button or the element following the button.

- An HTML web page is created with the left hand navigation occurring in the HTML after the main body content, and styled with CSS to appear on the left hand side of the page. This is done to allow focus to move to the main body content first without requiring tabindex attributes or JavaScript. While this example passes the Success Criterion, it is not necessarily true that all CSS positioning would. More complex positioning examples may or may not preserve meaning and operability
- The following example fails to meet the Success Criterion : A company's website includes a form that collects marketing data and allows users to subscribe to several newsletters published by the company. The section of the form for collecting marketing data includes fields such as name, street address, city, state or province, and postal code. Another section of the form includes several checkboxes so that users can indicate newsletters they want to receive. However, the tab order for the form skips between fields in different sections of the form, so that focus moves from the name field to a checkbox, then to the street address, then to another checkbox.

While this example passes the Success Criterion, it is not necessarily true that all CSS positioning would. More complex positioning examples may or may not preserve meaning and operability

2.4.4 Link Purpose (In Context) (Level A)

Source: WCAG 2.2 / Understanding 2.4.4 — Copyright © 2023 World Wide Web Consortium

The purpose of each link can be determined from the link text alone or from the link text together with its programmatically determined link context , except where the purpose of the link would be ambiguous to users in general .

In brief

Goal

Users understand what each link will do.

What to do

Provide descriptive names or context for all links.

Why it's important

People with visual and cognitive disabilities can navigate more easily.

Intent of Link Purpose (In Context)

The intent of this success criterion is to help users understand the purpose of each link so they can decide whether they want to follow the link. Whenever possible, provide link text that identifies the purpose of the link without needing additional context.

Assistive technology has the ability to provide users with a list of links that are on the web page. Link text that is as meaningful as possible will aid users who want to choose from this list of links. Meaningful link text also helps those who wish to tab from link to link. Meaningful links help users choose which links to follow without requiring complicated strategies to understand the page.

The text of, or associated with, the link is intended to describe the purpose of the link. In cases where the link takes one to a

document or a web application, the name of the document or web application would be sufficient to describe the purpose of the link (which is to take you to the document or web application). Note that it is not required to use the name of the document or web application; other things may also describe the purpose of the link.

[Success Criterion 2.4.2 Page Titled](#) deals with the titles of pages. Here also, the name of a document or web application

being presented on the page would be sufficient to describe the purpose of the page.

Having the link and the title agree, or be very similar, is good practice and provides

continuity between the link 'clicked on' and the web page that the user lands on.

In some situations, authors may want to provide part of the description of the link in logically related text that provides the context for the link. In this case the user should be able to identify the purpose of the link without moving focus from the link. In other words, they can arrive on a link and find out more about it without losing their place. This can be achieved by putting the description of the link in the same sentence, paragraph, list item, or table cell as the link, or in the table header cell for a link in a data table, because these are directly associated with the link itself. Alternatively, authors may choose to use an ARIA technique to associate additional text on the page with the link.

This context will be most usable if it precedes the link. (For instance, if you must use ambiguous link text, it is better to put it at the end of the sentence that describes its destination, rather than putting the ambiguous phrase at the beginning of the sentence.) If the description follows the link, there can be

confusion and difficulty for screen reader users who are reading through the page in order (top to bottom).

It is a best practice for links with the same destination to have consistent text

(and this is a requirement per [Success Criterion 3.2.4 Consistent Identification](#) for pages in a set). It is also a best practice for links with different purposes

and destinations to have different link text.

A best practice for links to conforming alternate versions is to ensure that the link text to the conforming alternate version indicates in link text that the page it leads to represents the more accessible version. This information may also be provided in text - the goal is to ensure that the end user knows what the purpose of the link is.

The success criterion includes an exception for links for which the purpose of the link cannot be determined from the information on the web page. In this situation, the person with the disability is not at a disadvantage; there is no additional context available to understand the link purpose. However, whatever amount of context is available on the web page that can be used to interpret the purpose of the link must be made available in the link text or programmatically associated with the link to satisfy the success criterion.

There may be situations where the purpose of the link is supposed to be unknown or obscured. For instance, a game may have links identified only as door #1, door #2, and door #3. This link text would be sufficient because the purpose of the links is to create suspense for all users.

See also [2.4.9 Link Purpose \(Link Only\)](#) .

Benefits of Link Purpose (In Context)

- This success criterion helps people with motion impairment by letting them skip links that they are not interested in, avoiding the keystrokes needed to visit the referenced content and then returning to the current content.
- People with cognitive limitations will not become disoriented by multiple means of navigation to and from content they are not interested in.
- People with visual disabilities will be able to determine the purpose of a link by exploring the link's context.

Examples of Link Purpose (In Context)

A link contains text that gives a description of the information at that URI

A page contains the sentence "There was much bloodshed during the Medieval period of history." Where "Medieval period of history" is a link.

A link is preceded by a text description of the information at that URI

A page contains the sentence "Learn more about the Government of Ireland's Commission on Electronic Voting at Go Vote!" where "Go Vote!" is a link.

Both an icon and text are included in the same link

An icon of a voting machine and the text "Government of Ireland's Commission of Electronic Voting" are combined to make a single link. The alt text for the icon is null, since the purpose of the link is already described by the text of the link next to the icon.

A list of book titles

A list of books is available in three formats: HTML, PDF, and mp3 (a recording of a person reading the book). To avoid hearing the

title of each book three times (once for each format), the first link for each book is the title of the book, the second link says "PDF" and the third says, "mp3."

News article summaries

A web page contains a collection of news articles. The main page lists the first few sentences of each article, followed by a "Read more" link. A screen reader command to read the current paragraph provides the context to interpret the purpose of the link.

Resources for Link Purpose (In Context)

- [Using Link Titles to Help Users Predict Where They Are Going](#)
- [WebAIM Techniques for Hypertext Links](#)

2.4.5 Multiple Ways (Level AA)

Source: [WCAG 2.2](#) / [Understanding 2.4.5](#) — Copyright © 2023 World Wide Web Consortium

More than one way is available to locate a web page within a set of web pages except where the web page is the result of, or a step in, a process .

In brief

Goal

Users can get to content in multiple ways.

What to do

Provide at least two options for reaching the same content.

Why it's important

Not everyone can navigate content in the same way.

Intent of Multiple Ways

The intent of this success criterion is to make it possible for users to locate content in a manner that best meets their needs. Users may find one technique easier or more comprehensible to use than another.

Even small sites should provide users some means of orientation. For a three or four page site, with all pages linked from the home page, it may be sufficient simply to provide links from and to the home page where the links on the home page can also serve as a site map.

Benefits of Multiple Ways

- Providing an opportunity to navigate sites in more than one manner can help people find information faster. Users with visual impairments may find it easier to navigate to the correct

part of the site by using a search, rather than scrolling through a large navigation bar using a screen magnifier or screen reader. A person with cognitive disabilities may prefer a table of contents or site map that provides an overview of the site rather than reading and traversing through several web pages. Some users may prefer to explore the site in a sequential manner, moving from web page to web page in order to best understand the concepts and layout.

- Individuals with cognitive limitations may find it easier to use search features than to use a hierarchical navigation scheme that may be difficult to understand.
- For readers of digital publications like EPUBs, being able to scan and navigate to the sections of a work using a table of contents makes it easier both to locate content and to find their position again. Providing an index allows readers to locate specific information they are interested in by topic. In mixed print and digital environments, such as education, having the ability to navigate to static page break locations allows readers to coordinate their position within a work with others.

Examples of Multiple Ways

A search mechanism

A large food processing company provides a site containing recipes created using its products. The site provides a search mechanism to search for recipes using a particular ingredient. In addition, it provides a list box that lists several categories of foods. A user may type "soup" in to the search engine or may select "soup" from the list box to go to a page with a list of recipes made from the company's soup products.

Links between web pages

A local hair salon has created a website to promote its services. The site contains only five web pages. There are links on each web page to sequentially move forward or backward through the web pages. In addition, each web page contains a list of links to reach each of the other web pages.

Where content is a result of a process or task - Funds transfer confirmation

An on-line banking site allows fund transfer between accounts via the web. There is no other way to locate the confirmation of fund transfer until the account owner completes the transfer.

Where content is a result of a process or task - Search engine results

A search engine provides the search results based on user input. There is no other way to locate the search results except to perform the search process itself.

2.4.6 Headings and Labels (Level AA)

Source: [WCAG 2.2](#) / [Understanding 2.4.6](#) — Copyright © 2023 World Wide Web Consortium

Headings and labels describe topic or purpose.

In brief

Goal

A page's content is described in headings and labels

What to do

Provide descriptive headings and labels

Why it's important

People can orient themselves, especially those with cognitive or visual disabilities.

Intent of Headings and Labels

The intent of this success criterion is to help users understand what information is contained in web pages and how that information is organized. When headings are clear and descriptive, users can find the information they seek more easily, and they can understand the relationships between different parts of the content more easily. Descriptive labels help users identify specific components within the content.

Labels and headings do not need to be lengthy. A word, or even a single character, may suffice if it provides an appropriate cue to finding and navigating content.

Labels of form controls are usually text-based. In some cases, images can serve as descriptive labels without additional text. In these cases, authors should ensure that the image and its use as a label (in context) are widely understood.

Note that the same image can be interpreted differently in different contexts. However, it can still be considered descriptive if its use is commonly understood in each context. For example, when accompanying a text field, a loupe or magnifying glass icon with text alternative of "Search" is commonly interpreted as indicating the field is for entering and submitting a search query.

Placed on or near another image, a loupe or magnifying glass icon is commonly interpreted as a means to view a magnified version of the image (for instance, acting as a mechanism to zoom into the image, or opening a full-sized image in a new window).

This success criterion requires that if headings or labels are provided, they be descriptive. This success criterion does not require headings or labels; labels for inputs are covered separately by [3.3.2 Labels or Instructions](#) . This success criterion also

does not require that content acting as a heading or label be correctly marked up or

identified — that aspect is covered separately by [1.3.1 Info and Relationships](#) . It is possible for content

to pass this success criterion (providing descriptive content that acts as headings or labels) while failing

Success Criterion 1.3.1 (if the headings or labels aren't correctly marked up/identified). Conversely,

it is also possible for content to pass Success Criterion 1.3.1 (with headings or labels correctly

marked up or identified), while failing this success criterion (if those headings or labels are inaccurate or insufficiently clear).

Further, in the case of labels, this success criterion does not take into consideration whether or not

alternative methods of providing an accessible name for form controls and inputs have been

used — that aspect is covered separately by [4.1.2 Name, Role, Value](#) . It is possible

for controls and inputs to have an appropriate accessible name (e.g. using `aria-label="..."`)

and therefore pass Success Criterion 4.1.2, but to still fail this success criterion (if the label is inaccurate or insufficiently clear or descriptive).

This success criterion does not require the use of labels; however, it does require that if labels are present, they must be accurate and sufficiently clear or descriptive. Please see [3.3.2 Labels or Instructions](#) for more information on the use of labels.

Benefits of Headings and Labels

- Descriptive headings are especially helpful for users who have disabilities that make reading slow and for people with limited short-term memory. These people benefit when section titles make it possible to predict what each section contains.
- Form input controls with labels that clearly and accurately describe the content that is expected to be entered helps users know how to successfully complete the form.
- When headings and labels are also correctly marked up and identified in accordance with [1.3.1 Info and Relationships](#) , this success criterion

helps people who use screen readers by ensuring that labels and headings are clearer when

presented in a different format — for example, in an automatically generated list of

headings, a table of contents, or when jumping from heading to heading within a page.

Examples of Headings and Labels

A news site

The home page of a news site lists the headlines for the top stories of the hour. Under each heading are the first 35 words of the story and a link to the full article. Each headline gives a clear and accurate idea of the article's subject.

A guide on how to write well

A guide on writing contains the following section titles: How To Write Well , Cut Out Useless Words , Identify Unnecessary Words , and so on. The section headings are clear and concise and the structure of the information is accurately reflected in the structure of the headings.

Consistent headings in different articles

A website contains papers from a conference. Submissions to the conference are required to have the following organization: Summary , Introduction , [other sections unique to this article], Conclusion , Author Biography , Glossary , and Bibliography . The title of each web page clearly identifies the article it contains, creating a useful balance between the uniqueness of the articles and the consistency of the section headings.

A form asking for the name of the user

A form asks for the name of the user. It consists of two input fields to ask for the first and last name. The first field is labeled First name , the second is labeled Last name .

A search field labeled by a magnifying glass icon

A search text input is followed by a button containing a magnifying glass icon that activates the search function. The icon has the string "search" as programmatically determinable label.

Resources for Headings and Labels

- How Users Read on the Web A study showing that most users scan web pages rather than reading them word by word.
- Applying Writing Guidelines to Web Pages A report on the effects of making websites concise, easy to scan, and objective.

2.4.7 Focus Visible (Level AA)

Source: [WCAG 2.2](#) / [Understanding 2.4.7](#) — Copyright © 2023 World Wide Web Consortium

Any keyboard operable user interface has a mode of operation where the keyboard focus indicator is visible.

In brief

Goal

Users know which element has keyboard focus.

What to do

Ensure each item receiving focus has a visible indicator.

Why it's important

Without a focus indicator, sighted keyboard users cannot operate the page.

Intent of Focus Visible

The purpose of this success criterion is to help a person know which element has the keyboard focus.

“Mode of operation” accounts for user agents which may not always show a focus indicator, or only show the focus indicator when the keyboard is used. User agents may optimize when the focus indicator is shown, such as only showing it when a keyboard is used. Authors are responsible for providing at least one mode of operation where the focus is visible. In most cases there is only one mode of operation so this success criterion applies. The focus indicator must not be time limited, when the keyboard focus is shown it must remain.

There may be situations where mouse/pointer users could also benefit from having a visible focus indicator, even though they did

not set focus to an element using the keyboard. As a best practice, consider still providing an explicit focus indicator for these cases.

Note that a keyboard focus indicator can take different forms. While Focus Visible does not specify what that form is, 2.4.13 Focus Appearance (Level AAA) provides guidance on creating a consistent, visible indicator.

Benefits of Focus Visible

- This success criterion helps anyone who relies on the keyboard to operate the page, by letting them visually determine the component on which keyboard operations will interact at any point in time.
- People with attention limitations, short term memory limitations, or limitations in executive processes benefit by being able to discover where the focus is located.

Examples of Focus Visible

- When text fields receive focus, a vertical bar is displayed in the field, indicating that the user can insert text, OR all of the text is highlighted, indicating that the user can type over the text.
- When a user interface control receives focus, a visible border is displayed around it.

Resources for Focus Visible

- [Styling form controls with CSS, revisited](#)

2.4.8 Location (Level AAA)

Source: [WCAG 2.2](#) / [Understanding 2.4.8](#) — Copyright © 2023 World Wide Web Consortium

Information about the user's location within a set of web pages is available.

In brief

Goal

Users know where they are in a set of pages.

What to do

Use breadcrumbs, site maps, or other indicators to give context.

Why it's important

Location indicators reduce confusion for people with cognitive disabilities.

Intent of Location

The intent of this success criterion is to provide a way for the user to orient herself within a set of web pages, a website, or a web application and find related information.

Benefits of Location

- This success criterion is helpful for people with a short attention span who may become confused when following a long series of navigation steps to a web page. It is also helpful when a user follows a link directly to a page deep within a set of web pages and needs to navigate that website to understand the content of that page or to find more related information.

Examples of Location

Links to help user determine their location in a site

A research group is part of an educational department at a university. The group's home page links to the department home page and the university's home page.

A breadcrumb trail

A portal website organizes topics into categories. As the user navigates through categories and subcategories, a breadcrumb trail shows the current location in the hierarchy of categories. Each page also contains a link to the portal home page.

Resources for Location

- [The Sound of the Accessible Title Tag Separator](#)

2.4.9 Link Purpose (Link Only) (Level AAA)

Source: [WCAG 2.2](#) / [Understanding 2.4.9](#) — Copyright © 2023 World Wide Web Consortium

A mechanism is available to allow the purpose of each link to be identified from link text alone,

except where the purpose of the link would be ambiguous to users in general .

In brief

Goal

Users understand what each link will do.

What to do

Provide descriptive names for all links.

Why it's important

Descriptive link text is more understandable for all users, especially when using assistive technology.

Intent of Link Purpose (Link Only)

The intent of this success criterion is to help users understand the purpose of each

link in the content, so they can decide whether they want to follow it. Best practice

is that links with the same destination would have the same descriptions, but links

with different purposes and destinations would have different descriptions (see also [Success Criterion 3.2.4 Consistent Identification](#) which calls for consistency in identifying components that have the same functionality).

Because the purpose of a link can be identified from its link text, links can be understood

when they are out of context, such as when the user agent provides a list of all the

links on a page.

The text in the link is intended to describe the purpose of the link. In cases where the link takes one to a document or a web application, the name of the document or web application would be sufficient to describe the purpose of the link (which is to take you to the document or web application). Note that it is not required to use the name of the document or web application; other things may also describe the purpose of the link.

[Success Criterion 2.4.2 Page Titled](#) deals with the titles of pages.

Here also, the name of a document or web application

being presented on the page would be sufficient to describe the purpose of the page.

Having the link and the title agree, or be very similar, is good practice and provides

continuity between the link 'clicked on' and the web page that the user lands on.

The success criterion includes an exception for links for which the purpose of the link cannot be determined from the information on the web page. In this situation, the person with the disability is not at a disadvantage; there is no additional context available to understand the link purpose. However, whatever amount of context is available on the web page that can be used to interpret the purpose of the link must be made available in the link text to satisfy the success criterion.

The word "mechanism" is used to allow authors to either make all links fully understandable out of context by default or to provide a way to make them this way. This is done because for some pages, making the links all unambiguous by themselves makes the pages easier for some users and harder for others. Providing the ability to make the links unambiguous (by them selves) or not provides both users with disabilities with the ability to use the page in the format that best meets their needs.

For example: A page listing 100 book titles along with links to download the books in HTML, PDF, DOC, TXT, MP3, or AAC might ordinarily be viewed as the title of the book as a link with the words "in HTML" after it. then the sentence "Also available in: " followed by a series of short links with text of "HTML", "PDF", "DOC", "TXT", "MP3", and "AAC". At Level 3, some users could opt to view the page this way - because they would find the page harder to understand or slower to use if the full title of the book were included in each of the links. Others could opt to view the page with the full title as part of each of the links so that each link was understandable in itself. Both the former and the latter groups could include people with visual or cognitive disabilities that used different techniques to browse or that had different types or severities of disability.

Benefits of Link Purpose (Link Only)

- This success criterion helps people with motion impairment by letting them skip web pages that they are not interested in, avoiding the keystrokes needed to visit the referenced content and then return to the current content.
- People with cognitive limitations will not become disoriented by extra navigation to and from content they are not interested in.

- People with visual disabilities will benefit from not losing their place in the content when they return to the original page. The screen reader's list of links is more useful for finding information because the target of the links are described.

Examples of Link Purpose (Link Only)

Both an icon and text are included in the same link

An icon of a voting machine and the text "Government of Ireland's Commission of Electronic Voting" are combined to make a single link.

A list of book titles

A list of books is available in three formats: HTML, PDF, and mp3 (a recording of a person reading the book). The title of the book is followed by links to the different formats. The rendered text for each link is just the format type, but the text associated with each link includes the title as well as the format; for instance, "Gulliver's Travels, MP3."

Resources for Link Purpose (Link Only)

- [Using Link Titles to Help Users Predict Where They Are Going](#)
- [WebAIM Techniques for Hypertext Links](#)

2.4.10 Section Headings (Level AAA)

Source: WCAG 2.2 / Understanding 2.4.10 — Copyright © 2023 World Wide Web Consortium

Section headings are used to organize the content.

"Heading" is used in its general sense and includes titles and other ways to add a heading to different types of content.

This success criterion covers sections within writing, not user interface components . User interface components are covered under Success Criterion 4.1.2 .

In brief

Goal

Users understand how content is organized in sections.

What to do

Where content is organized in sections, provide section headings.

Why it's important

People can orient themselves, especially those with cognitive or visual disabilities.

Intent of Section Headings

The intent of this success criterion is to provide headings for sections of a web page, when the page is organized into sections. For instance, long documents are often divided into a variety of chapters, chapters have subtopics, etc. When such sections exist, they need to have headings that introduce them. This clearly indicates the organization of the content, facilitates navigation within the content, and provides mental "handles" that aid in comprehension of the content. Other page elements may complement headings to improve presentation (e.g., horizontal

rules and boxes), but visual presentation is not sufficient to identify document sections.

This provision is included at Level AAA because it cannot be applied to all types of content and it may not always be possible to insert headings. For example, when posting a pre-existing document to the web, headings that an author did not include in the original document cannot be inserted. Or, a long letter would often cover different topics, but putting headings into a letter would be very strange. However, if a document can be broken up into sections with headings, it facilitates both understanding and navigation.

Benefits of Section Headings

- People who are blind will know when they have moved from one section of a web page to another and will know the purpose of each section.
- People with some learning disabilities will be able to use the headings to understand the overall organization of the page content more easily.
- People who navigate content by keyboard will be able to jump the focus from heading to heading, enabling them to find quickly content of interest.
- In pages where content in part of the page updates, headings can be used to quickly access updated content.

Examples of Section Headings

- A menu contains different sections for different courses. Each section has a heading: Appetizers, Salad, Soup, Entree, Dessert.
- A web application contains a settings page that is divided into groups of related settings. Each section contains a heading describing the class of settings.

Resources for Section Headings

- [WebAIM: Semantic Structure](#)
- [Heading Tags](#)

2.4.11 Focus Not Obscured (Minimum) (Level AA)

Source: [WCAG 2.2](#) / [Understanding 2.4.11](#) — Copyright © 2023 World Wide Web Consortium

When a user interface component receives keyboard focus, the component is not entirely hidden due to author-created content.

Where content in a configurable interface can be repositioned by the user, then only the initial positions of user-movable content are considered for testing and conformance of this success criterion.

Content opened by the *user* may obscure the component receiving focus. If the user can reveal the focused component without advancing the keyboard focus, the component with focus is not considered visually hidden due to author-created content.

In brief

Goal

Keep the focused item visible.

What to do

Ensure when an item gets keyboard focus, it is at least partially visible.

Why it's important

People who can't use a mouse need to see what has keyboard focus.

Intent of Focus Not Obscured (Minimum)

The intent of this success criterion is to ensure that the item receiving keyboard focus is always partially visible in the user's viewport. For sighted people who rely on a keyboard (or on a device that operates through the keyboard interface, such as a switch or voice input), knowing the current point of focus is

critical. The component with focus signals the interaction point on the page. Where users cannot see the item with focus, they may not know how to proceed, or may even think the system has become unresponsive.

In recognition of the complex responsive designs common today, this AA criterion allows for the component receiving focus to be *partially* obscured by other author-created content. A partly obscured component can still be very visible, although the more of it that is obscured, the less easy it is to see. For that reason, authors should attempt to design interactions to reduce the degree and frequency with which the item receiving focus is partly obscured. For best visibility, *none* of the component receiving focus should be obscured. This preferred outcome is covered by the AAA criterion [Focus Not Obscured \(Enhanced\)](#) .

Typical types of content that can overlap focused items are sticky footers, sticky headers, and non-modal dialogs. As a user tabs through the page, these layers of content can obscure the item receiving focus, along with its focus indicator.

A notification implemented as sticky content, such as a cookie banner, will fail this success criterion if it entirely obscures a component receiving focus. Ways of passing include making the banner modal so the user has to dismiss the banner before navigating through the page, or using [scroll padding](#) so the banner does not overlap other content. Notifications that do not require user action could also meet this criterion by closing on loss of focus.

Another form of obscuring can occur where light boxes or other semi-opaque effects overlap the item with focus. While less than 100 percent opacity is not causing the component to be entirely obscured , such semi-opaque overlaps may cause a failure of [1.4.11 Non-text Contrast](#) . When a focus indicator can be covered by a

semi-opaque component, the ability of the focus indicator to pass 1.4.11 Non-text Contrast should be evaluated (and pass) while the focus indicator is under the semi-opaque component. The intention in both situations is that the component receiving focus should never be obscured to the point a user cannot tell which item has focus.

This criterion evaluates the focused **component** , rather than the focus **indicator** . The component itself does not include the focus indicator when checking that "the component is not entirely hidden" - unless the focus indicator is inside the component, or focus is indicated by a change to the component itself. Although users benefit from both the component and the focus indicator (if external to the component) not being obscured when tracking the focus, for the purposes of this criterion only checking the component provides a clearer metric. However, if the focus indicator is fully obscured, it would likely fail [2.4.7 Focus Visible](#) .

User-movable content

This SC contains a note regarding content that can be repositioned. If users can move content regions, then they can potentially position the movable content such that it obscures other content that may receive focus. In such a case, the author is only responsible for ensuring that the movable content *in its initial position* does not obscure the item receiving focus.

This note is intended to accommodate a common interaction in complex applications such as authoring tools, where the main editing region (also called a canvas) can be enhanced by displaying toolbars or other panels, which can be repositioned around the canvas. It is possible to design such toolbars so they do *not* obscure focus. Authors are encouraged to do so, as well as pursue techniques which ensure equitable keyboard use of such toolbars. However, in recognition of the complexities involved in responsive

design as well as in supporting the ability to transform the text size and spacing of content, only the starting position of such movable panels is assessed.

User-opened content

This SC contains a note regarding content that is opened or disclosed by the user. One example of such content is a menu button opened by a user that opens a list of choices over pre-existing content on the screen. Such content can obscure other information on the screen, but it does not obscure an item receiving keyboard focus, because the new content doesn't stay open through a change of focus. However, authors may create user-opened content that is intentionally designed to persist until closed by the user, such as a chat window. Such persistent content has the potential to fail Focus Not Obscured (Minimum). Various types are described in this section. All can be designed so that they pass this success criterion.

This section only applies to content that the user actively discloses. Content pre-positioned by the author (such as a sticky footer), or content that appears without direct user initiation, such as system warnings, must not prevent the item receiving focus from being immediately visible in the viewport. Also, this note is not intended to apply to disclosures that are by convention non-persistent. As discussed in the following sub-section, an open dropdown that does not close when no longer focused is not following this convention.

A number of components on the web open (or disclose) additional content (on activation or on focus) intended for immediate user interaction or information. This new content is often on top of other content, obscuring it. Examples of such components are menu items, select element items, combobox lists (and other dropdown items), date picker calendars, and tooltips. The

common trait of all these components is that they are not expected to persist after being acted on or once they are no longer the primary point of user interaction. Such non-persistent disclosures do not fail this SC since they do not obscure the item with focus. However, if an author allows such components to **persist** after the user has 1) activated one of the opened items or 2) moved the focus away from the triggering item and the additional content, it is at risk of failing this criterion by obscuring the item with focus.

Some disclosure patterns provide a mechanism for the user to open additional content that remains open until intentionally closed by the user. Accordions are a simple example of such a pattern. Chatbots and expandable side navigation are more complex examples. All of these patterns can be implemented so they are not at risk of failing this SC. Some possible approaches are:

- When the additional content appears, it displaces existing content. An accordion is an example of this. When an accordion is opened, the disclosed content shifts existing content further down the page. Since the new content does not obscure existing content, it cannot obscure the item with focus.
- ****When the additional content appears, existing content reflows.**** The pop-out sidebar on the [WCAG standard](#) is an example of this pattern. When the side menu is activated, it opens a new section of information along the left side of the page. The main content area is reduced horizontally to accommodate the new content, and the existing content reflows to fit in the thinner space. As a result, there is no overlapping content between the two sections; the item receiving focus, whether in the left navigation or in the main content, will not be obscured by the other section. <figure id="">

<figcaption>Placeholder for future illustration pair showing an abstract representation of a sidebar reflowing a thinner main content region</figcaption>

</figure>

- ****When the additional content is opened, it takes focus and the tab ring is constrained to the new content until it is dismissed.**** This modality is somewhat like a dialog, in that a user cannot navigate beyond the opened content by keyboard without dismissing it first (typically by pressing Esc). However, unlike in a modal dialog, in some implementations a pointer user may be able to interact with content outside the opened section without dismissing it. Since this pattern potentially creates an inequitable experience between keyboard and pointer users, it should be used cautiously. That said, it does prevent the opened content from obscuring the keyboard focus in the main content, and thus should pass this SC. This is described and demonstrated in a short video in the Knowbility article in the reference section, under the section heading Keep keyboard focus in the slide-out navigation until it's closed .

<figure id="">

<figcaption>Placeholder for future illustrations showing an abstract representation of a side panel where the tab ring is restricted to the panel</figcaption>

</figure>

- The disclosure expands into an area of the page containing no other content. Many pages are designed with wide margins, providing significant white space into which new content can be opened. Many chatbots and toast notifications are designed to 'slide up' into the right unpopulated side of a page. Where authors are careful to ensure content is not obscured at each

breakpoint in a responsive design, no obscuring of other operable content need occur.

- When focus leaves the additional content, the additional content is automatically hidden or collapsed, or the content can be hidden or collapsed by use of a dedicated keyboard command (for example, the Escape key .) This is very similar to patterns discussed previously under Non-persistent opened information. A distinguishing factor can be that the user's last point of interaction in the disclosure is preserved (it persists) even though it may be hidden until a user returns. Some trees and side navigation patterns behave this way.

In recognition of more complex interfaces and user needs there is a note: Content opened by the user may obscure the component receiving focus . If the user can bring the item with focus into view using a method without having to navigate back to the user-opened content to dismiss it, this criterion would be passed. For example, keyboard actions that may allow the item with focus to be revealed include:

- using the Escape key to dismiss the obscuring content;
- using keys to scroll the content in the viewport to reveal the item with focus;
- issuing a key to move between overlays.

For example:

- A user opens a chat interface, which is a popover non-modal dialog. This results in some content of the underlying page being fully obscured. The user navigates away from the chat interface by use of the tab key, focusing onto a link that has been fully obscured by the dialog. The user presses the Escape key to close the chat interface, which un-obscures the link.

- A user expands a fixed-position page feedback component at the bottom of a web page. They then use their keyboard to navigate to a link that's fully obscured by the expanded component and press the down arrow or space key on their keyboard to scroll the content on the page, un-obscuring the link.
- A user opens a web-based multi-user authoring application. An overlay appears displaying a list of people who have contributed to the document. The user tabs through the list of contributors and activates one of them. The application displays a new overlay, which obscures the first one, that displays that person's recent contributions. The user presses the F6 key to toggle the stacking order of the two overlays.

Modal dialogs

A properly constructed modal dialog will always pass this SC. Even if it appears directly on top of an item with focus, the dialog takes focus on appearance, and thus the item receiving focus -- the dialog or one of its components -- is visible. A properly constructed modal maintains that focus and prevents interaction outside the modal until it is dismissed.

A dialog-like overlay that does not take focus on appearance and does not either constrain interaction to the overlay or dismiss itself on loss of focus (thus allowing focus to exit into the content behind it) will be at risk of failing this SC, where it is positioned such that it can obscure other focusable items.

Benefits of Focus Not Obscured (Minimum)

- Sighted users who rely on a keyboard interface to operate the page will be able to see the component which gets keyboard focus. Such users include those who rely on a keyboard or on devices which use the keyboard interface, including speech

input, sip-and-puff software, onscreen keyboards, scanning software, and a variety of assistive technologies and alternate keyboards.

- People with limited or low vision, who may primarily use a pointer for screen orientation and repositioning, nonetheless benefit from a visible indication of the current point of keyboard interaction, especially where magnification reduces the overall viewing portion of the screen.
- People with attention limitations, short term memory limitations, or limitations in executive processes benefit by being able to discover where the focus is located.

Examples of Focus Not Obscured (Minimum)

- A page has a sticky footer (attached to the bottom of the viewport). When tabbing down the page the focused item is not completely visually obscured by the footer because content in the viewport scrolls up to always display the item with keyboard focus using scroll padding.
- A page has a full-width cookie approval dialog. The dialog is modal, preventing access to the other controls in the page until it has been dismissed. Focus is not obscured because the major portion of the cookie approval dialog remains on screen (until selections are made and submitted), and so the major portion of the keyboard focus indicator remains visible.
- A notification is implemented as a sticky header and the keyboard focus is moved to the notification so at least part of the focus indicator is in view. The notification disappears when it loses focus so it does not obscure any other controls, and part of the prior keyboard focus indicator is visible.

Resources

- [Let's Focus on Slide-Out Navigation](#)

2.4.12 Focus Not Obscured (Enhanced) (Level AAA)

Source: [WCAG 2.2](#) / [Understanding 2.4.12](#) — Copyright © 2023 World Wide Web Consortium

When a user interface component receives keyboard focus, no part of the component is hidden by author-created content.

In brief

Goal

Don't cover any part of the item with focus.

What to do

Ensure when an item gets keyboard focus, it is fully visible.

Why it's important

People who can't use a mouse need to see what has keyboard focus.

Intent of Focus Not Obscured (Enhanced)

The intent of this success criterion is to ensure that the item receiving keyboard focus is always visible in the user's viewport. For sighted people who rely on a keyboard (or on a device that operates through the keyboard interface, such as a switch or voice input), knowing the current point of focus is critical. The component with focus signals the interaction point on the page. Where users cannot see the item with focus, they may not know how to proceed, or may even think the system has become unresponsive.

Typical types of content that can overlap focused items are sticky footers, sticky headers, and non-modal dialogs. As a user tabs through the page, these layers of content can hide the item receiving focus, along with its focus indicator.

A notification implemented as sticky content, such as a cookie banner, will fail this success criterion if it partially covers a component receiving focus. Ways of passing include making the banner modal so the user has to dismiss the banner before navigating through the page, or using scroll padding so the banner does not overlap other content. Notifications that do not require user action could also meet this criterion by closing on loss of focus.

Another form of obscuring can occur where light boxes or other semi-opaque effects overlap the item with focus. This form of obscuring is **not** in scope for this success criterion. While less than 100 percent opacity is not causing the component to be visually hidden, such semi-opaque overlaps may cause a failure of [1.4.11 Non-text Contrast](#) and/or [2.4.13 Focus Appearance](#). When a focus indicator can be covered by a semi-opaque component, the focus indicator should be assessed against 1.4.11 Non-text Contrast and 2.4.13 Focus Appearance. The intention in both situations is that the component receiving focus should never be obscured to the point a user cannot tell which item has focus.

This criterion evaluates the focused **component**, rather than the focus **indicator**. The component itself does not include the focus indicator when checking that "no part of the component is hidden" - unless the focus indicator is inside the component, or focus is indicated by a change to the component itself. Although users benefit from both the component and the focus indicator (if external to the component) not being obscured when tracking the focus, for the purposes of this criterion only checking the component provides a clearer metric. However, if the focus indicator is fully obscured, it would likely fail [2.4.7 Focus Visible](#).

Benefits of Focus Not Obscured (Enhanced)

- Sighted users who rely on a keyboard interface to operate the page will be able to see the component which gets keyboard focus. Such users include those who rely on devices which use the keyboard interface, including speech input, sip-and-puff software, on-screen keyboards, scanning software, and a variety of assistive technologies and alternate keyboards.
- People with limited or low vision but who rely upon a pointing device (for viewport orientation and repositioning) benefit from a clearly visible indication of the current point of keyboard interaction, especially where magnification reduces the overall useable portion of content.
- People with attention limitations, short term memory limitations, or limitations in executive processes benefit by being able to more easily discover where the focus is located.

Examples of Focus Not Obscured (Enhanced)

- A page has a sticky footer (attached to the bottom of the viewport). When tabbing down the page the focused item is not at all obscured by the footer because content in the viewport scrolls up to always display the item with keyboard focus using scroll padding.
- A page has a large (30% wide) cookie approval dialog. The dialog is modal, preventing access to the other controls in the page until it has been dismissed. Focus is not obscured because the cookie approval dialog (including the focus indicator) remains on screen until selections are made and submitted.
- A notification is implemented as a sticky header and the keyboard focus is moved to the notification. The notification disappears when it loses focus, and does not obscure any other

controls (including the focus indicator visible prior to the notification).

2.4.13 Focus Appearance (Level AAA)

Source: [WCAG 2.2](#) / [Understanding 2.4.13](#) — Copyright © 2023 World Wide Web Consortium

When the keyboard focus indicator is visible, an area of the focus indicator meets all the following:

- is at least as large as the area of a 2 CSS pixel thick perimeter of the unfocused component or sub-component, and
- has a contrast ratio of at least 3:1 between the same pixels in the focused and unfocused states.

Exceptions:

- The focus indicator is determined by the user agent and cannot be adjusted by the author, or
- The focus indicator and the indicator's background color are not modified by the author.

What is perceived as the user interface component or sub-component (to determine the perimeter) depends on its visual presentation . The visual presentation includes the component's visible content , border, and component-specific background. It does not include shadow and glow effects outside the component's content, background, or border.

Examples of sub-components that may receive a focus indicator are menu items in an opened drop-down menu, or focusable cells in a grid.

Contrast calculations can be based on colors defined within the technology (such as HTML, CSS, and SVG). Pixels modified by user agent resolution enhancements and anti-aliasing can be ignored.

In brief

Goal

Make it easier to spot the keyboard focus.

What to do

Use a focus indicator of sufficient size and contrast.

Why it's important

Many people can't see small changes in visual appearance, including older people.

Intent of Focus Appearance

The purpose of this success criterion is to ensure a keyboard focus indicator is clearly visible and

discernible. Focus Appearance is closely related to [2.4.7 Focus Visible](#) and [1.4.11 Non-text Contrast](#) .

Focus Visible requires that a visible focus indicator exists while a component has keyboard focus; Focus

Appearance defines a minimum level of visibility. Where Non-text Contrast requires a component to have

adequate contrast against the background in each of its states, Focus Appearance requires sufficient

contrast for the focus indicator itself.

For sighted people with mobility impairments who use a keyboard or a device that utilizes the keyboard interface (such as a switch or voice input), knowing the current point of focus is very important. Visible focus must also meet the needs

of users with low vision, who may also rely on the keyboard.

A keyboard focus indicator can take different forms. This Success Criterion encourages the use of a solid outline around the focused user interface component, but allows other types of indicators that are at least as large.

This Understanding document will elaborate on the minimum area requirement, color contrast requirements, and finally list some user agent exceptions.

Minimum area

The first part of the success criterion specifies a minimum area for the focus indicator:

> is at least as large as the area of a 2 CSS pixel thick perimeter of the unfocused component or sub-component

This only specifies a minimum area for the focus indicator. It does not require that the focus indicator literally be a 2 CSS pixel thick outline, only that the indicator be at least that large.

However, the simplest way to meet the size requirement is to use a focus indicator which is a solid 2 CSS pixel thick perimeter.

A CSS pixel is what developers use in CSS declarations like “width: 200px”. It is

device-independent and not to be confused with device pixels which vary depending on the physical pixel

density. The rest of this document notates CSS pixels as “px”.

The easiest and most common way to meet this requirement is to use a solid outline around the component. The outline must be at least 2px thick. The following illustration shows a **minimally** thick focus indicator, where a 2px thick band of white pixels making up the page background around an example button have been altered to black.

For non-rectangular components, the "perimeter" definition allows authors to use either of the following types of outline:

- a line which solidly encloses a shape, or
- a line which solidly encloses the minimum bounding box of a shape

For example, a star-shaped button may use either a focus indicator that follows the shape of the star or a focus indicator that follows the bounding box of the star. In the following examples, the same three stars have already been selected, and focus is on the third star. The first example uses a focus indicator which matches the star shape of the focused star. The second uses a rectangular indicator.

Offsetting indicators slightly from the focused component, as in the examples above, is not required to meet the minimum area requirement of the success criterion, but it can help make indicators more visible. In CSS, the `outline` and `outline-offset` properties are commonly used to achieve this.

The smallest possible 2 CSS pixel thick indicator that is still a "perimeter" is a solid line that appears inside the component against the component's outer edge, for example by using a CSS `border` property. Indicators that are inset further within the component (not directly against the component's outer edge) need to be thicker than 2 CSS pixels to meet the minimum size requirement.

Note that different Non-text Contrast requirements may apply depending on whether the focus indicator

is offset from, inset into, or against the edge of the component. See the Relationship with Non-text Contrast section

below.

This success criterion does not require that focus indicators be solid outlines. Other shapes may be used so long as they meet the minimum area requirement.

The minimum area of the focus indicator for a control is the area of a 2 CSS pixel thick perimeter of the control (or its minimum bounding box) in the control's unfocused state. For example, if a control is a rectangle 90px wide and 30px tall, the area of a 2 CSS pixel thick perimeter is difference between the areas of:

- A 92px by 32px rectangle (1px larger on all sides), and
- A 88px by 28px rectangle (1px smaller on all sides)

This results in a minimum area of $(92px \times 32px) - (88px \times 28px) = 480px^2$.

Some general formulas for 2 CSS pixel thick perimeters of common shapes are:

Rectangle with width and height

$$4h + 4w$$

Circle with radius r

$$4\pi r$$

Rounded rectangle with width, height, and border radius r

$$4h + 4w - (16 - 4\pi)r$$

If you need to use complex mathematics to work out if a focus indicator is large enough, it is probably a sign that you should use a larger indicator instead. The bigger the visible change when an item receives focus, the easier it is for someone to see.

The following 2 examples use a 90px wide by 30px tall button, with a minimum area requirement of $480px^2$:

Prefer using focus indicator techniques that scale with both the width and height of the focused control. Otherwise, if controls change size across different variations of a page (for example, in a responsive design), the indicator might meet the area requirement in some variations but not others. For example, in the above figure, if the width of the two highlight rectangles did not scale as the button grew wider, it would stop meeting the minimum area requirement if the button needed to grow any wider to accommodate a longer button label.

Another way of achieving the area requirement is to alter the appearance of the entire component, for instance by changing its color – provided that the new color has a contrast ratio of at least 3:1 against the original color. This can be effective in a set of closely placed buttons. The following example demonstrates this with 5 rating stars; the center star is filled in with a darker color to indicate focus. However, it is much more difficult to detect such a focus indicator when components are not near each other and so cannot be easily compared. For users using magnification, even components relatively close together may be difficult to compare, so it is not considered a best practice.

If an inline link is broken over multiple lines, some methods of creating a focus indicator create different results by browsers. CSS `outline` separately surrounds each part of a link that breaks across multiple lines. It is by far the most common CSS technique for focus indication, and produces a result that satisfies the minimum bounding box definition since each part is solidly bound. CSS `border` will split the perimeter across the parts of the link, which results in an unenclosed border for each line of the link. The minimum bounding box definition states that link focus can be assessed as if the link was all on one line, so a 2px thick border is also considered to meet the minimum area requirement.

Therefore, where the contrast requirements are met, each of these methods can produce a sufficient focus indicator.

Change of contrast

The second part of the Success Criterion's indicator requirements states that an area of the indicator:

- > has a contrast ratio of at least 3:1 between the same pixels in the focused and unfocused states

This requirement measures the change of contrast between the same pixels in different states. This is different from the Text Contrast and Non-text Contrast Success Criteria, which measure the contrast between different adjacent pixels in a single state at a time.

3:1 is the **minimum** allowable change-of-contrast ratio, but the greater the change of contrast between states, the easier it is for users to see the focus indicator. Authors are encouraged to make the change-of-contrast ratio as great as possible.

The following illustration shows a **minimally** contrasting focus indicator, where some of the white pixels making up the page background have been altered to a mid-grey that has a 3:1 contrast ratio with the original white. Authors are encouraged to exceed the minimum focus appearance. For instance, the dark blue lines in figures 2 and 3 are much more visible.

When a component changes to include a focus indicator, that change can be measured as a change of color contrast. For example, if a yellow outline is added to a button on a blue background, the change of color is from blue to yellow. This change can be measured whether the focus indicator is on the background around the component, or the background within the component.

If a control receiving focus changes its background (fill color) to a color that contrasts less than 3:1 with the original background, that would not pass the change of contrast.

If the background change is sufficient, it is a method of passing the criterion.

It is not necessary for the *entire* focus indicator to have a 3:1 change of contrast. It is sufficient for just a part of the indicator to meet the change of contrast requirement, so long as the contrasting part of the indicator meets the minimum area requirement.

When calculating whether a focus indicator meets the minimum area requirement, only the part of the indicator which meets the change-of-contrast requirement should be included in the calculation.

If a focus indicator has a gradient, the principle is to measure the contrast of the changed area, and ignore any part of the gradient which has less than a 3:1 change-of-contrast ratio.

If you eliminate the area which has less than 3:1 change-of-contrast, you can calculate the area of the remaining parts of the indicator to determine whether the indicator meets the minimum area requirement.

Some of the examples in this document are screen-captured images of elements. Due to loss of resolution in these images, the actual pixel color may not match the original. As such, they are intended to be used for illustrative purposes, and should not be inspected on a pixel-by-pixel basis for sufficient contrast.

Some designs have pages with a non-solid background image covering the whole (or part) of the page or make use of parallax scrolling effects which result in a near-infinite number of color

combinations if a page is scrolled and/or changes are made to the viewport size.

If the contrast of background colors that change are close enough to need to be tested for each

combination then they would likely not meet the user need of people with low vision in certain

scroll combinations and would likely fail in certain combinations as well. In these cases it would

be an easy solution to use a two-color focus indicator or some other mechanism to indicate focus such as a solid box with a border

to guarantee there is sufficient contrast across variations of background images or background

gradients.

It is possible to use visual patterns such as strips switching places to disguise a change of focus indicator. However, this is not considered a visible indicator.

Relationship with Non-text Contrast

Focus indicators are visual information required to identify a state of a user interface component. That

means that they are subject to [1.4.11 Non-text Contrast](#) , in addition to [2.4.13 Focus Appearance](#).

In combination with [2.4.7 Focus Visible](#) , [1.4.11 Non-text Contrast](#) requires that the visual focus indicator for a component

must have sufficient contrast against the adjacent colors when the component is focused, except where

the appearance of the component is determined by the user agent and not modified by the author.

The difference between the contrast requirements in Focus Appearance and Non-text Contrast is:

- Focus Appearance requires that focus indicators have a change of contrast between focused and non-focused states.
- Non-text Contrast requires that focus indicators have adjacent contrast between the indicator (in the focused state) and adjacent non-indicator colors.

Additionally, Non-text Contrast does not establish any size requirement and has slightly different rules for when exceptions are allowed.

See the Relationship with Focus Visible section of [Understanding 1.4.11 Non-text Contrast](#) for more details and examples.

Component keyboard focus

The preamble to this success criterion is "When a user interface component has keyboard focus..." The *keyboard focus* is the point of interaction for someone using a keyboard. For environments with a keyboard-operable interface, the keyboard focus can be moved around the interface in order to interact with different components. Whichever component is being interacted with has focus.

WCAG defines *user interface component* as "a part of the content that is perceived by users as a single control for a distinct function." Because different users may perceive controls differently, there is a potential for some variation when interpreting what constitutes both a *single control* and a *distinct function*. This is particularly the case when something visually presents in a way that may differ from how it is programmatically

created under the covers. Where there is not a native HTML component upon which to base designs, there can be great variations in how the components and their focus indicators are portrayed. Further, some components have sub-components that can take focus, such as the menu items on a menu.

Nonetheless, consistent results from different testers were obtained for this success criterion by using the focus indicator itself as the gauge of what constitutes the component being interacted with. For complex components, the three typical focus indicators are as follows:

- Focus indicator around only the whole component
- Focus indicators around both the component and subcomponent
- Focus indicator around only the subcomponent

Each of these will be discussed, using a tablist as a familiar complex component.

When the focus indicator is shown only around the whole tablist, the user is guided to considering the

tablist as a single user component. The tab items within it are visually distinguished between selected

and unselected states (and visual indicators of selection state must meet the criteria given in [1.4.11 Non-text Contrast](#)).

Having a focus indicator *only* around the whole is possible where there is no need to have a selected sub-component while another sub-component has focus. For a tablist which synchronizes its tab panel content with whatever tab is active, only one tab item can be selected at a time, and since whatever tab is selected is considered active, a separate focus indicator is redundant.

Result: the group focus indicator must meet the requirements of this success criterion.

A radio button group and a star-rating widget , which each use only a whole-component focus indicator, provide working examples of

different complex components that pass the primary requirements of this success criterion. In the star

ratings example, users can increment the rating by 1/2 stars. Not only is a focus indicator for each 1/2

star unnecessary, but it would actually be difficult to achieve without making the interaction

confusing.

For a tablist which does not keep its tab panel content synchronized with whatever tab is selected, there needs to be a focus indicator for the tab item subcomponent. This is because the tab item with focus may be different than the selected item.

The user can navigate to the tablist, which in this implementation has a focus rectangle around the whole tablist as well as one around a tab item (conventionally the item that is currently selected). The focus around the whole is helpful in cueing a keyboard user that this is a complex component that has its own interaction. The user can then move focus between the unselected and selected tab items -- each of which in turn has its own focus indicator -- before activating one, which then makes it selected as well as focused, and updates the tab panel to match.

In this scenario, either the group focus indicator or the sub-component indicator must meet the requirements of this success criterion. To avoid being overly prescriptive, the success criterion allows authors to choose which makes the most sense. Generally, if

a sub-component focus is necessary, it should be assessed instead of the group indicator.

Result: the focus indicator for the tab item meets the requirements of this success criterion. The tablist focus indicator does not need to meet the requirements.

A slider to pick colors provides a working example of a different complex component that predominantly shows

focus for the subcomponent. In this case, the thumb slider sub-component has a focus indicator of

sufficient size and contrast to pass the sufficient area calculation. There is also a subtle focus

around the whole slider component, but it does not need to be assessed to pass this success criterion.

The same unsynchronized tablist can also be implemented as something which only shows focus on the tab items and not on the whole. The behavior is the same as in the prior example, but there is never a focus indicator placed around the tablist. This interaction is acceptable, but it is not best practice since it demands more understanding from the user with less information. For example, some visual cues for the tablist and tab items (and tab panel) may not be clear. As well, keyboard users may not initially understand the expected keyboard interaction.

Result: the focus indicator for the tab item must meet the requirements of this Success Criterion, judging focus with both selected and unselected tab items.

A functional example of sub-component-only tab focus has an indicator that is large enough (at

least four times the shortest side) with sufficient contrast to pass the focus area language of this success criterion.

Some pages contain very large editing regions, such as web implementations of word processors and code

editors. Unlike a `textarea` element, which is a user interface component, these large

editing regions do not typically meet the definition of user interface components ;

they are not "perceived by users as a single control for a distinct function."

Providing focus indicators around such editing regions may still be beneficial to some; however, where

the region is not perceived as a single control, it is not covered by this success criterion. The web

page will still need to provide a insertion point (caret indicator) in such editing regions in order to

meet the requirements of [2.4.7 Focus Visible](#) .

Some non-operable elements can take focus (such as a heading element that is the target of a skip link). However, the preamble of this success criterion refers to user interface components; it is only when the element with focus is operable by keyboard that this success criterion applies.

Exceptions

There are two situations where the focus appearance does not need to be assessed:

- the focus indicator cannot be adjusted by the author

- the author has not modified the effects of the user agent default

> The focus indicator is determined by the user agent and cannot be adjusted by the author

Some components or technologies may not allow the author to adjust the focus indicator. This is the case with HTML `select` elements (both single and multi-select), where the visual treatments for selection and focus cannot be adjusted by the author. In this case the Success Criterion does not apply.

> The focus indicator and the indicator's background color are not modified by the author

If the focus indicator and the background behind the focus indicator are not modified by the author, the success criterion does not apply.

The intent of this exception is to reduce burden on authors by allowing them to rely on the default indicators provided by user agents (browsers). If all user agents provided good focus indicators, authors would be able to concentrate efforts on other accessibility considerations. Unfortunately, browser default focus indicators vary by component, browser, and across devices and operating systems, and the default focus indicators in some browsers can be difficult to see (such as a 1px dotted outline). For this reason, most authors override browser defaults in order to overcome these deficiencies and create a more uniform user experience, regardless of browser.

Some browser makers are improving their default focus indicators to make them more visible. As more browsers adopt defaults that meet the primary bullets of this Success Criterion, authors will be able to achieve improved focus indicators without customization.

Browser default focus indicators can be made *more* difficult to see if the author modifies the pixels directly adjacent to the indicator (commonly referred to as its background), such as by positioning a component on top of an image or gradient background, or altering the page's default white background color, for instance using a blue background in combination with a browser's blue default indicator. For this reason, where the author alters the pixels directly adjacent to the default focus indicator, the user agent exception does not apply, and the author will need to verify they meet the size and contrast requirements of this success criterion.

Altering the `body` element's `background-color` attribute is one way of altering the pixels directly adjacent to the indicator in most implementations. However, specifying a value of white (`#FFFFFF`) does not nullify this

exception since, as established in the third note of the contrast ratio definition, the

default ("unspecified") color is assumed to be white.

As well, if the browser provides an indicator *within* a component by default, then authors can potentially reduce the visibility by changing the component color (which in such a scenario is the background color for the focus indicator). For example, if the default indicator on a button uses a colored inner border, authors can negatively affect the focus appearance by making the button or its unfocused border color a similar-luminosity color. For this reason, this user agent exception can only be met if the author both does not modify the default focus indicator *and* does not modify its background.

Benefits of Focus Appearance

- This success criterion helps anyone who relies on the keyboard to operate the page, by letting them visually determine the component on which keyboard operations will interact at any point in time.
- People with attention limitations, short term memory limitations, or limitations in executive processes benefit by being able to discover where the focus is located.

Examples of Focus Appearance

- When links receive focus, an outline is displayed around the link that contrasts with the background adjacent to the link.
- When buttons receive focus, an outline is displayed within the button (around the text) that contrasts with the button's background.
- When text fields receive focus, an outline is displayed around the field, indicating that the input has focus.
- When radio buttons receive focus, an outline is displayed around the control, indicating that the input has focus.

Related Resources

- [A guide to designing accessible focus indicators](#) by Sara Soueidan
- [Focus appearance - testing version 3](#)
- [Focus visible testing \(Feb 2022\)](#)
- [Avoid Default Browser Focus Styles](#) by Adrian Roselli

Guideline 2.5: Input Modalities

[Source: W3C WCAG 2.2 —

<https://www.w3.org/TR/WCAG22/#guideline25>]

Make it easier for users to operate functionality through various inputs beyond keyboard.

2.5.1 Pointer Gestures (Level A)

Source: [WCAG 2.2](#) / [Understanding 2.5.1](#) — Copyright © 2023 World Wide Web Consortium

All functionality that uses multipoint or path-based gestures for operation can be operated with a single pointer without a path-based gesture, unless a multipoint or path-based gesture is essential .

This requirement applies to web content that interprets pointer actions (i.e., this does not apply to actions that are required to operate the user agent or assistive technology).

In brief

Goal

Let users operate touchscreens with one finger and reduced gestures.

What to do

Provide single-point operation for all functions.

Why it's important

Not everyone can perform complex and multi-touch gestures.

Intent of this Success Criterion

The intent of this success criterion is to ensure that content can be controlled with a range of pointing devices, abilities, and assistive technologies. Some people cannot perform gestures in a precise manner, or they may use a specialized or adapted input device such as a head pointer, eye-gaze system, or speech-controlled mouse emulator. Some pointing methods lack the capability or accuracy to perform multipoint or path-based gestures.

A **path-based gesture** involves an interaction where the movement of the pointer along a specific path matters. Examples

of a **path-based gesture** include swiping/"flicking" (which is only recognized when the user moves in a mostly straight line from the start-point to the end-point), having to move a pointer strictly along a specific path, and other gestures which trace a prescribed path such as drawing a specific shape. Such paths may be drawn with a finger or stylus on a touchscreen, graphics tablet, or trackpad, or with a mouse, joystick, or similar pointer device.

For many gestures, the exact start- and end-points themselves are not relevant – what matters is that the overall gesture (mostly) follows a particular path. For instance, if a page implemented a horizontal "flick" gesture to move to the previous or next page, the exact point where the user starts the "flick" or ends it is irrelevant, as long as the "flick" gesture is horizontal, is of sufficient length, and is performed above a specific speed.

Examples of **multipoint** gestures include a two-finger pinch/spread zoom, a split tap where one finger rests on the screen and a second finger taps, or a two- or three-finger tap or swipe. Users may find it difficult or impossible to accomplish these if they type and point with a single finger or stick.

Authors must ensure that their content can be operated without multipoint or path-based gestures. Multipoint or path-based gestures can be used so long as the functionality can also be operated by another method, such as a tap, click, double tap, double click, long press, or click & hold.

An exception is made for functionality that is inherently and necessarily based on complex paths or multipoint gestures. For example, entering your signature may be inherently path-based (although acknowledging something or confirming your identity need not be).

This success criterion applies to gestures in the author-provided content, *not* gestures defined by the operating system, user agent, or assistive technology. Examples of operating system gestures on touchscreen devices would be swiping down from the top of the screen to see system notifications, and gestures to operate built-in assistive technologies. Examples of user agent-implemented gestures on touchscreen devices would be horizontal flicking motions implemented by browsers for navigating back or forward within the page history, or vertical flicking gestures to "fling" page content (or content inside natively scrollable containers) to scroll it. For assistive technology, examples include gestures used to operate a screen reader on a touchscreen device, such as swiping left or right to move the screen reader's cursor to the previous or next element.

This success criterion *does not apply* to gestures that involve dragging in any direction, not restricted or limited to a specific path. However, such gestures do require fine motor control. Authors are encouraged to provide non-dragging methods – for instance, a drag and drop operation could also be achieved by selecting an item (with a click/tap and keyboard interaction) and then selecting its destination as a second step. These cases are covered in Success Criterion 2.5.7 Dragging Movements .

The difference between path-based Pointer Gestures and Dragging

In a path-based gesture, what matters is how the pointer moves: its path or direction (and sometimes its speed). For some gestures, even the exact start- and end-points may not matter.

Dragging movements, on the other hand, generally involve "picking up" or "grabbing" an object with a pointer (such as mouse cursor or a finger) and moving it to some other position.

This type of movement from the start-point (where the object is first picked/grabbed) to the end-point (where it is dropped/released) that does not require the user to follow any particular path, direction, or speed, is therefore not a path-based gesture.

While horizontal/vertical range sliders may initially appear to require strict path-based movement along their track, the vast majority of sliders (either provided natively by user agents, or implemented by the content using scripting) use "pointer capture" – once the user has "grabbed" a slider thumb, they can "drift" from the horizontal or vertical track of the slider while changing the value.

Unless the slider is implemented in a way that forces users to precisely follow the path of the track, and straying from the track "loses" the user's grip on the slider thumb, operating a slider is *only* considered a dragging movement, rather than a path-based gesture.

In the rare instances where a slider *has* been implemented in a way where straying from the track "loses" the user's grip on the slider thumb, it is considered to be *both* a dragging movement (as the action does involve grabbing the slider thumb, moving it, and then releasing it) *and* a path-based gesture at the same time. In those cases, the slider may fail the requirements of both this criterion and Success Criterion 2.5.7 Dragging Movements .

The same rationale can be applied to components such as draggable carousels – these only count as requiring a path-based gesture if they expect the user to perform a precise movement along a path in order to move the content.

Relationship to keyboard accessibility requirements

While content authors generally need to provide keyboard operability as alternative for pointer operation, including complex gestures (see Success Criteria [2.1.1 Keyboard](#) and [2.1.3 Keyboard \(No Exception\)](#)), keyboard operability is not sufficient to conform to this success criterion. That is because some users rely entirely on pointing devices, or find simple pointer inputs much easier to perform and understand than alternatives. For example, a user relying on a head-pointer would find clicking a control to be much more convenient than activating an on-screen keyboard to emulate a keyboard shortcut, and a person who has difficulty memorizing a series of keys (or gestures) may find it much easier to simply click on a labeled control. Therefore, if one or more pointer-based mechanisms are supported, then their benefits should be afforded to users through simple, single-point actions alone.

Benefits

- Users who cannot (accurately) perform path-based pointer gestures - on a touchscreen, or with a mouse - will have alternative means for operating the content.
- Users who cannot perform multi-pointer gestures on a touchscreen (for instance, because they are operating the touchscreen with an alternative input such as a head pointer) will have a single-pointer alternative for operating the content.
- Users who may not understand the custom gesture interaction intended by the author will be able to rely on simple, frequently used gestures, such as basic clicking or tapping, to interact. This can be especially beneficial for users with cognitive or learning disabilities.

Examples

- A website includes a map view that supports the pinch/spread gesture to zoom into the map content. As a single-pointer alternative, the map also includes plus/minus buttons to zoom in and out.
- A news site has a horizontal carousel with different news item teasers. The site implements custom code that detects fast horizontal swiping/flicking motion to move to the previous / next items. It also offers forward and backward buttons on the left and right edges of the carousel, to move the carousel to the previous and next item with a simple click/tap. These buttons can be visible (for instance, as large arrow icons) or visually hidden (but still operable with a pointer).

2.5.2 Pointer Cancellation (Level A)

Source: [WCAG 2.2](#) / [Understanding 2.5.2](#) — Copyright © 2023 World Wide Web Consortium

For functionality that can be operated using a single pointer , at least one of the following is true:

No Down-Event

The down-event of the pointer is not used to execute any part of the function;

Abort or Undo

Completion of the function is on the up-event , and a mechanism is available to abort the function before completion or to undo the function after completion;

Up Reversal

The up-event reverses any outcome of the preceding down-event;

Essential

Completing the function on the down-event is essential .

Functions that emulate a keyboard or numeric keypad key press are considered essential.

This requirement applies to web content that interprets pointer actions (i.e., this does not apply to actions that are required to operate the user agent or assistive technology).

In brief

Goal

Reduce accidental activation of controls by mouse or touch.

What to do

Make pointer cancellation predictable and consistent.

Why it's important

Make it easier for anyone to recover from something they didn't mean to do.

Intent

The intent of this success criterion is to make it easier for users to prevent accidental or erroneous pointer input. People with various disabilities can inadvertently initiate touch or mouse events with unwanted results. Each of the following subsections roughly aligns with the bullets of this Success Criterion, and outlines a means of allowing users to cancel pointer operations.

Up-Event activation or completion

The most accessible way to incorporate pointer cancellation is to make activation occur on the up-event.

Up-event activation refers to the activation of a target when the pointer is released. In a touchscreen interaction, when the finger touches a target, the up-event activation only occurs when the finger is lifted while still being within the target boundary. Similarly in mouse interaction, the up-event occurs when the mouse button is released while the cursor is still within the boundary of the initial target set when the mouse button was pressed.

Authors can reduce the problem of users inadvertently triggering an action by using generic platform activation/click events that activate functionality on the up-event. For example, the `click` event in JavaScript triggers on release of the primary mouse button, and is an example of an implicit up-event. Despite its name, the `click` event is device-independent and also works for touch and keyboard interaction.

The preference for up-events is implicit in the success criterion wording of the first bullet: The down-event of the pointer is not used to execute any part of the function. Authors meet the first bullet by using only the up-event.

Up-Event Abort or Undo

Where the interaction is equivalent to a simple "click", up-event activation has a built-in ability to cancel. There is a distinction between when someone touches a screen and when they remove their finger. Similarly, in mouse interaction, there is a difference between pressing and releasing the mouse button. When activation occurs only as the pointer is released, users have the opportunity to Abort (cancel) the activation. Users who have difficulty accurately using a mouse or touchscreen benefit greatly from this basic behavior. They normally receive visual feedback when an item is pressed. If they discover they have selected the wrong item, they can cancel the action by moving their pointer or finger away from the target before releasing.

For more complex interactions, such as drag and drop, the down- and up-events may initiate and end a series of actions to complete a process. For example, with drag and drop, the item may be:

- selected with a press (down-event),
- moved to a new location, while still being depressed, and
- released (up-event) to conclude the drop action.

In such a complex action, the need for an Abort or Undo function increases. Designers may elect to confirm the move through something like a confirmation dialog or an undo button, giving the user the ability to Undo the process just completed. Alternatively, the ability to Abort the action can be achieved if, before completing step 3, the user returns the selected item to its original location and concludes the process there. If other parts of the

screen disallow a move, the user can conclude the drag and drop there, effectively nullifying the operation.

Up Reversal

In other interactions, the down-event may trigger a behavior which can be reversed when the up-event concludes. Examples of this include press-and-hold actions such as where a transient popup appears (or a video plays) when the user presses on an object (down-event), but the popup (or video) disappears as soon as the user releases the pointer (up-event). Since the up-event reverses the preceding down event, the user is returned to their prior point, and has effectively cancelled the operation.

Down-Event

Completing the function on the down-event is only permitted when it is essential that the up-event not be used.

The most prevalent essential down-event activation occurs in keyboard emulation. On a physical keyboard, keys by default activate on the down-event -- a letter appears when the key is pressed. If a software keyboard emulator tried to override this expected behavior by making letters appear when the key is released, the behavior would be unexpected and would adversely affect expected interaction.

Note that a keyboard has a built-in Backspace or Delete button, which effectively provides an Undo option. Undo is not a requirement of the down-event Essential exception; however, providing an easy way for users to undo any action is a recommended practice (and may be a functional necessity), even where it is not a requirement of this success criterion.

Other examples where the timing of an activation is essential and requires the down-event would be:

- An activity that emulates a physical on-press trigger, such as when playing an on-screen piano keyboard. Activation on the up-event would significantly alter the desired behavior.
- A program for shooting skeets where waiting for the "up" event would invalidate the precise timing necessary for the activation.

Benefits

- Makes it easier for all users to recover from hitting the wrong target.
- Helps people with visual disabilities, cognitive limitations, and motor impairments by reducing the chance that a control will be accidentally activated or an action will occur unexpectedly, and also ensures that where complex controls are activated, a means of Undoing or Aborting the action is available.
- Individuals who are unable to detect changes of context are less likely to become disoriented while navigating a site.

Examples

- For interface elements that have a single tap or long press as input, the corresponding event is triggered when the finger is lifted inside that element.
- A drag-and-drop interface allows users to sort vertically stacked cards by picking up one card with the pointer (down-event), move it to a new position, and insert it at the new location when the pointer is released (up-event). Releasing the pointer outside the drop target area reverts the action, i.e., it moves the card back to the old position before the interaction started.

2.5.3 Label in Name (Level A)

Source: [WCAG 2.2](#) / [Understanding 2.5.3](#) — Copyright © 2023 World Wide Web Consortium

For user interface components with labels that include text or images of text , the name contains the text that is presented visually.

A best practice is to have the text of the label at the start of the name.

In brief

Goal

The visual label for controls is a trigger for speech activation.

What to do

Where practical, make the control's text label and name match.

Why it's important

People who operate with voice interaction use the visible labels in their commands.

Intent

The intent of this success criterion is to ensure that the words which visually label a component are also the words associated with the component programmatically. This helps ensure that people with disabilities can rely on visible labels as a means to interact with the components.

Most controls are accompanied by a visible text **label** . Those same controls have a programmatic **name** , also known as the accessible name . Users typically have a much better experience if the words and characters in the visible label of a control match or are contained within the accessible name. When these match, speech-input users (i.e., users of speech recognition applications)

can navigate by speaking the visible text labels of components, such as menus, links, and buttons, that appear on the screen. Sighted users who use text-to-speech (e.g., screen readers) will also have a better experience if the text they hear matches the text they see on the screen.

Note that where a visible text label does not exist for a component, this success criterion does not apply to that component.

Where text labels exist and are properly linked to the user interface components through established authoring practices, the label and name will normally match. When they don't match, speech-input users who attempt to use the visible text label as a means of navigation or selection (e.g., "move to Password") will be unsuccessful. The speech-based navigation fails because the visible label spoken by the users does not match (or is not part of) the accessible name that is enabled as a speech-input command. In addition, when the accessible name is different from the visible label, it may function as a hidden command that can be accidentally activated by speech-input users.

Mismatches between visible labels and programmatic names for controls are even more of an issue for speech-input and text-to-speech users who also have cognitive challenges. Mismatches create an extra cognitive load for speech-input users, who must remember to say a speech command that is different from the visible label they see on a control. It also creates extra cognitive load for a text-to-speech user to absorb and understand speech output that does not match the visible label.

Note that when a user interface component lacks an accessible name — a failure of [4.1.2 Name, Role, Value](#) — and has a visible text label, then it also fails this success criterion.

Identifying label text for components

In order for the label text and accessible name to be matched, it is first necessary to determine which text on the screen should be considered a label for any given control. There are often multiple text strings in a user interface that may be relevant to a control. However, there are reasons why it is best to conservatively interpret the label as being only the text in close proximity.

Conventionally the label for user interface components is the adjacent text string. The typical positioning for left to right languages is:

- immediately to the left of comboboxes, dropdown lists, text inputs, and other widgets (or in the absence of left-side labels, immediately above and aligned with the left edge of each input)
- immediately to the right of checkboxes and radio buttons
- inside buttons and tabs or immediately below icons serving as buttons

The rationale for some of these conventions is explained in [G162: Positioning labels to maximize predictability of relationships](#) .

It is important to bias towards treating only the adjacent text as a label because liberal interpretations of what constitutes a text label can jeopardize the value of this success criterion (SC) by lessening predictability. Isolating the label to the single string in close proximity to the component makes it easier for developers, testers, and end users to identify the label targeted for evaluation in this SC. Predictable interpretation of labeling allows users of speech recognition technologies to interact with the element via its conventionally positioned label, and allows users of screen reading technologies to enjoy consistency between the nearby visible label and the announced name of the component.

Note that placeholder text within an input field is not considered an appropriate means of providing a label. The HTML Standard states The placeholder attribute should not be used as an alternative to a label . However, it is worth noting that "label" in that statement is in code brackets and links to the `label` element. For the purposes of this Label in Name Success Criterion, "label" is not used in such a programmatic sense but is simply referring to a text string in close visual proximity to a component. As such, in the absence of any other nearby text string (as described in the preceding list), if an input contains placeholder text, such text may be a candidate for Label in Name. This is supported both through the accessible name calculation (discussed later) and from the practical sense that where a visible label is not otherwise provided, it is likely that a speech-input user may attempt to use the placeholder text value as a means of interacting with the input.

Text labels "express something in human language"

For the purposes of this SC, text should not be considered a visible label if it is used in a symbolic manner, rather than directly expressing something in human language as per the definition of text in WCAG. For example, [1.4.5 Images of Text](#) describes considerations for "symbolic text characters." In the images of text example "B", "I", and "ABC" appear on icons in a text editor, where they are meant to symbolize the functions for Bold, Italics, and Spelling, respectively. In such a case, the accessible name should be the function the button serves (e.g., "Spell check" or "Check spelling"), not the visible symbolic characters. A similar text editor is shown in the figure below.

Likewise, where an author has used a greater-than symbol (">") to mimic the appearance of the right-facing arrow, the text does not convey something in human language. It is a symbol, in this

scenario likely meant to mimic the icons used for a "Play" button or a "Next" arrow.

This criterion is primarily aimed at users of speech recognition / voice control. Capitalization (upper- or lowercase) and most punctuation are generally ignored when a user speaks a label as a means of interacting with a control. For this reason, differences in capitalization and punctuation are not relevant when evaluating this criterion.

In the following example, while it is certainly not an error to include the colon and capitalization in the accessible name, a computed name of "first name" is considered equivalent to "First Name:" and passes the requirements of this criterion. A text input with a label of "First Name:" above it. `<p>First Name:</p> <input`

`type="text" ... aria-label="first name">` Likewise, in the following example, while the visible label for the button is "Next...", having "Next" as the accessible name passes. A large button labelled "Next...". `<button ... aria-`

`label="Next">Next...</button>` There are situations where capitalization can impact meaning – for instance, where the use of uppercase letters indicates an abbreviation. While in the context of speech recognition / voice control this won't matter, the difference may be relevant for screen reader users. As a best practice, while normatively this success criterion is "case-insensitive", it's still recommended that authors match the case/capitalization of the visible label and the accessible name in these cases.

Mathematical expressions are an exception to the previous subsection about symbolic characters. Math symbols can be used as labels; for example "A>B" and "11×3=33" and convey meaning.

The label should not be overwritten in the accessible name, and substitutions of words where a formula is used should be avoided

since there are multiple ways to express the same equation. For example, making the name "eleven multiplied by three is equivalent to thirty-three" might mean a user who said "eleven times three equals thirty-three" may not match. It is best to leave the formulas as used in the label and count on the user's familiarity with their speech software to achieve a match. Further, converting a mathematical formula label into an accessible name that is a spelled-out equivalent may create issues for translation. The name should match the label's formula text. Note that a consideration for authors is to use the proper symbol in the formula. For instance 11×3 (with a lower or upper case letter "X"), $11 * 3$ (with the asterisk symbol), and 11×3 (with the `×` symbol) are all easy for sighted users to interpret as meaning the same formula, but may not all be matched to "11 times 3" by speech recognition / voice control software. The proper operator symbol (in this case the `×` symbol) should be used.

Accessible Name and Description Computation specification

It is important to understand how the accessible name is derived. The [Accessible Name and Description Computation 1.1](#) and the [HTML Accessibility API Mappings 1.0](#) describe how the accessible name is computed, including which attributes are considered in its calculation, and in what order of preference. If a component has multiple possible attribute values that could be used for its accessible name, only the most preferred of those values will be computed. None of the other, less preferred values will be part of the name. For the most part, existing established programmatic relationships between labels and controls are reinforced by the specification.

Other text displayed on the screen that is correctly coded to meet [1.3.1 Info and Relationships](#) is **not** normally factored into the calculation for the accessible name of a UI component without

author intervention (via ARIA labeling techniques). The most common of these are:

- headings and instructions
- group labels for sets of components (i.e., used with legend / fieldset or with role of group or radiogroup)

Such textual information may constitute part of the component's *description* . So from both a programmatic viewpoint, and from the conservative tactic of only considering a label to be "adjacent text," neither headings, instructions, nor group 'labels' should normally be considered **labels** for the purpose of this success criterion.

It is important to note that the specification allows authors to override the name calculated through native semantics. Both `aria-label` and `aria-labelledby` take precedence in the name calculation, overriding the visible text as the accessible name even when the visible text label is programmatically associated with the control. For this reason, when a visible label already exists, `aria-label` should be avoided or used carefully, and `aria-labelledby` should be used as a supplement with care.

Finally, `aria-describedby` is not included in the Accessible Name computation (instead it is part of the Accessible Description computation). By convention, text associated with a control through `aria-describedby` is announced immediately after the accessible name by screen readers. Therefore, the context of headings, instructions, and group labels can be provided through the accessible description to assist users of screen readers without affecting the experience of those who navigate using speech recognition software.

Note: The term "parenthetical" in this section is describing text that literally appears within a set of round brackets (). It is not used in the more abstract sense of "information related to".

It is important to mention parenthetical text in labels in the context of accessible name versus description. In common usage, text in parentheses is considered secondary but relevant to meaning. Users of speech recognition would not typically announce text in parentheses as part of the input name. For that reason, parenthetical text may be optionally considered a description and left out of the accessible name.

However, where parenthetical information provides important context, such as indication of a required field or limitations on what is allowed for input, this information must be provided programmatically in some other way to the user if that information is not included as part of the accessible name. For example, "Name (required)" and "Date (YYYY-MM-DD)" could accept "Name" and "Date" as the accessible names. However, in order to pass [1.3.1 Info and Relationships](#), authors would need to programmatically surface both the required state and the limit on the allowed data formatting (in this example, eight integers fitting the YYYY-MM-DD pattern). The "required" state could be surfaced through the HTML `required` attribute or by using `aria-required="true"`. The allowed data formatting could be surfaced by being referenced using the `aria-describedby` attribute.

Benefits

- Speech-input users can directly activate controls on a page with fewer surprising changes of focus.
- Text-to-speech users will have a better experience because the labels that they hear match the visible text labels that they see on the screen.

Examples

- Accessible name matches visible label: The accessible name and visible label of a control match.
- Accessible name starts with visible label: The accessible name "Search for a value" begins with the text of the visible label, "Search".

Related Resources

- [Accessible Name and Description Computation](#)
- [Accessible Name and Description Computation in HTML Accessibility API Mappings 1.0](#)

2.5.4 Motion Actuation (Level A)

Source: [WCAG 2.2](#) / [Understanding 2.5.4](#) — Copyright © 2023 World Wide Web Consortium

Functionality that can be operated by device motion or user motion can also be operated by user interface components and responding to the motion can be disabled to prevent accidental actuation, except when:

Supported Interface

The motion is used to operate functionality through an accessibility supported interface;

Essential

The motion is essential for the function and doing so would invalidate the activity.

In brief

Goal

Content is not dependent on a user's ability to move a device.

What to do

Don't rely solely on device motion to control page content.

Why it's important

Some people cannot hold or move a device steadily.

Intent of this Success Criterion

The intent of this success criterion is to ensure that functions triggered by moving a device (for example, shaking or tilting) or by gesturing towards the device (so that sensors like a camera can pick up and interpret the gesturing), can also be operated by more conventional user interface components.

This criterion concerns content that interprets motion such as explicit eye, face, or hand gestures to a camera, or tilting and shaking a device, as an input. It does not cover the movement of users through space as registered by geolocation sensors or beacons, or events observed by the device other than intentional input gestures and movements by the user. The "Supported Interface" exception directly exempts incidental user motion – such as the movement necessary to type on/operate a keyboard, move/click a mouse, the hand and finger movements used to operate a touchscreen, or any movement necessary to operate assistive technology.

Devices often have sensors that can act as inputs, such as accelerometer and gyroscope sensors on a phone or tablet device. These sensors can allow the user to control something by simply changing the orientation or moving the device in particular ways. In other situations, web content can interpret user gestures via the camera or other sensors to actuate functions. For example, shaking the device might issue an "Undo" command, or a gentle hand wave might be used to move forward or backward in a sequence of pages. Some users with disabilities are not able to operate these device sensors (either not at all, or not precisely enough) because the device is on a fixed mount (perhaps a wheelchair) or due to motor impairments. Therefore, functionality offered through motion must also be available by another mechanism.

In addition, some users may accidentally activate sensors due to tremors or other motor impairments. The user must have the ability to turn off motion actuation to prevent such accidental triggering of functions. Applications may be able to meet this requirement by supporting operating system settings which allow the user to disable motion detection at the system level.

There is an exception where motion is essential for the function or not using motions or gestures would invalidate the activity. Some applications are specifically created to use device sensor data. Examples of content that are exempt from this requirement include a pedometer that relies on device motion to count steps.

Benefits

- This success criterion helps people who may be unable to perform particular motions (such as tilting, shaking, or gesturing) because the device may be mounted or users may be physically unable to perform the necessary movement. This success criterion ensures that users can still operate all functionality by other means such as touch or via assistive technologies.
- Other users will benefit in situations where they are unable to move their devices.

Examples of Success Criterion 2.6.1

- A user can choose an application setting which turns off Shake to Undo and other motion-activated features.
- After text is input in a field, shaking a device shows a dialog offering users to undo the input. A cancel button next to the text field offers the same functionality.
- A user can tilt a device to advance to the next or a previous page. Buttons are also provided to perform the same function.
- A user can move or pan a device to change the view in an interactive photo. A control is also available to perform these same functions.
- A user can gesture towards the device to navigate content. Controls are also available to navigate.

Related Resources

- [Detecting device orientation](#)
- [DeviceMotionEvent](#)

2.5.5 Target Size (Enhanced) (Level AAA)

Source: [WCAG 2.2](#) / [Understanding 2.5.5](#) — Copyright © 2023 World Wide Web Consortium

The size of the target for pointer inputs is at least 44 by 44 CSS pixels except when:

Equivalent

The target is available through an equivalent link or control on the same page that is at least 44 by 44 CSS pixels;

Inline

The target is in a sentence or block of text ;

User Agent Control

The size of the target is determined by the user agent and is not modified by the author;

Essential

A particular presentation of the target is essential to the information being conveyed.

In brief

Goal

Controls can be operated more easily, especially on touch screens.

What to do

Make custom targets at least 44 by 44 pixels.

Why it's important

Some people cannot tap small objects.

Intent

The intent of this success criterion is to help users who may have trouble activating a small target because of hand tremors, limited

dexterity, or other reasons. If the target is too small, it may be difficult to aim at the target. Mice and similar pointing devices can be hard to use for these users, and a larger target will help them greatly in having positive outcomes on the web page.

Touch is particularly problematic as it is an input mechanism with coarse precision. Users lack the same level of fine control as on inputs such as a mouse or stylus. A finger is larger than a mouse pointer, and generally obstructs the user's view of the precise location on the screen that is being touched/activated.

The issue can be further complicated for responsive and mobile sites that need to accommodate different types of fine and coarse inputs (for example, a site that can be accessed on a traditional desktop or laptop with a mouse, as well as on a tablet or mobile phone with a touch screen).

While this criterion defines a minimum target size, it is recommended that larger sizes are used to reduce the possibility of unintentional actions. This is particularly relevant if any of the following are true:

- the control is used frequently;
- the result of the interaction cannot be easily undone;
- the control is positioned where it will be difficult to reach, or is near the edge of the screen;
- the control is part of a sequential task.

The targets on a screen can have different purposes and uses, and this success criterion specifies how each is to be handled.

Equivalent targets: If there is more than one target on a screen that performs the same action, only one of the targets need to meet the target size of 44 by 44 CSS pixels.

Inline: The success criterion does not apply to inline targets in sentences or blocks of text. Content is often designed to reflow based on the screen width available. This is known as responsive design and makes it easier to read since you do not need to scroll both horizontally and vertically. In reflowed content, the targets can appear anywhere on a line and can change position based on the width of the available screen. Since targets can appear anywhere on the line, the size cannot be larger than the available text and spacing between the sentences or paragraphs, otherwise the targets could overlap. It is for this reason targets which are contained within one or more sentences are excluded from the target size requirements.

If the target is the full sentence and the sentence is not in a block of text, then the target needs to be at least 44 by 44 CSS pixels.

A footnote or an icon within or at the end of a sentence is considered to be part of a sentence and therefore are excluded from the minimum target size.

User Agent Control: If the size of the target is not modified by the author through CSS or other size properties, then the target does not need to meet the target size of 44 by 44 CSS pixels.

Essential: If the target is required to be a particular target size and cannot be provided in another way, while changing it would essentially change the information or functionality of the content, then the target does not need to meet the target size of 44 by 44 CSS pixels.

Benefits

- Users who use a mobile device where touch screen is the primary mode of interaction
- Users with mobility impairments, such as hand tremors

- Users who use a mobile device in environments where they are exposed to shaking such as public transportation
- Users who find fine motor movements difficult
- Users who access a device using one hand
- Users with large fingers, or who are operating the device with only a part of their finger or knuckle
- Users who have low vision may better see the target

Examples

Example 1: Buttons

Three buttons are on-screen and the touch target area of each button is 44 by 44 CSS pixels.

Example 2: Equivalent target

Multiple targets are provided on the page that perform the same function. One of the targets is 44 by 44 CSS pixels. The other targets do not have a minimum touch target of 44 by 44 CSS pixels.

Example 3: Text Link in a paragraph

Links within a paragraph of text have varying touch target dimensions. Links within paragraphs of text do not need to meet the 44 by 44 CSS pixels requirements.

Example 4: Text Link in a sentence

A text link that is in a sentence is excluded and does not need to meet the 44 by 44 CSS pixel requirement. If the text link is the full sentence, then the text link target touch area does need to meet the 44 by 44 CSS pixels.

Example 5: Footnote

A footnote link at the end of a sentence does not need to meet the 44 by 44 CSS pixels requirements. The footnote at the end of the sentence is considered to be part of the sentence.

Example 6: Help icon

A help icon within or at the end of a sentence does not need to meet the 44 by 44 CSS pixels requirements. The icon at the end of the sentence is considered to be part of the sentence.

Resources

- [Apple touch target size recommendations](#)
- [Windows UWP Guidelines for touch targets](#)
- [Google Material Design Touch targets](#)
- [web.dev Accessible tap targets](#)
- [Human Fingertips to Investigate the Mechanics of Tactile Sense \(PDF\)](#)
- [One-Handed Thumb Use on Small Touchscreen Devices](#)

2.5.6 Concurrent Input Mechanisms (Level AAA)

Source: [WCAG 2.2](#) / [Understanding 2.5.6](#) — Copyright © 2023 World Wide Web Consortium

Web content does not restrict use of input modalities available on a platform except where the restriction is essential , required to ensure the security of the content, or required to respect user settings.

In brief

Goal

Users can choose different ways of inputting content.

What to do

Do not prevent users from switching their mode of input.

Why it's important

People may not be able to work using just one input method.

Intent

The intent of this success criterion is to ensure that people can use and switch between different modes of input when interacting with web content. Users may employ a variety of input mechanisms when interacting with web content. These may be a combination of mechanisms such as a keyboard or keyboard-like interfaces and pointer devices like a mouse, stylus or touchscreen.

Even though a device may have a primary input mechanism, the user may choose to employ alternative input mechanisms when interacting with the device. For example, the primary mechanism for mobile phones and tablets is the touchscreen. The user of these devices may choose to use a paired mouse or external keyboard as an alternative to using the touchscreen.

Users should be able to switch input mechanisms at any point should the user determine that certain tasks and interactions are more easily accomplished by using an alternative input mechanism. Content must not limit the user's interaction to any particular input mechanism unless the restriction is essential, or is required to ensure the security of the content or to respect user settings.

Note: A touch-typing web application, which teaches users how to touch-type on a keyboard and/or measures their proficiency and speed, would be an example of an essential limitation to a particular input mechanism.

Benefits

- Users can interact with web content with whichever input mechanism is preferred and available to them.
- Users may switch between input mechanisms when they desire or the circumstances require it.
- Users are allowed to add and remove input mechanisms at any point, where supported by the operating system.

Examples

- A user with mobility impairment pairs a mouse and keyboard to a mobile phone with a touchscreen. The phone can thereafter be operated by those input devices and the content does not accept the touchscreen as the only input mechanism.
- On a touch-enabled laptop with coarse precision, people who have difficulty activating a small target because of hand tremors, limited dexterity or other reasons are still able to interact with content using their keyboard and trackpad.
- A user starts interacting with a page using a desktop keyboard, and then attaches a secondary touch-enabled monitor. Content can be operated using this newly added input mechanism and

does not assume that the keyboard, the first input mechanism it detected, is the only one in use.

- A speech input user navigates content using voice commands which translate to simulate mouse (and keyboard) commands. When talking with a colleague, however, the user turns speech recognition off and uses the mouse instead.
- A user opens a menu with a mouse, and then navigates between the menu items with arrow keys.

Resources

- [W3C Pointer Events - Level 2](#)
- [Patrick H. Lauke - Detecting touch: it's the 'why', not the 'how'](#)
- [Chris Wilson / Paul Kinlan: Touch And Mouse - Together Again For The First Time](#)
- [W3C Touch Events - Level 2: Interaction with Mouse Events and click](#)
- [W3C CSS Media Queries Level 4: Interaction Media Features](#)

2.5.7 Dragging Movements (Level AA)

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All functionality that uses a dragging movement for operation can be achieved by a single pointer without dragging, unless dragging is essential or the functionality is determined by the user agent and not modified by the author.

This requirement applies to web content that interprets pointer actions (i.e., this does not apply to actions that are required to operate the user agent or assistive technology).

In brief

Goal

Don't rely on dragging for user actions.

What to do

For any action that involves dragging, provide a simple pointer alternative.

Why it's important

Some people cannot use a mouse to drag items.

Intent of Dragging Movements

The intent of this success criterion is to ensure functionality that uses a dragging movement has another single pointer mode of operation without the need for the dexterity required to drag elements.

Some people cannot perform dragging movements in a precise manner. Others use a specialized or adapted input device, such as a trackball, head pointer, eye-gaze system, or speech-controlled mouse emulator, which may make dragging cumbersome and error-prone.

When an interface implements functionality that uses dragging movements, users perform four discrete actions:

- tap or click to establish a start-point, then
- press and hold that contact while...
- performing a repositioning of the pointer, before...
- releasing the pointer at the end-point.

Not all users can accurately press and hold that contact while also repositioning the pointer. An alternative method must be provided so that users with mobility impairments who use a pointer (mouse, pen, or touch contact) can use the functionality.

For example, while a range slider is operated by dragging the slider thumb, an alternative pointer method to change the value is to click/tap anywhere on the slider track to move the thumb to that position.

This requirement is separate from keyboard accessibility because people using a touchscreen device may not use a physical keyboard. Keyboard specific interactions such as tabbing or arrow keys may not be possible when encountering a drag and drop control. Note, however, that providing a text input can be an acceptable single-pointer alternative to dragging. For example, an input beside a slider could allow any user to enter a precise value for the slider. In such a situation, the on-screen keyboard that appears for touch users offers a single-pointer means of entering an alphanumeric value.

This criterion *does not* apply to scrolling and dragging gestures enabled by the user agent, as it's the user agent's responsibility to provide an accessibility-supported mechanism for these. In browsers operated with a mouse, users generally scroll content by dragging the browser's scrollbar. In touchscreen browsers, users generally scroll by "dragging" the page on the screen. Similarly,

most touchscreen browsers provide a "drag to refresh" gesture to refresh/reload a page. In all these cases, the functionality is provided by the browser, rather than implemented by the content – so this criterion *does not* apply. This criterion also *does not* apply to the use of techniques such as CSS `overflow` that make a section of content scrollable, as the actual scrolling mechanism even in these situations is provided by the user agent. The criterion *does* apply if content actively suppresses the user agent's own scrolling functionality and/or implements its own scrolling mechanism – in these cases, the scrolling/dragging gesture is interpreted and processed by the content itself, and thus falls under the responsibility of the content author.

Relationship to keyboard accessibility requirements

Success Criteria [2.1.1 Keyboard](#) and [2.1.3 Keyboard \(No Exception\)](#) require dragging features to be keyboard accessible. However, achieving keyboard equivalence for a dragging operation does not automatically meet this success criterion, unless that equivalent keyboard operation also provides controls that can be clicked or tapped with a pointer. It is possible to create an interface that works with dragging and keyboard controls, but still does not work using only clicks or taps. While many designs can be created for a dragging alternative which address both keyboard accessibility and operability by single pointer operation, the two requirements are evaluated independently.

Distinguishing dragging movements from path-based pointer gestures

This success criterion applies to dragging movements in general, which involve the user "grabbing" an element and moving it to another position. Once the pointer engages with a target to pick up/grab it, the direction of the dragging movement does not factor into the interaction at all. [Success Criterion 2.5.1 Pointer Gestures](#) is concerned with gestures that are path-based, as well as multi-

point gestures. For pointer gestures, the direction of the pointer movement matters. However, if an action involves **both** a dragging movement (such as grabbing a slider thumb, moving it, and then releasing it) **and** a path-based gesture (if the slider requires the user to exactly follow its track, or otherwise the user's grip on the slider thumb is "lost"), it may fail against the requirements of **both** success criteria.

Alternatives for dragging movements on the same page

Where functionality can be executed via dragging movements and an equivalent option exists that allows for single-pointer access without dragging, this success criterion is satisfied. It does not have to be the same component, so long as the functionality is equivalent. An example is a color wheel where a color can be changed by dragging an indicator. In addition, text fields for the numerical input of color values allow the definition of a color without requiring dragging movements. (Note that a text input is considered device agnostic; although the purpose is to enter characters, text entry can take place through voice, pointer or keyboard.)

Benefits of Dragging Movements

- Users who struggle with performing dragging movements can still operate an interface with a pointer interface.

Examples of Dragging Movements with alternatives

- A map allows users to drag the view of the map around, and the map has up/down/left/right buttons to move the view as well.
- A sortable list of elements may, after tapping or clicking on a list element, provide adjacent controls for moving the element up or down in the list by simply tapping or clicking on those controls.

- A task board that allows users to drag and drop items between columns also provides an additional pop-up menu after tapping or clicking on items for moving the selected element to another column by tapping or clicking on pop-up menu entries.
- A radial control widget (color wheel) where the value can be set by dragging the marker for the currently selected color to another position, also allows picking another color value by tapping or clicking on another place in the color wheel.
- A range slider control widget, where the value can be set by dragging the visual indicator (thumb) showing the current value, allows tapping or clicking on any point of the slider track to change the value and set the thumb to that position.
- A widget where you can drag a gift to one person in a photo of a group of people also has a menu alternative where users can select the person that should receive the gift from the menu.
- A graphical interface allows the user to draw a selection rectangle on an image by first setting one corner of the rectangle on the pointer down-event, dragging the opposite corner with the pointer, and then setting that second corner position on the pointer up-event. As an non-drag alternative, the user can enable a selection mode, click/tap the first corner, then click/tap the opposite corner, without the need to keep the pointer pressed and dragging.
- In an online quiz, users are expected to draw a connecting line between each item in one column and its respective counterpart item in a second column. Users who can't (or don't want to) draw those lines manually can also just click/tap on one item in the first column, then click/tap on an item in the second column, and the connecting line is drawn automatically.

- A kanban widget with several vertical columns representing states in a defined process allows the user drag elements to move them to another column. The user can also accomplish this by selecting the element with a single tap or click, and then activating an arrow button to move the selected element.
- A news site has a horizontal carousel with different news teasers that can be dragged to move items into view. It also offers forward and backward buttons on the left and right of the carousel, to move the carousel to the previous and next item with a simple click/tap. These buttons can be visible (for instance, as large arrow icons) or visually hidden (but still operable with a pointer).

2.5.8 Target Size (Minimum) (Level AA)

Source: WCAG 2.2 / Understanding 2.5.8 — Copyright © 2023 World Wide Web Consortium

The size of the target for pointer inputs is at least 24 by 24 CSS pixels , except when:

Spacing

Undersized targets (those less than 24 by 24 CSS pixels) are positioned so that if a 24 CSS pixel diameter circle is centered on the bounding box of each, the circles do not intersect another target or the circle for another undersized target;

Equivalent

The function can be achieved through a different control on the same page that meets this criterion;

Inline

The target is in a sentence or its size is otherwise constrained by the line-height of non-target text;

User Agent Control

The size of the target is determined by the user agent and is not modified by the author;

Essential

A particular presentation of the target is essential or is legally required for the information being conveyed.

Targets that allow for values to be selected spatially based on position within the target are considered one target for the purpose of the success criterion. Examples include sliders, color pickers displaying a gradient of colors, or editable areas where you position the cursor.

For inline targets the line-height should be interpreted as perpendicular to the flow of text. For example, in a language displayed vertically, the line-height would be horizontal.

In brief

Goal

Make controls easier to activate.

What to do

Ensure targets meet a minimum size or have sufficient spacing around them.

Why it's important

Some people with physical impairments cannot click small buttons that are close together.

Intent of Target Size (Minimum)

The intent of this success criterion is to help ensure targets can be easily activated without accidentally activating an adjacent target. Users with dexterity limitations and those who have difficulty with fine motor movement find it difficult to accurately activate small targets when there are other targets that are too close. Providing sufficient size, or sufficient spacing between targets, will reduce the likelihood of accidentally activating the wrong control.

Disabilities addressed by this requirement include hand tremors, spasticity, and quadriplegia. Some people with disabilities use specialized input devices instead of a computer mouse or trackpad. Typically these types of input device do not provide as much accuracy as mainstream pointing devices. Meeting this requirement also ensures that touchscreen interfaces are easier to use.

This success criterion defines a **minimum size** and, if this can't be met, a **minimum spacing**. It is still possible to have very small, and difficult to activate, targets and meet the requirements of this Success Criterion, provided that the targets don't have any adjacent targets that are too close. However, using larger target sizes will help many people use targets more easily. **As a best practice** it is recommended to at least meet the **minimum size** requirement of the Success Criterion, regardless of spacing. For important links/controls, consider aiming for the stricter [2.5.5 Target Size \(Enhanced\)](#).

Exceptions

The requirement is for targets to be at least 24 by 24 CSS pixels in size. There are five exceptions:

- **Spacing:** Undersized targets (those less than 24 by 24 CSS pixels) are positioned so that if a 24 CSS pixel diameter circle is centered on the bounding box of each, the circles do not intersect another target or the circle for another undersized target.
- **Equivalent:** In cases where a target does not have a size equivalent to 24 by 24 CSS pixels, but there is another control that can achieve the underlying function that does meet the requirements of this success criterion, the target can be excepted based on the "Equivalent" exception.
- **Inline:** The success criterion does not apply to inline targets in sentences, or where the size of the target is constrained by the line-height of non-target text. This exception is allowed because text reflow based on viewport size makes it impossible for authors to anticipate where links may be positioned relative to one another. Applying this success criterion when multiple links are embedded in blocks of smaller text often results in an

undesirable design. It is more important to set the line height to a value that improves readability.

- User agent control: Browsers have default renderings of some controls, such as the days of the month calendar in an `<input type="date">`. As long as the author has not modified the user agent default size, the target size for a User agent control is excepted.
- Essential: If the size and spacing of the targets is fundamental to the information being conveyed, the Essential exception applies. For example, in digital maps, the position of pins is analogous to the position of places shown on the map. If there are many pins close together, the spacing between pins and neighboring pins will often be below 24 CSS pixels. It is essential to show the pins at the correct map location, therefore the Essential exception applies. A similar example is an interactive data visualization where targets are necessarily dense. Another example is where jurisdictions legally require online forms to replicate paper forms, which can impose constraints on the size of targets. In such jurisdictions, any legal requirement to replicate small targets can be considered essential. When the "Essential" exception is applicable, authors are strongly encouraged to provide equivalent functionality through alternative means to the extent practical.

Size requirement

For a target to be "at least 24 by 24 CSS pixels", it must be conceptually possible to draw a solid 24 by 24 CSS pixel square, aligned to the horizontal and vertical axis such that the square is completely within the target (does not extend outside the target's area).

The 24 by 24px square has to be aligned with the page, although the target itself could be skewed.

If a target is not large enough to allow for a 24 by 24px square to be drawn inside it, it is considered *undersized* , and does not pass the size requirement of the success criterion. However, if it has sufficient space around it without adjacent targets, it may still pass the criterion thanks to the spacing exception (below).

The requirement is independent of the zoom factor of the page; when users zoom in, the CSS pixel size of elements does not change. This means that authors cannot meet it by claiming that the target will have enough spacing or sufficient size if the user zooms into the page.

The requirement does not apply to targets while they are obscured by content displayed as a result of a user interaction or scripted behavior of content, for example:

- interacting with a combobox shows a dropdown list of suggestions
- activating a button displays a modal dialog
- content displays a cookie banner after page load
- content displays a "Take a survey" dialog after some period of user inactivity

The requirement does however apply to targets in any new content that appears on top of other content.

While the success criterion primarily helps touch users by providing target sizing to prevent accidental triggering of adjacent targets, it is also useful for mouse or pen users. It reduces the chances of erroneous activation due to either a tremor or reduced precision, whether because of reduced fine motor control or input imprecision.

Spacing

When the minimum size for a target is not met, spacing can at least improve the user experience. There is less chance of accidentally activating a neighboring target if a target is not immediately adjacent to another. Touchscreen devices and user agents generally have internal heuristics to identify which link or control is closest to a user's touch interaction - this means that sufficient spacing between targets can work as effectively as a larger target size itself.

When a target is smaller than 24 by 24 CSS pixels, it is *undersized*. In this case, we check if it at least has sufficient *spacing* by drawing an imaginary 24 CSS pixel diameter circle over the undersized target, centered on the target's bounding box. For rectangular targets, the bounding box coincides with the target itself – thus, the circle is placed on the center of the target. If the target is *not* rectangular – for instance, if the target is clipped, has rounded corners, or if it's a more complex clickable SVG shape – we need to first determine the bounding box, and then find the box's center. Note that for concave shapes, the center of the bounding box may be outside of the target itself. Now, we center the circle on the center of the bounding box.

We repeat this process for all adjacent undersized targets. To determine if an undersized target has sufficient spacing (to pass this Success Criterion's spacing exception), we check that the 24 CSS pixel diameter circle of the target does not intersect another target or the circle of any other adjacent undersized targets.

The following example shows three versions of a horizontal row of six icon-based buttons:

- In the top row, the dimensions of each target are 24 by 24 CSS pixels, passing this success criterion.

- In the second row, the same targets are only 20 by 20 CSS pixels, but have a 4 CSS pixel space between them – as the target size is below 24 by 24 CSS pixels, these need to be evaluated against the Success Criterion's spacing exception, and they pass.
- In the last row, the targets are again 20 by 20 CSS pixels, but have no space between them – these fail the spacing exception. This is because the imaginary 24 CSS pixel diameter circles over the targets would intersect.

The next two illustrations show sets of buttons which are only 16 CSS pixels tall. In the first set, there are no targets immediately above or below the buttons, so they pass. In the second illustration, there are further buttons, and they have been stacked on top of one another, resulting in a fail.

The following two illustrations show how menu items can be adjusted to properly meet this requirement. In the first illustration, the About us menu has been activated, showing that each of the menu item targets has a 24 CSS pixel height (text and padding), and so passes. In the second illustration, the same menu is expanded, but the menu items only achieve 18 CSS pixels in height, and so fail.

The following example has one large target (an image that links to a new page related to that image) and a very small second target (a control with a magnifier icon to open a zoomed-in preview, possibly in a modal, of the image).

In the top row, the small target overlaps - or, to be more technically accurate, *clips* - the large target. The small target itself has a size of 24 by 24 CSS pixels, so passes. In the second row, we see that if the second target is any smaller – in this case 16 by 16 CSS pixels – it fails the criterion, as the imaginary circle with a 24

CSS pixel diameter we draw over the small target will intersect the large target itself.

In the following example, we have the same two targets – a large target and a small target. This time, the small target touches/abuts the large target. If the small target is smaller than 24 by 24 CSS pixels, the imaginary circle with a 24 CSS pixel diameter we draw over the small target will intersect the large target itself, failing the requirement. The undersized target must be spaced further away from the large target until its circle doesn't intersect the large target.

Users with different disabilities have different needs for control sizes. It can be beneficial to provide an option to increase the active target area without increasing the visible target size. Another option is to provide a mechanism to control the density of layout and thereby change target size or spacing, or both. This can be beneficial for a wide range of users. For example, users with visual field loss may prefer a more condensed layout with smaller sized controls while users with other forms of low vision may prefer large controls.

User agent control

This success criterion has an exception for the size of targets determined by a user agent and not modified by the author. An example of this kind of target is a browser's scrollbars. If a scrollbar's dimensions have been modified by the author then there must be a passing amount of spacing between the scrollbar and the content of the page. The following example shows a passing and a failing design.

Benefits of Target Size (Minimum)

Having targets with sufficient size - or at least sufficient target spacing - can help all users who may have difficulty in confidently

targeting or operating small controls. Users who benefit include, but are not limited to:

- People who use a mobile device where the touch screen is the primary mode of interaction;
- People using mouse, stylus or touch input who have mobility impairments such as hand tremors;
- People using a device in environments where they are exposed to shaking such as public transportation;
- Mouse users who have difficulty with fine motor movements;
- People who access a device using one hand;
- People with large fingers, or who are operating the device with only a part of their finger or knuckle.

Examples of Target Size (Minimum)

- Three buttons are on-screen and the target size of each button is 24 by 24 CSS pixels. Since the target size itself is 24 by 24 CSS pixels, no additional spacing is required, the success criterion passes.
- A row of buttons, each of which has a horizontal width of more than 24 CSS pixels, a height of only 20 CSS pixels, and vertical margin of 4 CSS pixels above and below the row of buttons. Since there is sufficient spacing both above and below the row of buttons, the success criterion passes.
- Links within a paragraph of text have varying target dimensions. Links within paragraphs of text do not need to meet the 24 by 24 CSS pixels requirements, so the success criterion passes.

Resources

- [Target size study for one-handed thumb use on small touchscreen devices](#)

Principle 3: Understandable

[Source: W3C WCAG 2.2 —

<https://www.w3.org/TR/WCAG22/#principle3>]

Information and the operation of the user interface must be understandable.

Guideline 3.1: Readable

[Source: W3C WCAG 2.2 —

<https://www.w3.org/TR/WCAG22/#guideline31>]

Make text content readable and understandable.

3.1.1 Language of Page (Level A)

Source: WCAG 2.2 / Understanding 3.1.1 — Copyright © 2023 World Wide Web Consortium

The default human language of each web page can be programmatically determined .

In brief

Goal

Assistive technology can determine the language of a page.

What to do

Indicate the predominant language on a page.

Why it's important

People using assistive technology get information in the correct language.

Intent of Language of Page

The intent of this success criterion is to ensure that content developers provide information in the web page that user agents need to present text and other linguistic content correctly. Both assistive technologies and conventional user agents can render text more accurately when the language of the web page is identified. Screen readers can load the correct pronunciation rules. Visual browsers can display characters and scripts correctly. Media players can show captions correctly. As a result, users with disabilities will be better able to understand the content.

The default human language of the web page is the default text-processing language

as discussed in Internationalization Best Practices: Specifying Language in XHTML & HTML Content . When a web page uses several languages, the default text-processing language is

the language which is used most. (If several languages are used equally, the first

language used should be chosen as the default human language.)

For multilingual sites targeting Conformance Level A, the Working Group strongly encourages developers to follow Success Criterion 3.1.2 Language of Parts as well even though that is a Level AA success criterion.

Benefits of Language of Page

This success criterion helps:

- people who use screen readers or other technologies that convert text into synthetic speech;
- people who find it difficult to read written material with fluency and accuracy, such as recognizing characters and alphabets or decoding words;
- people with certain cognitive, language and learning disabilities who use text-to-speech software
- people who rely on captions for synchronized media.

Examples of Language of Page

Example 1. A web page with content in two languages

A web page produced in Germany and written in HTML includes content in both German and English, but most of the content is in German. The default human language is identified as German (de) by the lang attribute on the html element.

Resources for Language of Page

- [Internationalization Best Practices: Specifying Language in XHTML & HTML Content](#)
- [Declaring language in HTML](#)

3.1.2 Language of Parts (Level AA)

Source: [WCAG 2.2](#) / [Understanding 3.1.2](#) — Copyright © 2023 World Wide Web Consortium

The human language of each passage or phrase in the content can be programmatically determined except for proper names, technical terms, words of indeterminate language, and words or phrases that have become part of the vernacular of the immediately surrounding text .

In brief

Goal

Assistive technology can identify the languages used within a page.

What to do

Indicate when words are in a different language.

Why it's important

People using assistive technology get information in the correct language.

Intent of Language of Parts

The intent of this success criterion is to ensure that user agents can correctly present phrases, passages, and in some cases words written in multiple languages. This makes it possible for user agents and assistive technologies to present content according to the presentation and pronunciation rules for that language. This applies to graphical browsers as well as screen readers, braille displays, and other voice browsers.

Both assistive technologies and conventional user agents can render text more accurately if the language of each passage of text is identified. Screen readers can use the pronunciation rules of the language of the text. Visual browsers can display characters and

scripts in appropriate ways. This is especially important when switching between languages that read from left to right and languages that read from right to left, or when text is rendered in a language that uses a different alphabet. Users with disabilities who know all the languages used in the web page will be better able to understand the content when each passage is rendered appropriately.

When no other language has been specified for a phrase or passage of text, its human

language is the default human language of the web page (see [Success Criterion 3.1.1 Language of Page](#)).

So the human language of all content in single language documents can be programmatically determined.

In certain situations or modes of operation, screen readers and other assistive technologies will announce an element using its computed accessible name . As an example, this currently happens in most commonly available screen readers when accessing the isolated list of links in the current page.

Accessible names are "flat" strings, and don't contain any structure or information. Despite being correctly denoted in the underlying code or markup, changes in the natural language of content are not exposed to assistive technologies in these cases. As a result, assistive technologies won't switch pronunciation rules or voice/language packs for the specified language(s). While there may be workarounds for certain simple cases – such as moving the `lang` language attribute to the element itself, rather than indicating a change of language on a child nodes inside the element – these may not work in all user agent/assistive technology combinations.

As these problems are caused by limitations of user agents and assistive technologies, and are outside of an author's control, they *don't fail* the requirements of this success criterion. When evaluating content against this success criterion, auditors can therefore not solely rely on the spoken output from assistive technologies, but must verify whether or not changes in natural language have been identified correctly in the underlying code or markup.

Individual words or phrases in one language can become part of another language. For example, "rendezvous" is a French word that has been adopted in English, appears in English dictionaries, and is properly pronounced by English screen readers. Hence a passage of English text may contain the word "rendezvous" without specifying that its human language is French and still satisfy this success criterion. Frequently, when the human language of text appears to be changing for a single word, that word has become part of the language of the surrounding text. Because this is so common in some languages, single words should be considered part of the language of the surrounding text unless it is clear that a change in language was intended. If there is doubt whether a change in language is intended, consider whether the word would be pronounced the same (except for accent or intonation) in the language of the immediately surrounding text.

Most professions require frequent use of technical terms which may originate from a foreign language. Such terms are usually not translated to all languages. The universal nature of technical terms also facilitate communication between professionals.

Some common examples of technical terms include: Homo sapiens, Alpha Centauri, hertz, and habeas corpus.

Identifying changes in language is important for a number of reasons:

- It allows braille translation software to follow changes in language, e.g., substitute control codes for accented characters, and insert control codes necessary to prevent erroneous creation of Grade 2 braille contractions.
- Speech synthesizers that support multiple languages will be able to speak the text in the appropriate accent with proper pronunciation. If changes are not marked, the synthesizer will try its best to speak the words in the default language it works in. Thus, the French word for car, "voiture" would be pronounced "voyture" by a speech synthesizer that uses English as its default language.
- Marking changes in language can benefit future developments in technology, for example users who are unable to translate between languages themselves will be able to use machines to translate unfamiliar languages.
- Marking changes in language can also assist user agents in providing definitions using a dictionary.

Benefits of Language of Parts

This success criterion helps:

- people who use screen readers or other technologies that convert text into synthetic speech;
- people who find it difficult to read written material with fluency and accuracy, such as recognizing characters and alphabets, decoding words, and understanding words and phrases;
- people with certain cognitive, language and learning disabilities who use text-to-speech software;
- people who rely on captions to recognize language changes in the soundtrack of synchronized media content.

A German phrase in an English sentence

In the sentence, "He maintained that the DDR (German Democratic Republic) was just a 'Treppenwitz der Weltgeschichte'," the German phrase 'Treppenwitz der Weltgeschichte' is marked as German. Depending on the markup language, English may either be marked as the language for the entire document except where specified, or marked at the paragraph level. When a screen reader encounters the German phrase, it changes pronunciation rules from English to German to pronounce the word correctly.

Alternative language links

An HTML web page includes links to versions of the page in other languages (e.g., Deutsch , Français , Nederlands , Catalan , etc.). The text of each link is the name of the language, in that language. The language of each link is indicated via a `lang` attribute.

```
<ul>
  <li><a href="..." lang="de">Deutsch</a></li>
  <li><a href="..." lang="it">Italiano</a></li>
  <li><a href="..." lang="fr">Français</a></li>
  ...
  <li><a href="..." lang="zh-hant">繁體中文</a></li>
</ul>
```

"Podcast" used in a French sentence

Because "podcast" is part of the vernacular of the immediately surrounding text in the following excerpt, "À l'occasion de l'exposition "Energie éternelle. 1500 ans d'art indien", le Palais des Beaux-Arts de Bruxelles a lancé son premier podcast. Vous pouvez télécharger ce podcast au format M4A et MP3 ", no indication of language change is required.

The element's content and attribute values are in different languages

This example assumes that the page's default content is in English. The link's `title` attribute is in English, but the nested `span` element that contains the word Español has a `lang="es"` attribute.

```
<a title="Spanish" href="qa-html-language-declarations-es.html"><span lang="es">Español</span></a>
```

Resources for Language of Parts

- [HTML - The lang and xml:lang attributes](#) .
- [Language tags in HTML and XML](#) .
- [Authoring HTML: Language declarations](#) .
- [Declaring language in HTML](#)

3.1.3 Unusual Words (Level AAA)

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A mechanism is available for identifying specific definitions of words or phrases used in an unusual or restricted way , including idioms and jargon .

In brief

Goal

Users can identify and learn what unusual words mean.

What to do

Provide definitions for technical jargon and unusual terms.

Why it's important

More people, especially those with cognitive disabilities, can understand the meaning of content.

Intent of Unusual Words

Certain disabilities make it difficult to understand non-literal word usage and specialized words or usage. Certain disabilities make it difficult to understand figurative language or specialized usage. Providing mechanisms that help clarify these situations is vital for these audiences.

In accordance with the example provided in the definition for used in an unusual or restricted way ,

authors *can* assume that users will use context as a way to disambiguate words that may

otherwise be ambiguous. Similarly, authors can take into account the intended target audience

of a web page when evaluating whether or not particular words are used in an unusual or restricted way

that may not be clear to this audience. Authors are encouraged to satisfy this success criterion for

specialized information intended for non-specialist readers, even when claiming only A or AA conformance.

This criterion does not exempt words that *may be confusing to everybody* – such as new, made-up, or invented words.

Benefits of Unusual Words

This success criterion may help people with cognitive, language and learning disabilities who:

- have difficulty decoding words
- have difficulty understanding words and phrases

It would also help people with visual disabilities who:

- lose visual context when zoomed-in with a screen magnifier

Examples of Unusual Words

Text that includes a definition for a word used in an unusual way

Organize the list or "cascade" of dictionaries and other resources so that the definition search will find the intended definitions instead of displaying definitions from other sources in the "cascade." (The "cascade" lists the dictionaries and other reference materials in the order most likely to bring up the right definition. This controls the order to follow when searching for definitions.)

Including definitions in the glossary

WCAG 2 uses the word "text" in a specific way. Thus, when the word "text" is used within WCAG 2 it is linked to the definition of "text" provided in a glossary within the same web page.

The specific definition of a word is provided at the bottom of the page

The internal link from the word to the corresponding definition is also provided within the page.

Resources for Unusual Words

The inclusion of a product or vendor name in the list below does not constitute an endorsement by the Accessibility Guidelines Working Group or the Web Accessibility Initiative of the World Wide Web Consortium. This list is provided simply for convenience and to give users an idea of what resources may be available.

- Free bilingual dictionaries for a number of languages are available from the [Freedict.org website](http://Freedict.org) . The dictionaries are of uneven quality and size as noted on the site. Retrieved

9 April 2005.

- [The WorldStar Free Dictionaries, Translators and Search Engines](#) site provides access to free on-line dictionaries and search engines in many languages.

Retrieved 18 November 2005.

- More dictionaries are at [your dictionary](#) , freelang.com (in English, Spanish and French!) and many other places.

3.1.4 Abbreviations (Level AAA)

Source: [WCAG 2.2](#) / [Understanding 3.1.4](#) — Copyright © 2023 World Wide Web Consortium

A mechanism for identifying the expanded form or meaning of abbreviations is available.

In brief

Goal

Users can identify and learn what abbreviations mean.

What to do

Provide the expanded form of abbreviations to users.

Why it's important

Some people, including those with cognitive disabilities, may not understand the shortened form of words.

Intent of Abbreviations

The intent of this success criterion is to ensure that users can access the expanded form of abbreviations.

If an abbreviation has become part of the language , it is exempt from this requirement.

For instance laser was an initialism (and many years ago commonly presented in capital letters as LASER). Now it is most commonly used as a word in its own right, and marking it as an abbreviation (expanding it to Light Amplification through the Stimulated Emission of Radiation) would likely be confusing and unhelpful for people.

If an abbreviation is listed in a common dictionary as a word, that is reasonable grounds to claim it has become part of the language .

Another signal that an abbreviation can be considered to have become part of the language are well-known company/organization names or product trademarks, such as IBM or NASA, that are generally used as words in their own right.

Beyond this, the criterion does not exempt abbreviations even if they are common or familiar.

Benefits of Abbreviations

This success criterion may help people who:

- have difficulty decoding words;
- rely on screen magnifiers (magnification may reduce contextual cues);
- have limited memory;
- have difficulty using context to aid understanding.

Abbreviations may confuse some readers in different ways:

- Some abbreviations do not look like normal words and cannot be pronounced according to the usual rules of the language. For example, the English word "room" is abbreviated as "rm," which does not correspond to any English word or phoneme. The user has to know that "rm" is an abbreviation for the word "room" in order to say it correctly.
- Sometimes, the same abbreviation means different things in different contexts. For example, in the English sentence "Dr. Johnson lives on Boswell Dr.," the first "Dr." is an abbreviation for "Doctor" and the second instance is an abbreviation for the word "Drive" (a word that means "street"). Users must be able to understand the context in order to know what the abbreviations mean.
- Some acronyms spell common words but are used in different ways. For example, "JAWS" is an acronym for a screen reader

whose full name is "Job Access with Speech." It is also a common English word referring to the part of the mouth that holds the teeth. The acronym is used differently than the common word.

- Some acronyms sound like common words but are spelled differently. For example, the acronym for Synchronized Multimedia Integration Language, S M I L, is pronounced like the English word "smile."

It would also help people with visual disabilities who:

- Lose context when zoomed-in with a screen magnifier

Examples of Abbreviations

An abbreviation whose expansion is provided the first time the abbreviation appears in the content

The name, "World Wide Web Consortium," appears as the first heading on the organization's home page. The abbreviation, "W3C," is enclosed in parentheses in the same heading.

A dictionary search form

A website includes a search form provided by an on-line acronym service. Users enter an acronym and the form returns a list of possible expansions from the sources that it searched.

A medical website

A medical website provides information for both doctors and patients. The site includes a set of cascading dictionaries; a very specialized medical dictionary is first, followed by a second medical dictionary for the general public. The cascade also includes a list of acronyms and abbreviations that are unique to the site, and finally there is a standard dictionary as well. The standard dictionary at the end of the list provides definitions for most words in the text. The specialized medical dictionary yields

definitions of unusual medical terms. Definitions for words that appear in more than one dictionary are listed in the order of the cascade. The meaning of acronyms and abbreviations is provided by the list of acronyms and abbreviations.

Expanded forms of Abbreviations

The expanded form of each abbreviation is available in a programmatically determinable manner. User agents that speak the text can use the expanded form to announce the abbreviation. Other user agents might make the expanded form available as a tooltip or as contextual help for the abbreviation.

Resources for Abbreviations

- [Acronym finder](#) - You can search with the exact acronym, the beginning of the acronym, wildcards

and reverse lookup.

- [Abbreviations.com](#) .

3.1.5 Reading Level (Level AAA)

Source: WCAG 2.2 / Understanding 3.1.5 — Copyright © 2023 World Wide Web Consortium

When text requires reading ability more advanced than the lower secondary education level after removal of proper names and titles, supplemental content , or a version that does not require reading ability more advanced than the lower

secondary education level, is available.

In brief

Goal

Users can get a simplified version of complex information.

What to do

When text information becomes complex, create a more easily understood version.

Why it's important

More people, especially those with cognitive disabilities, can understand the meaning of content.

Intent of Reading Level

Content should be written as clearly and simply as possible. The intent of this Success Criterion is:

- to ensure that additional content is available to aid the understanding of difficult or complex text;
- to establish a testable measure indicating when such additional content is required.

This success criterion helps people with reading disabilities while also allowing authors to publish difficult or complex web content. Text difficulty is described in terms of the level of education

required to read the text. Education levels are defined according to the International Standard Classification of Education [[UNESCO]], which was created to allow international comparison among systems of education.

Difficult or complex text may be appropriate for most members of the intended audience (that is, most of the people for whom the content has been created). But there are people with disabilities, including reading disabilities, even among highly educated users with specialized knowledge of the subject matter. It may be possible to accommodate these users by making the text more readable. If the text cannot be made more readable, then supplemental content is needed. Supplemental content is required when text demands reading ability more advanced than the lower secondary education level—that is, more than nine years of school. Such text presents severe obstacles to people with reading disabilities and is considered difficult even for people without disabilities who have completed upper secondary education.

Reading disabilities such as dyslexia make it difficult to recognize written or printed words and associate them with the correct sounds. This is called "decoding" the text. Decoding must be automatic in order for people to read fluently. The act of decoding text word by word consumes much of the mental energy that most people are able to use for understanding what they read. Text that uses short, common words and short sentences is easier to decode and usually requires less advanced reading ability than text that uses long sentences and long or unfamiliar words.

The education level required to read text content (also called "readability") is measured by analyzing selected passages of text from the web page. If the web page includes text written for different purposes or different styles are used, the selected passages include samples of the types of content in the web page

and the different styles in which the content is written. (In many cases, the web page contains only one kind of text content—e.g., technical documentation, a legal notice, marketing material, etc.—and all the content uses the same style.)

Educators can also measure the education level required to read text content. For example, qualified teachers can evaluate text according to local education standards to determine if it requires reading ability beyond what is expected for students in the last year of lower secondary education.

Because people's names, the names of cities or other proper names cannot be changed to shorter names with fewer syllables, and because doing so or just referring to everyone by pronouns can make sentences harder to understand, the success criterion specifies that proper names can be ignored or removed from the text before assessing whether it meets the reading ability requirement. Titles refer to the name of documents, books, movies, etc. Titles are removed or ignored for the analysis because changing the words in titles might make the titles easier to read but would make it impossible to understand the item to which the title refers. This would make it harder to read and understand the content. (e.g., a book, academic paper, article, etc.). Therefore, titles are also exempted specifically.

When a web page contains multiple languages, a readability result should be calculated for each language that constitutes at least 5% of the content and that is used in full sentences or paragraphs (not just individual words or phrases). The overall readability of the page should be judged on the language that yields the worst readability results.

The readability of content may also be determined by applying a readability formula to the selected passage. Many (though not all) readability formulas base their calculations on passages of 100

words. Such formulas have been developed for many languages. The number of words in the passage is counted and the length of the words is determined by counting either the number of syllables or the number of characters. Most readability formulas also count the number and length of sentences. The average length of words and sentences in the content is then used to calculate a readability score. (Some languages, such as Japanese, may include multiple scripts within the same passage. Readability formulas for these languages may use the number and length of such "runs" in their calculations.) The result may be presented as a number (for example, on a scale from 0-100) or as a grade level. These results can then be interpreted using the education levels described in the International Standard Classification of Education. Readability formulas are available for at least some languages when running the spell checkers in popular software if you specify in the options of this engine that you want to have the statistics when it has finished checking your documents.

Levels of education

Primary education

First 6 years of school

Lower secondary education

7-9 years of school

Upper secondary education

10-12 years of school

Advanced education

More than 12 years of school

Adapted from International Standard Classification of Education (UNESCO) .

According to the Open Society Mental Health Initiative, the concept of Easy to Read cannot be universal, and it will not be possible to write a text that will suit the abilities of all people with literacy and comprehension problems. Using the clearest and simplest language appropriate is highly desirable, but the Accessibility Guidelines Working Group could not find a way to test whether this had been achieved. The use of reading level is a way to introduce testability into a success criterion that encourages clear writing. Supplementary content can be a powerful technique for people with some classes of cognitive disability.

Benefits of Reading Level

This success criterion may help people who:

- Have difficulty comprehending and interpreting written language (e.g., articles, instructions, or newspapers in text or braille), for the purpose of obtaining general knowledge or specific information

Examples of Reading Level

A scientific journal including readable summaries of complex research articles

A scientific journal includes articles written in highly technical language aimed at specialists in the field. The journal's Table of Contents page includes a plain-language summary of each article. The summaries are intended for a general audience with eight years of school. The metadata for the journal uses the Dublin Core specification to identify the education level of the articles' intended audience as "advanced," and the education level of the intended audience for the summaries as "lower secondary education."

Medical information for members of the public

A medical school operates a website that explains recent medical and scientific discoveries. The articles on the site are written for people without medical training. Each article uses the Dublin Core metadata specification to identify the education level of the intended audience as "lower secondary education" and includes the Flesch Reading Ease score for the article. A link on each page displays the education level and other metadata. No supplemental content is required because people who read at the lower secondary education level can read the articles.

An e-learning application

An on-line course about Spanish cultural history includes a unit on Moorish architecture. The unit includes text written for students with different reading abilities. Photographs and drawings of buildings illustrate architectural concepts and styles. Graphic organizers are used to illustrate complex relationships, and an audio version using synthetic speech is available. The metadata for each version describes the academic level of the content and includes a readability score based on formulas developed for Spanish-language text. The learning application uses this metadata and metadata about the students to provide versions of instructional content that match the needs of individual students.

Science information that requires a reading ability at the lower secondary education level

The text below (116 words total) requires a reading ability of grade 4.2 in the United States according to the Flesch-Kincaid formula. In the US, grade 4.2 is at the primary education level. Science is about testing — and about looking closely. Some scientists use microscopes to take a close look. We're going to use a simple piece of paper. Here's what you do: Print this page and cut out the square to make a "window" one inch square. (It's easiest to fold

the page in half before you cut.) Choose something interesting: a tree trunk, a leaf, flower, the soil surface, or a slice of soil from a shovel. Put your window over the thing and look at it closely. Take your time — this is not a race. To help you see more details, draw a picture of what's inside your square. Now let's think about what you've found. (Source: Howard Hughes Medical Institute "CoolScience for Kids" Project)

Resources for Reading Level

- The [Plain Language Association International \(PLAIN\)](#) website provides many useful resources to help writers produce documents that communicate clearly in a variety of cultural and rhetorical contexts.
- [The US Government's plain language website](#) provides general information about plain language as well as information about use

of plain language in US Government documents, including legal requirements

- [The Plain English Campaign website](#) provides useful information and guidance for authors writing in English.
- [Juicy Studio's Readability Test](#) analyzes the readability of all rendered content.
- [Entry for Audience Education Level. Using Dublin Core – Dublin Core Qualifiers](#)
- [IMS Learner Information Packaging Model Information Specification, Table 6.3 "accessibility" learner information data structure detailed description](#)
- [Leesbaar Nederlands](#) ("Readable Dutch") contains guidelines for writing text that is accessible for people

with a reading disability. These guidelines address language, content and design.

- European Easy-to-Read Guidelines
- Flesch-Kincaid Readability Test

3.1.6 Pronunciation (Level AAA)

Source: [WCAG 2.2](#) / [Understanding 3.1.6](#) — Copyright © 2023 World Wide Web Consortium

A mechanism is available for identifying specific pronunciation of words where meaning of the

words, in context, is ambiguous without knowing the pronunciation.

In brief

Goal

Users can identify the pronunciation of ambiguous words.

What to do

Indicate how to pronounce a word, where its meaning is otherwise unclear.

Why it's important

Some people, including those with cognitive disabilities, may not understand the meaning of content.

Intent of Pronunciation

The intent of this success criterion is to help people who are blind, people who have low vision, and people with reading disabilities to understand content in cases where meaning depends on pronunciation. Often words or characters have different meanings, each with its own pronunciation. The meaning of such words or characters can usually be determined from the context of the sentence. However, for more complex or ambiguous sentences, or for some languages, the meaning of the word cannot be easily determined or determined at all without knowing the pronunciation. When the sentence is read aloud and the screen reader reads the word using the wrong pronunciation, it can be

even more difficult to understand than when read visually. When words are ambiguous or indeterminate unless the pronunciation is known, then providing some means of determining the pronunciation is needed.

For example, in the English language heteronyms are words that are spelled the same but have different pronunciations and meanings, such as the words desert (abandon) and desert (arid region). If the proper pronunciation can be determined from the context of the sentence, then nothing is required. If it cannot then some mechanism for determining the proper pronunciation would be required. Additionally, in some languages certain characters can be pronounced in different ways. In Japanese, for example, there are characters like Han characters (kanji) that have multiple pronunciations. Screen readers may speak the characters incorrectly without the information on pronunciation. When read incorrectly, the content will not make sense to users.

Benefits of Pronunciation

This success criterion may help people who:

- have difficulty decoding words
- have difficulty using context to aid understanding
- use technologies that read the words aloud

Examples of Pronunciation

Giving the reading of a person's name

Web content in Japanese provides kana (Japanese phonetic syllabary characters) written next to Han characters (kanji) show the pronunciation of a person's name. The kana is provided to users in parentheses right after the word. Giving the reading of the words written in Han characters (kanji) allows both sighted users and screen readers to read/pronounce the words correctly. Note

that screen readers will speak the word twice: the Han characters (kanji) that can be pronounced in a wrong way are read first and then kana is spoken in order to provide the correct reading.

Showing the reading of the words by ruby element

Web content using HTML provides kana (phonetic syllabary characters) written above the characters to show the reading (pronunciation) of the words by using the ruby element.

Providing sound files of the pronunciation

A document includes some words whose meaning cannot be determined without knowing the correct pronunciation. Each word is linked to a sound file that gives the correct pronunciation. Users can select these links to find out how to pronounce the words.

Including pronunciation information in the glossary

A web page includes a glossary section. Some items in the glossary include pronunciation information as well as definitions. Words in the content whose meaning cannot be determined without knowing their pronunciation are linked to the appropriate entries in the glossary.

Text that includes pronunciation information for characters shared by several languages but pronounced differently in each language

A Japanese university website includes several short phrases quoted from scholarly texts in Chinese and Korean. The quotations are written using the same script as the Japanese text. Pronunciation information is provided to show the correct reading of the Chinese and Korean characters.

For Japanese, the ruby element is used for showing the "reading" rather than "pronunciation."

Guideline 3.2: Predictable

[Source: W3C WCAG 2.2 —

<https://www.w3.org/TR/WCAG22/#guideline32>]

Make web pages appear and operate in predictable ways.

3.2.1 On Focus (Level A)

Source: [WCAG 2.2](#) / [Understanding 3.2.1](#) — Copyright © 2023 World Wide Web Consortium

When any user interface component receives focus, it does not initiate a change of context .

In brief

Goal

Content can be navigated more predictably.

What to do

Do not change a user's context when items get focus.

Why it's important

Content that behaves predictably is especially important to people with disabilities.

Intent of On Focus

The intent of this success criterion is to ensure that functionality is predictable as visitors navigate their way through a document. Any component that is able to trigger an event when it receives focus must not change the context. Examples of changing context when a component receives focus include, but are not limited to:

- forms submitted automatically when a component receives focus;
- new windows launched when a component receives focus;
- focus is changed to another component when that component receives focus;

Focus may be moved to a control either via the keyboard (e.g. tabbing to a control) or the mouse (e.g. clicking on a text field). Moving the mouse over a control does not move the focus unless

scripting implements this behavior. Note that for some types of controls, clicking on a control may also activate the control (e.g. button), which may, in turn, initiate a change in context.

What is meant by "component" here is also sometimes called "user interface element" or "user interface component".

Benefits of On Focus

- This success criterion helps people with visual disabilities, cognitive limitations, and motor impairments by reducing the chance that a change of context will occur unexpectedly.

Examples of On Focus

Example 1: A dropdown menu

A dropdown menu on a page allows users to choose between jump destinations. If the person uses the keyboard to move down to a choice and activates it (with a spacebar or enter key) it will jump to a new page. However, if the person moves down to a choice and either hits the escape or the tab key to move out of the pulldown menu – it does not jump to a new screen as the focus shifts out of the dropdown menu.

Example of a Failure: A help dialog

When a field receives focus, a help dialog window describing the field and providing options opens. As a keyboard user tabs through the web page, the dialog opens, moving the keyboard focus away from the control every time the user attempts to tab past the field.

3.2.2 On Input (Level A)

Source: [WCAG 2.2](#) / [Understanding 3.2.2](#) — Copyright © 2023 World Wide Web Consortium

Changing the setting of any user interface component does not automatically cause a change of context unless the user has been advised of the behavior before using the component.

In brief

Goal

Content can be operated more predictably.

What to do

Forewarn users if their context will change based on their input.

Why it's important

Content that behaves predictably is especially important to people with disabilities.

Intent of On Input

The intent of this success criterion is to ensure that entering data or selecting a form control has predictable effects. Changing the setting of any user interface component is changing some aspect in the control that will persist when the user is no longer interacting with it. So checking a checkbox, entering text into a text field, or changing the selected option in a list control changes its setting, but activating a link or a button does not. Changes in context can confuse users who do not easily perceive the change or are easily distracted by changes. Changes of context are appropriate only when it is clear that such a change will happen in response to the user's action.

This success criterion covers changes in context due to changing the setting of a control. Clicking on links or buttons is activating a control, not changing the setting of that control.

What is meant by "component" and "user interface component" here is also sometimes called "user interface element".

Benefits of On Input

- This success criterion helps users with disabilities by making interactive content more predictable. Unexpected changes of context can be so disorienting for users with visual disabilities or cognitive limitations that they are unable to use the content.
- Individuals who are unable to detect changes of context are less likely to become disoriented while navigating a site. For example: Individuals who are blind or have low vision may have difficulty knowing when a visual context change has occurred, such as a new window popping up. In this case, warning users of context changes in advance minimizes confusion when the user discovers that the back button no longer behaves as expected.
- Some individuals with low vision, with reading and intellectual disabilities, and others who have difficulty interpreting visual cues may benefit from additional cues in order to detect changes of context.

Examples of On Input

- A form is provided for creating calendar entries in a web-based calendaring and scheduling application. Along with the standard fields for subject, time and location, a select dropdown allows the user to choose the type of calendar entry to create. The calendar entry type can be meeting, appointment or reminder. If the user selects the meeting option, additional fields are displayed on the page for entering

the meeting participants. Different fields appear if the reminder option is chosen. Because only parts of the entry change and the overall structure remains the same, the basic context remains for the user.

- A form contains fields representing US phone numbers. All of the numbers have a three digit area code followed by a three digit prefix and finally a four digit number, and each part of the phone number is entered into a separate field. When the user completes the entry of one field the focus automatically moves to the next field of the phone number. This behavior of phone fields is described for the user at the beginning of the form.

3.2.3 Consistent Navigation (Level AA)

Source: WCAG 2.2 / Understanding 3.2.3 — Copyright © 2023 World Wide Web Consortium

Navigational mechanisms that are repeated on multiple web pages within a set of web pages occur in the same relative order each time they are repeated, unless a change is initiated by the user.

In brief

Goal

Content can be navigated more predictably.

What to do

Consistently order navigation that repeats across multiple pages.

Why it's important

Content that behaves predictably is especially important to people with disabilities.

Intent of Consistent Navigation

The intent of this success criterion is to encourage the use of consistent presentation and layout for users who interact with repeated content within a set of web pages and need to locate specific information or functionality more than once. Individuals with low vision who use screen magnification to display a small portion of the screen at a time often use visual cues and page boundaries to quickly locate repeated content. Presenting repeated content in the same order is also important for visual users who use spatial memory or visual cues within the design to locate repeated content.

It is important to note that the use of the phrase "same order" in this section is not meant to imply that sub-navigation menus cannot be used or that blocks of secondary navigation or page

structure cannot be used. Instead, this success criterion is intended to assist users who interact with repeated content across web pages to be able to predict the location of the content they are looking for and find it more quickly when they encounter it again.

Users may initiate a change in the order by using adaptive user agents or by setting preferences so that the information is presented in a way that is most useful to them.

Benefits of Consistent Navigation

- Ensuring that repeated components occur in the same order on each page of a site helps users become comfortable that they will be able to predict where they can find things on each page. This helps users with cognitive limitations, users with low vision, users with intellectual disabilities, and also those who are blind.

Examples of Consistent Navigation

A consistently located control

A search field is the last item on every web page in a site. Users can quickly locate the search function.

An expanding navigation menu

A navigation menu includes a list of seven items with links to the main sections of a site. When a user selects one of these items, a list of sub-navigation items is inserted into the top-level navigation menu.

Consistently positioned skip navigation controls

A "skip navigation" (or "skip to main content") link is included as the first link on every page in a website. The link allows users to quickly bypass heading information and navigational content and begin interacting with the main content of a page.

Skip to navigation link

Navigational content is consistently located at the end of each page in a set of web pages. A "skip to navigation" link is consistently located at the beginning of each page so that keyboard users can easily locate it when needed.

Resources for Consistent Navigation

- Detweiler, M.C. and Omanson, R.C. (1996), Ameritech Web Page User Interface Standards and Design Guidelines.
- [IBM: User experience design - Navigation](#) .

3.2.4 Consistent Identification (Level AA)

Source: WCAG 2.2 / Understanding 3.2.4 — Copyright © 2023 World Wide Web Consortium

Components that have the same functionality within a set of web pages are identified consistently.

In brief

Goal

Actions are more predictable across pages.

What to do

Identify repeating functions consistently.

Why it's important

Consistently identified actions are especially important to people with disabilities.

Intent of Consistent Identification

The intent of this success criterion is to ensure consistent identification of functional

components that appear repeatedly within a set of web pages. A strategy that people

who use screen readers use when operating a website is to rely heavily on their familiarity

with functions that may appear on different web pages. If identical functions have

different labels (or, more generally, a different accessible name)

on different web pages, the site will be considerably more difficult

to use. It may also be confusing and increase the cognitive load for people with cognitive

limitations. Therefore, consistent labeling will help.

This consistency extends to the text alternatives. If icons or other non-text items have the same functionality, then their text alternatives should be consistent as well.

If there are two components on a web page that both have the same functionality as a component on another page in a set of web pages, then all 3 must be consistent. Hence the two on the same page will be consistent.

While it is desirable and best practice always to be consistent within a single web page, 3.2.4 Consistent Identification only addresses consistency within a set of web pages where something is repeated on more than one page in the set.

Benefits of Consistent Identification

- People who learn functionality on one page on a site can find the desired functions on other pages if they are present.
- When non-text content is used in a consistent way to identify components with the same functionality, people with difficulty reading text or detecting text alternatives can interact with the web without depending on text alternatives.
- People who depend on text alternatives can have a more predictable experience. They can also search for the component if it has a consistent label on different pages.

Examples of Consistent Identification

Example 1: Document Icon

A document icon is used to indicate document download throughout a site. The text alternative for the icon always begins with the word "Download," followed by a shortened form of the document title. Using different text alternatives to identify

document names for different documents is a consistent use of text alternatives.

Example 2: Check Mark

A check mark icon functions as "approved", on one page but as "included" on another. Since they serve different functions, they have different text alternatives.

Example 3: Consistent references to other pages

A website publishes articles on-line. Each article spans multiple web pages and each page contains a link to the first page, the next page and the previous page of the article. If the references to the next page read "page 2", "page 3", "page 4" etcetera, the labels are not the same but they are consistent. Therefore, these references are not failures of this success criterion.

Example 4: Icons with similar functions

An e-commerce application uses a printer icon that allows the user to print receipts and invoices. In one part of the application, the printer icon is labeled "Print receipt" and is used to print receipts, while in another part it is labeled "Print invoice" and is used to print invoices. The labeling is consistent ("Print x"), but the labels are different to reflect the different functions of the icons. Therefore, this example does not fail the success criterion.

Example 5: Save icon

A common "save" icon is used through out the site where page save function is provided on multiple web pages.

Example 6: Icon and adjacent link to same destination

An icon with alt text and a link are next to each other and go to the same location.

The best practice would be to group them into one link as per . However if they are visually positioned one above the other but separated in the

source, this may not be possible. To meet the Success Criterion, the link text for

these two links need only be consistent, not identical. But best practice is to have

identical text so that when users encounter the second one, it is clear that it goes

to the same place as the first.

Example 7: Example of a Failure

A submit "search" button on one web page and a "find" button on another web page both have a field to enter a term and list topics in the website related to the term submitted. In this case, the buttons have the same functionality but are not labeled consistently.

Example 8: Failure primarily impacting assistive technology users

Two buttons with the same functionality visually have the same text, but have been given different aria-label="..." accessible names. For users of assistive technologies, these two buttons will be announced differently and inconsistently.

3.2.5 Change on Request (Level AAA)

Source: [WCAG 2.2](#) / [Understanding 3.2.5](#) — Copyright © 2023 World Wide Web Consortium

Changes of context are initiated only by user request or a mechanism is available to turn off such changes.

In brief

Goal

Users have full control of major content changes.

What to do

Provide ways for users to trigger or turn off changes of context.

Why it's important

Content that behaves predictably is especially important to people with disabilities.

Intent of Change on Request

The intent of this success criterion is to encourage design of web content that gives users full control of changes of context. This success criterion aims to eliminate potential confusion that may be caused by unexpected changes of context, such as launching new windows, or automatic submission of forms after selecting an item from a list. Such unexpected changes of context may cause difficulties for people with motor impairments, people with low vision, people who are blind, and people with certain cognitive limitations.

Some types of change of context are not disruptive to some users, or actively benefit some users. For example, single-switch users rely on context changes that are animated by the system, and the preferences of low-vision users may vary depending on how much of the content they can see at once and how much of the session

structure they can retain in working memory. Some types of content, such as slide shows, require the ability to change context in order to provide the intended user experience. Content that initiates changes of context automatically only when user preferences allow can conform to this success criterion.

It is possible for more than one change of context to occur simultaneously. For example, clicking on a link which opens a new window or tab is an example of two separate changes of context related to the change in content and to the change in the viewport (window or tab). The change in the content in this case is initiated by user request when they click on the link, but *opening a new window or tab* cannot be regarded as user-initiated if it happens without the user explicitly requesting it.

Benefits of Change on Request

- Individuals who are unable to detect changes of context or who may not realize that the context has changed are less likely to become disoriented if they initiate the change themselves. For example: individuals who are blind or have low vision may have difficulty knowing when a visual context change has occurred, such as a new window popping up. In this case, warning users of context changes in advance minimizes confusion when the user discovers that the back button no longer behaves as expected.
- Some individuals with low vision, with reading and intellectual disabilities, and who have difficulty interpreting visual cues may benefit from additional cues in order to detect changes of context.
- People with certain cognitive limitations do not get confused if automatic redirects are performed by the web server instead of the browser.

Examples of Change on Request

An "update now" button

Instead of automatically updating the content, the author provides an "Update now" button that requests a refresh of the content.

An automatic redirection

Users are automatically redirected from an old page to a new page in such a way that they never realize the redirect has occurred.

Resources for Change on Request

- Use standard redirects: don't break the back button! (W3C

QA Tip).

- RFC 9110: HTTP Semantics 15.4. Redirection 3xx .

3.2.6 Consistent Help (Level A)

Source: [WCAG 2.2](#) / [Understanding 3.2.6](#) — Copyright © 2023 World Wide Web Consortium

If a web page contains any of the following help mechanisms , and those mechanisms are repeated on multiple web pages within a set of web pages , they occur in the same order relative to other page content, unless a change is initiated by the user:

- Human contact details;
- Human contact mechanism;
- Self-help option;
- A fully automated contact mechanism.

Help mechanisms may be provided directly on the page, or may be provided via a direct link to a different page containing the information.

For this success criterion, "the same order relative to other page content" can be thought of as how the content is ordered when the page is serialized. The visual position of a help mechanism is likely to be consistent across pages for the same page variation (e.g., CSS breakpoint). The user can initiate a change, such as changing the page's zoom or orientation, which may trigger a different page variation. This criterion is concerned with relative order across pages displayed in the same page variation (e.g., same zoom level and orientation).

In brief

Goal

Make it easier to find help and support.

What to do

Put help in the same place when it is on multiple pages.

Why it's important

People who need help can find it more easily if it's in the same place.

Intent of Consistent Help

The intent of this success criterion is to ensure users can find help for completing tasks on a website, when it is available. When the placement of the help mechanism is kept consistent across a set of pages, users looking for help will find it easier to identify. This is distinct from interface-level help, such as contextual help, features like spell checkers, and instructional text in a form.

Locating the help mechanism in a consistent location across pages makes it easier for users to find it. For example, when a mechanism or link is located in the header of one web page, it will be easier to find if it is in the header of other pages. The help mechanism, such as a contact phone number, may be provided directly on the page, or it may also be a direct link to a contact page. Regardless of which approach is used, the mechanism must be located in the same relative order on each page within the set of pages.

When testing this Success Criterion, it is the help item which is relative to the rest of the content. When testing a page, other content that is present across the set of web pages and is before the help item should be before the help item on this page. Items which are after the help item on other pages should be after the help item on this page.

If the help item is visually in a different location, but in the same serial order, that is not helpful from a user's point of view, but it would not fail this criterion.

When having problems completing a task on a website (or part of a website, what we call a set of web pages), people with some types

of disabilities may not be able to work through the issue without further help. Issues could include difficulty:

completing a form, or finding a document or page which provides information required to complete a task.

Without help, some users may abandon the task. They may also fail to correctly complete a task, or they may require assistance from people who do not necessarily keep private information secure.

While it is recommended to consistently implement a help mechanism across a set of web pages, this criterion specifically pertains to pages that do include such a mechanism. Therefore, the absence of a help mechanism on certain pages within a set does not constitute a violation.

Limitations and Exceptions

It is not the intent of this success criterion to require authors to provide help or access to help. The Criterion only requires that *when* one of the listed forms of help is available across multiple pages that it be in a consistent location. It does not require authors to provide help information on PDFs or other static documents that may be available for viewing/download from the web pages. PDFs and other static documents are not considered part of the "set of web pages" from which they are downloaded.

It is also not the intent of this success criterion to require a human be available at all times. Ideally, if the human contact is not available during certain hours or certain days then information would be provided so the user can tell when it will be available.

This success criterion only requires help mechanisms to be consistent *within* a particular set of web pages. Some complex websites consist of multiple different sets of web pages with

different purposes. For example, a web-based spreadsheet application might have one set of pages for editing spreadsheets and a separate set of pages for marketing the application. This success criterion would allow the different sets of web pages to use different help mechanism locations. However, it is best if help mechanisms are located as consistently as possible even among different related sets of web pages.

This success criterion contains an exception when "a change is initiated by the user." This exception is intended to cover cases where a user performs an action with the intent of changing the display or layout of a page, such as changing the zoom level, orientation, or viewport size. Help mechanism locations may change in response to such a user-initiated change; as the criterion's second note clarifies, "this criterion is concerned with relative order across pages displayed in the same page variation (e.g., same zoom level and orientation)."

This exception allows the location in a smaller viewport to be different than in a larger viewport. However, it is best if the mechanism or link is consistent across a set of web pages. A consistent location, both visually and programmatically, is the most usable.

This exception is *not* intended to treat every action that a user might initiate as a "change"; to qualify for the exception, the user must be initiating an action that would reasonably be expected to change the relative order of components within a page. For example, merely navigating between pages within a set of web pages is not a "change initiated by the user" for the purposes of this exception. Similarly, logging into or out of a page would not typically qualify, unless logging in would present the user with a distinct set of web pages .

Help Mechanisms

Typical help mechanisms include:

- Human contact details such as a phone number, email address, hours of operation.
- Human contact mechanism such as a messaging system, chat client, contact form, social media channel.
- Self-help option such as an up-to-date Frequently Asked Questions, How Do I page, Support page.
- A fully automated contact mechanism such as a chatbot.

The order of the types of help listed in the success criterion does not imply priority.

Support for people with cognitive and learning disabilities

This section is not required by the Consistent Help success criterion, but provides advice related to Making Content Usable for People with Cognitive and Learning Disabilities .

The human contact details enable users to connect with the organization or the part of the organization that can assist with the content. For example, an online jobs / recruitment portal may provide a contact method for the team that supports the recruitment portal and not a catch-all for the entire company. Each layer of contact added prolongs the time before the user will receive help.

The human contact mechanism enables a person to express what they are looking for using their own words. For some with cognitive disabilities, this may be the best way for them to find an answer to their problem.

For pages for which no human support is available it helps if a self-help option says that no human support is available. Self-help options can go beyond allowing the user to search within the site.

Contextual help is still recommended (see [Success Criterion 3.3.5 Help](#) for more information), but a self-help option provides a single location that makes it easier for people with cognitive disabilities to understand what help is available without having to hunt for it. While some people may easily be able to identify that no support would be available for a particular type of website, this may not be apparent to some users with disabilities.

Chatbots can work for many people, and particularly for people with cognitive disabilities if they:

- recognize misspelled words,
- provide human contact details if the chatbot is unable to provide a satisfactory response after 3 attempts, and
- can be dismissed with a single interaction, and recalled using a link or button.

This criterion does not require that a site provide a help mechanism. However, when help is available:

- People who may have difficulty locating help are more likely to find it and complete their task.
- Users that experience cognitive fatigue or cognitive shut down will be able to reserve their energy for the task, instead of using it to find support.
- Enabling users (especially those with cognitive disabilities) to find solutions while expressing their question using their own words (for example by interacting with a chatbot) increases their chances of success for completing a task.

Self help methods beyond the site, such as using internet search to find the contact information for an organization, can be too difficult. Further, the user's disability may make it more difficult to find the help available (such as a "contact us" link, phone number,

or support page) if the information is not consistently present within a few interactions (e.g., displayed in the header, or via a menu). In addition, for some users with disabilities, struggling to complete a task on a site may cause additional cognitive challenges when searching for help within the site.

When a user is quickly able to find help, they are able to complete the task even if they encounter challenges.

Benefits of Consistent Help

- People who may have difficulty locating help are more likely to find it when it is consistently located.

Examples of Consistent Help

- On-line job application: Some of the application questions may be hard for new job seekers to understand even after reading the contextual help. For example, the form may request their identification number, but they may have several and not know which one to enter. Consistently located contact information will enable them to use phone or email so they can get an answer to their question.
- Medical appointment scheduling form: When the service a patient is trying to book is not easily findable within the interface, they may need human help. A consistently located messaging option (chat client) enables them to quickly interact with a staff person that can help, without requiring them to manage a second interface.
- Finding a specific policy or procedure: An employee who needs to complete a work task may have difficulty locating the specific policy or procedure document on their employer's website. A consistently located "How Do I" page may include the information that enables them to independently complete this task.

Resources

- Cognitive Accessibility Gap Analysis Topic 6: Familiar Interface
- Making Content Usable for People with Cognitive and Learning Disabilities 4.8.5 Make it Easy to Find Help and Give Feedback

Guideline 3.3: Input Assistance

[Source: W3C WCAG 2.2 —

<https://www.w3.org/TR/WCAG22/#guideline33>]

Help users avoid and correct mistakes.

3.3.1 Error Identification (Level A)

Source: WCAG 2.2 / Understanding 3.3.1 — Copyright © 2023 World Wide Web Consortium

If an input error is automatically detected, the item that is in error is identified and the error is described to the user in text.

In brief

Goal

Users know an error exists and what is wrong.

What to do

Provide descriptive notification of errors.

Why it's important

Flagging errors helps people with reduced sight and cognitive disabilities resolve them.

Intent of Error Identification

The intent of this success criterion is to ensure that users are aware that an error

has occurred and can determine what is wrong. In the case of an unsuccessful form submission,

it is not sufficient to only re-display the form without providing any hint that the submission failed.

The error must be indicated in text .

This SC requires that users be provided with information about the nature of the error, including the identity of the item in error. What the user should do to correct the item in error is covered by

[3.3.3 Error Suggestion](#) . Often, the error description can be phrased so that it meets both Success Criteria 3.3.1 Error Identification and 3.3.3 Error Suggestion at the same time. For instance, "Email is not valid" would pass 3.3.1, but "Please provide a valid email address in the format name@domain.com" also conveys how it can be fixed and passes both.

An "input error" includes:

- information that is required by the web page but omitted by the user, or
- information that is provided by the user but that falls outside the required data format or allowed values.

For example:

- the user fails to enter the proper abbreviation in a state, province, or region field;
- the user enters a state abbreviation that is not a valid state;
- the user enters a non existent zip or postal code;
- the user enters a birth date 2 years in the future;
- the user enters alphabetic characters or parentheses into their phone number field that only accepts numbers;
- the user enters a bid that is below the previous bid or the minimum bid increment.

If a user enters a value that is too high or too low, and the coding on the page automatically

changes that value to fall within the allowed range, the user's error would still

need to be described to them as required by the success criterion. Such an error description

telling the person of the changed value would meet both this success criterion (Error

Identification) and [3.3.3 Error Suggestion](#) .

The identification and description of an error can be combined with programmatic information

that user agents or assistive technologies can use to identify an error and provide

error information to the user. For example, certain technologies can specify that

the user's input must not fall outside a specific range, or that a form field is required.

This type of programmatic information is not required for this success criterion, but may be covered

by other criteria such as [4.1.2 Name, Role, Value](#) .

It is perfectly acceptable to indicate the error in other ways such as through the use of an image, color, or other visual indicator, in addition to the text description.

This criterion does not mandate any particular way in which errors should be displayed. Depending on the situation, it may be more suitable for all errors to be listed at the start or before a form. In other cases, it may be more appropriate to show errors inline, with error messages next to the specific fields that are in error. Errors could also be listed in an alert, or dialog. This criterion does not cover which of these methods should be used - the only requirement is for errors to be presented to users in text or a text alternative.

See also [3.3.3 Error Suggestion](#) .

User agent native HTML form validation

When using native HTML client-side form validation ,

user agents will automatically prevent the submission of incomplete or invalid forms, and display generic error messages to the user.

The user agent will generally set focus back to the first form field that is in error, and as a result scroll the page

so that the field in error and the generated error message will be visible in the viewport.

In most common user agent and screen reader combinations, the screen reader will announce the error message and the programmatic name of the focused field. While this meets the requirements of this success criterion, it should be noted that there are several disadvantages related to this approach:

- Depending on the user agent, the message may not be permanent, or fail to scroll with the page.
- Depending on the user agent, even if a user has zoomed-in (magnified) the content, the error messages will not appear magnified, as the text in the validation message will be displayed at the same size as the user agent interface; the message may be too small for users to read.
- The default HTML validation error messages are generally quite generic, and they may not provide sufficiently helpful or specific suggestions to the user

that would conform to [3.3.3 Error Suggestion](#) .

- If several errors are present, only the first error message is exposed; once the user has provided an input that conforms to the type of field, and resubmits the form, the next error (if

present) will be exposed. This means that repeated resubmissions and corrections may be required.

As these problems relate to user agent behavior, developers will need to carefully consider if native browser validation is accessibility supported .

Benefits of Error Identification

- Providing information about input errors in text allows users who are blind, have low vision, or have color vision deficiency to perceive the fact that an error occurred.
- This success criterion may help people with cognitive, language, and learning disabilities who have difficulty understanding the specific reason why a form submission failed (in cases where this is not already made obvious by the nature of the form).

Examples of Error Identification

Identifying errors in a form submission

An airline website offers a special promotion on discounted flights. The user is asked to complete a simple form that asks for personal information such as name, address, phone number, seating preference and email address. If any of the fields of the form are either not completed or completed incorrectly, an alert is displayed notifying the user which field or fields were missing or incorrect. This success criterion does not mean that color or text styles cannot be used to indicate errors. It simply requires that errors also be identified using text.

Providing multiple cues

The user fails to fill in two fields on the form. In addition to describing the error and providing a unique character to make it

easy to search for the fields, the fields are highlighted in yellow to make it easier to visually search for them as well.

This success criterion does not mean that color or text styles cannot be used to indicate errors. It simply requires that errors also be identified using text.

3.3.2 Labels or Instructions (Level A)

Source: [WCAG 2.2](#) / [Understanding 3.3.2](#) — Copyright © 2023 World Wide Web Consortium

Labels or instructions are provided when content requires user input.

In brief

Goal

Users know what information to enter.

What to do

Provide labels or instructions for inputs.

Why it's important

Everyone, especially those with cognitive disabilities, will know how to respond.

Intent of Labels or Instructions

The intent of this success criterion is to have content authors present instructions or labels that identify the controls in a form so that users know what input data is expected. In the case of radio buttons, checkboxes, comboboxes, or similar controls that provide users with options, each option must have an appropriate label so that users know what they are actually selecting. Instructions or labels may also specify data formats for data entry fields, especially if they are out of the customary formats or if there are specific rules for correct input. Content authors may also choose to make such instructions available to users only when the individual control has focus especially when instructions are long and verbose.

The intent of this success criterion is not to clutter the page with unnecessary information but to provide important cues and

instructions that will benefit people with disabilities. Too much information or instruction can be just as harmful as too little. The goal is to make certain that enough information is provided for the user to accomplish the task without undue confusion or navigation.

Note that the majority of form control labels are text-based. Using images as labels meets the requirements of the criterion, but care should be taken to ensure that the images are widely understood by the intended target audience. Authors may consider providing additional hints, such as text-based tooltips or supplementary text, to support clarity when using image-based labels.

This success criterion does not require that labels or instructions be correctly marked up,

identified, or associated with their respective controls — that aspect is covered separately by [1.3.1 Info and Relationships](#). It is possible for content

to pass this success criterion (providing relevant labels and instructions) while failing

Success Criterion 1.3.1 (if the labels or instructions aren't correctly marked up, identified, or associated).

Further, this success criterion does not take into consideration whether or not alternative methods of

providing an accessible name or description for form controls and inputs has been used — that aspect is

covered separately by [4.1.2 Name, Role, Value](#). It is possible

for controls and inputs to have an appropriate accessible name or description (e.g. using `aria-label="..."`)

and therefore pass Success Criterion 4.1.2, but to still fail this success criterion (if the labels or instructions

aren't presented to all users, not just those using assistive technologies).

This success criterion does not apply to links or other controls (such as an expand/collapse widget, or similar interactive components) that are not associated with data entry.

While this success criterion requires that controls and inputs have labels or instructions, whether or

not labels (if used) are accurate, sufficiently clear, or descriptive is covered separately by [2.4.6 Headings and Labels](#) .

The use of "requires" in this criterion's normative wording does not mean that the criterion only applies to *required* form fields. It is used here as a synonym for "accepts", "expects", or "allows". The criterion applies to all form fields, whether they're required or optional.

Benefits of Labels or Instructions

- Providing labels and instructions (including examples of expected data formats) helps all users — but particularly those with cognitive, language, and learning disabilities — to enter information correctly.
- Providing labels and instructions (including identification of required fields) can prevent users from making incomplete or incorrect form submissions, which prevents users from having to navigate once more through a page/form in order to fix submission errors.

Examples of Labels or Instructions

- A field which asks the user to enter the two character abbreviation for a US state has a link next to it which will pop up an alphabetized list of state names and the correct abbreviation.
- A field for entering a date has text instructions to indicate the correct format for the date.
- On one website, a field with the text label of "username" is provided for someone to create a username to login to a website. On another website, there are strict rules about what characters can be used to create a username. On this website additional instructions would need to accompany the field to prevent users from encountering unnecessary errors.
- A website provides a global search field in the header of the site. Any term can be entered, so there are no instructions needed, but the field needs a cue to communicate its purpose. Commonly, such search field will be paired with a "loupe" or "magnify glass" search icon, serving as its visible label, if not also doubling as the visual identifier for the button that submits the search query.
- To enter their name, users are provided with two separate text fields. Rather than having a single label "Name" (which would appear to leave the second text field unlabelled), each field is given an explicit label — "Given Name" and "Family Name".
- A U.S. phone number separates the area code, exchange, and number into three fields. Parentheses surround the area code field, and a dash separates the exchange and number fields. While the punctuation provides visual clues to those familiar with the U.S. telephone number format, the punctuation is not sufficient to label the fields. The single "Phone number" label also cannot label all three fields. To address this, the three fields are grouped in a fieldset with the legend "Phone

number". Visual labels for the fields (beyond the punctuation) cannot be provided in the design, so invisible labels are provided with the "title" attribute to each of the three fields. The value of this attribute for the three fields are, respectively, "Area Code", "Exchange", and "Number".

3.3.3 Error Suggestion (Level AA)

Source: WCAG 2.2 / Understanding 3.3.3 — Copyright © 2023 World Wide Web Consortium

If an input error is automatically detected and suggestions for correction are known, then the suggestions

are provided to the user, unless it would jeopardize the security or purpose of the

content.

In brief

Goal

Users get suggestions on how to resolve errors.

What to do

Where errors are detected, suggest known ways to correct them.

Why it's important

People can address errors faster and with reduced effort.

Intent of Error Suggestion

The intent of this success criterion is to ensure that users receive appropriate suggestions for correction of an input error if it is possible. The definition of "input error" says that it is "information provided by the user that is not accepted" by the system. Some examples of information that is not accepted include information that is required but omitted by the user and information that is provided by the user but that falls outside the required data format or allowed values.

Success Criterion 3.3.1 Error Identification provides for notification of errors. However, persons with cognitive limitations may find it difficult to understand how to correct the errors.

People with visual disabilities may not be able to figure out exactly how to correct the error. In the case of an unsuccessful form submission, users may abandon the form because they may be unsure of how to correct the error even though they are aware that it has occurred.

The content author may provide the description of the error, or the user agent may provide the description of the error based on technology-specific, programmatically determined information.

Benefits of Error Suggestion

- Providing information about how to correct input errors allows users who have learning disabilities to fill in a form successfully.
- Users who are blind or have impaired vision understand more easily the nature of the input error and how to correct it.
- People with motion impairment can reduce the number of times they need to change an input value.

Examples of Error Suggestion

Additional Help for Correcting An Input Error

The result of a form that was not successfully submitted describes an input error in place in the page along with the correct input and offers additional help for the form field that caused the input error.

Suggestions from a Limited Set of Values

An input field requires that a month name be entered. If the user enters "12," suggestions for correction may include: A list of the acceptable values, e.g., "Choose one of: January, February, March, April, May, June, July, August, September, October, November, December." The conversion of the input data interpreted as a different month format, e.g., "Do you mean 'December'?"

3.3.4 Error Prevention (Legal, Financial, Data) (Level AA)

Source: [WCAG 2.2](#) / [Understanding 3.3.4](#) — Copyright © 2023 World Wide Web Consortium

For web pages that cause legal commitments or financial transactions for the user to occur, that modify or delete user-controllable data in data storage systems, or that submit user test responses, at least one of

the following is true:

Reversible

Submissions are reversible.

Checked

Data entered by the user is checked for input errors and the user is provided an opportunity to correct them.

Confirmed

A mechanism is available for reviewing, confirming, and correcting information before finalizing the submission.

In brief

Goal

Users can avoid submitting incorrect important information.

What to do

Provide ways for users to confirm, correct, or reverse important submissions.

Why it's important

People with disabilities may be more likely to make mistakes, or not notice them.

Intent of Error Prevention (Legal, Financial, Data)

The intent of this success criterion is to help users with disabilities avoid serious consequences as the result of a mistake when performing an action that cannot be reversed. For example, purchasing non-refundable airline tickets or submitting an order to purchase stock in a brokerage account are financial transactions with serious consequences. If users have made a mistake on the date of air travel, they could end up with a ticket for the wrong day that cannot be exchanged. If users made a mistake on the number of stock shares to be purchased, they could end up purchasing more stock than intended. Both of these types of mistakes involve transactions that take place immediately and cannot be altered afterwards, and can be very costly. Likewise, it may be an unrecoverable error if users unintentionally modify or delete data stored in a database that they later need to access, such as their entire travel profile in a travel services website. When referring to modification or deletion of 'user controllable' data, the intent is to prevent mass loss of data such as deleting a file or record. It is not the intent to require a confirmation for each save command or the simple creation or editing of documents, records or other data.

Users with disabilities may be more likely to make mistakes. People with reading disabilities may transpose numbers and letters, and those with motor disabilities may hit keys by mistake. Providing the ability to reverse actions allows users to correct a mistake that could result in serious consequences. Providing the ability to review and correct information gives the user an opportunity to detect a mistake before taking an action that has serious consequences.

User-controllable data is user-viewable data that the user can change and/or delete through an intentional action. Examples of the user controlling such data would be updating the phone

number and address for the user's account, or deleting a record of past invoices from a website. It does not refer such things as internet logs and search engine monitoring data that the user can't view or interact with directly.

Benefits of Error Prevention (Legal, Financial, Data)

- Providing safeguards to avoid serious consequences resulting from mistakes helps users with all disabilities who may be more likely to make mistakes.

Examples of Error Prevention (Legal, Financial, Data)

Order confirmation

A web retailer offers on-line shopping for customers. When an order is submitted, the order information—including items ordered, quantity of each ordered item, shipping address, and payment method—are displayed so that the user can inspect the order for correctness. The user can either confirm the order or make changes.

Stock sale

A financial services website lets users buy and sell stock online. When a user submits an order to buy or sell stock, the system checks to see whether or not the market is open. If it is after hours, the user is alerted that the transaction will be an after-hours transaction, is told about the risks of trading outside of regular market hours, and given the opportunity to cancel or confirm the order.

3.3.5 Help (Level AAA)

Source: [WCAG 2.2](#) / [Understanding 3.3.5](#) — Copyright © 2023 World Wide Web Consortium

Context-sensitive help is available.

In brief

Goal

Users can avoid making mistakes.

What to do

Provide help to users on the function currently being performed.

Why it's important

People with cognitive or other disabilities can complete their tasks more easily.

Intent of Help

The intent of this success criterion is to help users avoid making mistakes. Some users with disabilities may be more likely to make mistakes than users without disabilities. Using context-sensitive help, users find out how to perform an operation without losing track of what they are doing.

Context-sensitive help only needs to be provided when the label is not sufficient to describe all functionality. The existence of context-sensitive help should be obvious to the user and they should be able to obtain it whenever they require it.

The content author may provide the help text, or the user agent may provide the help text based on technology-specific, programmatically determined information.

Benefits of Help

- Assistance for text input helps individuals with writing disabilities and people with reading and intellectual disabilities who often have difficulty writing text in forms or other places that need text input.
- Additionally, these kinds of assistance help people who are aging and have the same difficulty in text input and/or mouse operation.

Examples of Help

on-line job application

Some of the questions may be hard for new job seekers to understand. A help link next to each question provides instructions and explanations for each question.

3.3.6 Error Prevention (All) (Level AAA)

Source: WCAG 2.2 / Understanding 3.3.6 — Copyright © 2023 World Wide Web Consortium

For web pages that require the user to submit information, at least one of the following is true:

Reversible

Submissions are reversible.

Checked

Data entered by the user is checked for input errors and the user is provided an opportunity to correct them.

Confirmed

A mechanism is available for reviewing, confirming, and correcting information before finalizing the submission.

In brief

Goal

Users can avoid submitting incorrect information.

What to do

Provide ways for users to confirm, correct, or reverse any submissions.

Why it's important

People with disabilities may be more likely to make mistakes, or not notice them.

Intent of Error Prevention (All)

The intent of this success criterion is to help users with disabilities avoid consequences

that may result from making a mistake when submitting form data. This criterion builds

on [Success Criterion 3.3.4 Error Prevention \(Legal, Financial, Data\)](#) in that it applies to all forms that require users to submit information.

Users with disabilities may be more likely to make mistakes and may have more difficulty detecting or recovering from mistakes. People with reading disabilities may transpose numbers and letters, and those with motor disabilities may hit keys by mistake. Providing the ability to reverse actions allows users to correct a mistake. Providing the ability to review and correct information gives the user an opportunity to detect a mistake before taking an action.

Benefits of Error Prevention (All)

- Providing safeguards to avoid consequences resulting from mistakes helps users with all disabilities who may be more likely to make mistakes.

3.3.7 Redundant Entry (Level A)

Source: [WCAG 2.2](#) / [Understanding 3.3.7](#) — Copyright © 2023 World Wide Web Consortium

Information previously entered by or provided to the user that is required to be entered again in the same process is either:

- auto-populated, or
- available for the user to select.

Except when:

- re-entering the information is essential ,
- the information is required to ensure the security of the content, or
- previously entered information is no longer valid.

In brief

Goal

Make it easier for users to complete multi-step processes.

What to do

Don't ask for the same information twice in the same activity.

Why it's important

Some people with cognitive disabilities have difficulty remembering what they entered before.

Intent of Redundant Entry

The intent of this success criterion is to ensure that users can successfully complete multi-step processes. It reduces cognitive effort where information is asked for more than once during a process. It also reduces the need to recall information provided in a previous step.

Information that is required to be remembered for input can pose a significant barrier to users with cognitive or memory difficulties. All users experience a natural gradual mental fatigue as they proceed through steps in a process. This fatigue is accelerated by the stress of recalling information from short-term working memory. Users with learning, and cognitive disabilities are highly susceptible to mental fatigue.

Requiring people to recall information previously entered can cause them to give up or re-enter the same information incorrectly. The autocomplete feature of browsers is not considered sufficient because it is the content (the website) that needs to provide the stored information for a redundant entry, or avoid asking for the same information again.

This success criterion does not add a requirement to store information between sessions. A process is defined on the basis of an activity and is not applicable when a user returns after closing a session or navigating away. However, a process can run across different domains, so if a check-out process includes a 3rd party payment provider, that would be in scope.

The term "available to select" is not prescriptive. The term allows authors to develop techniques where auto-population is not possible. It can include allowing the user to:

- select and populate a field, including from a drop-down;
- select text from the page and copy it into an input;
- tick a checkbox to populate inputs with the same values as previously entered (e.g., my billing address is the same as my shipping address).

Data which is "available to select" would need to be on the same page. Ideally, it would be visible by default and closely associated

with the input where the data is required. However, it could be elsewhere on a page, including within a show/hide component.

This success criterion does not apply if data is provided by the user with a different method, such as uploading a resume in a document format.

This success criterion does not impact [Accessible Authentication \(Minimum\)](#) , for which allowing auto-filling of passwords is a sufficient technique. In that case the filling is performed by the user's browser. Redundant Entry is asking for the website content to make the previous entry available, but not between sessions or for essential purposes such as asking for a password.

This criterion does not include requirements or exceptions specific to privacy or personally identifiable information (PII), but when implementing techniques such as auto-population, authors should ensure data protection when storing information even temporarily during a process. It is possible to eliminate redundant entry in ways that do not introduce additional privacy risks, but it is also possible that a poor implementation (for meeting this criterion) could leak additional PII.

Exceptions

There are exceptions for:

- Essential uses of input re-entry for things like memory games which would be invalidated if the previous answers were supplied.
- Security measures such as preventing a password string from being shown or copied. When creating a password, it should be a unique and complex string and therefore cannot be validated by the author. If the system requires the user to manually create a password that is not displayed, having users re-validate their new string is allowed as an exception.

- When the previously entered information is no longer valid, it can be requested that the user enter that information again.

Benefits of Redundant Entry

- Users with cognitive disabilities experience short-term, working memory difficulty. Not having to repeatedly remember particular information reduces stress and the likelihood of mistakes.
- Users who experience difficulty forming new memories, recalling information, and other functions related to cognition can complete processes without having to unnecessarily rely on their memory.
- Users with mobility impairments, for example using switch control or voice input, benefit from a reduced need for text entry.

Examples of Redundant Entry

- A form requests the user's corporate identification number (ID) in the first step of a process to purchase a new computer. In the 3rd step the user is asked to confirm that the computer will belong to the user (rather than a colleague), and re-shows the ID. It allows the user to change the ID, but defaults to the previously entered one.
- A form on an e-commerce website allows the user to confirm that the billing address and delivery address are the same address.
- A user submits a checkout form with an incorrect credit card number in it. The page updates, showing an error message. Submitted information, such as credit card number, is not cleared from the form.
- A search results page pre-fills the search input with the previously entered search term in the same process.

Resources

- Cognitive Accessibility Gap Analysis Topic 3: Entering Data, Error Prevention, & Recovery
- Making Content Usable for People with Cognitive and Learning Disabilities 4.5.4 Design Forms to Prevent Mistakes

3.3.8 Accessible Authentication (Minimum) (Level AA)

Source: [WCAG 2.2](#) / [Understanding 3.3.8](#) — Copyright © 2023 World Wide Web Consortium

A cognitive function test (such as remembering a password or solving a puzzle) is not required for any step in an authentication process unless that step provides at least one of the following:

Alternative

Another authentication method that does not rely on a cognitive function test.

Mechanism

A mechanism is available to assist the user in completing the cognitive function test.

Object Recognition

The cognitive function test is to recognize objects.

Personal Content

The cognitive function test is to identify non-text content the user provided to the website.

"Object recognition" and "Personal content" may be represented by images, video, or audio.

Examples of mechanisms that satisfy this criterion include: support for password entry by password managers to reduce memory need, and copy and paste to reduce the cognitive burden of re-typing.

In brief

Goal

Make logins possible with less mental effort.

What to do

Don't make people solve, recall, or transcribe something to log in.

Why it's important

Some people with cognitive disabilities cannot solve puzzles, memorize a username and password, or retype one-time passcodes.

Intent of Accessible Authentication (Minimum)

The purpose of this success criterion is to ensure there is an accessible, easy-to-use, and secure method for users to authenticate when logging into an existing account. As the most prevalent form of authentication, websites commonly rely on usernames and passwords to log in. However, memorizing a username and password places a very high or impossible burden upon people with certain cognitive disabilities, as do additional steps often added to authentication processes. For instance, the need to transcribe a one-time verification code or requiring a puzzle to be solved.

While websites can use the recognition of objects or of non-text content provided by the user to meet this Success Criterion, such techniques do not fully support the cognitive accessibility community and should be avoided if possible. Refer to [Accessible Authentication \(Enhanced\)](#) for guidance to be more inclusive and accessible.

This success criterion is focused on authentication of existing users. It does *not* cover creation of a username or initiation of an account. For many websites, establishing an initial username and credentials may not differ greatly from logging in with that username. The techniques used to satisfy this criterion (particularly allowing pasting into inputs and not relying on transcription) can also reduce the cognitive burden in account creation. However, the focus of the success criterion is on reducing

the ongoing need for users to recall previously supplied information each time they log in or otherwise authenticate to an account.

Cognitive Function Tests

Remembering a site-specific password is a cognitive function test . Such tests are known to be problematic for many people with cognitive disabilities. Whether it is remembering random strings of characters, or a pattern gesture to perform on a touch screen, cognitive function tests will exclude some people. When a cognitive function test is used, at least one other authentication method must be available which is not a cognitive function test.

Some CAPTCHA systems have an audio alternative of the visible text. If the user needs to transcribe this audio, it cannot be used to meet the Alternative exception.

If there is more than one step in the authentication process, such as with multi-factor authentication, all steps need to comply with this success criterion to pass. There needs to be a path through authentication that does not rely on cognitive function tests.

Being able to recover or change the email and password is an important part of authentication. If the user is authenticating with alternative information in order to recover their account, there needs to be a method that is not a cognitive function test.

Many organizations are required to use 2-factor authentication that combines independent sources to confirm a user's identity. These sources can consist of combining authentication through:

- knowledge (e.g., password, letters in a passphrase or memorized swipe path);
- possession (e.g., a verification code generated or received on a device, or scanning of a QR code on an external device);

- biometrics (e.g., fingerprint scanning, facial recognition or keystroke dynamics).

Most knowledge-based authentication methods rely on a cognitive function test, so mechanisms to assist users must be available. When authentication relies on performing an action on a separate device, it should be possible to complete the action without the need to transcribe information. It may not be possible to know what device-based authentication methods are available to a user; offering a choice of methods can allow them to choose the path that most suits them.

Authentication Approaches

Websites can employ username (or email) and password inputs as an authentication method if the author enables the user agent (browsers and third-party password managers) to fill in the fields automatically. Generally, if the login form meets [Success Criterion 1.3.5 Input Purpose](#), and the form controls have an appropriate accessible name in accordance with [Success Criterion 4.1.2 Name, Role, Value](#), the user agent should be able to reliably recognize the fields and automatically fill them in. However, if the user agent is actively blocked from filling in the fields (for instance, by a script), then the page would not pass this criterion because it prevents the mechanism from working.

Copy and paste

Copy and paste can be relied on to avoid transcription. Users can copy their login credentials from a local source (such as a standalone third-party password manager) and paste it into the username and password fields on a login form, or into a web-based command line interfaces asking for a password. Blocking people from pasting into authentication fields, or using a different format between the copied text and the input field (for example,

"Enter the 3rd, 4th, and 6th character of your password"), would force the user to transcribe information and therefore fail this criterion, unless another method is available.

Two-factor authentication systems (verification codes)

Beyond usernames and passwords, some sites may use two-factor authentication, asking the user to enter a verification code (also called a passcode or one-time password). A service that requires *manual* transcription of a verification code is not compliant. As with usernames and passwords, it must be possible for a user to at least paste the code (such as from a standalone third-party password manager, text message application, or software-based security key), or to allow user agents to fill in the fields automatically.

There are scenarios where a verification code must be received or generated on a secondary device. For example, authenticating in a web browser on a laptop requires a verification code that is sent as an SMS text message to a mobile phone. However, in most cases, it is possible for the code to then be sent directly to the primary device, where it can then be copied and pasted (for example, by copying the code on the secondary device and emailing it to the primary device, or through the use of a shared cross-device clipboard where copying content on the secondary device makes it available to paste on the primary device). Evaluating whether or not the code can be seamlessly transferred from the secondary device to the primary device is *outside of the scope* for this success criterion. For the purpose of evaluating web content that relies on authentication using these types of secondary device systems, it is assumed that provisions are in place that make the code available in the user's clipboard. Evaluating this criterion therefore only requires verification that the web content does allow pasting the clipboard content in the related authentication challenge field.

Note that two-factor systems that do not rely on codes — including hardware authentication devices (such as YubiKey), secondary applications (either on the same primary device, or on a secondary device) that expect the user to confirm that it is indeed them trying to log in, and authentication methods provided by the user's operating system (such as Windows Hello, or Touch ID/Face ID on macOS and iOS) — are *not* a cognitive function test .

Object Recognition

If a CAPTCHA is used as part of an authentication process, there must be a method that does not include a cognitive function test, unless it meets the exception. If the test is based on something the website has set such as remembering or transcribing a word, or recognizing a picture the website provided, that would be a cognitive functional test. Recognizing objects, or a picture the user has provided is a cognitive function test; however, it is excepted at the AA level.

An object in this context means the general English definition ("a material thing that can be seen and touched") and can include vehicles and animals. If the test goes beyond recognition (e.g. multiply the number cats by the number of dogs), that does not meet the exception.

Some forms of object recognition may require an understanding of a particular culture. For example, taxis can appear differently in different locales. This is an issue for many people, including people with disabilities, but it is not considered an accessibility-specific issue.

Some CAPTCHAs and cognitive function tests used for authentication may only appear in certain situations, such as when ad blockers are present, or after repeated incorrect password entry. This criterion applies when these tests are used regardless

of whether they are used every time or only triggered by specific scenarios.

There are a number of technologies that can be employed to prevent scripted abuse of the authentication process.

- 1.1.1. Rate-limited Access
- 1.1.2. Client Geo-Location
- 1.1.3. Private Client Authentication

None of these systems are 100% effective. However, they may reduce the likelihood of a CAPTCHA being displayed.

Personal Content

Personal content is sometimes used as a second factor for authentication. For example, as part of account creation the user would upload a picture, and when logging in they would be asked to select that picture from several possible alternatives. Care must be taken to provide adequate security in this case, since non-legitimate users might be able to guess the correct personal content when presented with a choice.

Text-based personal content does not qualify for this exception as it relies on recall (rather than recognition), and transcription (rather than selecting an item). Whilst picture-based personal content will still be a barrier for some people, text based versions tend to be a much larger barrier.

Hiding characters

Another factor that can contribute to cognitive load is hiding characters when typing. Although this criterion requires that users do not have to type in (transcribe) a password, there are scenarios where that is necessary such as creating a password to be saved by a password manager. Providing a feature to optionally show a

password can improve the chance of success for some people with cognitive disabilities or those who have difficulties with accurately typing.

Benefits of Accessible Authentication (Minimum)

People with cognitive issues relating to memory, reading (for example, dyslexia), numbers (for example, dyscalculia), or perception-processing limitations will be able to authenticate irrespective of the level of their cognitive abilities.

Examples of Accessible Authentication (Minimum)

The examples of this success criterion are the same as the [Accessible Authentication \(Enhanced\)](#) examples.

- A website uses a properly marked up username (or email) and password fields as the login authentication (meeting [Success Criterion 1.3.5 Input Purpose](#) and [Success Criterion 4.1.2 Name, Role, Value](#)). The user's browser or integrated third-party password manager extension can identify the purpose of the inputs and automatically fill in the username and password.
- A website does not block paste functionality. The user is able to use a third-party password manager to store credentials, copy them, and paste them directly into a login form.
- A website uses WebAuthn so the user can authenticate with their device instead of username/password. The user's device could use any available modality. Common methods on laptops and phones are facial-scan, fingerprint, and PIN (Personal Identification Number). The website is not enforcing any particular use; it is assumed a user will set up a method that suits them.
- A website offers the ability to login with a third-party provider using the OAuth method.

- A website that requires two-factor authentication allows for multiple options for the 2nd factor, including a USB-based method where the user simply presses a button to enter a time-based token.
- A website that requires two-factor authentication displays a QR code which can be scanned by an app on a user's device to confirm identity.
- A website that requires two-factor authentication sends a notification to a user's device. The user must use their device's authentication mechanism (for example, user-defined PIN, fingerprint, facial recognition) to confirm identity.

Resources

- [Cognitive Accessibility Gap Analysis Topic 1: Authentication and Safety](#)
- [Cognitive Accessibility Issue Papers 4. Web Security and Privacy Technologies](#) and [Web Security and Privacy Technologies](#)
- [Making Content Usable for People with Cognitive and Learning Disabilities 4.7.1 Provide a Login that Does Not Rely on Memory or Other Cognitive Skills](#)
- [WebAuthN specification](#)
- [WebAuthN Demo site](#)
- [Web Authentication API on MDN](#)
- [OAuth on Wikipedia](#)
- ["Let them paste passwords", from the UK's National Cyber Security Centre \(archived\)](#)
- [NIST SP 800-63 Digital Identity Guidelines \(Second Public Draft of Revision 4\) / SP 800-63B Authentication & Authenticator Management](#)

3.3.9 Accessible Authentication (Enhanced) (Level AAA)

Source: [WCAG 2.2](#) / [Understanding 3.3.9](#) — Copyright © 2023 World Wide Web Consortium

A cognitive function test (such as remembering a password or solving a puzzle) is not required for any step in an authentication process unless that step provides at least one of the following:

Alternative

Another authentication method that does not rely on a cognitive function test.

Mechanism

A mechanism is available to assist the user in completing the cognitive function test.

In brief

Goal

Make logins possible with less mental effort.

What to do

Don't make people recognize objects or user-supplied images and media to login.

Why it's important

Some people with cognitive disabilities can't do puzzles, including identifying objects and non-text information they previously supplied.

Intent of Accessible Authentication (Enhanced)

The purpose of this success criterion is to ensure there is an accessible, easy-to-use, and secure method to log in, access content, and undertake tasks. This criterion is the same as

[Accessible Authentication \(Minimum\)](#) but without the exceptions for objects and user-provided content.

Any required step of the authentication process:

- cannot display a selection of images, videos, or audio clips, where users must choose which image they provided;
- cannot display a selection of images, where users must choose the images which contain a specific object, such as a car.

Benefits of Accessible Authentication (Enhanced)

The benefits of this success criterion are the same as [Accessible Authentication \(Minimum\)](#) .

People with cognitive issues relating to memory, reading (for example, dyslexia), numbers (for example, dyscalculia), or perception-processing limitations will be able to authenticate irrespective of the level of their cognitive abilities.

Examples of Accessible Authentication (Enhanced)

The examples of this success criterion are very similar to the [Accessible Authentication \(Minimum\)](#) examples.

- A website uses a properly marked up username (or email) and password fields as the login authentication (meeting [Success Criterion 1.3.5 Input Purpose](#) and [Success Criterion 4.1.2 Name, Role, Value](#)). The user's browser or integrated third-party password manager extension can identify the purpose of the inputs and automatically fill in the username and password.
- A website does not block paste functionality. The user is able to use a third-party password manager to store credentials, copy them, and paste them directly into a login form.
- A website uses WebAuthn so the user can authenticate with their device instead of username/password. The user's device

could use any available modality. Common methods on laptops and phones are facial-scan, fingerprint, and PIN (Personal Identification Number). The website is not enforcing any particular use; it is assumed a user will set up a method that suits them.

- A website offers the ability to login with a third-party provider using the OAuth method.
- A website that requires two-factor authentication allows for multiple options for the 2nd factor, including a USB-based method where the user simply presses a button to enter a time-based token.
- A website that requires two-factor authentication displays a QR code which can be scanned by an app on a user's device to confirm identity.
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Resources

- [Cognitive Accessibility Gap Analysis Topic 1: Authentication and Safety](#)
- [Cognitive Accessibility Issue Papers 4. Web Security and Privacy Technologies](#) and [Web Security and Privacy Technologies](#)
- [Making Content Usable for People with Cognitive and Learning Disabilities 4.7.1 Provide a Login that Does Not Rely on Memory or Other Cognitive Skills](#)
- [Security and Privacy Technologies issue paper from the Cognitive Task Force .](#)
- [WebAuthN specification .](#)
- [Web Authentication API on MDN .](#)

- WebAuthN Demo site .
- OAuth on Wikipedia .
- "Let them paste passwords", from the UK's National Cyber Security Centre (archived)
- NIST SP 800-63 Digital Identity Guidelines (Second Public Draft of Revision 4) / SP 800-63B Authentication & Authenticator Management

Principle 4: Robust

[Source: W3C WCAG 2.2 —

<https://www.w3.org/TR/WCAG22/#principle4>]

Content must be robust enough that it can be interpreted by a wide variety of user agents, including assistive technologies.

Guideline 4.1: Compatible

[Source: W3C WCAG 2.2 —

<https://www.w3.org/TR/WCAG22/#guideline41>]

Maximize compatibility with current and future user agents, including assistive technologies.

4.1.1 Parsing (Obsolete and removed) (Level A)

Source: WCAG 2.2 / Understanding 4.1.1 — Copyright © 2023 World Wide Web Consortium

This criterion was originally adopted to address problems that assistive technology had directly parsing HTML. Assistive technology no longer has any need to directly parse HTML. Consequently, these problems either no longer exist or are addressed by other criteria. This criterion no longer has utility and is removed.

In brief

Goal

Assistive technology can properly present page content.

What to do

Create web pages according to specifications.

Why it's important

People can browse web content more easily with their assistive technology.

Intent of Parsing

This criterion has been removed from WCAG 2.2.

The intent of this success criterion was to ensure that user agents, including assistive technologies, can accurately interpret and parse content. Since WCAG 2.0 was published, the specifications (such as HTML) and browsers have improved their handling of parsing errors. It is also the case that assistive technology used to do their own parsing of markup, but now rely on the browser. For that reason this success criterion has been removed. Many issues that would have failed this criterion will fail [Info and Relationships](#) or

[Name, Role, Value](#) . Other issues are excepted by the "except where the specification allow these features" part of the criterion.

The following content is left for historical purposes to show the original intent.

> Success Criterion 4.1.1 Parsing (Level A): In content implemented using markup languages, elements have complete start and end tags, elements are nested according to their specifications, elements do not contain duplicate attributes, and any IDs are unique, except where the specifications allow these features. Start and end tags that are missing a critical character in their formation, such as a closing angle bracket or a mismatched attribute value quotation mark are not complete.

The intent of this success criterion is to ensure that user agents, including assistive technologies, can accurately interpret and parse content. If the content cannot be parsed into a data structure, then different user agents may present it differently or be completely unable to parse it. Some user agents use "repair techniques" to render poorly coded content.

Since repair techniques vary among user agents, authors cannot assume that content will be accurately parsed into a data structure or that it will be rendered correctly by specialized user agents, including assistive technologies, unless the content is created according to the rules defined in the formal grammar for that technology. In markup languages, errors in element and attribute syntax and failure to provide properly nested start/end tags lead to errors that prevent user agents from parsing the content reliably. Therefore, the success criterion requires that the content can be parsed using only the rules of the formal grammar.

The concept of "well formed" is close to what is required here. However, exact parsing requirements vary amongst markup

languages, and most non XML-based languages do not explicitly define requirements for well formedness. Therefore, it was necessary to be more explicit in the success criterion in order to be generally applicable to markup languages. Because the term "well formed" is only defined in XML, and (because end tags are sometimes optional) valid HTML does not require well formed code, the term is not used in this success criterion.

With the exception of one success criterion ([1.4.4 Resize Text](#) , which specifically mentions that the effect specified by the success criterion must

be achieved without relying on an assistive technology) authors can meet the Success

Criteria with content that assumes use of an assistive technology (or access features

in use agents) by the user, where such assistive technologies (or access features

in user agents) exist and are available to the user.

Benefits of Parsing

- Ensuring that web pages have complete start and end tags and are nested according to specification helps ensure that assistive technologies can parse the content accurately and without crashing.

4.1.2 Name, Role, Value (Level A)

Source: [WCAG 2.2](#) / [Understanding 4.1.2](#) — Copyright © 2023 World Wide Web Consortium

For all user interface components (including but not limited to: form elements, links and components generated by scripts), the name and role can be programmatically determined ; states , properties, and values that can be set by the user can be programmatically set ; and notification of changes to these items is available to user agents , including assistive technologies .

This success criterion is primarily for web authors who develop or script their own user interface components. For example, standard HTML controls already meet this success criterion when used according to specification.

In brief

Goal

People using assistive technology understand all components.

What to do

Give components correct names, roles, states, and values.

Why it's important

Assistive technology only works well when code is done properly.

Intent of Name, Role, Value

The intent of this success criterion is to ensure that Assistive Technologies (AT) can gather appropriate information about, activate (or set) and keep up to date on the status of user interface controls in the content.

When standard controls from accessible technologies are used, this process is straightforward. If the user interface elements are

used according to specification the conditions of this provision will be met. (See examples of Success Criterion 4.1.2 Name, Role, Value below)

If custom controls are created, however, or interface elements are programmed (in code or script) to have a different role and/or function than usual, then additional measures need to be taken to ensure that the controls provide important and appropriate information to assistive technologies and allow themselves to be controlled by assistive technologies.

What roles and states are appropriate to convey to assistive technology will depend

on what the control represents. Specifics about such information are defined by other

specifications, such as WAI-ARIA , or the

relevant platform standards. Another factor to consider is whether there is sufficient accessibility support with assistive technologies to convey the information as specified.

A particularly important state of a user interface control is whether or not it has focus. The focus state of a control can be programmatically determined, and notifications about change of focus are sent to user agents and assistive technology. Other examples of user interface control states are whether or not a checkbox or radio button has been selected, or whether a collapsible tree view or accordion is expanded or collapsed.

Success Criterion 4.1.2 Name, Role, Value requires a programmatically determinable name for all user interface components. Names may be visible or invisible. Occasionally, the name needs to be visible, in which case it is identified as a label.

Refer to the definition of name and label in the glossary for more information.

Benefits of Name, Role, Value

- Providing role, state, and value information on all user interface components enables compatibility with assistive technology, such as screen readers, screen magnifiers, and speech recognition software, used by people with disabilities.

Examples of Name, Role, Value

Accessible APIs

A Java applet uses the accessibility API defined by the language.

Resources for Name, Role, Value

- [Web Accessibility Initiative - Accessible Rich Internet Applications \(ARIA\)](#)
- [ARIA in HTML](#)
- [Accessible Name and Description Computation 1.2](#)
- [ARIA Authoring Practices Guide](#)

4.1.3 Status Messages (Level AA)

Source: [WCAG 2.2](#) / [Understanding 4.1.3](#) — Copyright © 2023 World Wide Web Consortium

In content implemented using markup languages, status messages can be programmatically determined through role or properties such that they can be presented to the user by assistive technologies without receiving focus.

In brief

Goal

Make users aware of important changes in content.

What to do

Let assistive technology notify users about status changes that don't take focus.

Why it's important

People who do not see messages need to be informed about them.

Intent of this Success Criterion

The intent of this success criterion is to make users aware of important changes in content that are not given focus, and to do so in a way that doesn't unnecessarily interrupt their work.

The intended beneficiaries are blind and low vision users of assistive technologies with screen reader capabilities. An additional benefit is that assistive technologies for users with cognitive disabilities may achieve an alternative means of indicating (or even delaying or suppressing) status messages, as preferred by the user.

The scope of this success criterion is specific to changes in content that involve status messages. A status message is a defined term in

WCAG. There are two main criteria that determine whether something meets the definition of a status message:

- the message provides information to the user on the success or results of an action, on the waiting state of an application, on the progress of a process, or on the existence of errors;
- the message is not delivered via a change in context.

Information can be added to pages which does not meet the definition of a status message. For example, the list of results obtained from a search are not considered a status update and thus are not covered by this success criterion. However, brief text messages displayed *about* the completion or status of the search, such as "Searching...", "18 results returned" or "No results returned" would be status updates if they do not take focus. Examples of status messages are given in the section titled Status Message Examples below.

This success criterion specifically addresses scenarios where new content is added to the page without changing the user's context. Changes of context, by their nature, interrupt the user by taking focus. They are already surfaced by assistive technologies, and so have already met the goal to alert the user to new content. As such, messages that involve changes of context do not need to be considered and are not within the scope of this success criterion. Examples of scenarios that add new content by changing the context are given in the section titled Examples of Changes That Are Not Status Messages below.

Benefits

- When appropriate roles or properties are assigned to status messages, the new content is spoken by screen readers in such a way as to assist blind and low vision users. Most sighted users can observe text peripherally added to the viewport.

Such content provides additional information without affecting the user's current point of regard. The ability of an assistive technology to announce such new important text content allows more users to benefit from an awareness of the information in an equivalent manner.

- Assigning proper roles or properties to status messages provides possible future uses and personalization opportunities, such as the potential to be exploited by assistive technologies created for users with some cognitive disabilities. Where page authors elect to design additions to the screen which do not change the user's context (i.e., take focus), the information is arguably of less importance than something presented using a modal dialog, which must be acknowledged by the user. As such, depending on the user's preferences, an assistive technology may choose to delay, suppress, or transform such messages so a user is not unnecessarily interrupted; or conversely the assistive technology may highlight such messages where the user finds it optimal to do so.

Status Message Examples

- After a user presses a Search button, the page content is updated to include the results of the search, which are displayed in a section below the Search button. The change to content also includes the message "5 results returned" near the top of this new content. This text is given an appropriate role for a status message. A screen reader announces, "Five results returned".
- After a user presses an Add to Shopping Cart button, a section of content near the Shopping Cart icon adds the text "5 items". A screen reader announces "Five items" or "Shopping cart, five items".

- After a user enters incorrect text in an input called Postal Code, a message appears above the input reading "Invalid entry". The screen reader announces, "Invalid entry" or "Postal code, invalid entry".
- After a user activates a process, an icon symbolizing 'busy' appears on the screen. The screen reader announces "application busy".
- An application displays a dynamic progress bar to indicate the status of an upgrade. The screen reader provides intermittent announcements of the progress.
- After a user submits a form, text is added to the existing form which reads, "Your form was successfully submitted." The screen reader announces the same message.
- After a user unsuccessfully fills in a form because some of the data is in the incorrect format, text is added to the existing form which reads "5 errors on page". The screen reader announces the same message.
- After a user puts a photo in an album in an online photo app, the app dynamically displays a popup message of "Saved in 'Wedding' album", which is also read by a screen reader.

Examples of Status Messages that Do Not Add New Text to the Screen

This success criterion was intentionally worded to apply primarily when visible text is added to (or becomes visible on) the page. The reason for this is that where new text is displayed, it is intended to be visible to all users. By providing a programmatic means of ensuring the text is also surfaced through assistive technologies, the success criterion provides the same information to users who cannot or may not see it. However, not all changes to content involve the addition of text to the screen. The following are all considerations relevant to this success criterion:

- Non-displayed text specific to AT users;

- Modification of status text;
- Removal of status text; and
- Non-textual status content, such as images.

There may be cases where the addition of visible text does not by itself convey sufficient information to the user of assistive technology. For example, the proximity of new content to other pieces of information on the screen may provide a visual context that is lacking in the text alone.

In such cases, authors may wish to designate additional content for inclusion in the status message, including non-displayed text which can be provided to the assistive technologies, for added context. Important considerations regarding the appropriate use of such techniques are further discussed in the Sufficient Techniques.

If a status message persists on the page, modifications to this text are usually equivalent to a new status message. An example would be a shopping cart which updates text from reading "0 items" to "3 items". Typical methods of writing such changes in the page content result in the entire modified text string being considered a new change, and thus read by assistive technologies. However, where only the number in this string was coded as an updated chunk of content, the resulting experience for screen reader users could be to only hear "three", which may not be sufficient information to provide context for the user. In such situations, marking the entire "3 items" string as the status text would normally be a better solution. See Sufficient Techniques for more discussion, including the use of `aria-atomic`. In this case it would also be a courtesy to add offscreen text such as "in shopping cart" to the message.

In situations where status text is entirely removed, its absence may itself convey information about the status. The most obvious example of this is where a message is displayed that the system is "busy" or "waiting". For a sighted user, when this text disappears, it is normally an indication that the state is now available.

However non-sighted users would be unaware of this change, unless the end of the waiting state results in a change of context for the user. Where updating the visible message (e.g., to "system available") is not feasible, the use of a non-visible status message, such as "system available", ensures equivalent status information is provided. See [Sufficient Techniques](#) for more discussion.

Changes in content are not restricted to text changes. Where an icon or sound indicates a status message, this information will be surfaced by the screen reader through a combination of two things: 1) existing WCAG requirements governing text alternatives (under [Success Criterion 1.1.1 Non-Text Content](#)), and 2) the requirement of this current success criterion to supply an appropriate role.

Examples of Changes That Are Not Status Messages

The following examples identify situations where no additional author action is necessary. All cases are excepted from this success criterion since they do not meet the definition of "status messages."

- An author displays an error message in a dialog. Since the dialog takes focus, it is defined as a change of context and does not meet the definition of a status message. As a result of taking focus, the new change of context is already announced by the screen reader, and thus does not need to be included in the scope of this success criterion.

- Content is exposed or hidden when a user interacts with a user interface component, for example expanding components such as a menu, select, accordion or tree, or selecting a different tab item in a tablist. None of the resulting changes to content meet the definition of status messages. Further, all components that meet the definition of a user interface component already have requirements specified under 4.1.2 Name, Role, Value , including the need to make notifications of changes to values and states available to user agents, including assistive technologies. As a result, changes in state, such as "expanded" or "collapsed," would be announced by the screen reader, and thus the user would be alerted to the 'addition' or 'removal' of content. As such, such content does not need to be addressed by this success criterion.
- After a user completes a survey question which indicates they are unhappy, a series of new questions are added to the page about customer satisfaction. The new inputs do not meet the definition of status message. They do not "provide information to the user on the success or results of an action, on the waiting state of an application, on the progress of a process or on the existence of errors," and so are not required to meet this success criterion. Creating a status message about these questions being added, or notifying the user in advance that content changes may take place based on the user's response, are best practices but are not requirements in this scenario.

Other uses of live regions or alerts

Live regions and alerts can be usefully applied in many situations where a change of content takes place which does not constitute a status message, as defined in this success criterion. However, there is a risk of making an application too "chatty" for a screen reader user. User testing should be carried out to ensure the

appropriate level of feedback is achieved. The Advisory Techniques provide examples of how alerts or live regions can enhance the user experience.

The purpose of this success criterion is not to force authors to generate new status messages. Its intent is to ensure that when status messages *are* displayed, they are programmatically identified in a way that allows assistive technologies to present them to the user.

Part Two: The W3C Accessibility Ecosystem

[Editorial — Jamal Mazrui — organization, selection, and framing of this part]

The sections in this part draw from current, published W3C Recommendations or completed W3C Group Notes, reproduced under the W3C Document License.

The Essential Components of Web Accessibility

Source: *Essential Components of Web Accessibility*, W3C WAI.

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<https://www.w3.org/WAI/fundamentals/components/>

Web accessibility depends on multiple components working together. A failure in any one component creates barriers even when others are fully accessible.

Web content is the information and functionality in a web page or application. WCAG addresses web content directly.

Authoring tools are software and services developers use to create web content: HTML editors, content management systems, website builders, and word processors with web export. ATAG addresses authoring tools.

User agents are browsers, media players, and other applications that render web content. UAAG addresses user agents.

Assistive technologies are software or hardware used by people with disabilities: screen readers, braille displays, alternative keyboards, switch access, voice recognition, and eye tracking.

W3C WAI publishes a diagram of these relationships at <https://www.w3.org/WAI/fundamentals/components/> (text description at <https://www.w3.org/WAI/fundamentals/components/examples/>) . WCAG addresses content; ATAG addresses authoring tools; UAAG addresses user agents; ARIA provides the semantic foundation. All rest on the base of HTML, CSS, SVG, and other web technical specifications.

WAI-ARIA: Accessible Rich Internet Applications

Source: *WAI-ARIA Overview*, W3C WAI.

<https://www.w3.org/WAI/standards-guidelines/aria/>

WAI-ARIA 1.2, W3C Recommendation, 6 June 2023. Copyright © 2023 World Wide Web Consortium. <https://www.w3.org/TR/wai-aria-1.2/>

WAI-ARIA defines **roles**, **states**, and **properties** that communicate the purpose, behavior, and current state of custom interface components to assistive technologies — filling semantic gaps that HTML alone cannot cover in dynamic applications.

WAI-ARIA is referenced throughout WCAG 2.2, particularly for SC 1.3.1, SC 4.1.2, and SC 4.1.3. The cardinal rule: "No ARIA is better than bad ARIA." Native HTML semantics should always be preferred; ARIA supplements when insufficient.

WAI-ARIA 1.2 is the current W3C Recommendation (June 2023). WAI-ARIA 1.3 is a Working Draft in progress — not yet a standard.

The **ARIA Authoring Practices Guide (APG)** provides patterns for over 30 widget types with keyboard interaction guidance and code examples:

<https://www.w3.org/WAI/ARIA/apg/>

The ARIA suite also includes: Core Accessibility API Mappings (Core-AAM), Accessible Name and Description Computation (accname), and ARIA in HTML (W3C Recommendation, December 2021, with ongoing updates).

Authoring Tool Accessibility Guidelines (ATAG 2.0)

Source: *ATAG Overview*, W3C WAI.

<https://www.w3.org/WAI/standards-guidelines/atag/>

ATAG 2.0, W3C Recommendation, 24 September 2015. Copyright © 2015 World Wide Web Consortium.

<https://www.w3.org/TR/ATAG20/>

Part A requires the authoring tool's own interface to be accessible (perceivable, operable, understandable, robust).

Part B requires the tool to support production of accessible content: prompting for alt text, preserving accessibility information, providing checking features, and making accessible-authoring workflows as functional as non-accessible ones.

Many web accessibility failures originate with authoring tools that lack the means to create accessible content. ATAG addresses this structurally. W3C encourages applying ATAG 2.0 with reference to WCAG 2.2 as the current standard.

User Agent Accessibility Guidelines (UAAG 2.0)

Source: *UAAG Overview*, W3C WAI.

<https://www.w3.org/WAI/standards-guidelines/uaag/>

UAAG 2.0, W3C Note, 15 December 2015. Copyright © 2015 World Wide Web Consortium. <https://www.w3.org/TR/UAAG20/>

UAAG 2.0 explains how browsers, media players, and other user agents should be accessible. Some needs — text zoom, color customization, disabling animations, keyboard access — depend on what browsers do, not what content authors do. UAAG 2.0 remains applicable guidance; the UAAG Working Group closed in 2016 but the specification is current.

Accessibility Conformance Testing (ACT) Rules Format 1.1

Source: *ACT Overview*, W3C WAI.

<https://www.w3.org/WAI/standards-guidelines/act/>

ACT Rules Format 1.1, W3C Recommendation, 4 February 2026. Copyright © 2026 World Wide Web Consortium. <https://www.w3.org/TR/act-rules-format/>

ACT Rules Format 1.1 (published February 2026) is a W3C Recommendation defining a standard way to document and share accessibility test procedures. ACT Rules enable transparency and harmonization of testing methods across automated tools and manual methodologies. Each rule specifies applicability, expectations, and mapping to an accessibility requirement such as a WCAG success criterion.

A published library of ACT Rules for WCAG is maintained at <https://www.w3.org/WAI/standards-guidelines/act/rules/>

The Relative Luminance and Contrast Ratio Formulas

Source: *WCAG 2.2 Specification*, Appendix A. Copyright © 2023 World Wide Web Consortium.

<https://www.w3.org/TR/WCAG22/>

Three WCAG 2.2 success criteria reference contrast ratios: SC 1.4.3 (Contrast Minimum, Level AA), SC 1.4.6 (Contrast Enhanced, Level AAA), and SC 1.4.11 (Non-text Contrast, Level AA).

Relative Luminance Formula

For sRGB components RsRGB, GsRGB, BsRGB (each normalized to 0–1):

For each component C:

```
if C <= 0.04045: C_linear = C / 12.92
else:              C_linear = ((C + 0.055) / 1.055) ^ 2.4
```

```
L = 0.2126 * R_linear + 0.7152 * G_linear + 0.0722 * B_linear
```

Contrast Ratio

```
ratio = (L1 + 0.05) / (L2 + 0.05)    [where L1 >= L2]
```

Range: 1:1 (no contrast) to 21:1 (black on white).

WCAG Requirements

- SC 1.4.3 (AA): normal text $\geq 4.5:1$; large text (18pt or 14pt bold) $\geq 3:1$.
- SC 1.4.6 (AAA): normal text $\geq 7:1$; large text $\geq 4.5:1$.
- SC 1.4.11 (AA): UI components and graphical objects $\geq 3:1$ against adjacent colors.

Applying WCAG to Non-Web Technologies (WCAG2ICT)

Source: *WCAG2ICT*, W3C Group Note, 8 October 2024. Copyright © 2024 World Wide Web Consortium.
<https://www.w3.org/TR/wcag2ict-22/>

WCAG2ICT provides authoritative guidance on applying WCAG 2 to non-web ICT: desktop software, mobile applications, kiosk systems, and non-web documents. Most WCAG 2.2 success criteria apply with no changes or minor interpretation adjustments. A small number do not apply to non-web contexts, including SC 2.4.1 Bypass Blocks, SC 2.4.5 Multiple Ways, SC 3.2.3 Consistent Navigation, and SC 3.2.4 Consistent Identification.

WCAG2ICT has been a key reference for Section 508 (US), EN 301 549 (Europe/EAA), and the 2024 U.S. DOJ rule on state/local government website accessibility. The October 2024 update covers all WCAG 2.2 success criteria.

Web Accessibility Laws and Policies

Source: *Web Accessibility Laws & Policies*, W3C WAI.
<https://www.w3.org/WAI/policies/>

United States — The ADA (enforced via DOJ) and Section 508 (federal ICT procurement) both reference WCAG. A 2024 DOJ rule under Title II explicitly requires WCAG 2.1 Level AA for state/local government entities.

European Union — The European Accessibility Act (in force 28 June 2025) and the EU Web Accessibility Directive (2016) reference EN 301 549, which incorporates WCAG 2.1 and is expected to be updated to WCAG 2.2.

United Kingdom — Public Sector Bodies Accessibility Regulations 2018 require WCAG 2.1 Level AA.

ISO — WCAG 2.2 is approved as ISO/IEC 40500:2025, available free from ISO.

For a comprehensive current listing see

<https://www.w3.org/WAI/policies/>

Conformance and Accessibility Statements

Source: *Understanding Conformance*, W3C WAI.

<https://www.w3.org/WAI/WCAG22/Understanding/conformance>

To conform to WCAG 2.2 at a given level, a page must satisfy all success criteria at that level and all lower levels. Five conformance requirements apply: (1) claimed level is met; (2) applies to full pages; (3) applies to complete processes; (4) only accessibility-supported technologies are relied upon; (5) non-supported technologies must not block access.

W3C provides a free **Accessibility Statement Generator** at <https://www.w3.org/WAI/planning/statements/> and the **WCAG-EM** evaluation methodology at <https://www.w3.org/WAI/test-evaluate/conformance/wcag-em/>

About

[Editorial — Jamal Mazrui]

Jamal Mazrui is a software developer and accessibility specialist with extensive experience in assistive technologies and web standards. He is blind and uses screen readers including JAWS and NVDA on Windows, and VoiceOver on iOS.

This compendium was produced through a collaborative process in which Jamal directed the work and Claude AI (Anthropic) performed the technical extraction, organization, and cleanup of content from the W3C WCAG source repository. The substantive

content is entirely the work of the W3C Accessibility Guidelines Working Group, reproduced under the W3C Document License.

The source code for this project is published at
https://github.com/jamalmazrui/wcag_in_a_book

Resources

WCAG 2.2 Specification — <https://www.w3.org/TR/WCAG22/>

Understanding WCAG 2.2 —
<https://www.w3.org/WAI/WCAG22/Understanding/>

Techniques for WCAG 2.2 —
<https://www.w3.org/WAI/WCAG22/Techniques/>

How to Meet WCAG 2.2 (Quick Reference) —
<https://www.w3.org/WAI/WCAG22/quickref/>

WCAG 2 Overview — <https://www.w3.org/WAI/standards-guidelines/wcag/>

WAI-ARIA Overview — <https://www.w3.org/WAI/standards-guidelines/aria/>

ARIA Authoring Practices Guide —
<https://www.w3.org/WAI/ARIA/apg/>

ATAG Overview — <https://www.w3.org/WAI/standards-guidelines/atag/>

UAAG Overview — <https://www.w3.org/WAI/standards-guidelines/uaag/>

ACT Rules Overview — <https://www.w3.org/WAI/standards-guidelines/act/>

WCAG2ICT — <https://www.w3.org/TR/wcag2ict-22/>

Essential Components of Web Accessibility —
<https://www.w3.org/WAI/fundamentals/components/>

Web Accessibility Laws & Policies —
<https://www.w3.org/WAI/policies/>

Accessibility Statement Generator —
<https://www.w3.org/WAI/planning/statements/>

Accessibility Guidelines Working Group —
<https://www.w3.org/groups/wg/ag/>

Web Accessibility Initiative (WAI) —
<https://www.w3.org/WAI/>

Appendix: WCAG 2.2 Glossary

Source: *WCAG 2.2 Specification*, Appendix A. Copyright © 2023 World Wide Web Consortium.

<https://www.w3.org/TR/WCAG22/#glossary>

The following terms are defined in the normative WCAG 2.2 specification. Where a success criterion uses a defined term, the definition applies.

abbreviation

shortened form of a word, phrase, or name where the abbreviation has not become part of the language This includes initialisms and acronyms where: initialisms are shortened forms of a name or phrase made from the initial letters of words or syllables contained in that name or phrase Note: Not defined in all languages. SNCF is a French initialism that contains the initial letters of the Société Nationale des Chemins de Fer , the French national railroad. ESP is an initialism for extrasensory perception. acronyms are abbreviated forms made from the initial letters or parts of other words (in a name or phrase) which may be pronounced as a word

NOAA is an acronym made from the initial letters of the National Oceanic and Atmospheric Administration in the United States. Some companies have adopted what used to be an initialism as their company name. In these cases, the new name of the company is the letters (for example, Ecma) and the word is no longer considered an abbreviation.

accessibility supported

supported by users' assistive technologies as well as the accessibility features in browsers and other user agents To qualify as an accessibility-supported use of a web content technology (or feature of a technology), both 1 and 2 must be satisfied for a web content technology (or feature): The way that the web content technology is used must be supported by users' assistive technology (AT). This means that the way that the technology is used has been tested for interoperability with users' assistive technology in the human language(s) of the content, AND The web content technology must have accessibility-supported user agents that are available to users. This means that at least one of the following four statements is true: The technology is supported natively in widely-distributed user agents that are also accessibility supported (such as HTML and CSS); OR The technology is supported in a widely-distributed plug-in that is also accessibility supported; OR The content is available in a closed environment, such as a university or corporate network, where the user agent required by the technology and used by the organization is also accessibility supported; OR The user agent(s) that support the technology are accessibility supported and are available for download or purchase in a way that: does not cost a person with a disability any more than a person without a disability and is as easy to find and obtain for a person with a disability as it is for a person without disabilities. The Accessibility Guidelines Working Group and the W3C do not specify which or how much support by

assistive technologies there must be for a particular use of a web technology in order for it to be classified as accessibility supported. (See Level of Assistive Technology Support Needed for "Accessibility Support" .) Web technologies can be used in ways that are not accessibility supported as long as they are not relied upon and the page as a whole meets the conformance requirements, including Conformance Requirement 4 and Conformance Requirement 5 . When a web technology is used in a way that is "accessibility supported," it does not imply that the entire technology or all uses of the technology are supported. Most technologies, including HTML, lack support for at least one feature or use. Pages conform to WCAG only if the uses of the technology that are accessibility supported can be relied upon to meet WCAG requirements. When citing web content technologies that have multiple versions, the version(s) supported should be specified. One way for authors to locate uses of a technology that are accessibility supported would be to consult compilations of uses that are documented to be accessibility supported. (See Understanding Accessibility-Supported Web Technology Uses .) Authors, companies, technology vendors, or others may document accessibility-supported ways of using web content technologies. However, all ways of using technologies in the documentation would need to meet the definition of accessibility-supported web content technologies above.

alternative for time-based media

document including correctly sequenced text descriptions of time-based visual and auditory information and providing a means for achieving the outcomes of any time-based interaction A screenplay used to create the synchronized media content would meet this definition only if it was corrected to accurately represent the final synchronized media after editing.

ambiguous to users in general

the purpose cannot be determined from the link and all information of the web page presented to the user simultaneously with the link (i.e., readers without disabilities would not know what a link would do until they activated it) The word guava in the following sentence "One of the notable exports is guava" is a link. The link could lead to a definition of guava, a chart listing the quantity of guava exported or a photograph of people harvesting guava. Until the link is activated, all readers are unsure and the person with a disability is not at any disadvantage.

ASCII art

picture created by a spatial arrangement of characters or glyphs (typically from the 95 printable characters defined by ASCII)

assistive technology

hardware and/or software that acts as a user agent , or along with a mainstream user agent, to provide functionality to meet the requirements of users with disabilities that go beyond those offered by mainstream user agents Functionality provided by assistive technology includes alternative presentations (e.g., as synthesized speech or magnified content), alternative input methods (e.g., voice), additional navigation or orientation mechanisms, and content transformations (e.g., to make tables more accessible). Assistive technologies often communicate data and messages with mainstream user agents by using and monitoring APIs. The distinction between mainstream user agents and assistive technologies is not absolute. Many mainstream user agents provide some features to assist individuals with disabilities. The basic difference is that mainstream user agents target broad and diverse audiences that usually include people with and without disabilities. Assistive technologies target narrowly defined populations of users with specific disabilities. The assistance

provided by an assistive technology is more specific and appropriate to the needs of its target users. The mainstream user agent may provide important functionality to assistive technologies like retrieving web content from program objects or parsing markup into identifiable bundles. Assistive technologies that are important in the context of this document include the following: screen magnifiers, and other visual reading assistants, which are used by people with visual, perceptual and physical print disabilities to change text font, size, spacing, color, synchronization with speech, etc. in order to improve the visual readability of rendered text and images; screen readers, which are used by people who are blind to read textual information through synthesized speech or braille; text-to-speech software, which is used by some people with cognitive, language, and learning disabilities to convert text into synthetic speech; speech recognition software, which may be used by people who have some physical disabilities; alternative keyboards, which are used by people with certain physical disabilities to simulate the keyboard (including alternate keyboards that use head pointers, single switches, sip/puff and other special input devices.); alternative pointing devices, which are used by people with certain physical disabilities to simulate mouse pointing and button activations.

audio

the technology of sound reproduction Audio can be created synthetically (including speech synthesis), recorded from real world sounds, or both.

audio description

narration added to the soundtrack to describe important visual details that cannot be understood from the main soundtrack alone Audio description of video provides information about actions, characters, scene changes, on-screen text, and other visual

content. In standard audio description, narration is added during existing pauses in dialogue. (See also extended audio description .) Where all of the important video information is already provided in existing audio , no additional audio description is necessary. Also called "video description" and "descriptive narration."

audio-only

a time-based presentation that contains only audio (no video and no interaction)

blinking

switch back and forth between two visual states in a way that is meant to draw attention See also flash . It is possible for something to be large enough and blink brightly enough at the right frequency to be also classified as a flash.

blocks of text

more than one sentence of text

CAPTCHA

initialism for "Completely Automated Public Turing test to tell Computers and Humans Apart" CAPTCHA tests often involve asking the user to type in text that is displayed in an obscured image or audio file. A Turing test is any system of tests designed to differentiate a human from a computer. It is named after famed computer scientist Alan Turing. The term was coined by researchers at Carnegie Mellon University.

captions

synchronized visual and/or text alternative for both speech and non-speech audio information needed to understand the media content Captions are similar to dialogue-only subtitles except captions convey not only the content of spoken dialogue, but also equivalents for non-dialogue audio information needed to

understand the program content, including sound effects, music, laughter, speaker identification and location. Closed Captions are equivalents that can be turned on and off with some players. Open Captions are any captions that cannot be turned off. For example, if the captions are visual equivalent images of text embedded in video. Captions should not obscure or obstruct relevant information in the video. In some countries, captions are called subtitles. Audio descriptions can be, but do not need to be, captioned since they are descriptions of information that is already presented visually.

changes of context

major changes that, if made without user awareness, can disorient users who are not able to view the entire page simultaneously. Changes in context include changes of: user agent ; viewport ; focus; content that changes the meaning of the web page. A change of content is not always a change of context. Changes in content, such as an expanding outline, dynamic menu, or a tab control do not necessarily change the context, unless they also change one of the above (e.g., focus). Opening a new window, moving focus to a different component, going to a new page (including anything that would look to a user as if they had moved to a new page) or significantly re-arranging the content of a page are examples of changes of context.

cognitive function test

New A task that requires the user to remember, manipulate, or transcribe information. Examples include, but are not limited to: memorization, such as remembering a username, password, set of characters, images, or patterns. The common identifiers name, email, and phone number are not considered cognitive function tests as they are personal to the user and consistent across

websites; transcription, such as typing in characters; use of correct spelling; performance of calculations; solving of puzzles.

conformance

satisfying all the requirements of a given standard, guideline or specification

conforming alternate version

version that conforms at the designated level, and provides all of the same information and functionality in the same human language , and is as up to date as the non-conforming content, and for which at least one of the following is true: the conforming version can be reached from the non-conforming page via an accessibility-supported mechanism , or the non-conforming version can only be reached from the conforming version, or the non-conforming version can only be reached from a conforming page that also provides a mechanism to reach the conforming version In this definition, "can only be reached" means that there is some mechanism, such as a conditional redirect, that prevents a user from "reaching" (loading) the non-conforming page unless the user had just come from the conforming version. The alternate version does not need to be matched page for page with the original (e.g., the conforming alternate version may consist of multiple pages). If multiple language versions are available, then conforming alternate versions are required for each language offered. Alternate versions may be provided to accommodate different technology environments or user groups. Each version should be as conformant as possible. One version would need to be fully conformant in order to meet conformance requirement 1 . The conforming alternative version does not need to reside within the scope of conformance, or even on the same website, as long as it is as freely available as the non-conforming version. Alternate versions should not be confused with supplementary content ,

which support the original page and enhance comprehension. Setting user preferences within the content to produce a conforming version is an acceptable mechanism for reaching another version as long as the method used to set the preferences is accessibility supported. See Understanding Conforming Alternate Versions

content

information and sensory experience to be communicated to the user by means of a user agent , including code or markup that defines the content's structure , presentation , and interactions

context-sensitive help

help text that provides information related to the function currently being performed Clear labels can act as context-sensitive help.

contrast ratio

$(L_1 + 0.05) / (L_2 + 0.05)$, where L_1 is the relative luminance of the lighter of the colors, and L_2 is the relative luminance of the darker of the colors. Contrast ratios can range from 1 to 21 (commonly written 1:1 to 21:1). Because authors do not have control over user settings as to how text is rendered (for example font smoothing or anti-aliasing), the contrast ratio for text can be evaluated with anti-aliasing turned off. For the purpose of Success Criteria 1.4.3 and 1.4.6, contrast is measured with respect to the specified background over which the text is rendered in normal usage. If no background color is specified, then white is assumed. Background color is the specified color of content over which the text is to be rendered in normal usage. It is a failure if no background color is specified when the text color is specified, because the user's default background color is unknown and cannot be evaluated for sufficient contrast. For the same reason, it is a failure if no text color is specified when a background color is specified. When

there is a border around the letter, the border can add contrast and would be used in calculating the contrast between the letter and its background. A narrow border around the letter would be used as the letter. A wide border around the letter that fills in the inner details of the letters acts as a halo and would be considered background. WCAG conformance should be evaluated for color pairs specified in the content that an author would expect to appear adjacent in typical presentation. Authors need not consider unusual presentations, such as color changes made by the user agent, except where caused by authors' code.

correct reading sequence

any sequence where words and paragraphs are presented in an order that does not change the meaning of the content

CSS pixel

visual angle of about 0.0213 degrees A CSS pixel is the canonical unit of measure for all lengths and measurements in CSS. This unit is density-independent, and distinct from actual hardware pixels present in a display. User agents and operating systems should ensure that a CSS pixel is set as closely as possible to the CSS Values and Units Module Level 3 reference pixel `[[css3-values]]`, which takes into account the physical dimensions of the display and the assumed viewing distance (factors that cannot be determined by content authors).

down-event

platform event that occurs when the trigger stimulus of a pointer is depressed The down-event may have different names on different platforms, such as "touchstart" or "mousedown".

dragging movement

New an operation where the pointer engages with an element on the down-event and the element (or a representation of its

position) follows the pointer until an up-event Examples of draggable elements include list items, text elements, and images.

emergency

a sudden, unexpected situation or occurrence that requires immediate action to preserve health, safety, or property

essential

if removed, would fundamentally change the information or functionality of the content, and information and functionality cannot be achieved in another way that would conform

extended audio description

audio description that is added to an audiovisual presentation by pausing the video so that there is time to add additional description This technique is only used when the sense of the video would be lost without the additional audio description and the pauses between dialogue/narration are too short.

flash

a pair of opposing changes in relative luminance that can cause seizures in some people if it is large enough and in the right frequency range See general flash and red flash thresholds for information about types of flash that are not allowed. See also blinking .

focus indicator

New pixels that are changed to visually indicate when a user interface component is in a focused state

functionality

processes and outcomes achievable through user action

general flash and red flash thresholds

a flash or rapidly changing image sequence is below the threshold (i.e., content passes) if any of the following are true: there are no more than three general flashes and / or no more than three red flashes within any one-second period; or the combined area of flashes occurring concurrently occupies no more than a total of .006 steradians within any 10 degree visual field on the screen (25% of any 10 degree visual field on the screen) at typical viewing distance where: A general flash is defined as a pair of opposing changes in relative luminance of 10% or more of the maximum relative luminance (1.0) where the relative luminance of the darker image is below 0.80; and where "a pair of opposing changes" is an increase followed by a decrease, or a decrease followed by an increase, and A red flash is defined as any pair of opposing transitions involving a saturated red Exception: Flashing that is a fine, balanced, pattern such as white noise or an alternating checkerboard pattern with "squares" smaller than 0.1 degree (of visual field at typical viewing distance) on a side does not violate the thresholds. For general software or web content, using a 341 x 256 pixel rectangle anywhere on the displayed screen area when the content is viewed at 1024 x 768 pixels will provide a good estimate of a 10 degree visual field for standard screen sizes and viewing distances (e.g., 15-17 inch screen at 22-26 inches). This resolution of 75 - 85 ppi is known to be lower, and thus more conservative than the nominal CSS pixel resolution of 96 ppi in CSS specifications. Higher resolutions displays showing the same rendering of the content yield smaller and safer images so it is lower resolutions that are used to define the thresholds. A transition is the change in relative luminance (or relative luminance/color for red flashing) between adjacent peaks and valleys in a plot of relative luminance (or relative luminance/color for red flashing) measurement against time. A flash consists of two opposing transitions. The new working definition in the field for

"pair of opposing transitions involving a saturated red" (from WCAG 2.2) is a pair of opposing transitions where, one transition is either to or from a state with a value $R/(R + G + B)$ that is greater than or equal to 0.8, and the difference between states is more than 0.2 (unitless) in the CIE 1976 UCS chromaticity diagram. [[ISO_9241-391]] Tools are available that will carry out analysis from video screen capture. However, no tool is necessary to evaluate for this condition if flashing is less than or equal to 3 flashes in any one second. Content automatically passes (see #1 and #2 above).

human language

language that is spoken, written or signed (through visual or tactile means) to communicate with humans See also sign language .

idiom

phrase whose meaning cannot be deduced from the meaning of the individual words and the specific words cannot be changed without losing the meaning Idioms cannot be translated directly, word for word, without losing their (cultural or language-dependent) meaning. In English, "spilling the beans" means "revealing a secret." However, "knocking over the beans" or "spilling the vegetables" does not mean the same thing. In Japanese, the phrase " さじを投げる " literally translates into "he throws a spoon," but it means that there is nothing he can do and finally he gives up. In Dutch, " Hij ging met de kippen op stok " literally translates into "He went to roost with the chickens," but it means that he went to bed early.

image of text

text that has been rendered in a non-text form (e.g., an image) in order to achieve a particular visual effect This does not include

text that is part of a picture that contains significant other visual content. A person's name on a nametag in a photograph.

informative

for information purposes and not required for conformance
Content required for conformance is referred to as " normative ."

input error

information provided by the user that is not accepted This includes: Information that is required by the web page but omitted by the user Information that is provided by the user but that falls outside the required data format or values

jargon

words used in a particular way by people in a particular field The word StickyKeys is jargon from the field of assistive technology/accessibility.

keyboard interface

interface used by software to obtain keystroke input A keyboard interface allows users to provide keystroke input to programs even if the native technology does not contain a keyboard. A touchscreen PDA has a keyboard interface built into its operating system as well as a connector for external keyboards. Applications on the PDA can use the interface to obtain keyboard input either from an external keyboard or from other applications that provide simulated keyboard output, such as handwriting interpreters or speech-to-text applications with "keyboard emulation" functionality. Operation of the application (or parts of the application) through a keyboard-operated mouse emulator, such as MouseKeys, does not qualify as operation through a keyboard interface because operation of the program is through its pointing device interface, not through its keyboard interface.

keyboard shortcut

alternative means of triggering an action by the pressing of one or more keys

label

text or other component with a text alternative that is presented to a user to identify a component within web content A label is presented to all users whereas the name may be hidden and only exposed by assistive technology. In many (but not all) cases the name and the label are the same. The term label is not limited to the label element in HTML.

large scale

with at least 18 point or 14 point bold or font size that would yield equivalent size for Chinese, Japanese and Korean (CJK) fonts
Fonts with extraordinarily thin strokes or unusual features and characteristics that reduce the familiarity of their letter forms are harder to read, especially at lower contrast levels. Font size is the size when the content is delivered. It does not include resizing that may be done by a user. The actual size of the character that a user sees is dependent both on the author-defined size and the user's display or user agent settings. For many mainstream body text fonts, 14 and 18 point is roughly equivalent to 1.2 and 1.5 em or to 120% or 150% of the default size for body text (assuming that the body font is 100%), but authors would need to check this for the particular fonts in use. When fonts are defined in relative units, the actual point size is calculated by the user agent for display. The point size should be obtained from the user agent, or calculated based on font metrics as the user agent does, when evaluating this success criterion. Users who have low vision would be responsible for choosing appropriate settings. When using text without specifying the font size, the smallest font size used on major browsers for unspecified text would be a reasonable size to assume for the font. If a level 1 heading is rendered in 14pt bold or higher

on major browsers, then it would be reasonable to assume it is large text. Relative scaling can be calculated from the default sizes in a similar fashion. The 18 and 14 point sizes for roman texts are taken from the minimum size for large print (14pt) and the larger standard font size (18pt). For other fonts such as CJK languages, the "equivalent" sizes would be the minimum large print size used for those languages and the next larger standard large print size.

legal commitments

transactions where the person incurs a legally binding obligation or benefit A marriage license, a stock trade (financial and legal), a will, a loan, adoption, signing up for the army, a contract of any type, etc.

link purpose

nature of the result obtained by activating a hyperlink

live

information captured from a real-world event and transmitted to the receiver with no more than a broadcast delay A broadcast delay is a short (usually automated) delay, for example used in order to give the broadcaster time to cue or censor the audio (or video) feed, but not sufficient to allow significant editing. If information is completely computer generated, it is not live.

lower secondary education level

the two or three year period of education that begins after completion of six years of school and ends nine years after the beginning of primary education This definition is based on the International Standard Classification of Education [[UNESCO]].

mechanism

process or technique for achieving a result The mechanism may be explicitly provided in the content, or may be relied upon to be

provided by either the platform or by user agents , including assistive technologies . The mechanism needs to meet all success criteria for the conformance level claimed.

media alternative for text

media that presents no more information than is already presented in text (directly or via text alternatives) A media alternative for text is provided for those who benefit from alternate representations of text. Media alternatives for text may be audio-only, video-only (including sign-language video), or audio-video.

minimum bounding box

New the smallest enclosing rectangle aligned to the horizontal axis within which all the points of a shape lie. For components which wrap onto multiple lines as part of a sentence or block of text (such as hypertext links), the bounding box is based on how the component would appear on a single line.

motion animation

addition of steps between conditions to create the illusion of movement or to give a sense of a smooth transition For example, an element which moves into place or changes size while appearing is considered to be animated. An element which appears or changes instantly, without transition/animation steps, is not using animation. Motion animation does not include changes – such as changes of color or opacity – that do not alter the perceived size, shape, position, or distance/depth of the element to the viewer.

name

text by which software can identify a component within web content to the user The name may be hidden and only exposed by assistive technology, whereas a label is presented to all users. In

many (but not all) cases, the label and the name are the same. This is unrelated to the name attribute in HTML.

navigated sequentially

navigated in the order defined for advancing focus (from one element to the next) using a keyboard interface

non-text content

any content that is not a sequence of characters that can be programmatically determined or where the sequence is not expressing something in human language This includes ASCII art (which is a pattern of characters), emoticons, leetspeak (which uses character substitution), and images representing text

normative

required for conformance One may conform in a variety of well-defined ways to this document. Content identified as "informative" or "non-normative" is never required for conformance .

on a full-screen window

on the most common sized desktop/laptop display with the viewport maximized Since people generally keep their computers for several years, it is best not to rely on the latest desktop/laptop display resolutions but to consider the common desktop/laptop display resolutions over the course of several years when making this evaluation.

paused

stopped by user request and not resumed until requested by user

perimeter

New continuous line forming the boundary of a shape not including shared pixels, or the minimum bounding box , whichever is shortest The perimeter calculation for a 2 CSS pixel perimeter around a rectangle is $4 h + 4 w$, where h is the height

and w is the width. For a 2 CSS pixel perimeter around a circle it is $4\pi r$.

pointer input

input from a device that can target a specific coordinate (or set of coordinates) on a screen, such as a mouse, pen, or touch contact See the Pointer Events definition for "pointer" [[pointerevents]].

prerecorded

information that is not live

presentation

rendering of the content in a form to be perceived by users

primary education level

six year time period that begins between the ages of five and seven, possibly without any previous education This definition is based on the International Standard Classification of Education [[UNESCO]].

process

series of user actions where each action is required in order to complete an activity Successful use of a series of web pages on a shopping site requires users to view alternative products, prices and offers, select products, submit an order, provide shipping information and provide payment information. An account registration page requires successful completion of a Turing test before the registration form can be accessed.

programmatically determined

determined by software from author-supplied data provided in a way that different user agents , including assistive technologies , can extract and present this information to users in different modalities Determined in a markup language from elements and attributes that are accessed directly by commonly available

assistive technology. Determined from technology-specific data structures in a non-markup language and exposed to assistive technology via an accessibility API that is supported by commonly available assistive technology.

programmatically determined link context

additional information that can be programmatically determined from relationships with a link, combined with the link text, and presented to users in different modalities In HTML, information that is programmatically determinable from a link in English includes text that is in the same paragraph , list item, or table cell as the link or in a table header cell that is associated with the table cell that contains the link. Since screen readers interpret punctuation, they can also provide the context from the current sentence, when the focus is on a link in that sentence.

programmatically set

set by software using methods that are supported by user agents , including assistive technologies

pure decoration

serving only an aesthetic purpose, providing no information, and having no functionality Text is only purely decorative if the words can be rearranged or substituted without changing their purpose. The cover page of a dictionary has random words in very light text in the background.

real-time event

event that a) occurs at the same time as the viewing and b) is not completely generated by the content A Webcast of a live performance (occurs at the same time as the viewing and is not prerecorded). An on-line auction with people bidding (occurs at the same time as the viewing). Live humans interacting in a virtual

world using avatars (is not completely generated by the content and occurs at the same time as the viewing).

region

perceivable, programmatically determined section of content In HTML, any area designated with a landmark role would be a region.

relationships

meaningful associations between distinct pieces of content

relative luminance

the relative brightness of any point in a colorspace, normalized to 0 for darkest black and 1 for lightest white For the sRGB colorspace, the relative luminance of a color is defined as $L = 0.2126 R + 0.7152 G + 0.0722 B$ where R, G and B are defined as: if $R_{sRGB} \leq 0.04045$ then $R = R_{sRGB}/12.92$ else $R = ((R_{sRGB}+0.055)/1.055)^{2.4}$ if $G_{sRGB} \leq 0.04045$ then $G = G_{sRGB}/12.92$ else $G = ((G_{sRGB}+0.055)/1.055)^{2.4}$ if $B_{sRGB} \leq 0.04045$ then $B = B_{sRGB}/12.92$ else $B = ((B_{sRGB}+0.055)/1.055)^{2.4}$ and R_{sRGB} , G_{sRGB} , and B_{sRGB} are defined as: $R_{sRGB} = R_{8bit}/255$ $G_{sRGB} = G_{8bit}/255$ $B_{sRGB} = B_{8bit}/255$ The "^" character is the exponentiation operator. (Formula taken from [[SRGB]].) Before May 2021 the value of 0.04045 in the definition was different (0.03928). It was taken from an older version of the specification and has been updated. It has no practical effect on the calculations in the context of these guidelines. Almost all systems used today to view web content assume sRGB encoding. Unless it is known that another color space will be used to process and display the content, authors should evaluate using sRGB colorspace. If using other color spaces, see Understanding Success Criterion 1.4.3 . If dithering occurs after delivery, then the source color value is used. For colors that are dithered at the source, the average values of the colors that are dithered should be used

(average R, average G, and average B). Tools are available that automatically do the calculations when testing contrast and flash. A separate page giving the relative luminance definition using MathML to display the formulas is available.

relied upon

the content would not conform if that technology is turned off or is not supported

role

text or number by which software can identify the function of a component within web content A number that indicates whether an image functions as a hyperlink, command button, or check box.

same functionality

same result when used A submit "search" button on one web page and a "find" button on another web page may both have a field to enter a term and list topics in the website related to the term submitted. In this case, they would have the same functionality but would not be labeled consistently.

same relative order

same position relative to other items Items are considered to be in the same relative order even if other items are inserted or removed from the original order. For example, expanding navigation menus may insert an additional level of detail or a secondary navigation section may be inserted into the reading order.

satisfies a success criterion

the success criterion does not evaluate to 'false' when applied to the page

section

a self-contained portion of written content that deals with one or more related topics or thoughts A section may consist of one or

more paragraphs and include graphics, tables, lists and sub-sections.

set of web pages

collection of web pages that share a common purpose and that are created by the same author, group or organization Examples include: a publication which is split across multiple web pages, where each page contains one chapter or other significant section of the work. The publication is logically a single contiguous unit, and contains navigation features that enable access to the full set of pages. an e-commerce website shows products in a set of web pages that all share the same navigation and identification. However, when progressing to the checkout process, the template changes; the navigation and other elements are removed, so the pages in that process are functionally and visually different. The checkout pages are not part of the set of product pages. a blog on a sub-domain (e.g. blog.example.com) which has a different navigation and is authored by a distinct set of people from the pages on the primary domain (example.com). Different language versions would be considered different sets of web pages.

sign language

a language using combinations of movements of the hands and arms, facial expressions, or body positions to convey meaning

sign language interpretation

translation of one language, generally a spoken language, into a sign language True sign languages are independent languages that are unrelated to the spoken language(s) of the same country or region.

single pointer

an input modality that only targets a single point on the page/screen at a time – such as a mouse, single finger on a touch

screen, or stylus Single pointer interactions include clicks, double clicks, taps, dragging motions, and single-finger swipe gestures. In contrast, multipoint interactions involve the use of two or more pointers at the same time, such as two-finger interactions on a touchscreen, or the simultaneous use of a mouse and stylus.

specific sensory experience

a sensory experience that is not purely decorative and does not primarily convey important information or perform a function Examples include a performance of a flute solo, works of visual art etc.

state

dynamic property expressing characteristics of a user interface component that may change in response to user action or automated processes States do not affect the nature of the component, but represent data associated with the component or user interaction possibilities. Examples include focus, hover, select, press, check, visited/unvisited, and expand/collapse.

status message

change in content that is not a change of context , and that provides information to the user on the success or results of an action, on the waiting state of an application, on the progress of a process , or on the existence of errors

structure

The way the parts of a web page are organized in relation to each other; and The way a collection of web pages is organized

style property

property whose value determines the presentation (e.g. font, color, size, location, padding, volume, synthesized speech prosody) of content elements as they are rendered (e.g. onscreen, via

loudspeaker, via braille display) by user agents

Style properties can have several origins:

- User agent default styles:** The default style property values applied in the absence of any author or user styles.
- Some web content technologies specify a default rendering, others do not;**
- Author styles:** Style property values that are set by the author as part of the content (e.g. in-line styles, author style sheets);
- User styles:** Style property values that are set by the user (e.g. via user agent interface settings, user style sheets)

supplemental content

additional content that illustrates or clarifies the primary content

- An audio version of a web page .
- An illustration of a complex process .
- A paragraph summarizing the major outcomes and recommendations made in a research study.

synchronized media

audio or video synchronized with another format for presenting information and/or with time-based interactive components, unless the media is a media alternative for text that is clearly labeled as such

target

region of the display that will accept a pointer action, such as the interactive area of a user interface component

If two or more targets are overlapping, the overlapping area should not be included in the measurement of the target size, except when the overlapping targets perform the same action or open the same page.

technology

mechanism for encoding instructions to be rendered, played or executed by user agents

As used in these guidelines "web technology" and the word "technology" (when used alone) both refer to web content technologies. Web content technologies may

include markup languages, data formats, or programming languages that authors may use alone or in combination to create end-user experiences that range from static web pages to synchronized media presentations to dynamic web applications. Some common examples of web content technologies include HTML, CSS, SVG, PNG, PDF, Flash, and JavaScript.

text

sequence of characters that can be programmatically determined , where the sequence is expressing something in human language

text alternative

Text that is programmatically associated with non-text content or referred to from text that is programmatically associated with non-text content. Programmatically associated text is text whose location can be programmatically determined from the non-text content. An image of a chart is described in text in the paragraph after the chart. The short text alternative for the chart indicates that a description follows. Refer to Understanding Text Alternatives for more information.

up-event

platform event that occurs when the trigger stimulus of a pointer is released The up-event may have different names on different platforms, such as "touchend" or "mouseup".

used in an unusual or restricted way

words used in such a way that requires users to know exactly which definition to apply in order to understand the content correctly The term "gig" means something different if it occurs in a discussion of music concerts than it does in article about computer hard drive space, but the appropriate definition can be determined from context. By contrast, the word "text" is used in a very specific way in WCAG 2, so a definition is supplied in the glossary.

user agent

any software that retrieves and presents web content for users Web browsers, media players, plug-ins, and other programs — including assistive technologies — that help in retrieving, rendering, and interacting with web content.

user inactivity

any continuous period of time where no user actions occur The method of tracking will be determined by the website or application.

user interface component

a part of the content that is perceived by users as a single control for a distinct function Multiple user interface components may be implemented as a single programmatic element. "Components" here is not tied to programming techniques, but rather to what the user perceives as separate controls. User interface components include form elements and links as well as components generated by scripts. What is meant by "component" or "user interface component" here is also sometimes called "user interface element". An applet has a "control" that can be used to move through content by line or page or random access. Since each of these would need to have a name and be settable independently, they would each be a "user interface component."

user-controllable

data that is intended to be accessed by users This does not refer to such things as Internet logs and search engine monitoring data. Name and address fields for a user's account.

video

the technology of moving or sequenced pictures or images Video can be made up of animated or photographic images, or both.

video-only

a time-based presentation that contains only video (no audio and no interaction)

viewport

object in which the user agent presents content The user agent presents content through one or more viewports. Viewports include windows, frames, loudspeakers, and virtual magnifying glasses. A viewport may contain another viewport (e.g., nested frames). Interface components created by the user agent such as prompts, menus, and alerts are not viewports. This definition is based on User Agent Accessibility Guidelines 1.0 Glossary [[UAAG10]].

visually customized

the font, size, color, and background can be set

web page

a non-embedded resource obtained from a single URI using HTTP plus any other resources that are used in the rendering or intended to be rendered together with it by a user agent Although any "other resources" would be rendered together with the primary resource, they would not necessarily be rendered simultaneously with each other. For the purposes of conformance with these guidelines, a resource must be "non-embedded" within the scope of conformance to be considered a web page. A web resource including all embedded images and media. A web mail program built using Asynchronous JavaScript and XML (AJAX). The program lives entirely at <http://example.com/mail>, but includes an inbox, a contacts area and a calendar. Links or buttons are provided that cause the inbox, contacts, or calendar to display, but do not change the URI of the page as a whole. A customizable portal site, where users can choose content to display from a set of

different content modules. When you enter "http://shopping.example.com/" in your browser, you enter a movie-like interactive shopping environment where you visually move around in a store dragging products off of the shelves around you and into a visual shopping cart in front of you. Clicking on a product causes it to be demonstrated with a specification sheet floating alongside. This might be a single-page website or just one page within a website.

WCAG in a Book — Understanding the Web Content Accessibility Guidelines

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<https://www.w3.org/TR/WCAG22/>